



A new method to establish NMR T_2 spectrum based on bimodal Gaussian density function: A case study of tight sandstone in East China Sea Basin

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ABSTRACT

The identification and evaluation of tight sandstone reservoirs has been very difficult due to their low porosity, low permeability, complex pore structure and strong heterogeneity. In order to evaluate the quality of tight sandstone reservoir, priority should be given to the quantitative evaluation of pore size. Nuclear magnetic resonance (NMR) log is frequently employed in pore size evaluation; however, due to its high technical costs and other limitations, it can only be conducted in very few wells and the wide application can never be found. In this study, 12 core samples from the tight sandstone in East China Sea Basin have been collected to conduct centrifugal NMR experiments. Based on the analysis of data obtained from experiments, it is discovered that the bimodal Gaussian density function is well fitted to the NMR T_2 spectrum and can be applied to the fitting of NMR T_2 spectrum. Furthermore, in the analysis of physical property parameters of reservoir and each individual parameter of bimodal Gaussian density function, it is indicated that the porosity and permeability are best correlated with them. Therefore, the empirical formula between porosity, permeability and the parameters of bimodal Gaussian density function were established to construct pseudo NMR T_2 spectrum by using bimodal Gaussian density function and it has been verified by the comparison between the result and experimental data. In order to achieve the more accurate and wide application of this method, the NMR T_2 spectrum versus proportion frequency curve was transformed into the pore radius R versus porosity component P_p curve before the fitting. When used in exploration field, the method can construct continuous pore size distribution curve which can be used for the quantitative evaluation of tight sandstone reservoir's quality. Compared with the method of NMR logging, this technique can be applied to the evaluation of pore structure of tight sandstone reservoirs more widely.

1. Introduction

The tight sandstone reservoir is defined as a reservoir with a porosity of less than 12% and a permeability of less than 1 mD (Kazemi, 1982; Guan, 1995; Surdam et al., 1997; Pang et al., 2003). The tight sandstone reservoir, known as an important supplement to the conventional oil and gas resources in the world today, has been attracting the attention of the world's giant oilfield enterprises, experts and scholars from its exploration to recent (Law and Curtis, 2002; Taylor et al., 2010). The tight sandstone reservoir is characterized by a low porosity, low permeability, strong heterogeneity, high water saturation, low resistivity, and complex gas-water relationship (Zeng and Li, 1994; Zou, 2012). Because of these characteristics of tight sandstone reservoir, the evaluation of it is confronted with many problems, and it is difficult for the conventional method to judge the pore structure of the

reservoir effectively (Kukul et al., 1983; Amiri et al., 2012).

The research on pore structure focuses on pore and throat, which include the size, distribution, morphology of pores and their connectivity and is of pivotal importance to evaluate the quality of reservoirs (Patrick et al., 2004). The current method is to conduct laboratory experiments on core samples, including thin section identification, scanning electron microscopy (SEM), X-ray computer tomography (CT), incremental pressure mercury injection (IPMI), nuclear magnetic resonance (NMR) and other methods. The pore structure can be characterized by these methods, such that thin section identification and SEM can give a direction observation to the two-dimensional image and size of pore and throat (Taylor et al., 2010; Golab et al., 2010; Bera et al., 2011; Shao et al., 2017); CT scanning can provide size and three-dimensional network of pores (Golab et al., 2010; Bera et al., 2011); IPMI reflects the size of pore and throat and their connectivity

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(Pittman, 1992; Yang et al., 2006; Comisky et al., 2007; Yu and Huang, 2009; Ziarani and Aguilera, 2012; Buryakovsky et al., 2012); NMR experiment mainly reflects the size and distribution of pores (Kleinberg, 1999; Ren et al., 2002), which carrying substantial weight in the evaluation of pore structure. However, these methods take long and costly and need a large number of core samples, making it impractical to conduct accurate description in the whole well interval (Shao et al., 2017; Eslami et al., 2013). The NMR log, which can be employed in the pore size evaluation in the whole well interval, can only be used in very few wells due to its high technical costs and other limitation which makes its application in all well difficult (Xiao et al., 2016; Kenyon, 1997; Prammer, 2004; Volokitin et al., 2001; Coates et al., 1999). Therefore, a method which is simpler and easier to be propagated is in desperate need to conduct pore structure evaluation continually.

For the sake of this, predecessors tried miscellaneous methods, of which the fitting of mercury injection capillary pressure (MICP) curve was frequently used (Xiao et al., 2016). The method aims to establish artificially fitted MICP curve, thereby conducting pore structure evaluation in single well continually. The method can be divided into two categories. One is that many scholars use the T_2 spectrum of NMR logging to convert them into pseudo MICP curve by the comparison of experimental data and mathematical transformation (Eslami et al., 2013; Volokitin et al., 2001; Glorioso et al., 2003; Hamada, 2009; Ouzzane et al., 2006; Arogun and Nwosu, 2011; Altunbay et al., 2001; Green et al., 2008; Looyestijn, 2001; Liu et al., 2003; He et al., 2005), including the transformation using a single scaling factor and a residual error analysis technique for the analysis of average saturation (Volokitin et al., 2001); the similar comparison method (Liu et al., 2003); and the segmentation power series method based on non-linear relation (He et al., 2005). These methods provided solutions to the issues from different perspectives. Although these were continuously ameliorated, the drawbacks of the methods have been still obvious. First of all, the prerequisite of this method is that it needs reliable NMR logging data, thus showing greater limitation that it can't be employed in the wells without NMR logging data. Secondly, the T_2 spectrum shape of NMR logging, which is affected by the properties and contents of fluids in the formation, can't reflect true pore structure in strata containing hydrocarbons, thereby making converted (MICP) curves insignificant (Volokitin et al., 2001; Ouzzane et al., 2006). The other is that some experts and scholars believed that the mercury injection pressure or mercury injection saturation for each measured points on the MICP curve is closely linked to physical property parameters of the reservoir. To make a fitting of them and to clarify the mathematical relationship and then pseudo MICP curve can be artificially fitted (Xiao et al., 2016; Liu et al., 2008; Yan et al., 2002). Zhong (1997), Liao et al. (2000), and Liu et al. (2008) established the empirical formula among pore, permeability and capillary pressure curve by using a large number of experimental data in different regions, and then fitted into the MICP curve; based on the principle and method of fractal geometry, Yan et al. (2002) used conventional physical property data to invert capillary pressure curve; Xiao et al. (2016) established an empirical formula between mercury injection saturation and porosity, and permeability after the classification of reservoir type by using J function. This method is more flexible than before, and can be widely used. However, the method has its shortcomings as follows; the artificial fitting of the curve is affected by mercury injection experiments, and the data points are constant and rare, thereby obtaining blunt curve shape which can't fully display the pore size distribution and pore structure. Moreover, high pressure fluid is used in the mercury injection experiment, which can easily damage the pore structure of the tight sandstone, resulting in the deviation of the result (Clarkson et al., 2013). In order to overcome shortcomings of above methods, based on the centrifugal core NMR experimental data of 12 core samples (Table 1) in the East China Sea Basin, and combined with porosity and permeability by formula fitting, the pore size distribution curve was established by using bimodal Gaussian density function. It is proved that the method is accurate and

Table 1

Physical properties of 12 core samples of X3-X6 well in East China Sea Basin.

Zone	Sample number	Depth(m)	Porosity (%)	Permeability (mD)
X3	1	3600.72	10.84	11.88
X3	2	3600.72	10.72	7.48
X4	3	3961.62	5.13	0.17
X4	4	3980.48	8.61	0.96
X4	5	3980.48	9.41	1.03
X6	6	4322.24	4.32	0.18
X3	7	3742.45	12.64	28.86
X3	8	3742.45	12.84	23.49
X3	9	3792.35	12.69	54.82
X3	10	3126.45	15.82	14.29
X3	11	3132.47	16.08	5.37
X3	12	3132.47	16.36	6.56

reliable, and can obtain continuous pore size distribution curve which can be used for quantitative evaluation of tight sandstone reservoir, thereby making up for the drawbacks of evaluation of tight sandstone reservoir.

2. Theory and methodology

Pore size distribution, which is associated with all the physical properties of the rock, is the core issue of reservoir characteristics of rocks (Archie, 1950). A number of models have been proposed to describe and simulate it, including a variety of density functions to simulate the pore size distribution and associated petrophysical parameters. For example, Hidajat et al. (2002) used the three-peak Weibull distribution to fit the NMR T_2 spectrum. Similarly, Genty et al. (2007) used three or more Gaussian components to fit the NMR T_2 spectrum. In fact, the Gaussian density function is often used to simulate pore size and particle size distribution in publicly reported literature (Spencer, 1963; Nimmo, 2004). Based on the previous study and combined with the experimental data, it is considered that the bimodal Gaussian density function model is the most suitable for the fitting of NMR T_2 spectrum of the tight sandstone (Fig. 1). On the basis of this, The NMR T_2 spectrum is assumed to be a superposed curve of two bimodal Gaussian density functions with six variables of T_2 .

The formula for the bimodal Gaussian density function curve (which is referred to Xu and Torres-Verdin, 2013) is as follows:

$$P(\log T_2, w_1, \log u_1, \log \sigma_1; w_2, \log u_2, \log \sigma_2)$$

$$P = w_1 * \frac{1}{\sqrt{2\pi} * \log \sigma_1} * e^{-\frac{(\log T_2 - \log u_1)^2}{2 * (\log \sigma_1)^2}} + w_2 * \frac{1}{\sqrt{2\pi} * \log \sigma_2} * e^{-\frac{(\log T_2 - \log u_2)^2}{2 * (\log \sigma_2)^2}} \quad (1)$$

$$w_1 + w_2 = S, \quad \text{and} \quad 0 < w_1, 2 < S \quad (2)$$

Where T_2 is the transverse relaxation time in ms; P is signal proportion, in dimensionless;

w_1 and w_2 are the weighting coefficients of a single Gaussian function, and the sum of the two is the whole function area S . w_2 represents the fraction of large pore volume, which mainly determines the flow capacity and permeability, w_1 represents the fraction of small pore volume, in which irreducible water are mainly retained in the centrifugal nuclear magnetic experiment, and it shows little contribution to the fluid flow.

$\log u_1$ and $\log u_2$ are the logarithmic radius averages of smaller pore volume and larger pore volume respectively; generally larger value indicates larger pore volume which contains larger flow channel and higher conductivity; $\log \sigma_1$ and $\log \sigma_2$ are the logarithmic standard deviations of smaller pore volume and larger pore volume, respectively; it indicates the heterogeneity of the pore size; larger values indicate poor

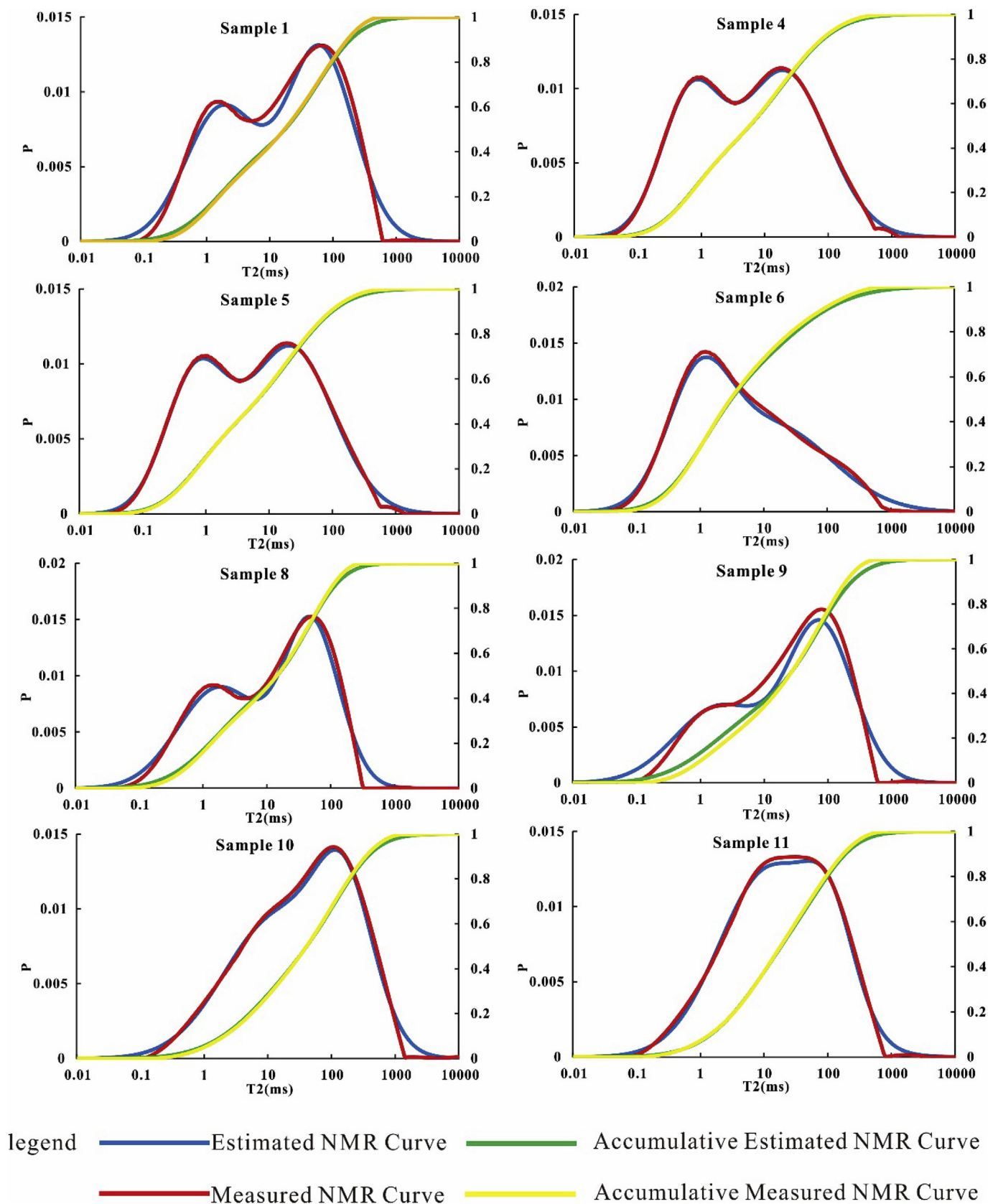


Fig. 1. Comparison between experimental NMR T_2 spectrum and fitted curves of 8 samples in East China Sea Basin.

pore sorting and higher tortuosity of pore network, resulting in a lower permeability under the same condition of pore medium radius and pore volume.

From formula (1), it can be seen that the whole bimodal Gaussian density function curve is a p -value curve about $\log T_2$ controlled by six values of w_1 , $\log u_1$, $\log \sigma_1$; w_2 , $\log u_2$, $\log \sigma_2$. The six parameters are

intimately related to pore structure properties of rocks. By the same token, parameters such as porosity and permeability are also the direct parameters that reflect the pore structure of rocks. If an experimental formula is established between physical properties (porosity, permeability) and the six parameters, then the fitted T_2 spectrum curve between $\log T_2$ and P can be established. And then it can be used to evaluate the pore size distribution of reservoir.

3. Techniques and procedures

In order to predict the NMR curve and establish the empirical formula between the basic parameters of the reservoir and the bimodal Gaussian density function, the calculation works were conducted according to the following procedures.

3.1. Procedures

The NMR data from 12 core samples of the East China Sea were used to derive the formula and the procedures are as follows:

- (1) To collect the core samples and data, and test the porosity and permeability of the core samples, and then carry out the NMR experiment under the condition of total water saturation in the samples. The measured NMR T_2 spectrum was established based on the experimental data.
- (2) According to the NMR T_2 spectrum, the parameters ($w_1, \log \mu_1, \log \sigma_1; w_2, \log \mu_2, \log \sigma_2$) of the bimodal Gaussian density function were fitted.
- (3) To establish the empirical formula between the parameters of the reservoir and the parameters of the bimodal Gaussian density function: the relationship between the parameters such as porosity and permeability, and the ratio of pore size, the mean square deviation of pore size, and peak location of pore size.
- (4) Using the empirical formula established in procedure 3, the fitted NMR T_2 spectrum curve is established by reservoir physical parameters.

Porosity and Permeability are major parameters employed in this method. The NMR T_2 spectrum curve can be fitted by data from analyzing physical properties of core samples in the coring layer, and permeability and porosity data from logging can be used in the layer without core.

3.2. Treatment and experiment of samples

A total of 12 cores were used in this test. The core was first placed in a drying oven and dried at 105 °C for 2 h. Then, the sample is placed in a high-pressure saturation device, first evacuated to -0.1 MPa for 1 h, then water-filled 20 MPa saturated device saturated for 15 h, the saturated reagent is the formation water, and the ratio is as follows (Table 2):

After entirely saturated by water, the MacroMR12-150H-I large-scale NMR imaging analyzer was used to conduct experiments on samples with total water saturation. The resonance frequency was 12.80 MHz, the magnet strength was 0.301 T, the coil diameter was 25 mm, the magnet temperature was 32 °C; the time of echo (TE) was

0.15 ms, echo number (NECH) 16000. The number of inversion was selected to be 200.

3.3. NMR T_2 spectrum fitting

A total of 12 samples from the East China Sea Basin were used to conduct nuclear magnetic experiment, and 8 samples of them were selected for curve fitting, however, the remaining four can be used for verification of reliability. At the completion of step 1, we obtained the saturated NMR T_2 spectrum which was employed in the fitting by using the bimodal Gaussian density function in *Origin* software. And the mean-square error is used as the cost function. When the mean square error is less than 5×10^{-7} , the fitting is considered a success. It is found that the fitting function has high fitting rate with the measured function (Fig. 1), with R^2 generally being more than 0.98. It indicates that the use of bimodal Gaussian density function can accurately fit the NMR T_2 spectrum curve and its attribute characteristics.

By extracting each parameters in the bimodal Gaussian density function ($w_1, \log \mu_1, \log \sigma_1; w_2, \log \mu_2, \log \sigma_2$) (Table 3), and parameters related to pore structure of reservoir were fitted, ultimately it is found that the parameters are intimately related with porosity and permeability. In order to demonstrate their correlation with porosity and permeability more clearly, the data points of each parameter of bimodal Gaussian density function and of porosity and permeability were casted in three dimensional spaces (Fig. 2). It is concluded that the parameters of the NMR curve are closely related to porosity and permeability, and the three can form a good linear relationship. The two parameters are further bimodal fitted to obtain the fitting formula in Table 4, and the fitting degree of each parameter is higher than 0.8. Among them, the curve inversion takes 200 points, the abscissa T_2 ranges from 0.01 ms to 10000 ms, $\log T_2$ range from $-2 \sim 4$, the interval between the points $\Delta \log T_2$ approximate to 0.0302; ordinate P (signal proportion) total 1. According to the formula (3), area S equals to 0.0302.

$$S = \int_{-\infty}^{\infty} P d \log T_2 \quad (3)$$

Combined with Fig. 2 and Table 4, it can be seen that the porosity, permeability have a good correlation between the parameters. By using the formula in Table 4, parameters of NMR T_2 curve can be obtained by combining porosity and permeability, thereby establishing more accurate fitted NMR T_2 curve.

For the remaining four samples, they were used to verify the reliability of the formula in Table 2. The four samples contain a porosity of 5.13%–16.36%, and a permeability of 0.17mD - 28.86mD. The distribution of porosity and permeability is defined as a tight sandstone reservoir and can therefore be used to verify the reliability of the method.

3.4. Reliability verifications

According to the formula in Table 4, the fitted NMR T_2 spectrum parameters of the rest four samples were obtained (Table 5), by which the fitted NMR T_2 spectrum curve was established. Then, we compared it with the measured curve (Fig. 3). It is illustrated in Fig. 3 that the fitting of NMR T_2 spectrum is in good agreement with measured curve, which can reflect the characteristics and properties of the measured curve, showing good results. Further, the R^2 values of the two range from 0.938 to 0.996, which reflects that the two are extremely close, with small deviation. Fig. 4 shows the convergence diagram of data points from experimental method and fitted curve method of the remaining four samples. The data corresponding to the X-axis is the signal proportion obtained from the NMR test, and the Y-axis corresponds to the values of signal proportion calculated by using formula in Table 4. The red line is a diagonal, which is used to indicate the deviation between the formula fit and the measured value. It can be seen that the

Table 2
Formulation of formation water (1 L).

Reagent	Quality (mg)
NaHCO ₃	1020
Na ₂ SO ₄	1500
CaCl ₂	150
NaCl	7330

Table 3
Pseudo-NMR pore size distribution curve (bimodal Gaussian density function) of the parameters.

Well name	Zone	Sample number	Depth(m)	φ (%)	K (mD)	$\log u_1$	$\log \sigma_1$	W_1	$\log u_2$	$\log \sigma_2$	W_2	R^2
Well A	X3	1	3600.72	10.84	11.88	0.243	0.577	0.013	1.798	0.534	0.017	0.988
	X4	4	3980.48	8.61	0.96	-0.162	0.476	0.011	1.306	0.678	0.019	0.999
		5	3980.48	9.41	1.03	-0.153	0.489	0.011	1.340	0.676	0.019	0.999
	X6	6	4322.24	4.32	0.18	-0.028	0.502	0.013	1.156	0.916	0.017	0.997
Well B	X3	8	3742.45	12.84	23.49	0.254	0.643	0.014	1.700	0.432	0.016	0.989
		9	3792.35	12.69	54.82	0.316	0.706	0.012	1.885	0.515	0.018	0.985
Well C	X3	10	3126.45	15.82	14.29	1.144	0.829	0.020	2.215	0.454	0.011	0.997
		11	3132.47	16.08	5.37	1.090	0.785	0.025	2.053	0.384	0.006	0.997

predicted and measured values are converged near the diagonal.
It can be seen from Figs. 3 and 4 that the method of fitting NMR T_2 spectrum curve with porosity and permeability is accurate and reliable. Once the method is applied to field operation, it is possible to establish a continuous stratigraphic fitting of the NMR T_2 spectrum data by employing accurately predicted permeability and porosity data of reservoir. Therefore, it can give a quantitative evaluation of the pore size distribution of tight sandstone reservoir.

4. Case analysis

The method is applied to the tight sandstone reservoir of East China Sea Basin for further verification. Fig. 5 is an example of D well. The distribution of the NMR pore size calculated by the above-mentioned method (bimodal Gaussian function fitting) is shown and the pore size distribution of the tight sandstone reservoir is quantitatively estimated. There are no NMR logging data in this well, but there are six depth

Table 4
Fitted formulas and fitting degree between each parameter of pseudo NMR pore distribution curve (bimodal Gaussian density function) parameters and physical properties (permeability, porosity).

Parameter	Formulae for predicting parameter	Correlation coefficient (R^2)
$\log u_2$	$\log u_2 = 0.78 + 0.047\log_{10}(K) + 0.078\varphi$	$R^2 = 0.9$
$\log u_1$	$\log u_1 = -1.217 - 0.319\log_{10}(K) + 0.157\varphi$	$R^2 = 0.827$
$\log \sigma_2$	$\log \sigma_2 = 0.902 - 0.027\log_{10}(K) - 0.03\varphi$	$R^2 = 0.801$
$\log \sigma_1$	$\log \sigma_1 = 0.272 - 0.022\log_{10}(K) + 0.032\varphi$	$R^2 = 0.812$
W_2	$W_2 = [0.207 + 0.09\log_{10}(K) - 0.01\varphi] / \varphi$	$R^2 = 0.81$
W_1	$W_1 = 0.0302 - W_2$	

points that have been sampled and subjected to NMR experiments, the formula in Table 4 can be used to fit the NMR pore size distribution and to compare it with the measured data at the sampling point, thereby verifying the result of the method. In Fig. 5, the first track shows that

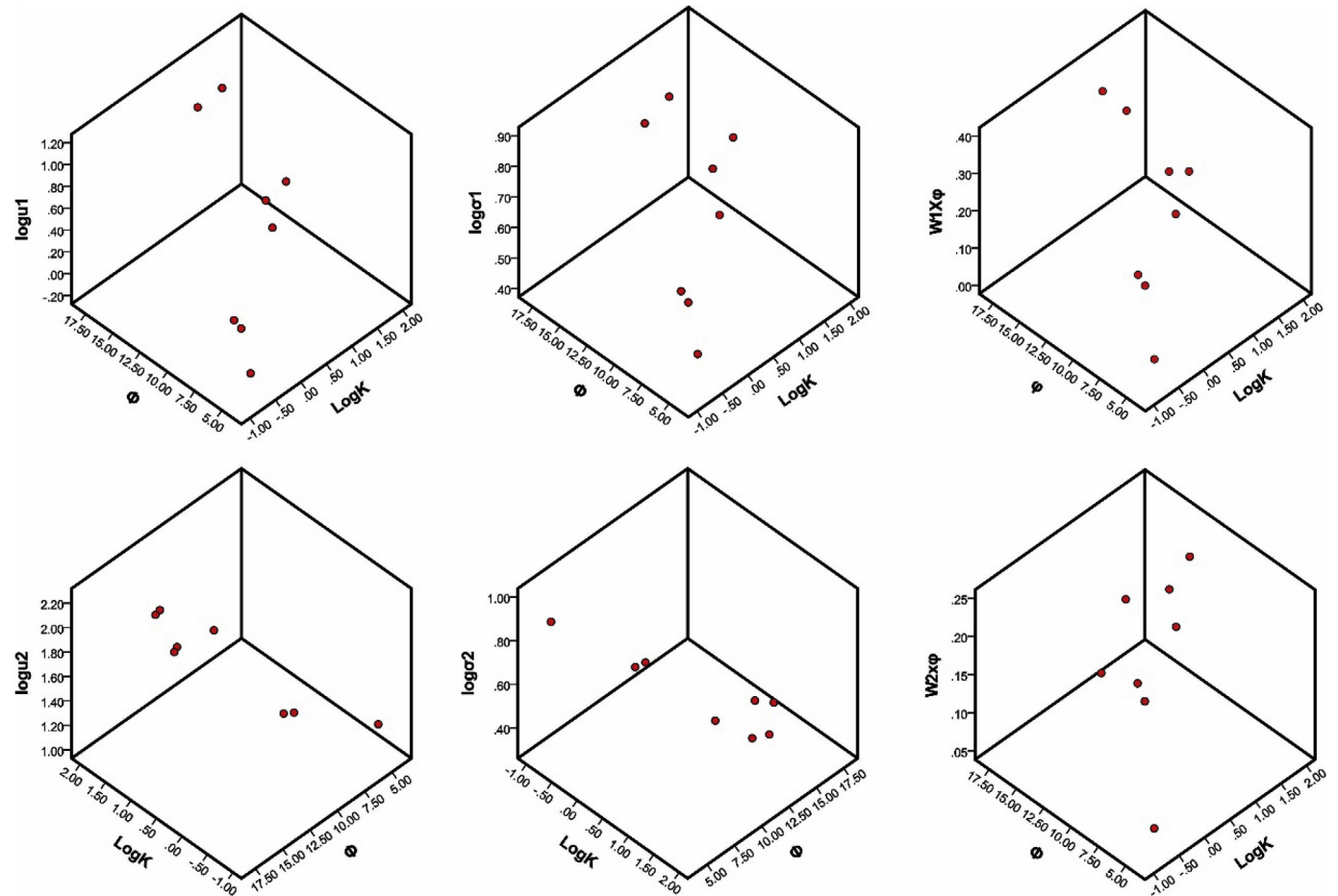


Fig. 2. Pseudo-NMR pore distribution curve (bimodal Gaussian density function) parameters and porosity, permeability 3D cast point map.

Table 5
Each parameter of pseudo-NMR pore distribution curve (bimodal Gaussian density function) of parameters for verification.

Well name	Zone	Sample number	Depth (m)	Φ (%)	K (m)	$\log u_1$	$\log \sigma_1$	W_1	$\log u_2$	$\log \sigma_2$	W_2	R^2
Well A	X3	2	3600.72	10.72	7.48	0.187	0.596	0.014	1.657	0.557	0.017	0.971
Well A	X4	3	3961.62	5.13	0.17	-0.162	0.453	0.013	1.143	0.769	0.017	0.940
Well B	X3	7	3742.45	12.64	28.86	0.302	0.644	0.013	1.835	0.483	0.017	0.938
Well C	X3	12	3132.47	16.36	6.56	1.091	0.778	0.024	2.094	0.389	0.006	0.996

the stratum is located in the X3 segment. The second track is the depth in meters. And the third track shows that the strata are mainly composed of light gray sandstone. The fourth track shows the natural gamma curve (GR), phase shift resistivity 40 inch spacing (P40H) and deep lateral resistivity (RD), all curves indicate the effective formation. The fifth track is the acoustic time difference (AC), density (DEN), and compensated neutron log (CNCF) used to estimate porosity. The sixth and seventh tracks are predicted porosity and permeability obtained by the conventional method with the practical core experimental data. It can be seen that the predicted values fit well with the experimental measurements and that the predicted porosity and permeability are credible. By using them to calculate NMR pore size distribution is reliable. The eighth track is the predicted NMR pore size distribution (PPSD). The ninth track is the comparison between the predicted NMR pore size distribution and the core measurement data. For the coring depth interval with NMR experiments, the green curve represents the predicted NMR curve (PPSD) and the red curve represents measured NMR curve (MPSD). From the comparison, it is indicated that the two are almost consistent. In the tenth track, in contrast, we compared the measured mean NMR radius (MMR from NMR) with the predicted mean NMR radius (PMR), and the measured value agrees well with the predicted value. It is proved that the fitted NMR pore size distribution

curve is in good agreement with core experimental results. The eleventh track is the mean throat radius (MMR from IPMI) measured in mercury injection experiment and the predicted mean NMR pore radius (PMR). And the difference between the two is generally 1 μm , which is caused by the fact that mercury injection mainly reflects breakthrough throat radius, while NMR reflects pore volume. It results in a lower value in the mercury injection experiment than that in NMR experiment. Meanwhile, the high pressure condition used in the mercury injection experiment can also affect the pore structure and clay minerals in tight sandstone, which may also be the origin of the differences between the two (Clarkson et al., 2013). However, although the two values are different, they show similar varying trend in the curve, indicating that the pseudo NMR curve can fully reflect the reservoir pore structure and quality of real strata. It proved the reliability of the fitted curve and feasibility of the method as well.

5. Discussion

According to the formula in Table 4, it is known to us that two parameters affecting the fitting curve are porosity and permeability. However, after accurately obtaining the permeability and porosity by logging, the error of the curve is mainly due to the differences between

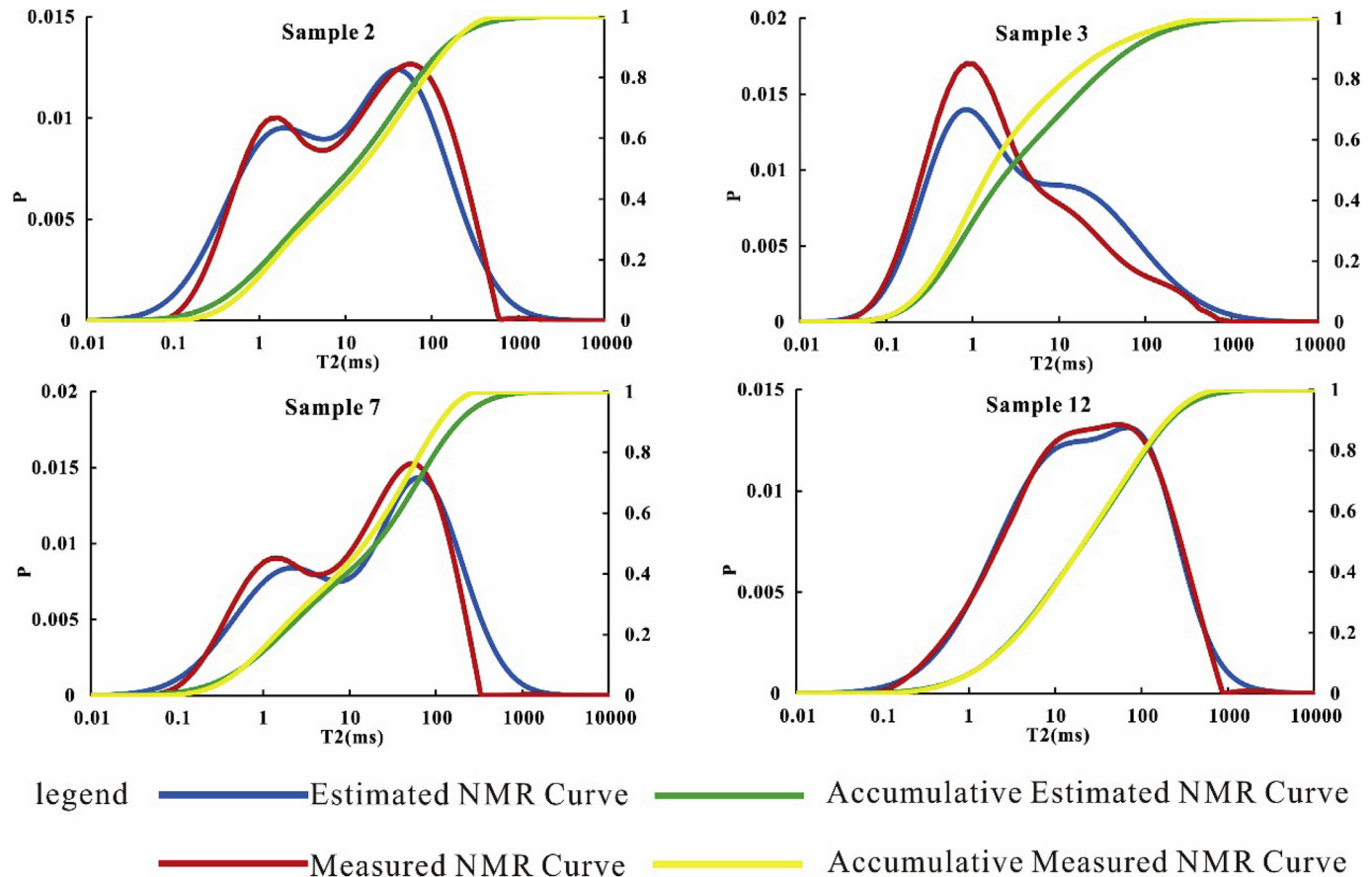


Fig. 3. Comparison of experimental results of NMR test curve and fitting curve of 4 samples in East China Sea Basin.

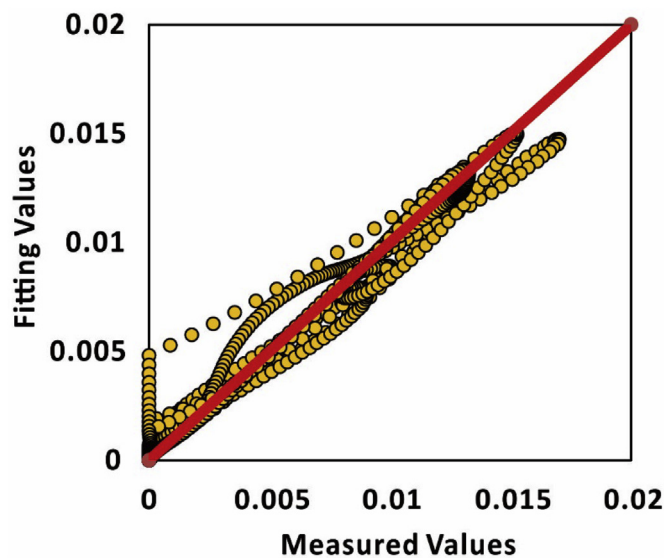


Fig. 4. Comparison of experimental results of nuclear magnetic test and fitting curve of 4 samples in East China Sea.

the measured curve and the bimodal Gaussian density function. For instance, the curve of most measured data showed a cliff-shaped decline in the right, however, due to the characteristics of bimodal Gaussian density function, the curve gradually slowed down in the right, resulting in greater differences in the right between the two. The fitting value is generally greater than the measured value. Meanwhile, due to equivalent area of the two curves, the slope of the values in the right of fitted curve was smaller and the curve was dispersive, generally resulting in a lower value of fitted value that measured value in the peak or ebb between two peaks. In Fig. 4, it can be seen that in the low value intervals (the left and right ends of the curve), the yellow dots were generally above the diagonal, indicating that fitted value was greater than measured value; in the high value intervals (the peak or ebb intervals between peaks), the yellow dots were generally below the diagonal, indicating that fitted value was smaller than measured value. It is indicated by calculation that the error of fitting curve for the samples ranges from 2.2% to 18.7%, with an average of 8.8%. The error of the larger modal in the right is 0.8%–9.2%, accounting for 11.4%–49.4% of the total error. The differences between this part of fitted curve and the measured data account for a larger part of the error, but we can see that the overall error is still minor. From Figs. 1 and 3, the fitted curve

accurately reflects the trend of the measured curve and a variety of parameters. And the error is in an acceptable range, thus verifying the accuracy and reliability of the fitting curve. And the method can be applicable.

In the above experiment, the experimental data of the samples are carried out under the same experimental conditions, so there is no need to make a correction between the curves. In the curve, the $NMR\ T_2$ spectrum was taken as the abscissa, and the signal proportion P was taken as the ordinate. However, when the samples in different regions under different experimental conditions are compared with each other, the $NMR\ T_2$ spectrum of the abscissa may also be shifted due to the different petrophysical properties, mineral composition, changes in testing instrument and inversion formulas. In this case, it might bring out a large deviation if the measured curves under different experimental conditions were fitted by using the formula in Table 4. Therefore, in order to make the convenient fitting of samples under different conditions and a contrastive observation of pore size distribution more intuitively, the $NMR\ T_2$ spectrum was converted into the pore radius R for convenient comparison between different samples by using previously mentioned empirical methods (including similar comparison method and minimum average saturation error method).

Variations in experimental conditions will not only affect T_2 spectrum in the abscissa, but also lead to the deviation of P in the ordinate. Under different experimental conditions, different numbers of inversion points were selected. When the number of points is small, the points are dispersed distribution, and the P is relatively concentrated. The ordinate signal proportion P is larger and the curve becomes higher. When the number of points is larger, the data points are in denser distribution, and the P is scattered, the ordinate P becomes smaller. Therefore, the curve became shorter. In this case, the shape of various data curve differs greatly and can't be fitted. Firstly, each curve was integrated to unify curve area (S), the P is converted into the one when the curve area was 1 (Equation (4)). And then the fitting is carried out. Meanwhile, taking the area P as the ordinate is not conducive to the comparison between samples with different porosities. After the normalization of curve area, the ordinate P was multiplied with porosity, and then it was converted into pore radius component (P_p). And it can be convenient in practical comparison.

$$S = \int_{-\infty}^{\infty} P d\log R = 1 \tag{4}$$

Fig. 6a shows the characteristics of the sample curve when different inversion points are selected. Fig. 6a illustrated the curve comparison of two samples, the red line represents that 200 points were selected and

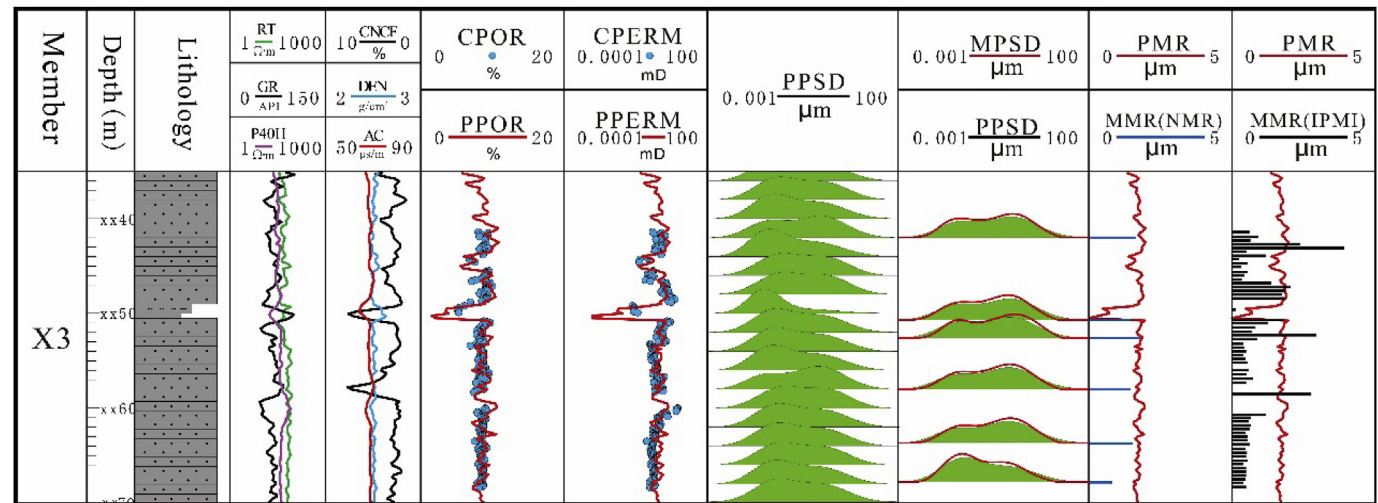


Fig. 5. Comparison of magnetic and magnetic curves obtained by two methods.

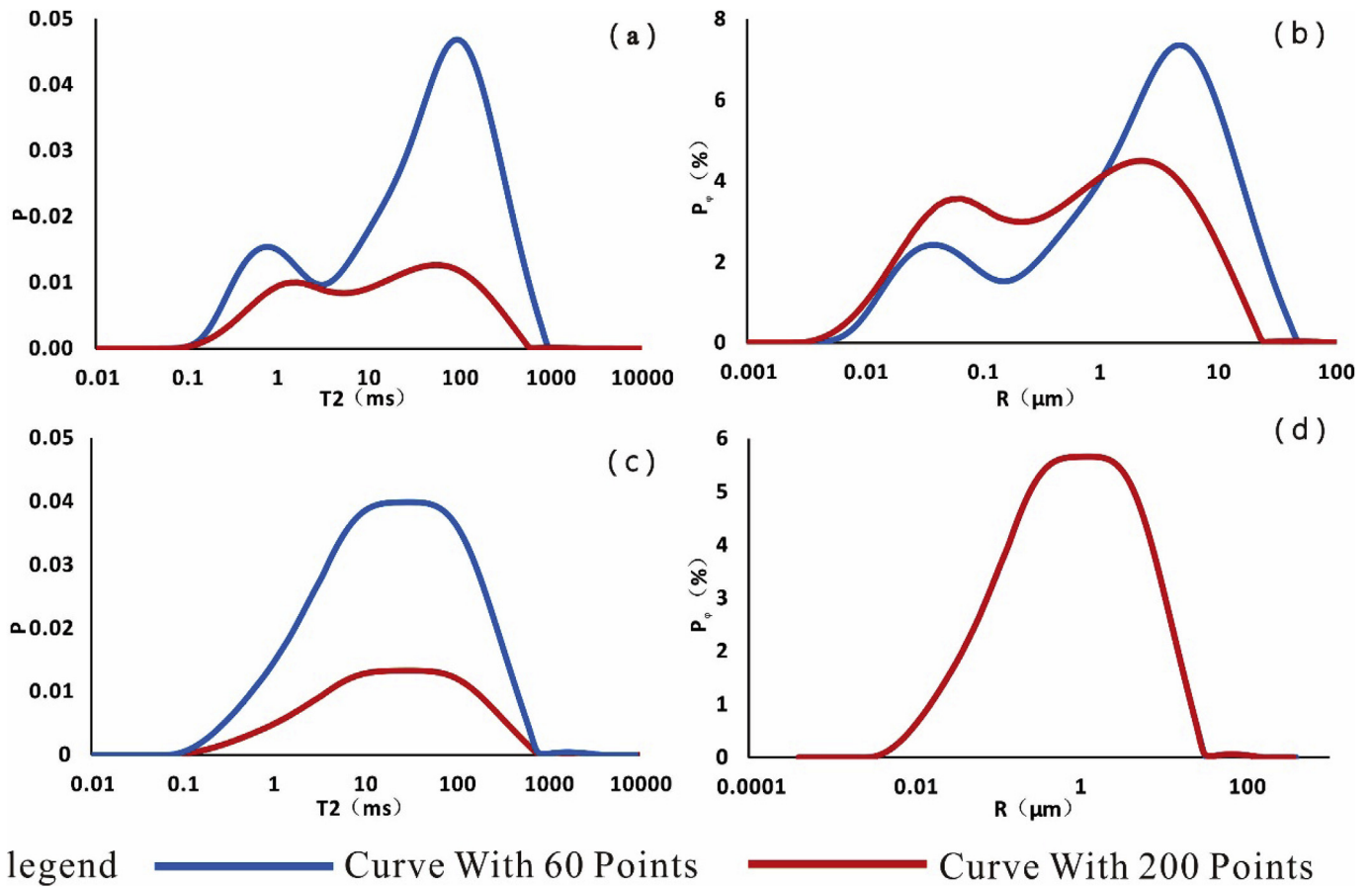


Fig. 6. Comparison between original NMR T2 spectrum and NMR T2 spectrum with ordinate conversion.

Table 6

Fitting formula and fitting degree between each parameter of pseudo-NMR pore distribution curve (bimodal Gaussian density function) and permeability, porosity.

Parameter	Formulae for predicting parameter	Correlation coefficient (R ²)
logu ₂	logu ₂ = -0.618 + 0.047log ₁₀ (K) + 0.078φ	R ² = 0.9
logu ₁	logu ₁ = -2.615 - 0.319log ₁₀ (K) + 0.157φ	R ² = 0.827
logσ ₂	logσ ₂ = 0.902 - 0.027log ₁₀ (K) - 0.03φ	R ² = 0.801
logσ ₁	logσ ₁ = 0.272 - 0.022log ₁₀ (K) + 0.032φ	R ² = 0.812
W ₂	W ₂ = [6.85 + 2.98log ₁₀ (K) - 0.3φ]/φ	R ² = 0.81
W ₁	W ₁ = 1 - W ₂	

the curve is shorter; the blue line represents that 60 points were selected and the curve is higher. There is great difference between the two samples, thus making the fitting difficult. Fig. 6c illustrates the comparison of curves from the same sample but with different inversion points, and the differences between the two are large. Therefore, under the condition of conversion of abscissa T_2 into the pore radius R , we integrated the areas of the two curves and unified to define them as 1. At this moment, the sum of w_1 and w_2 is 1 due to the properties of bimodal Gaussian density function fitting curve. Meanwhile, in order to facilitate the comparison between samples with different porosities, the ordinate is multiplied by the porosity and converted into the aperture component (Fig. 6b and 6d). Therefore, when fitting the sample data under different experimental conditions, the abscissa and ordinate are first unified, that is, the NMR T_2 spectrum is converted into the pore radius R , and then the curve area is normalized by integration. Finally, for the convenient division of samples with different porosities, the function was multiplied by the value of porosity as a whole. The fitting

curve formula can be converted into the following:

$$P\varphi(\log R; w_1, \log u_1, \log \sigma_1; w_2, \log u_2, \log \sigma_2)$$

$$P_\varphi = w_1 * \frac{1}{\sqrt{2\pi} * \log \sigma_1} * e^{-\frac{(\log R - \log u_1)^2}{2 * (\log \sigma_1)^2}} + w_2 * \frac{1}{\sqrt{2\pi} * \log \sigma_2} * e^{-\frac{(\log R - \log u_2)^2}{2 * (\log \sigma_2)^2}} \quad (5)$$

$$w_1 + w_2 = 1, \quad \text{and} \quad 0 < w_1, w_2 < 1 \quad (6)$$

Where R represents the pore radius in μm , φ represents porosity of samples, in%, and the rest parameters have the same meaning as formula 1. For converted curve, the abscissa is pore radius R , the ordinate is porosity component P_φ . That is, the T_2 - P diagram is converted into the R - P_φ diagram (Fig. 6), and then the parameters of bimodal Gaussian density function are determined by the above method to establish the formula for the sample curve fitting (Table 6). After determining the curve fitting formula, the porosity and permeability data can be used to establish the fitted NMR pore size distribution curve, to predict pore size distribution, and to evaluate pore structure of reservoir.

Lithologies of the target horizon in the study area vary slightly. And the rocks in the horizon experienced similar burial history and diagenesis, thereby making the method feasible and reliable in the study area. Meanwhile, the target reservoir with a relatively simple pore structure is characterized by few micro-pores and good correlation between permeability and porosity, that it is consistent with Hu et al. (2013) and Li (2015). If the geological backgrounds in other areas are similar that in the study area, this method is likely to be applicable. However, for reservoir dominated by micro-pores, it shows complex structure and relatively poor correlation between porosity and

permeability. This method might be applicable by combining with other parameters or fitting by categories that it is our next research direction.

6. Conclusions

Evaluation of the pore size is an indispensable part of the identification and evaluation of tight sandstone reservoir, and it is difficult to evaluate the tight sandstone reservoir effectively by conventional method. The complex pore structure can be quantitatively described by using the bimodal Gaussian density function. The parameters can be used to judge petrophysical properties of rocks, and the following conclusions were made by combining with the researches on the samples in East China Sea Basin.

- (1) By analyzing the NMR curve data, the bimodal Gaussian density function is selected to fit the NMR curve, showing a high fitting degree. For eight samples among them, it is above 0.98. Therefore, the NMR T_2 spectrum curve and its attribute characteristics can be accurately fitted by using bimodal Gaussian density function.
- (2) To establish the corresponding relationship between the parameters of the bimodal Gaussian density function and physical properties (permeability, porosity), further the accurate permeability and porosity data from logging can be employed to establish NMR T_2 spectrum curve. By using this method, there are high fitting degree and small errors between fitted curve and measured curve. And reliability of the method has been verified by NMR experimental data.
- (3) Under different experimental conditions and basic conditions of NMR curve, the deviation is large, making it difficult in fitting the curve. Therefore, before fitting, the experimental data need to be pretreated, converting T_2 - P curve into R - P_ϕ curve. After this treatment, the fitting curve is more accurate and more convenient to analyze the contrast.

The method can be widely used in the evaluation of tight sandstone reservoirs. In the absence of NMR logging, the method can achieved good results on the aspects such as pore size distribution, pore structure parameters and identification of effective tight reservoirs which can't be identified by conventional methods.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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Analysis of Porous Media and Objects of Cultural Heritage by Mobile NMR

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Berichte aus der Physik

Agnes Haber

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To my family...

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Chapter 1

Introduction

The development of unilateral mobile NMR sensors started in 1980 by Jasper Jackson, 35 years after the NMR signal was first successfully detected by Felix Bloch (Blümich 05a). Already 15 years after Jackson, the unilateral inside-out magnets were commercialized by the oil industry as well-logging tools (Jackson 80). The use of permanent magnets which generate an inhomogeneous magnetic field was important also for many other research areas, soft-matter science, environmental science, chemical engineering, etc.. The research in the mobile NMR field is still ongoing, with different applications (Casanova 11).

Why is this thesis focusing on porous media and cultural heritage? Because I think these research areas are vast, complex, interesting and influencing our everyday life. Porous media are in the walls of our house, in the streets on which we are walking, in the food that we are eating and in the cultural heritage objects that we inherited from our ancestors. NMR proved to be a good tool to probe the dynamics of fluids inside these porous systems from a microscopic to a macroscopic range over a long period of time. Moreover, single-sided NMR demonstrated not only once its efficiency in analyzing cultural heritage objects. One aim of this work is to contribute to the understanding of fluid dynamics in porous systems through 1D and 2D NMR relaxation measurements. In the case of the cultural heritage objects the aim is to help the conservators to identify suitable conservation strategies and to detect undocumented restoration efforts of the past.

After a short overview of the instrumentation used for obtaining the results presented in this thesis and the pulse sequences used in the experiments, the

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chapters following focus on the dynamics of water in porous media. Chapter 3 presents the theory of multi-site exchange NMR with focus on a simple two site exchange problem. This method correlates the relaxation time distribution at two different time intervals, observing in this way how the proton spins migrate during this time period from one environment to the other. Based on this theory, numerical simulations were done and compared to experimental results, which evidenced the exchange between the two sites. Moreover, the exchange rate was determined which provides information on the pore connectivity. Furthermore, another system was tested for the same reason but with more than two sites.

Chapter 4 focuses on water dynamics in natural porous systems, which are one fine sand sample and two silt loams samples with different compositions. The functions of each soil type are directly related to the structure of the pore system, which is characterized by the pore size distribution, pore connectivity, water content, and water dynamics. For this reason, the water dynamics is analyzed in both saturated and unsaturated systems. For these systems the focus was to evidence how the different structural compositions are influencing the dynamics and to investigate what can be learned from the relaxation time distributions in both one and two dimensions.

Chapter 5 presents the experimental results obtained for two different concrete types with different composition that were tested on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process is behaving totally different. This study was relevant for the industrial partner to compare the MRE and NMR techniques based on different analytical methods, and from our academic point of view to gain deeper understanding of complex porous systems. The use of the NMR-MOUSE showed to be appropriate, especially for the on-site measurements, where the measurements could be performed fast and non-invasively.

The last chapter focuses on cultural heritage objects. The first part deals with wall paintings and mosaics, located in the ancient Roman city of Herculaneum. Depth profiles were recorded from the Mosaic of Neptune and Amphitrite, and from the wall paintings located in the House with the Black Room and in the Villa of the Papyri. These measurements were made to identify the type of information available from unilateral NMR with the NMR-MOUSE and their significance to

the conservation and restoration of mosaics and wall paintings. In the second part two oil paintings (a Rembrandt self portrait and an anonymous Cologne painting) were examined with the aim to determine the stratigraphy of them and to find clues about the age of the paintings through the NMR measurements.

Mobile NMR turns out to be a simple but reliable and powerful tool to investigate moisture distributions and pore structures in porous media as well as the conservation state and history of objects of cultural heritage.

1. INTRODUCTION

Chapter 2

Nuclear Magnetic Resonance

2.1 Introduction

Nuclear magnetic resonance (NMR) is a physical resonance phenomenon, which occurs when atomic nuclei, that have a $I \neq 0$ nuclear spin, are placed in a static magnetic field B_0 and exposed to a B_1 oscillating magnetic field (Abragam 61, Ernst 87). Since the phenomenon's observation in 1945 by Purcell and Bloch, the variety and number of applications is still increasing. It is used for the analysis of matter with various applications in medicine, chemistry, materials science, geophysics, and chemical engineering (Ernst 87, Blümich 05a). In conventional NMR, massive, superconducting magnets with field strengths of up to 22 T are used to obtain high sensitivity. Among the drawbacks for these high-field magnets are: they cannot be transported easily, they need to be cooled with liquid helium and nitrogen, large samples do not fit the given magnet bore. This is why mobile permanent magnets were introduced to materials testing where the object of interest is positioned in the stray field outside the magnet. First applications in this mobile low-field were made in the early 1950s by the oil companies (Cooper 80, Burnett 80, Jackson 80).

2. NUCLEAR MAGNETIC RESONANCE

2.2 Relaxation

In the absence of an external magnetic field, the spins are uniformly distributed in space. When an static magnetic field is applied a bulk magnetization is built up along the field direction. The magnetization will precess along the field direction with the Larmor frequency ω_0 defined as:

$$\omega_0 = -\gamma B_0 \quad (2.1)$$

where γ is the gyromagnetic ratio and B_0 the applied static magnetic field strength. The stronger the magnetic field is, the faster the spins will precess.

When a radio-frequency field, B_1 , is applied orthogonal to the main magnetic field, B_0 , with the frequency ω_0 , the magnetization vector will flip conventionally from z -direction to the xy plane, but still precessing around the z axis. So for example if the B_1 field is induced on the x axis the B_0 field will flip from z -direction to the y direction. This precession will induce a current in the coil, the same coil which generated before the B_1 field. The current signal will decay with the time constant T_2 , the transverse relaxation time, because of the loss of the net magnetization in the xy plane. The initial amplitude of the signal is proportional to the number of the excited spins. This signal is called free induction decay (FID).

The relaxation process refers to the process in which the spins return to the thermodynamic equilibrium (the spins are relaxing back into their lowest energy state) and no transverse magnetization is present. This implies also that all the coherences disappear (Levitt 08). In NMR two major relaxation processes can occur: longitudinal relaxation (T_1) and transverse relaxation (T_2).

1. The longitudinal relaxation time is the time needed by the spins to realign with the axis of the B_0 magnetic field (lowest energy state) and give up all their excess energy to the lattice in order to restore the thermal equilibrium. The longitudinal relaxation time, T_1 , called also spin-lattice relaxation time characterizes the rate at which the z component of the net magnetization, M_z , recovers its initial value M_0 . It can be described by an exponential function:

$$M_z(t) = M_0(t)(1 - \exp(-t/T_1)) \quad (2.2)$$

2. The transverse relaxation time is the time in which the net magnetization in the transverse plane decays. In this time the spins get out of phase with respect to each other because they are experiencing slightly different precession frequencies, due to molecular interactions or magnetic field inhomogeneities. The transverse relaxation time, T_2 , called also spin-spin relaxation time, in a perfectly homogeneous B_0 magnetic field is described by an exponential function:

$$M_{xy}(t) = M_{xy,0}(t)\exp(-t/T_2) \quad (2.3)$$

2.2.1 Relaxation in porous media

The different relaxation times measured in porous media give important information about parameters that characterize the pore space, such as porosity, surface-to-volume ratio, permeability, formation factor, tortuosity, pore size distribution and pore connectivity (Barrie 00). This is possible because the relaxation times are sensitive to molecular motion and pore geometry. For a porous medium the dimensions of the pores are related to relaxation times by the Brownstein - Tarr equation (Brownstein 77, Brownstein 79). The T_1 and T_2 relaxation times can be written as:

$$\frac{1}{T_1} = \frac{1}{T_{1\text{bulk}}} + \rho_1 \frac{S}{V} \quad (2.4)$$

$$\frac{1}{T_2} = \frac{1}{T_{2\text{bulk}}} + \rho_2 \frac{S}{V} + \frac{D}{12}(\gamma G t_E)^2 \quad (2.5)$$

where T_1 and T_2 are the apparent relaxation times measured in seconds (s), ρ_1 and ρ_2 are the surface relaxivities measured in $\mu\text{ m/s}$, S/V is the surface-to-volume ratio measured in m^{-1} , D is the diffusion coefficient measured in m^2/s , γ is the gyromagnetic ratio measured in MHz/T , G is the magnetic field gradient strength measured in T/m and t_E is the echo time measured in s.

If the echo time, is chosen small enough, then the 3rd element in equation 2.5 can be neglected and the relaxation is dominated just by the surface relaxation. In this case, the relaxation rate of a fluid in a pore, with surface-to-volume S/V is given by (Casanova 11):

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1,2\text{bulk}}} + \rho_{1,2} \frac{S}{V} \quad (2.6)$$

2. NUCLEAR MAGNETIC RESONANCE

2.3 Magnet designs

In the following section the two magnet geometries will be presented, which were used in the experiments presented in the following chapters of this thesis. Both geometries used are made of permanent magnets. The magnetic field strength is in both cases low, which is preferable for characterizing porous media. The magnets can be divided into open and closed geometries (Blümich 11). In case of the open magnets, the investigated object is exposed to the stray field of the magnet and the NMR signal is acquired from a sensitive volume outside the magnet and inside the object. This principle was first used in the well-logging industry and become popular for materials testing with the invention of the NMR-MOUSE® (Eidmann 96, Blümich 08). Stray-field NMR does not limit the size of the object, but has low sensitivity. A large sensitive volume is obtained with homogeneous and weakly inhomogeneous fields inside closed magnets, where the most popular design is due to Halbach (Raich 04).

2.3.1 Profile NMR-MOUSE®

The original sensor consists of two permanent magnets mounted on a iron yoke, with opposite polarizations. In the gap between the two magnets an rf surface coil is placed as shown in figure 2.1. This coil is used for both, excitation and detection. This arrangement of magnet and coil creates a sensitive volume outside the sensor, from which the signal is received. In order to improve the sensitivity, newer geometries were made of four magnets with the rf coil placed in the gap between them. In this way a highly flat sensitive volume is achieved (Profile NMR-MOUSE) (Perlo 05). For the two Profile NMR-MOUSE devices used in this work, these sensitive volumes are a flat slice of about 1 cm respectively 5 cm diameter parallel to the surface of the sensor and allowing a penetration depth of 5 mm respectively 25 mm. For measurements, the NMR-MOUSE is mounted on a precision lift and its position is controlled by a computer. Then the assembly is positioned close to the object in such a way that the sensitive slice is parallel to the object. The measurement starts with the smallest distance between sensor and object at the largest measurement depth, and then the sensor is retracted

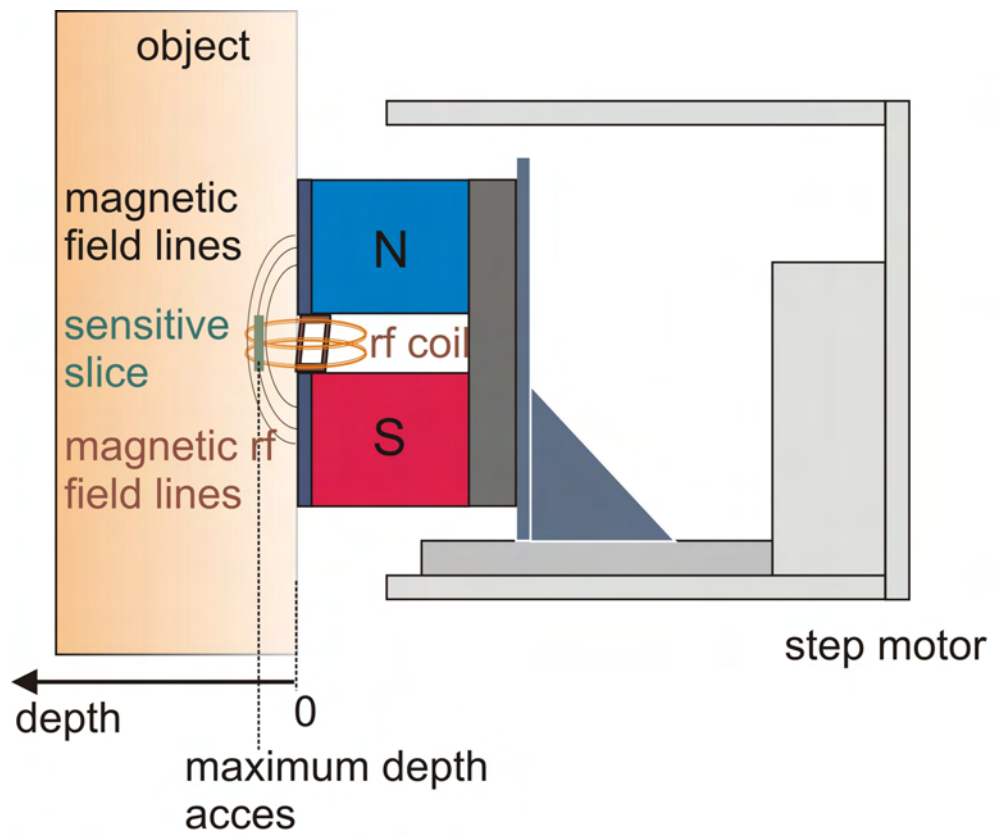


Figure 2.1: The NMR-MOUSE[®] for measuring depth profiles. Components and principle of operation. The magnet is positioned on a high precision lift, in order to record the signal at different depths in the sample.

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step by step. At each depth an NMR signal is received, from which a number is computed which determines the amplitude in a depth profile.

2.3.2 Halbach sensor

Following Klaus Halbach, a closed magnet with zero stray field outside and a comparatively homogeneous field inside transverse to the cylinder axis can be constructed from individual blocks of permanent magnets (Raich 04, Blümich 09). Such a Halbach magnet is composed of several rings with discrete magnet elements. The distance between the rings is adjusted to achieve axial field homogeneity. The magnet used for our experiments consists of two rings with 12 hexagonal bar magnets each (Fig. 2.2). The rings are separated by a 2.5 cm gap, which leads to a more homogeneous field along the magnet axis. The field strength is $B_0 = 0.51$ T and defines a proton resonance frequency of 21.85 MHz. The radio frequency coil was wound on a hollow Teflon cylinder with an inner diameter of 11 mm using five windings of copper sheet wire across a length of 5 cm. This allowed us to measure 10 mm thick samples.

2.4 Pulse sequences

In this section the pulse sequences are presented, by which the results presented in this thesis were obtained.

2.4.1 CPMG sequence

The most important and most frequently used pulse sequence is the CPMG named after its discoverers Carr, Purcell, Meiboom, Gill (Carr 54, Meiboom 58). It is used to measure the transverse relaxation time T_2 . A sketch of the sequence is presented in Fig. 2.3. It consists of a 90° rf pulse followed by a train of 180° pulses. The 180° pulse is double in amplitude but has the same length as the 90° pulse. In this way the frequency bandwidth is the same and the uniform slice selection is maintained. The time between two consecutive 180° is called echo time, t_E . With the help of the 180° pulses the magnetization, that started before to dephase due to B_0 inhomogeneities, is being refocused and an echo is formed. The envelope of

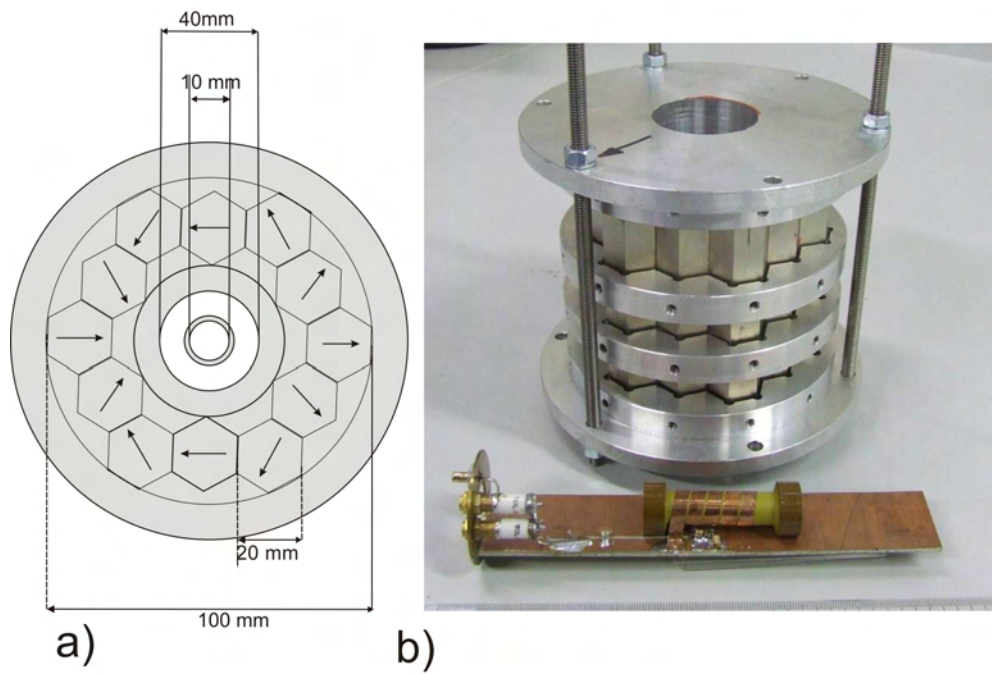


Figure 2.2: Halbach magnet geometry and setup. a) Sketch of one Halbach ring made of hexagonal magnet blocks with the positions of the blocks and the dimensions of the components. b) The magnet and the coil used for the measurements.

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the CPMG echo train is recorded (Blümich 05a). The echo amplitude decays with time. Its initial amplitude provides the proton density, which is proportional to

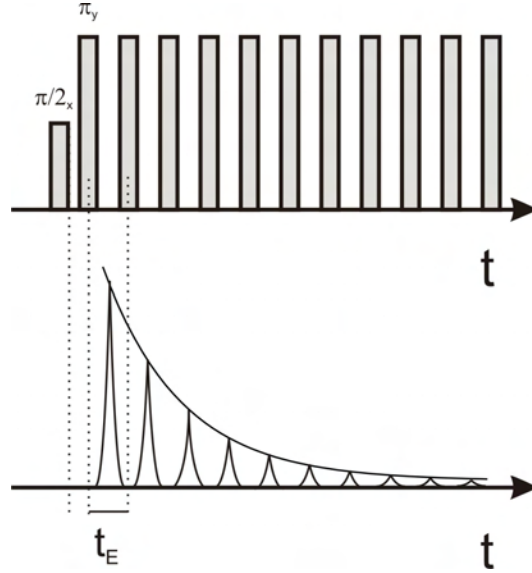


Figure 2.3: CPMG pulse sequence. Echoes can be generated with 180° impulses, the peaks of which stroboscopically sample the homogeneous-field decay. The echo envelope decays with time constant T_2 .

the content of hydrogen atoms in the sample. The decay-time constant T_2 varies with the mobility of the protons. It is proved that this is a robust method that can be used even in porous media with strong internal gradients (Song 02a).

The experimental CPMG signal $s(t)$ can be processed in different ways. A good approximation of the signal is a sum of exponential functions with relaxation times $T_{2eff,i}$

$$s(t) = \sum_i A_i \exp\left\{\frac{-t}{T_{2eff,i}}\right\} \quad (2.7)$$

where A_i is the relative spin density or mole fraction of components with the relaxation times $T_{2eff,i}$ in the sensitive volume if $s(t)$ is normalized to one. Another type of exponential function is the stretched function, where b is the shape parameter.

$$s(t) = A \exp\left\{-\frac{1}{b} \left\{\frac{-t}{T_{2eff,i}}\right\}^b\right\} \quad (2.8)$$

Often used is also the weighting parameter w . It is defined by the ratio of partial integrals over $s(t)$,

$$w = \frac{\int_{t_1}^{t_2} s(t) dt}{\int_{t_0}^{t_1} s(t) dt}. \quad (2.9)$$

2.4.2 T_1 saturation recovery

This sequence is used to measure the longitudinal relaxation time T_1 . It consist of a series of 90° pulses which will bring the magnetization in the xy plane and destroy it by totally dephasing it. Actually this is a 2D experiment ad it is used also for the T_1 - T_2 correlation experiments. During the recovery time τ_{re} the magnetization is restored on the z axis. The signal is read out with a CPMG sequence. So the longer the recovery time the more signal amplitude we obtain, until the magnetization is completely recovered and the maximum signal intensity is recorded. A sketch of the pulse sequence is shown in Fig. 2.4 For single-

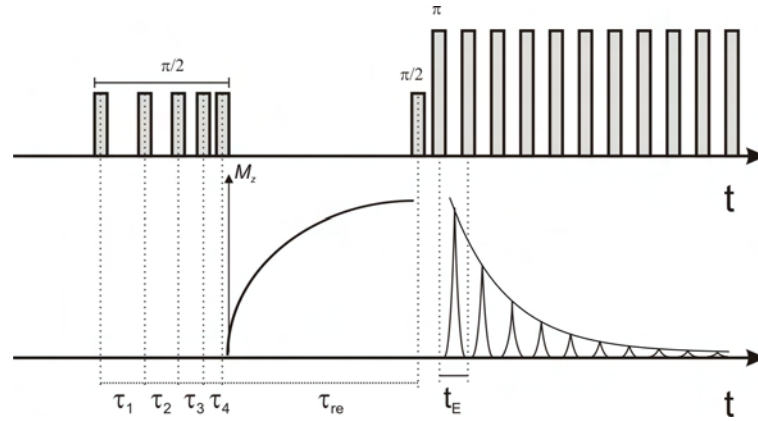


Figure 2.4: T_1 saturation recovery pulse sequence. In the first part of the sequence the net magnetization is destroyed. During the recovery time the magnetization is restored and read out with a CPMG sequence.

exponential relaxation the experimental signal is described as a function of the recovery time:

$$s(t) = A \left\{ 1 - \exp \left\{ \frac{-\tau_{re}}{T_1} \right\} \right\} \quad (2.10)$$

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The spin-lattice relaxation time give us information about the molecular mobility. Knowing the T_1 value, we can determine the recycle delay that we need to wait between two consecutive scans. Theoretically this is $5T_1$ and corresponds to the time in which 99 % of the magnetization recovers to the equilibrium state.

2.4.3 2D T_2 - T_2 exchange

The pulse sequence of the two-dimensional T_2 - T_2 relaxation exchange experiment (Washburn 06) is presented in Fig. 2.5. It has three main time periods: a T_2 encoding period for the indirectly detected relaxation dimension, a mixing time t_m , and an encoding period for read-out of the signal decay times after the mixing time. During the encoding periods, transverse relaxation is detected by the CPMG sequence. The time between the echoes, t_E , is kept at a constant value. The numbers of echoes used in the first encoding period is increased logarithmically so that the maximum first encoding time covers the same range as the second encoding time. The mixing times used between the two T_2 encoding times is varied. The second CPMG sequence is used for signal read out. This section and

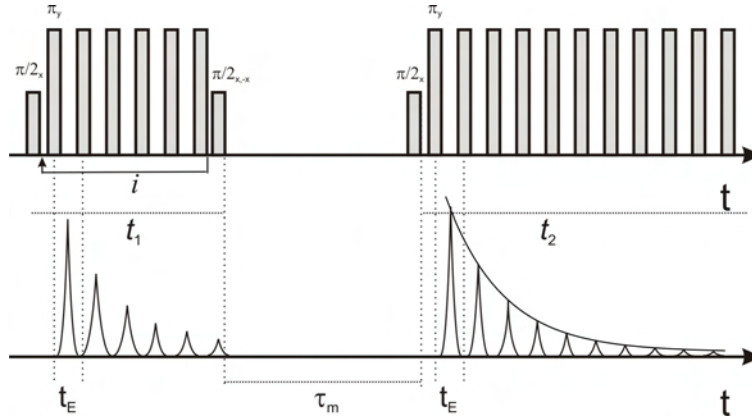


Figure 2.5: 2D T_2 - T_2 exchange pulse sequence. t_E is the echo time, which is kept fixed, and t_m is the mixing time, which extends the exchange time. The index i counts the 180° pulses in the indirect dimension. It is varied logarithmically in such a way that for maximum i , the evolution time in the indirect dimension is equal to the detection time in the direct dimension.

the theory to it will be discussed in detail in the next chapter.

2.4.4 Stimulated echo sequence

This sequence is used to measure the self-diffusion coefficient of the sample. The sketch of the pulse sequence is presented in Fig. 2.6. This sequence helps us to measure diffusion, D , as a function of the evolution time, Δ . It can be even measured at longer evolution times because in this time the magnetization is stored in the z direction and it decays with T_1 . The information obtained from

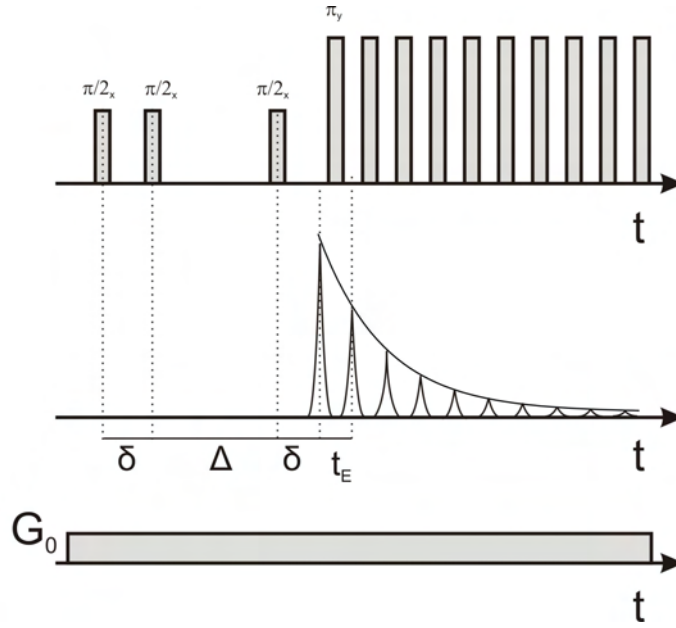


Figure 2.6: Stimulated echo sequence. In the first part the magnetization is stored in the z direction and the spins are allowed to diffuse. The signal is read out by a CPMG sequence. The more time the spins are allowed to diffuse the more the signal intensity will be attenuated.

the time depended diffusion coefficient is used in porous media to probe length scales. The measured signal is described by:

$$\ln \frac{I}{I_0} = -D\gamma^2 G^2 \delta^2 \left\{ \Delta + \frac{2}{3} \delta \right\} \quad (2.11)$$

where δ is the encoding time. This sequence is used also for the 2D D - T_2 correlation experiments, in which δ is varied.

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Chapter 3

Simple porous systems

Porous media have great influence in everyday life. They are present in different areas, starting from soil science and geology, continuing in medicine and biology and ending in civil engineering. The main idea is to understand the dynamics of the fluids captured inside the pores or moving from one pore environment to another. Once the dynamics is understood valuable informations are gained about the internal structure and surface properties. Moreover, with the help of the simulations and comparison with the experimental results, further helpful quantitative information can be gained for such porous systems. In order to gain deep understanding about the fluid dynamics in porous systems, first simple porous systems were tested and afterwards gradually more complex systems.

NMR can offer various solutions to solve the fluid dynamics problem inside a porous system. As an example local transport phenomena can be probed with different flow imaging techniques (Paciok 11). The well logging industry is using nowadays 2D relaxation correlation experiments to probe different fluid properties inside rocks down hole (Casanova 11). This shows that NMR techniques can probe microscopic as well as macroscopic phenomena and the changes inside the porous systems can be monitored from milliseconds up to days (Barrie 00). In this chapter fluid dynamics inside water saturated silica particles and water saturated aluminium oxide particles were tested using 2D relaxation exchange NMR experiments. Some of the results were published in the doctoral thesis of Maxime van Landeghem, as in this common project he was mainly focusing on the simulations and myself more on the experiments.

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3.1 Introduction

2D relaxation exchange NMR is a form of 2D NMR where molecular motion and transport are mapped by relaxation exchange. This NMR technique has a lot of applications, especially in porous media studies (Hürlimann 02, Washburn 06, Mitchell 07). This experiment is analogous to 2D Fourier Transform frequency exchange NMR but appears more appropriate for the study of exchange of liquids between pores since relaxivity is expected to depend on pore size while the chemical shift does not (McDonald 05).

2D FT frequency exchange NMR spectroscopy is able to discriminate different sites in terms of their chemical shifts in an NMR spectrum (Bain 03). Site in this case is a given magnetic environment. But, in order to be able to measure a chemical shift spectrum a highly homogeneous magnetic field is needed and the nuclear magnetic response of a molecule to a particular environment is not characterized only by a chemical shift but also by a rate of decay. Thus, the chemical shift scale can be replaced by a relaxation or a diffusion scale, so that the occupancy of the different sites is obtained not by the structure of the frequency distribution but instead by the distribution of relaxation rates or diffusion coefficients. This information can easily be obtained with simple magnets and inhomogeneous fields. This concept of relaxation exchange NMR is not new (Kroeker 86, Lee 93) but gained recent popularity with the development of a stable 2D inverse Laplace transformation (ILT) algorithm (Venkataramanan 02).

However, the interpretation of 2D exchange maps is not so simple. The reason is that the evolution times must be long enough to allow relaxation encoding. To simplify the problem, one would ideally want the exchange rate to be short enough so that exchange would be negligible during the relaxation encoding periods. This limit is called the slow exchange limit. The mixing time on the other hand must be long enough so that exchange occurs significantly. The applicability of the slow exchange limit during the evolution period requires that the system relaxes faster than it exchanges. This means, that relaxation can only be significant during the mixing period and thus that relaxation and exchange must be considered simultaneously at least during the mixing period. On the other hand, if the exchange rate is fast enough, the mixing period must be kept short enough

to avoid relaxation implying that significant exchange would occur during the relaxation encoding periods. The fact that it is relaxation which is probed during the encoding periods implies that exchange can never be restricted to the mixing period and must also be considered during the encoding periods. Consequently, contrary to the case of 2D Fourier exchange spectra, 2D relaxation exchange maps can never be interpreted as pure exchange maps that is measurements of the exchange probability densities. In the next sections the theory of multi-site exchange NMR is presented followed by the results obtained for two and multiple site exchange (Landeghem 10).

3.2 Multi-site exchange NMR

Time evolution

Generalizing the problem and considering a porous system with N sites, which means N different environments responding in different ways to a magnetic environment, $s(t)$ would be the NMR signal coming from the superposition of all the magnetization components.

$$s(t) = \sum_{i=1}^N M_i(t) \quad (3.1)$$

Considering the magnetization evolution just for one site $M_i(t)$ in the absence of exchange, it will follow the differential equation

$$\frac{d}{dt}[M_i(t) - M_i^{\text{eq}}] = -R_i[M_i(t) - M_i^{\text{eq}}], \quad (3.2)$$

where the relaxation rate, R , is $R_i = 1/T_1$ and the magnetization at equilibrium, M^{eq} , is $M_i^{\text{eq}} = M_i^0$ for longitudinal magnetization and $R_i = 1/T_2$ and $M_i^{\text{eq}} = 0$ for transverse magnetization. If we consider N magnetization components they will be brought together in the magnetization vector $\mathbf{M}(t)$ and equation 3.2 becomes:

$$\frac{d}{dt}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] = -\mathbf{R}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}], \quad (3.3)$$

where $\mathbf{R}$ is a diagonal matrix with the corresponding relaxation rates.

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If exchange occur between the different sites an off-diagonal exchange matrix $\mathbf{K}$, that will mix the magnetization components, needs to be introduced in equation 3.3.

$$\frac{d}{dt}[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] = -(\mathbf{R} + \mathbf{K})[\mathbf{M}(t) - \mathbf{M}^{\text{eq}}] \quad (3.4)$$

This equation will have the familiar solution

$$\mathbf{M}(t) - \mathbf{M}^{\text{eq}} = \exp\{-(\mathbf{R} + \mathbf{K})(t - t_0)\}[\mathbf{M}(t_0) - \mathbf{M}^{\text{eq}}] \quad (3.5)$$

The exchange matrix

The generalized exchange matrix $\mathbf{K}$ can be written as

$$\mathbf{K} = \begin{bmatrix} k_{11} & -k_{12} & \cdots & -k_{1n} \\ -k_{21} & k_{22} & \cdots & -k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -k_{n1} & -k_{n2} & \cdots & k_{nn} \end{bmatrix},$$

where k_{ij} is the exchange rate between sites i and site j . As $\mathbf{K}$ only mixes magnetization components and preserves the total magnetization, detailed mass balance requires that whatever amount of magnetization is lost in one component has to be recovered from other components:

$$\mathbf{K}\mathbf{M}^{\text{eq}} = 0 \quad (3.6)$$

For example, for magnetization component M_1 can be written

$$k_{11}M_1^{\text{eq}} - \sum_{j=2}^n k_{1j}M_j^{\text{eq}} = 0 \quad (3.7)$$

Likewise, whatever total amount of magnetization is received by one of the magnetization components from other magnetization components has to be restored to them. For example, for magnetization component M_1 ,

$$\sum_{j=2}^n k_{1j}M_j^{\text{eq}} = \sum_{i=2}^n k_{i1}M_1^{\text{eq}} \quad (3.8)$$

This will lead to

$$k_{11} - \sum_{i=2}^n k_{i1} = 0 \quad (3.9)$$

so that the column sums of $\mathbf{K}$ vanish. The kinetic matrix is determined by $(n-1)^2$ independent parameters. Note that the rates k_{ij} apply to the forward kinetics from state i to state j , so that the rates k_{ji} apply to the reverse kinetics from state j to state i . The associated equilibrium constant K_{ij} which is defined by the ratio $[M_j]/[M_i]$ of concentrations is given by

$$K_{ij} = \frac{[M_j]}{[M_i]} = \frac{k_{ij}}{k_{ji}} \quad (3.10)$$

Magnetization evolution in T_2 - T_2 exchange NMR experiments

The time evolution of the magnetization is examined for a T_2 - T_2 exchange experiment with a mixing period during which the magnetization experiences T_1 relaxation. The pulse sequence is presented in Fig. 2.5 in section 2.4.3. The magnetization after the mixing time t_m follows from Eqn. 3.5:

$$\mathbf{M}(t_m + t_1 + t_0) - \mathbf{M}^0 = \exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\}(\mathbf{M}(t_1 + t_0) - \mathbf{M}^0) \quad (3.11)$$

During the evolution period t_1 and the detection period t_2 the magnetization is given by

$$\mathbf{M}(t_1 + t_0) = \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) \quad (3.12)$$

and

$$\mathbf{M}(t_2 + t_m + t_1 + t_0) = \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_2)\}\mathbf{M}(t_m + t_1 + t_0) \quad (3.13)$$

The last three equations combine into

$$\begin{aligned} \mathbf{M}(t_2 + t_m + t_1 + t_0) = & \exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_2)\}[\exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\} \\ & \times (\exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) - \mathbf{M}^0) + \mathbf{M}^0] \end{aligned} \quad (3.14)$$

Statistical interpretation of peak intensities in the slow exchange limit

In 2D Fourier exchange NMR experiments, the fast exchange limit may apply during the mixing time so that relaxation can be neglected during t_m , while

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Table 3.1: Evolution operators for the three cases that can occur in 2D relaxation exchange NMR

	$U(t_1)$	$U(t_m)$	$U(t_2)$
Case 1	$\exp\{-[\mathbf{R}](t_1)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R}](t_2)\}$
Case 2	$\exp\{-[\mathbf{R} + \mathbf{K}](t_1)\}$	$\exp\{-[\mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_2)\}$
Case 3	$\exp\{-[\mathbf{R} + \mathbf{K}](t_1)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_m)\}$	$\exp\{-[\mathbf{R} + \mathbf{K}](t_2)\}$

the slow exchange limit applies during the evolution and detection times. This situation is never encountered in 2D relaxation exchange NMR. Only the following three cases may be encountered. The corresponding evolution operators are given in Table 3.1.

1. The exchange is slow during the evolution periods, which means that the time scale of exchange is long compared the time scale of relaxation, and pure relaxation encoding is possible during the evolution periods. Consequently, there is no fast exchange during the mixing period.
2. The exchange is fast compared to relaxation. Then the fast exchange limit is satisfied during the mixing time. In this case, the mixing period involves only exchange but not relaxation, while the evolution and detection periods encode both, relaxation and exchange.
3. The exchange rate is commensurate to the relaxation rates, and both must be considered during the evolution, mixing, and detection periods.

Only in the first case can the 2D relaxation cross peaks be analyzed in terms of exchange kinetics, although not easy. Being in the slow exchange limit during evolution, means that one can measure the relaxation rate R_i at time t and identify it with the magnetization of the corresponding site M_i . In this case, the slow exchange limit applies to the t_1 and t_2 encoding periods, and Eqn. 3.14 simplifies to

$$\begin{aligned}
 \mathbf{M}(t_2 + t_m + t_1 + t_0) = & \exp\{-[\mathbf{R}^{(2)}](t_2)\}[\exp\{-[\mathbf{R}^{(1)} + \mathbf{K}](t_m)\} \\
 & \times (\exp\{-[\mathbf{R}^{(2)} + \mathbf{K}](t_1 + t_0)\}\mathbf{M}(t_0) - \mathbf{M}^0) + \mathbf{M}^0]
 \end{aligned}
 \tag{3.15}$$

Since the relaxation matrix $\mathbf{R}^{(2)}$ is diagonal, the magnetization of a particular site i at the end of the evolution period is obtained as

$$M_i(t_1 + t_0) \approx \exp\{-R_i^{(2)}t_1\}M_i(t_0) = \exp\{-t_1/T_{2,i}\}M_i(t_0) \quad (3.16)$$

After the mixing time t_m , one obtains from Eqn. 3.5,

$$M_j(t_2 + t_m + t_1 + t_0) \approx \exp\{-R_j^{(2)}t_2\}M_j(t_m + t_1 + t_0) = \exp\{-t_2/T_{2,j}\}M_j(t_m + t_1 + t_0) \quad (3.17)$$

The resultant time domain signal is then approximated by

$$M_j(t_2 + t_m + t_1 + t_0) \approx \exp\{-t_2/T_{2,j}\} \left[\sum_i \{ [\exp\{-(\mathbf{R}^{(1)} + \mathbf{K})t_m\}]_{ji} (\exp\{-t_2/T_{2,i}\}M_i(t_0) - M_i^0) \} + M_j^0 \right] \quad (3.18)$$

After the 2D inversion over t_1 and t_2 cross peaks arise at coordinates $(T_{2,i}, T_{2,j})$ with amplitudes given by

$$M_{ij} = [\exp\{-(\mathbf{R}^{(1)} + \mathbf{K})t_m\}]_{ij} M_i(t_0) \quad (3.19)$$

Therefore, when this slow exchange limit is fulfilled for the encoding and detection periods t_1 and t_2 , the cross-peak amplitude represents the magnetization from site j that originated from site i before the mixing time t_m .

The other two cases are even more complex as magnetization is not conserved or follows multiple single relaxation paths. As a consequence, 2D relaxation exchange maps cannot be interpreted in terms of conditional probability densities, and the observed decay rates are not associated with single site relaxation rates (McDonald 05, Monteilhet 06).

3.3 The ILT problem

2D Laplace NMR did not evolve as fast as 2D Fourier NMR. This is because of the ill-defined ILT which inverts the data from the time domain into the relaxation rate domain. This problem was presented in papers by Venkataramanan and Song (Venkataramanan 02, Song 02b, Song 10) and finally became possible after Venkataramanan (Venkataramanan 02) proposed an ILT computing method

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that was not that consuming for the PC. The oldest algorithm used for ILT is the CONTIN program (Provencher 82a, Provencher 82b). The software that I used is the one developed by Prof. P.T. Callaghan but it is based on the algorithm of Venkataramanan (Venkataramanan 02, Godefroy 03).

Inside a porous medium different environments can be found which will influence the NMR signal in different ways (Casanova 11). This signal can be observed, in general terms, as a multi-exponential decay, $s(t_i)$. In case of the 1D experiments this will be:

$$s(t_i) = \sum_s A(T_s) \exp\left\{-\frac{t_i}{T_s}\right\} + \varepsilon_i, \quad (3.20)$$

where T_s is the relaxation times and $A(T_s)$ the relation time distribution of the studied system. ε_i represent here the noise of the signal.

The procedure to invert the data and to determine the distribution of relaxation times $A(T_s)$ consist in applying the non-negative least square algorithm described in (Lawson 74). This consists in minimization of:

$$\min\{\chi^2 = \sum_{i=1}^n (s(t_i) - \sum A(T_s) \exp\{-\frac{t_i}{T_s}\})^2\}. \quad (3.21)$$

However, because the real signal has some noise, too, in it, the determination of the distribution by a simple non-negative least square fit would lead to a multitude of peaks, in this way the problem is ill-posed. To avoid it, a regularization function is introduced (or the so called smoothing parameter)(Borgia 98).

$$\min\{\chi^2 = \sum_{i=1}^n (s(t_i) - \sum A(T_s) \exp\{-\frac{t_i}{T_s}\})^2 + \alpha^{-1} \sum_{s=1}^m (2A(T_s) - A(T_{s-1}) - A(T_{s+1}))^2\}, \quad (3.22)$$

where α represents the regularization parameter. This expression can be rewritten as:

$$Y = KX + E, \quad (3.23)$$

where X is the distribution that need to be determined, Y is the signal measured, K is the known matrix of the $\exp(-t_i/T_s)$. The least square function has now the form:

$$\chi^2 = \|KX + E\|^2 + \alpha^{-1} \|X\|^2. \quad (3.24)$$

In case of a 2D experiment, for example for a relaxation-relaxation correlation experiment, the signal measured follows:

$$s(t_1, t_2) = \sum A(T_1, T_2) \exp\{-\frac{t_1}{T_1}\} \exp\{-\frac{t_2}{T_2}\} + \varepsilon_{t_1, t_2}. \quad (3.25)$$

and in the matrix form it would be:

$$Y = K_1 X K_2' + E. \quad (3.26)$$

Here the K_1 and K_2 matrices are the $\exp(-t_{1i}/T_{1s})$ and $\exp(-t_{2i}/T_{2s})$. In order to get a 2D Laplace inversion of the data defined by Y , the matrix Y of size $m \times n$ needs to be reduced to a vector of length $m \times n$. This is obtained by using the singular value decomposition algorithm (Song 02b).

$$\begin{aligned} K_1 &= U_1 S_1 V_1' \\ K_2 &= U_2 S_2 V_2' \end{aligned} \quad (3.27)$$

where U and V are unitary matrices. The reduced matrix is determined according to:

$$\begin{aligned} \tilde{K}_1 &= U_1' K_1 \\ \tilde{K}_2 &= U_2' K_2 \\ \tilde{Y} &= U_1' Y U_2 \end{aligned} \quad (3.28)$$

Equation 3.26 can be rewritten:

$$\tilde{Y} = \tilde{K}_1 X \tilde{K}_2' + E. \quad (3.29)$$

This 2D form can be reduced to a 1D form after the tensor product of the matrices $\tilde{K}_1$ and $\tilde{K}_2$ and the data can be processed in the same way as for the 1D case mentioned above (Eq. 3.23, 3.24).

3.4 Results

Numerical simulations, which are presented in (Landeghem 10), were made for both 1D and 2D experiments. From the simulations, the following important conclusions can be made:

In case of 1D two site exchange, the influence of the exchange rate on peak positions and peak integrals, revealed the following:

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- when the exchange time becomes short compared to the relaxation times, one of the peaks grows at the expense of the other, until only a single peak is observed but its integral corresponds to the number of spins from both sites.
- when the exchange time is longer than the longest relaxation time the integrals provide the right population values. (In the case presented in (Landeghem 10) the exchange time was 10 times longer than the longest relaxation time).
- in the 1D relaxation distributions (after the ILT) the relaxation times extracted from the maxima of each distributions are just apparent relaxation times given by the eigenvalues of the $\mathbf{R} + \mathbf{K}$ matrix from eq. 3.4. Depending on whether or not it is fast exchange or slow exchange the distributions will be shifted on the relaxation time axis. It was shown that just for really slow exchange an unbiased estimation of the relaxation times and distributions was obtained.

In case of 2D two-site exchange, the following conclusions can be made:

- when the exchange is very fast, the exchange map exhibits a single diagonal peak in accordance with the 1D study. When the exchange time is increased and becomes comparable to the relaxation times, cross peaks appear as well as two distinct diagonal peaks and when the exchange time becomes longer than the relaxation times, the exchange is too slow to be seen and the cross peaks disappear.
- the peak intensities in the 2D maps vary with the mixing time t_m . The diagonal peaks decay and the cross-peaks grow exponentially with the mixing time t_m until the the longitudinal relaxation time influences this mixing period and the intensities start to decay towards 0.
- regarding the sensitivity of the ILT to noise (which sometimes give rise to artifacts in the 2D distribution maps), the simulations showed that the noise affects strongly the integral values. For only a 2% noise level in the NMR signal the relative deviations of the integral can go up to 10% and

also one of the diagonal peaks is shifting away from the diagonal for a noise level higher than 2%.

3.4.1 Two-site 2D relaxation exchange experiments

The results obtained from the simulations were compared to experimental data. For this purpose silica particles were synthesized according to the Stober reaction (Green 03, Stober 68). The reason for choosing these particles is the known and tested mono-disperse particle size distribution. The particles size was determined with a SEM to be 500 nm with a standard variation of 40 nm. These mono-disperse particles are good candidates for two site exchange. They just need to be water saturated and the two sites would be the bulk water between the particles and the water at the surface of the particles (Landeghem 10).

The experiments were performed in a Halbach magnet presented before in section 2.2. The field strength was $B_0 = 0.51$ T, defining a proton resonance frequency of 21.85 MHz. Figure 3.1 show the spin-lattice and spin-spin relaxation decays measured with a saturation recovery and a CPMG sequence. These pulse sequences were presented before in Fig. 2.4 and Fig. 2.3 in Section 2.4.2 and 2.4.1. This system of saturated silica particles exhibit two apparent T_2' s which

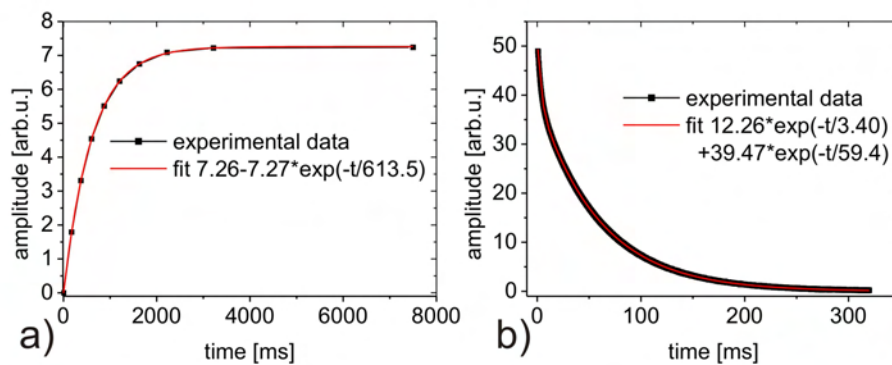


Figure 3.1: Water saturated silica particles. a) T_1 saturation curve and b) T_2 CPMG decay curve. Two apparent relaxation times can be distinguished.

correspond to two different sites, but has a single apparent T_1 . Taking into account

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the results from the 1D simulations we know that we are in the situation where the exchange time is small compared to the spin-lattice relaxation times.

T_1 - T_2 correlation maps were simulated and measured as a support material for the T_2 - T_2 exchange maps. The results are shown in Fig. 3.2. In the experimental

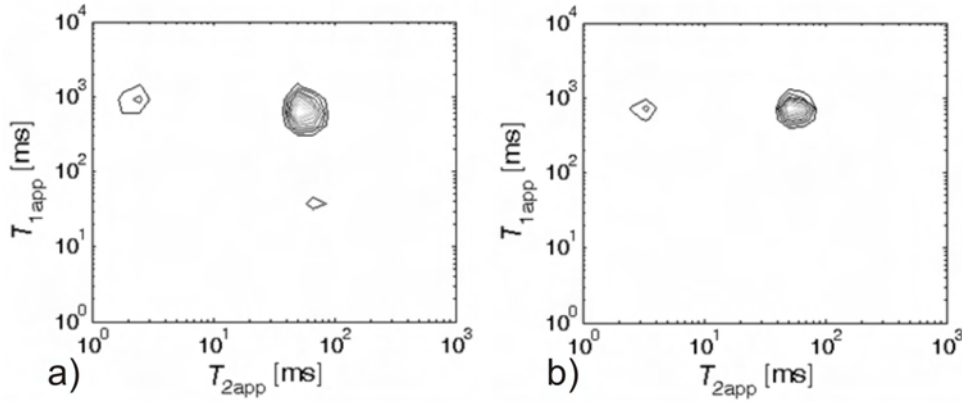


Figure 3.2: Experimental and simulated T_1 - T_2 correlation maps of water saturated silica particles. a) Experimental map and b) Simulated map. The experimental map shows a small peak which is attributed to noise in the experimental data.

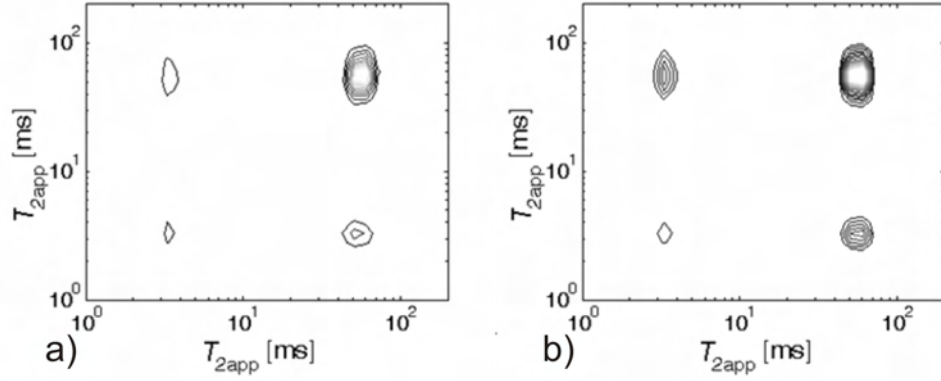
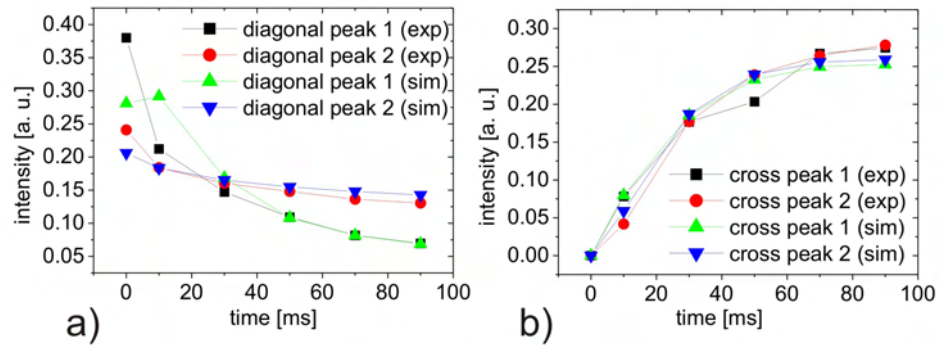
map a very small peak can be noticed which is not existent in simulated map. It is likely to be noise which gave raise to this artifact also because in the 1D fitted data just one component could be identified.

T_2 - T_2 relaxation exchange experiments were simulated and performed using the pulse sequence presented in Section 2.4.3. Six different mixing times were used (0, 10, 30, 50, 70 and 90 ms). The comparison between simulations and experiments is shown for one mixing time in Fig. 3.3. The parameters obtained by the simulations are summarized in Table 3.2.

Fig. 3.3a is obtained by inverting the experimental data and Fig. 3.3b by inverting the simulated data. The peaks of the simulated spectra are broader than the experimental ones. This is interpreted as an artifact due to regularization during the ILT. To verify the quality of the fitting, the peak integrals of both simulated and experimental exchange maps were compared. This evolution of the integrals over the mixing time is represented in Fig. 3.4. Except for a divergency for the diagonal peak 1 at low mixing time, the results present a pretty good

Table 3.2: Parameters obtained by the simulations.

Parameter	Value
M^A	5.5
M^B	18.4
T_2^A	3.9 ms
T_2^B	103.5 ms
T_1^A	610 ms
T_1^B	630 ms
k	$1/110 \text{ ms}^{-1}$

**Figure 3.3:** a) Experimental and b) Simulated T_2 - T_2 correlation maps for 70 ms mixing time. The agreement is evident.**Figure 3.4:** Comparison between the peak integrals obtained experimentally and the simulated ones for different mixing times. a) Integrals obtained for the diagonal peaks and b) Integrals obtained for the cross peaks. Good agreement is observed.

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agreement between the mixing time dependence of the peaks intensities observed experimentally and the ones simulated.

3.4.1.1 Temperature dependence

The same system of water saturated silica particles was tested for its behavior at different temperatures. For this purpose the NMR system was put into a temperature chamber and the evolution of the T_2 - T_2 exchange maps at three different temperatures was monitored for different mixing times. The three temperatures were 5, 15 and 25°C.

First, after the whole system was equilibrated for the corresponding temperature, at each temperature the spin-lattice relaxation time was measured. T_1 behaviour at the three different temperatures is shown in Fig. 3.5. From the

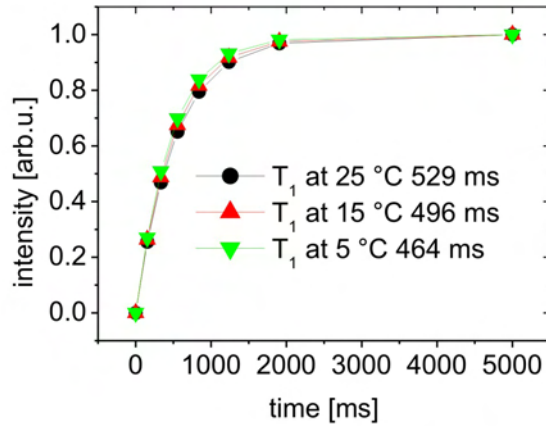


Figure 3.5: T_1 behavior at different temperatures. T_1 is increasing with temperature.

figure it is noticeable that the relaxation time is increasing with increasing temperature. This is a known fact (Levitt 08, Fukushima 81).

T_2 - T_2 relaxation exchange experiments were made at different temperatures for different mixing times, from 0 up to 160 ms. For 0 ms mixing time the ILT maps for all three temperatures were over plotted, and the 1D projection on both

relaxation time axis show the behavior of the spin-spin relaxation time with temperature. Figure 3.6 present the two main diagonal peaks, which are present

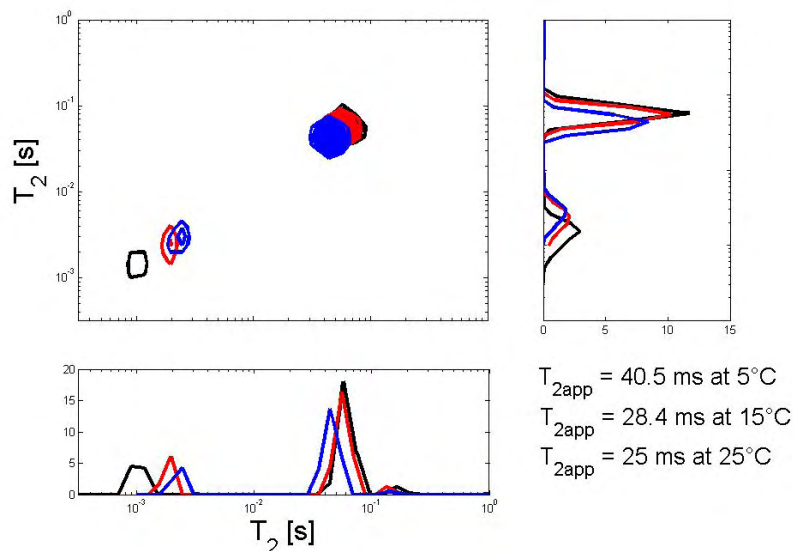


Figure 3.6: T_2 - T_2 correlation maps for different temperatures at 0 ms mixing time. The projection of the T_2 distributions on the relaxation axis show that T_{2app} is decreasing with increasing temperature.

at each temperature and their projections on each relaxation axis. It is observable that the T_{2app} is decreasing with increasing temperature. This may be due to higher molecular mobility. Higher molecular mobility gives a higher diffusion coefficient and this will induce a lower spin-spin relaxation time. The comparison between simulations and experiments is shown for 70 ms mixing time and at three different temperatures in Fig. 3.7. The peaks of the simulated spectra are broader than the experimental ones. This is interpreted as an artifact due to regularization during the ILT. The experimental spectra present an asymmetry for one of the cross peaks. This is due to noise artifacts, fact demonstrated (Landeghem 10). Nevertheless, good agreement between simulation and experimental data is observed. The parameters obtained by the simulations for the three different temperatures are summarized in Table 3.3.

To verify the quality of the ILT 2D correlation maps, the peak integrals of the exchange maps were compared. An overall evolution of the integrals over the

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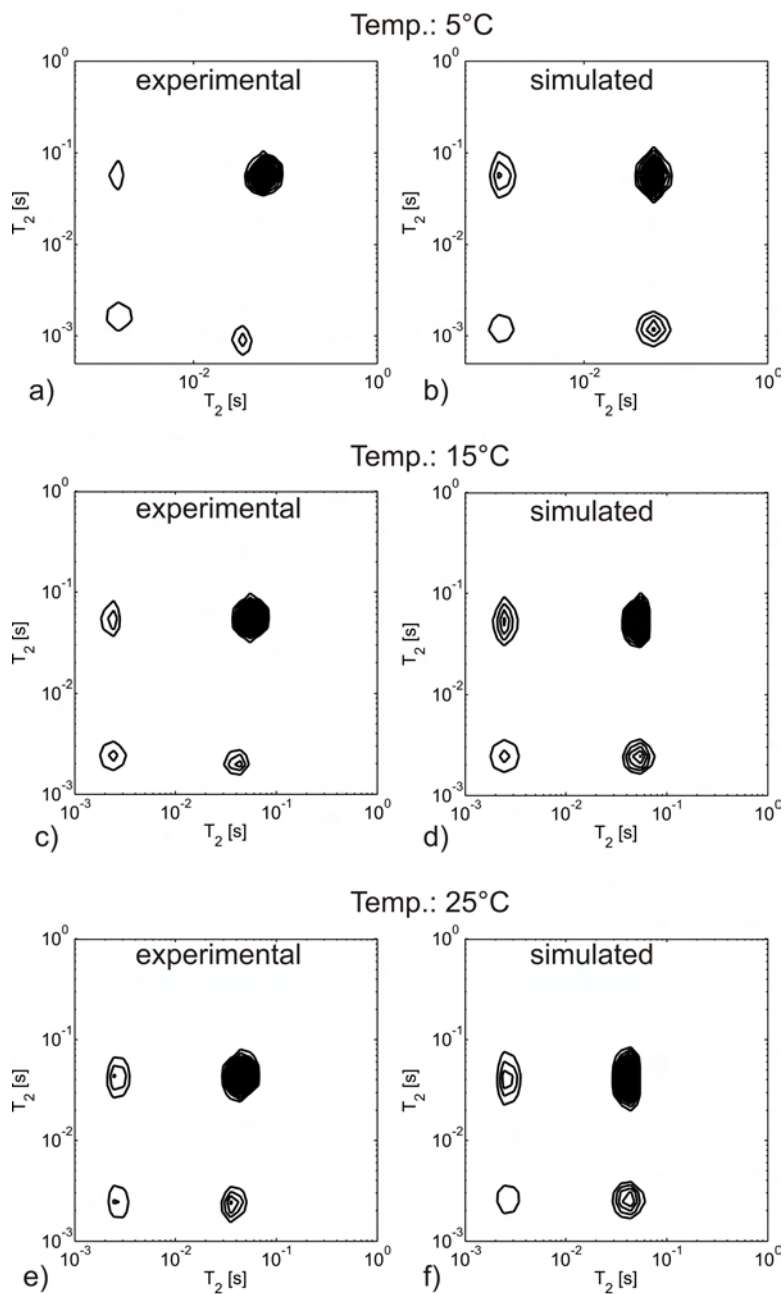


Figure 3.7: T_2 - T_2 correlation maps for different temperatures at 70 ms mixing time. a) Experimental 2D correlation map at 5°C. b) Simulated 2D correlation map at 5°C. c) Experimental 2D correlation map at 15°C. d) Simulated 2D correlation map at 15°C. e) Experimental 2D correlation map at 25°C. f) Simulated 2D correlation map at 25°C.

Table 3.3: Parameters obtained by the simulations for the three different temperatures.

Parameter	Value at		
	5°C	15°C	25°C
M^A	16	12	7.9
M^B	37	25	15.4
T_2^A	2 ms	2.7 ms	3 ms
T_2^B	130 ms	90 ms	70 ms
T_1^A	460 ms	500 ms	510 ms
T_1^B	470 ms	510 ms	540 ms
k	1/135 ms ⁻¹	1/121 ms ⁻¹	1/110 ms ⁻¹

mixing time at different temperatures is represented in Fig. 3.8. A detailed evo-

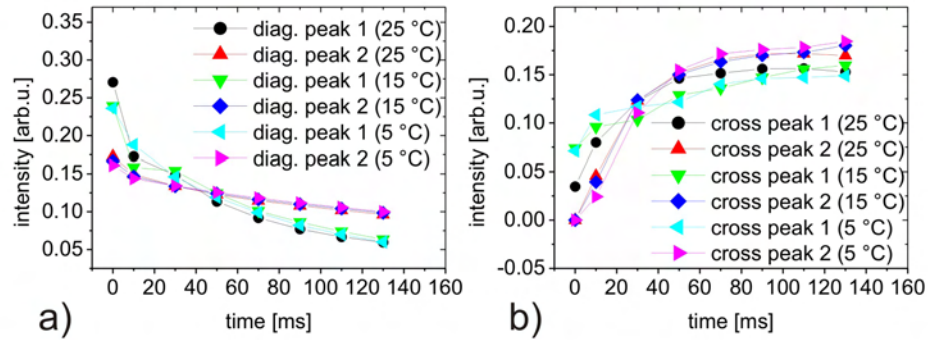


Figure 3.8: Comparison between the diagonal and cross peaks integrals for different mixing times at different temperatures. a) Diagonal peaks integrals at different temperatures. b) Cross peaks integrals behavior at different temperatures.

lution of the simulated and experimentally obtained peak integrals for different mixing times and at different temperatures is presented in Fig. 3.9. For all temperatures a slowly deviation of on cross peak integral evolution is observed. This is due to the noise artifact that influence the integral evolution (Landeghem 10). Good agreement between the simulations and experimentally obtained peak integrals is observed. From the dependence of the reaction rate, k , on temperature,

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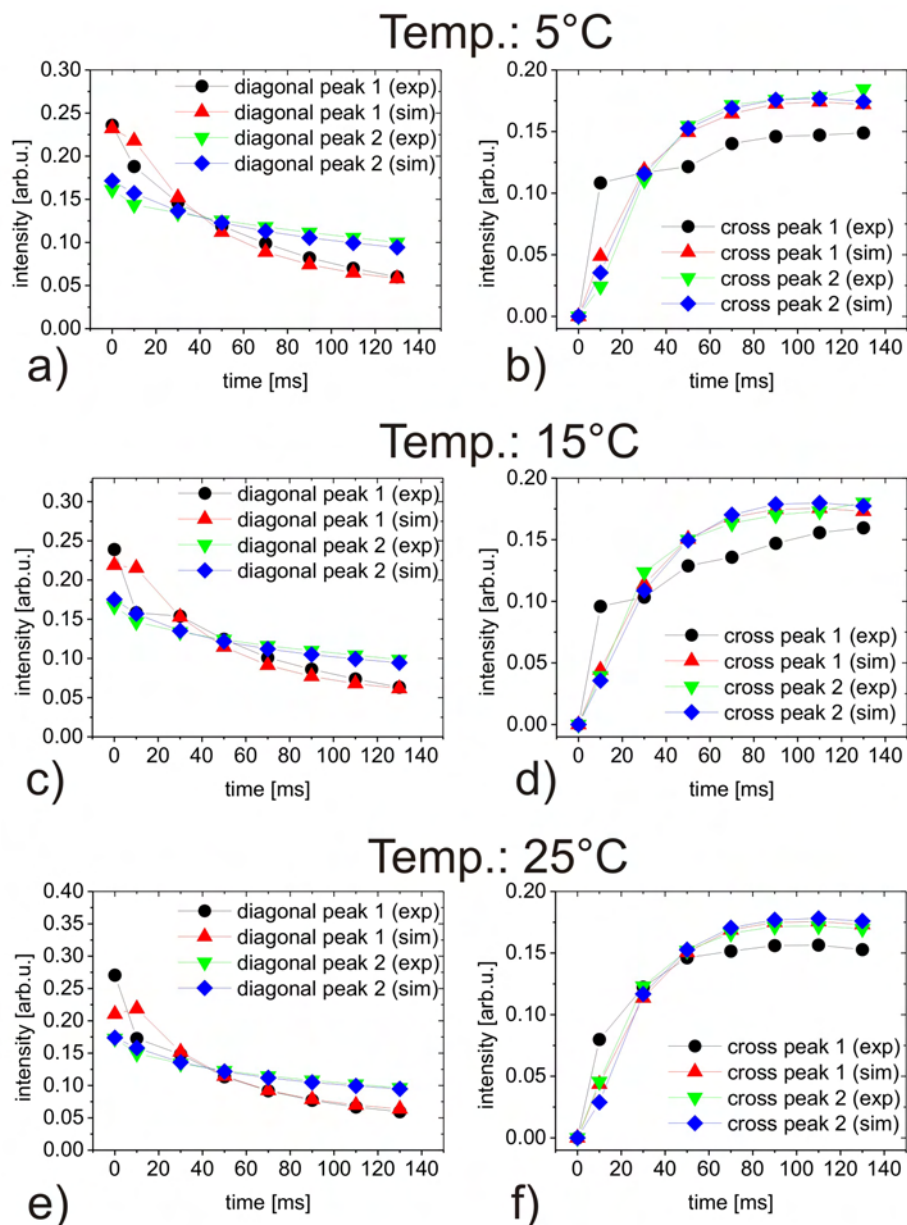


Figure 3.9: Comparison between the peak integrals obtained experimentally and simulated for different mixing times at different temperatures. a) Integrals obtained for the diagonal peaks at 5°C. b) Integrals obtained for the cross peaks at 5°C. c) Integrals obtained for the diagonal peaks at 15°C. d) Integrals obtained for the cross peaks at 15°C. e) Integrals obtained for the diagonal peaks at 25°C. f) Integrals obtained for the cross peaks at 25°C.

T , the activation energy can be estimated by the Arrhenius law (Laidler 65):

$$k = A \exp\left(-\frac{\Delta E}{RT}\right) \quad (3.30)$$

In this equation, A is the pre-exponential factor, R is the gas constant and ΔE is the activation energy (Atkins 02). The pre-exponential factor, is a statistical parameter which gives us information about the collision frequency. The dependence is plotted in Fig. 3.10. From the fit an activation energy of 7.059 kJ/mol and a pre-exponential factor of 1.036 ms^{-1} could be extracted. The value obtained for the activation energy is higher than the energy for dipole-dipole interaction and the Van-der-Waals interaction (2 kJ/mol) but lower than the energy for ion-dipole interaction and the hydrogen bond (15 respectively 20 kJ/mol) (Holleman 95, Moore 86). This activation energy is the energy that the hydrogen atom needs for moving from one environment to the other, in other words the diffusion activation energy (Stallons 01).

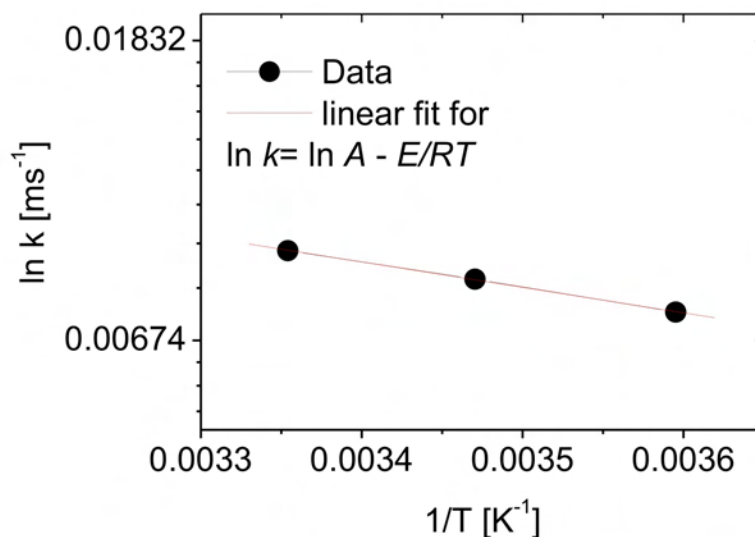


Figure 3.10: Dependence of the reaction rate on temperature. The activation energy can be extracted by the fit.

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3.4.2 Multi-site 2D relaxation exchange

There is a difference between two- and n -site exchange. For two-sites, the exchange maps are symmetric as mass balance requires the forward and backward exchange to be identical. When exchange takes place between more than two sites, detailed mass balance applies to the overall exchange but not to subsets consisting of only two of n sites, so that asymmetric exchange maps can arise. To obtain a symmetric map, all exchange times have to have the same value, whereas they differ strongly in the non-symmetric case.

To test this, another system was used, water saturated aluminium oxide. The substance was provided by Merck KGaA. Aluminium oxide is a hygroscopic, white powder used usual as adsorbent in chromatography in specifically defined pH ranges, with a mean pore size of 9 nm and particle size range of 60- 200 μm . T_2 - T_2 experiments were performed for different mixing times, from 0 to 150 ms.

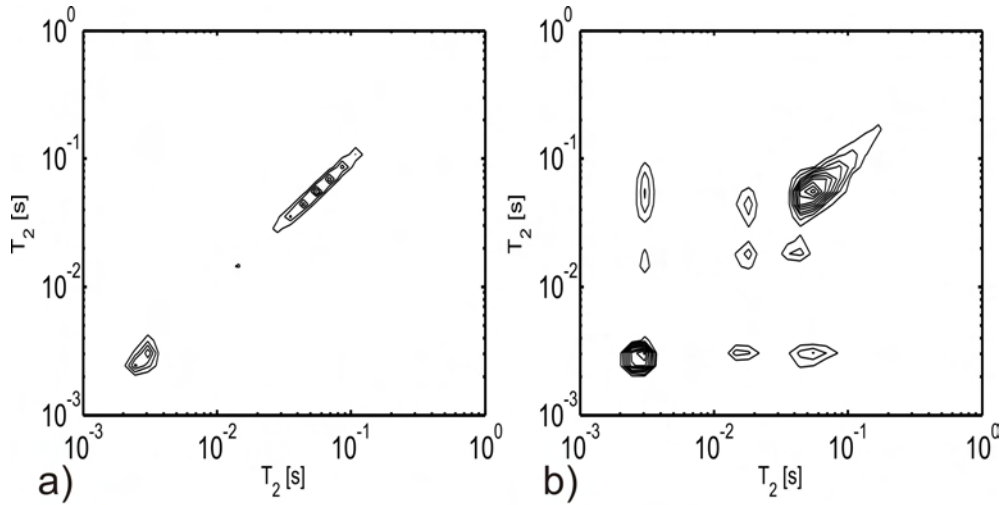


Figure 3.11: T_2 - T_2 exchange maps for different mixing times. a) Mixing time 1 ms, showing the multiple main peaks. b) Mixing time 40 ms showing a symmetric exchange map.

Figure 3.11 shows the 2D T_2 - T_2 exchange distributions for two different mixing times. For the left plot where mixing time is 1 ms, three, or possible more, main peaks without exchange between the different sites can be observed. The three components have $T_{2\text{app}}$ values of 3.4, 11.5 and 73.6 ms. We assume that the

shortest T_2 value arise from the water molecules inside the pores. The middle value arises from the water layers surrounding the Aluminium Oxide particle and the longest and broadest T_2 distribution arise from the bulk water. In the second plot, the mixing time used was 40 ms, and the exchange between the environments is quite noticeable. Water migration between different environments is observed.

To verify the quality of the ILT 2D correlation maps, the peak integrals of the exchange maps were compared. This evolution of the integrals over the mixing time is represented in Fig. 3.12. It shows how the intensity of the integrals of the

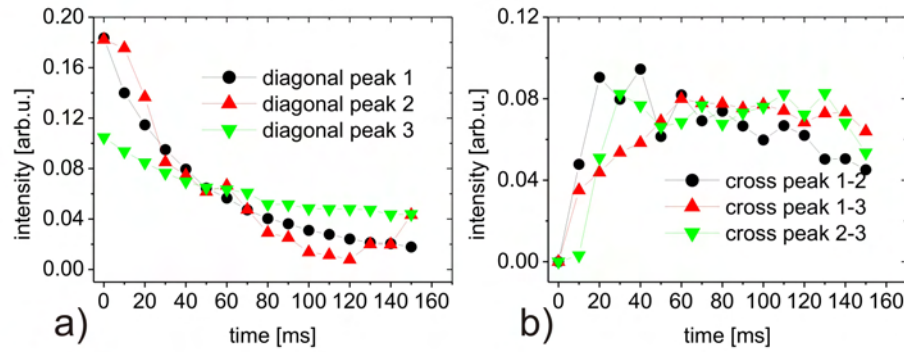


Figure 3.12: Evolution of the diagonal and cross peaks integrals for different mixing times. a) Evolution of diagonal peaks integrals for different mixing times. b) Evolution of cross peaks integrals for different mixing times.

diagonal peaks and off-diagonal peaks evolve with the mixing time. The intensity of the diagonal peaks decrease exponentially, and the intensity of the off-diagonal peaks are increasing with time, but after a time they start to decrease, this is because of the influence of the longitudinal relaxation time during the mixing time, fact that was demonstrated in the simulations.

To prove the accuracy of our dataset several other experiments were made. One of them was a drying profile for a 3 mm thick water saturated sample. The drying process was monitored during several hours with a Profile NMR-MOUSE with a proton resonance frequency of 17.1 MHz. The profiles shown in Fig. 3.13 exhibit a strange behavior at the beginning with water evaporating at the surface (surface being at 0 μm in this figure) and the capillary forces are moving the

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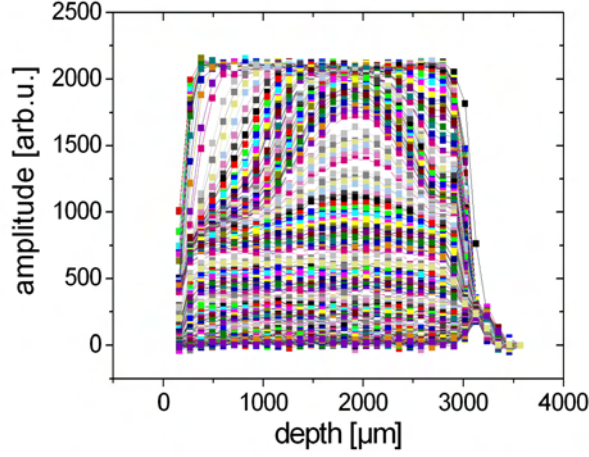


Figure 3.13: Drying evolution of the sample observed over several hours for a depth of 3 mm. At the beginning it shows a different behavior than after it is half dried.

water from the bottom to the top but keeping the intensity almost constant in the centre. After enough time the water in between particles evaporates and due to the capillary forces inside the particles, these remain filled. It leads to a constant profile with amplitude of about 700 arb. u.. At this time the sample is removed and introduced in the NMR tube to make an experiment similar to the one carried out for a saturated sample. A CPMG was measured on the partially dried sample and compared to the saturated sample.

Fig. 3.14 shows the comparison between the saturated and partially saturated sample. An Inverse Laplace Transformation was applied on these two exponential decays to see the T_2 relaxation distributions corresponding to the different environments. For the partially saturated sample we see only one distribution in comparison with the saturated sample. This shows that in the drying process the bulk and interstitial water vanished but the water inside the pore remained.

Another experiment that we made, to prove the water dynamics in the sample, was a 2D D - T_2 correlation experiment. This experiment was performed with the Profile NMR- MOUSE, with a proton resonance frequency of 18.1 MHz and a constant field gradient of 22 T/m. The correlation map is illustrated in Fig. 3.15. The correlation distributions are situated under the dotted line, which is the

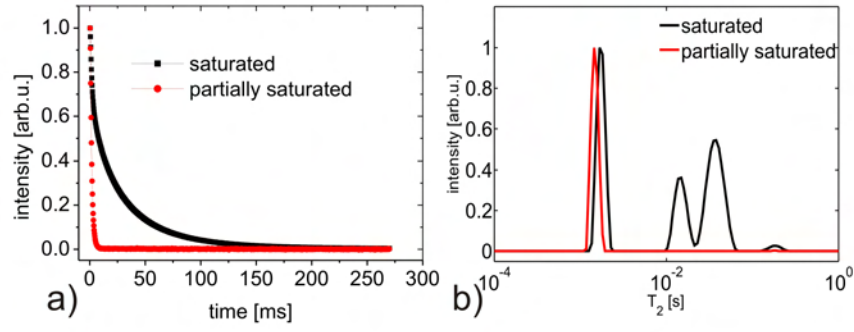


Figure 3.14: CPMG decays and the ILT distributions for the saturated and partially saturated aluminium oxide sample. a) CPMG decays showing that the spins relax much faster in the partially dried sample. b) ILT relaxation distributions for both samples. Only the water inside the pores remains.

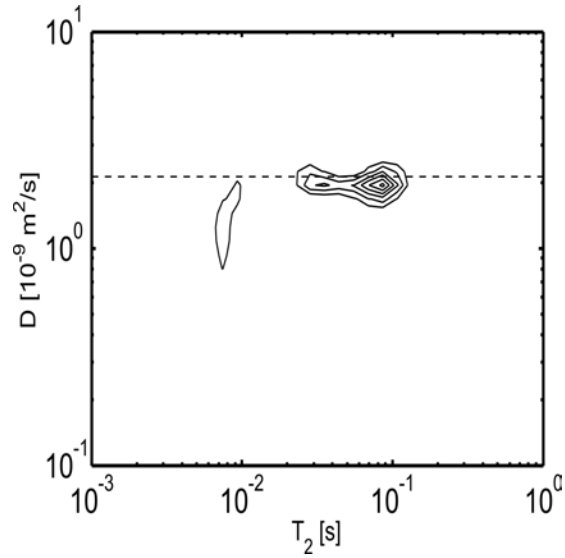


Figure 3.15: D - T_2 correlation map for the water saturated aluminium oxide system. Restricted diffusion is observable.

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line corresponding to the self-diffusion coefficient of water, $2.13 * 10^{-9} m^2/s$. This means that the water in the system is restricted, more restricted in the pores than in-between the particles and even the so considered bulk water. So the water that is the most restricted is situated inside the pores.

3.5 Conclusion

A general analytical solution for multi-site exchange NMR was presented, with a focus on a simple two-site exchange problem. Even though the theory allows studies on n -site systems, the more sites exist the more complicated gets the problem since we have more and more independent parameters. In order to understand the simple systems, experimental results were compared to simulated ones for a simple two-site system made of water saturated silica particles. T_1 - T_2 and T_2 - T_2 exchange correlation maps were compared. Moreover, the integrals of each relaxation distributions and their cross peaks were compared for different mixing times. The overlapping of the two evolutions obtained by simulations and experiments is observable. Complementary experiments were made at different temperatures for the same system. A temperature dependence of the main diagonal peaks integrals and the cross peaks integral could be noticed and compared with the simulation results. From the dependence of the reaction rate on temperature the activation energy could be extracted.

These 2D exchange experiments were tested further on a more complex system, a water saturated aluminium oxide system, which showed a more complex relaxation sites distribution. As the simulations showed that for more than two site systems, asymmetric or symmetric exchange maps can occur, for this system a symmetric 2D correlation map was obtained. In the same way as for the silica system, the peaks integral evolution over the mixing times was extracted. With these two types of information a multi-site system is expected, as the simulations showed that the maps can be symmetric just if the relaxation rates are equal. This is not the case here. In comparison with the simple system, the mixing time period is affected by the longitudinal relaxation time. This has the consequence that the integral values obtained for the cross peaks start to decrease towards

3.5 Conclusion

zero for too long mixing times. The restricted diffusion of the water molecules inside this sample is evidenced by the D - T_2 correlation map.

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Chapter 4

Natural porous systems

Soils possess a complex composition with high variability in their properties and appearance. To the most important soil function belong filtering of water on its way down to aquifers and controlling the supply of plant nutrients. They relate to the structure of the pore system, which is characterized by the pore size distribution, pore connectivity, water content, and water dynamics. The understanding of the water transport dynamics is important in water resource management in agriculture, industry, household and environment.

4.1 Introduction

Nuclear magnetic resonance has proven to be suitable to investigate water in natural porous media, since NMR methods are non-invasive and directly probe the substance of interest, water. In the past, such natural pore systems have mainly been characterized by one-dimensional NMR relaxometry (Dunn 02, Todoruk 03, Kleinberg 05, Schaumann 05, Pohlmeier 09, Stingaciu 09), leading to pore size distribution functions by using the well-established Brownstein–Tarr relationship in combination with a uniform surface relaxivity as scaling parameter (Hinedi 97, Barrie 00). One drawback of soil is its heterogeneity in pore size, pore geometry, mineral content, and organic material content. This complicates the interpretation of relaxation time distribution functions, which may contain modes resulting from different pore sizes confounded by modes resulting from surface relaxation effects. Since a few years new techniques are used which

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can characterize much better the porous system, the two- and three-dimensional inverse Laplace correlation NMR methods (Lee 93, Godefroy 03, McDonald 05, Washburn 06, Song 08, Arns 07). In the 2D exchange NMR experiments, the magnetization of a spin system is monitored successively in two different time periods, called the encoding and detection period separated by a so-called mixing, exchange, or storage time, t_m as shown in Fig. 2.5 in chapter 2.4.3. These methods yield information far more detailed than the respective one-dimensional experiments. At present, longitudinal relaxation time T_1 - T_2 , diffusion D - T_2 , T_2 - T_2 and D - D correlation methods are gaining popularity for studying artificial porous media and rocks (Song 02b, Washburn 06, Hürlimann 02).

In T_2 - T_2 maps, the transverse relaxation time distributions from the first and the second encoding periods are plotted vs. each other. We have determined that this simplistic view can be used for a qualitative analysis, although it is flawed by the fact that the encoding and mixing times are on the same order of magnitude so that a quantitative analysis needs to be based on computer simulations of the experimental data (Landeghem 10). In this simple picture, water molecules that remain in or return to the same pore-size class during the mixing interval contribute to peaks on the diagonal, while cross-peaks indicate water molecules that have exchanged between pores of different sizes. Performing such experiments in a low magnetic field of 2 MHz are advantageous because heterogeneous structures are probed, which gives rise to local distortions of the magnetic field with so-called internal magnetic-field gradients, which may be small or even negligible at low magnetic field.

For the first time, T_2 - T_2 exchange experiments were applied to water-saturated soil systems to probe the water dynamics in the pore space. This project is part of a transregional collaborative research project (SFB TR32) between four different universities from Nord-Rhine Westphalia. The overall aim to find patterns in soil-vegetation-atmosphere systems, and to combine the information from the microscopic scale up to the macroscopic scale, in order to be able to predict water, CO₂ and energy transfer between these different scales. Our research group was involved in characterization of flow, transport and structure of saturated and unsaturated soils at the pore to meter scale. For this purpose 1D and 2D

NMR relaxation experiments were performed. In the past, fast transverse relaxation times have been assigned to immobile and clay-bound water in porous rocks (Straley 97). The question is whether two-dimensional relaxometry can give information on whether such water is really immobile or exchanges after all.

4.2 Saturated soils

Three repacked soil materials were chosen that represent pure fine sand and two silt loams with different sand and silt contents. The fine sand was purchased from Quarzwerke GmbH, Frechen, Germany, and the soil materials were from Merzenhausen and Selhausen, Germany. The names refer to the locations where the materials are taken from. The textural composition of each sample is presented in Table 4.1. To have an idea how different the textural compositions can be, CT images are shown in Fig. 4.1. The material was air dried, sieved in order to remove stone particles and to obtain an aggregate size smaller than 2 mm, packed into 9-by 100-mm cuvettes, centrifuged for 30 min at $1000 \times g$, and finally saturated with tap water for 48 h.

Table 4.1: Soil texture according to the USDA and maximum volumetric water content (θ) of the investigated soil samples.

Soil type	Sand %	Silt %	Clay %	($\theta_{\max}$) cm^3/cm^3
FH31 fine sand	100	0	0	0.36
Selhausen silt loam	15	68	17	0.4
Merzenhausen silt loam	4	81	15	0.43

1D experiments

Initially, one-dimensional CPMG experiments were performed to obtain transverse relaxation decays. The parameters used for the experiments for each soil type are listed in Table 4.2. The raw data were fitted with multi-exponential decays and subsequently inverted by one-dimensional inverse Laplace transformation (Godefroy 03) to obtain a transverse relaxation time distribution (Fig.

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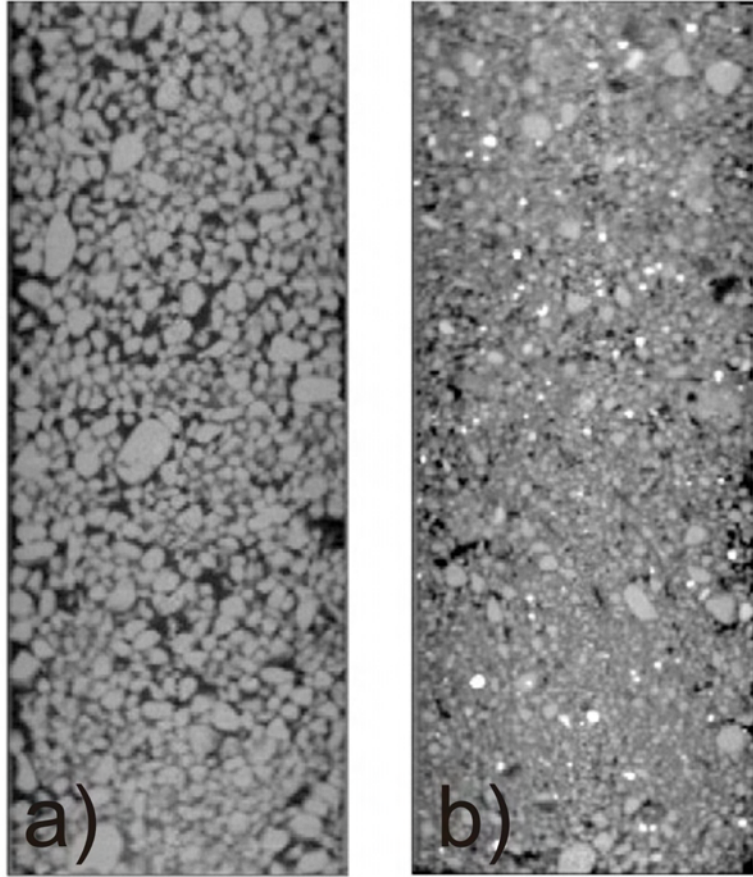


Figure 4.1: CT images of soil samples showing the different textural compositions. a) CT image for sand FH31. b) CT image for Merzenhausen silt loam. The resolution is 0.033 mm per pixel, the field of view is 6 by 15 mm. Data from Dr. Sabina Haber-Pohlmeier.

Table 4.2: NMR parameters used for the experiments and the obtained longitudinal as well as transverse relaxation times.

Soil type	NMR parameter			Relaxation times	
	t_E [ms]	nr. echoes	nr. scans	T_2 [ms]	T_1 [ms]
Sand FH31	0.5	7000	4	220/700	840/1650
Selhausen	0.2	600	8	1.3/7/40	21/114
Merzenhausen	0.1	500	8	0.3/2/8.6	4/16

4.2). Moreover, such experiments reveal information about pore connectivity. The relaxation times listed in Table 4.2 were obtained by fitting the raw data. The

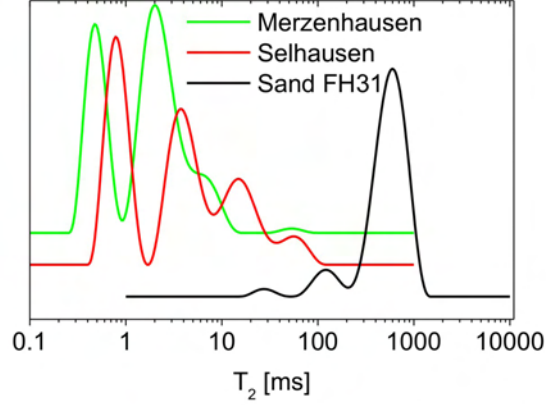


Figure 4.2: One-dimensional T_2 distribution functions obtained by inverse Laplace transformation for different water-saturated soil types : the black line is the FH31 fine sand, red line represents Selhausen silt loam, and the green line Merzenhausen silt loam.

longest T_2 relaxation distribution obtained for FH31 fine sand was also the simplest distribution, with maxima at 230 and 700 ms. The soil materials exhibited relaxation times much faster than those of the fine sand, covering ranges between 1 and 50 ms for the Selhausen soil and 0.5 to 10 ms for the Merzenhausen soil. Furthermore, the simple behavior found for the sand changed for the soils to multi-modal distribution functions. Comparing the relaxation times with the textural composition of each sample, it can be seen that with increasing silt and clay content, the T_2 relaxation times decreased significantly. The relaxation times decreased with smaller pore sizes under the assumption of a constant surface relaxivity (Barrie 00).

Another set of experiments was to check if the transverse relaxation is or not affected by the diffusion term, as shown in equation 4.1:

$$\frac{1}{T_{2,\text{app}}} = \frac{1}{T_{2,\text{bulk}}} + \rho_2 \frac{S}{V} + \frac{D}{12} (\gamma G t_E)^2 \quad (4.1)$$

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where, ρ_2 represents the surface relaxivity, S/V represents the surface to volume ratio, D the diffusion coefficient, G the gradient strength, γ the gyromagnetic constant and t_E the echo time. In principle, T_2 experiments will be done at different echo times, and plot $1/T_2$ against the echo time square, t_E^2 . A linear dependence is expected, and from the slope of the fit the gradient strength could be extracted, under the assumption that D is constant and can be estimated reliably. Working at low fields will help to reduce the internal gradient strength of the samples, as known that the higher the magnetic field strength the higher the internal gradient effect. This is due to the magnetic susceptibility contrast between the grain material and the pore fluid (Kleinberg 90). Based on a model by Glasel (Glasel 74), Kleinberg (Kleinberg 90) mentions the average gradient for a region close to a grain wall as

$$G = \mu_0 B_0 \Delta\chi / 4R_g, \quad (4.2)$$

where μ_0 is the magnetic permeability of vacuum, B_0 the applied magnetic field strength, $\Delta\chi$ the difference between the volume magnetic susceptibilities of the grain and the external fluid and R_g the radius of the grain. Hürlimann mentioned in his paper (Hürlimann 98) the upper limit for the local effective gradient as :

$$G_{\text{eff,max}} \approx (\frac{\gamma}{D_0})^{1/2} (\Delta\chi B_0)^{3/2} \quad (4.3)$$

The largest possible effective gradient is scaling with a power of 3/2. The reason is that the largest effective gradient comes from the smallest pore.

First the dependence was checked for bulk water. In this way the inhomogeneity of the magnet was tested. The result is shown in Fig. 4.3. A linear dependence is observed and the field gradient was estimated using eq. 4.1 to be 180 mT/m. The T_2 dependence for the sand FH31 is shown in Fig. 4.4. In Fig. 4.4a the relaxation distributions for different t_E are over plotted and in Fig. 4.4b the dependence of $1/T_2$ with t_E^2 is represented. In case of the logarithmic mean T_2 (marked as $T_{2\text{LM}}$ in the figure), the whole relaxation distribution has been taken into account, whereas for the other dependencies (marked as T_{2l}) the value at maximum height has been taken. In case of the relaxation distributions a shift of the longer T_2 component towards shorter values is observed with increasing echo time. The dependence of $1/T_2$ with t_E^2 presents a linear dependence for the

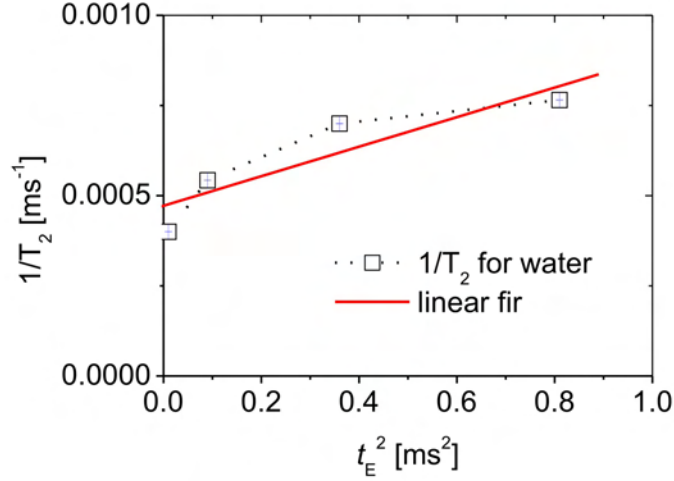


Figure 4.3: Influence of the echo time on the T_2 relaxation distribution for bulk water. The line between the points serves as a guide to the eye.

logarithmic mean T_{2LM} value and for the long T_{2l} value. The field gradient was estimated using eq. 4.1 to be 0.269 T/m for T_{2LM} and 0.227 T/m for T_{2l} . These values are not so high if we extract from them the value that already exists representing the field inhomogeneity.

In case of the Selhausen silt loam, Fig. 4.5, the same behavior is observed. But this time, the majority of the spins are trapped in the smaller pores, which is influencing the relaxation distribution function in a different way. Fig. 4.5a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.5b the dependence of $1/T_2$ with t_E^2 . For this type of soil, it looks like all the different T_2 components are shifting to higher values with increasing echo time, but still the longest T_2 value, T_{2l} obtained for the shortest echo time is not exceeded in either of the distributions. In case of the $1/T_2$ dependence with t_E^2 , a linear dependence is noticed only for the longest component, T_{2l} , whereas the other components show an exponential decay behavior, marked with T_{2sh} for the shortest component, T_{2m1} for first middle component, T_{2m2} the second middle component and T_{2LM} the relaxation distribution logarithmic mean value. The fastest decay rate is noticed again for the shortest T_2 component. The field

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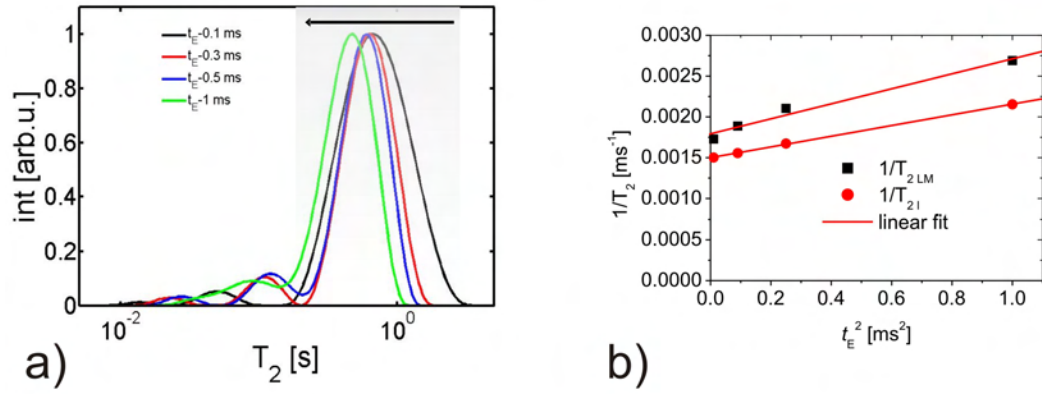


Figure 4.4: Sand FH31. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 .

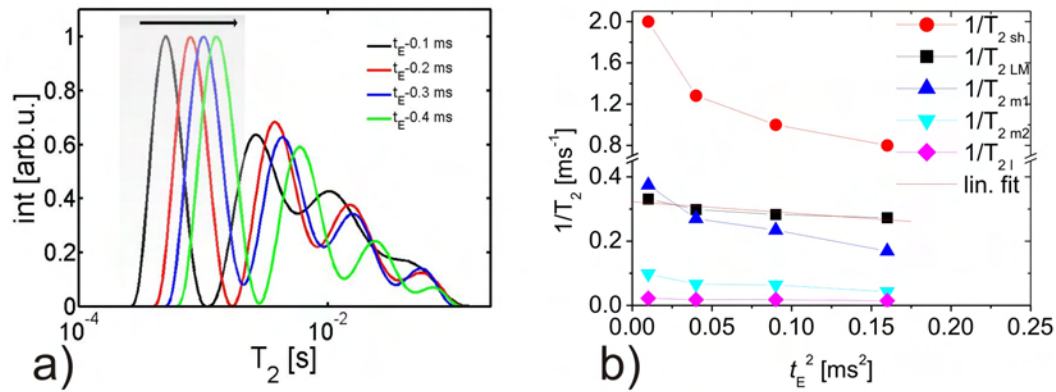


Figure 4.5: Selhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The lines connecting the points are just as a guide to the eye.

gradient was estimated using eq. 4.1 to be 5.25 T/m for T_{2LM} .

The T_2 dependence on t_E^2 for Merzenhausen silt loam is presented in Fig. 4.6. Fig. 4.6a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.6b the dependence of $1/T_2$ with t_E^2 . It behaves in the same way as for the Selhausen silt loam. In this case, all the components shift to higher T_2 values. The $1/T_2$ dependence with t_E^2 , show linear dependence just for the longest

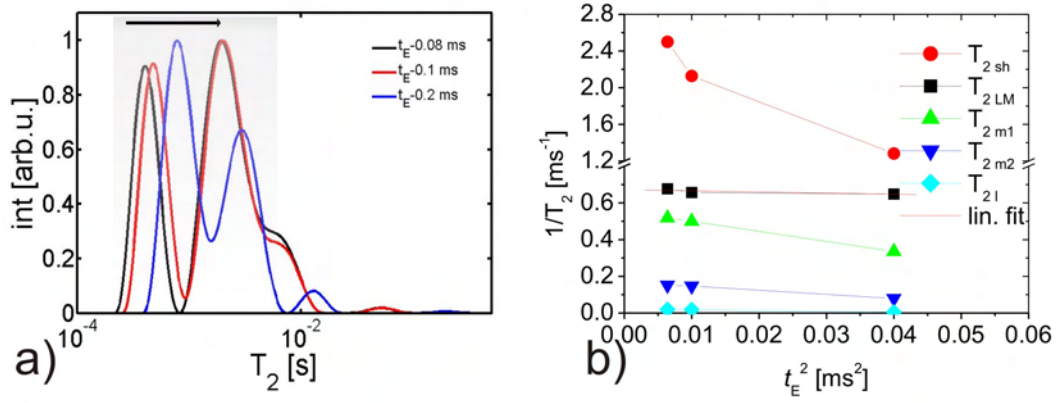


Figure 4.6: Merzenhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The lines connecting the points are just as a guide to the eye.

component, T_{2l} , and all the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 6.98 T/m for T_{2LM} .

From the results shown above, it can be seen that sand behaves differently than the other two soil types. Actually the sand behaves as we would expect it. The relaxation rate, $1/T_2$ present a linear dependence with t_E^2 as predicted from equation 4.1. Moreover the gradient strength extracted is not even double the value obtained for pure water measurements. Taking a closer look to the T_2 distribution function dependence with t_E , for the two silt loams, a difference is noticed. The distribution function components are shifting to higher values with increasing the echo time. This will affect of course also the $1/T_2$ dependence with t_E^2 . A linear dependence is noticed just for the long relaxation components. But

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this linear dependence has a negative slope, and not a positive as for sand. The shorter the relaxation components get, the more curved this dependence gets. A possible reason for this might be the non-linear behavior of the field. The CPMG signal is not able to refocus the effects coming from this quadratic field, this can be achieved with other pulse sequences (Blümich 05a). Another explanation would be the exchange of magnetization components for longer t_E^2 . This would explain the decrease in relaxation rates. This results make us understand that the texture and connectivity of sand is simple and different from that of the two silt loams. In the case of the sand we can assume that we have grains but in the case of the silt loams we find a more heterogeneous distribution, a sponge-like matrix with grains but mostly thin films.

2D experiments

To test if the two-dimensional exchange experiment and the data processing by the two-dimensional inverse Laplace transformation program worked reliably and to ascertain that the observed cross peaks were not artifacts from poor signal/noise ratios, a reference experiment with two stacked, individual, 5 mm NMR tubes was performed. Each tube was filled with different aqueous CuSO_4 solutions, doped water, which gave different T_2 values. Figure 4.7 shows the results from the relaxation–relaxation exchange experiments that were performed for mixing times of 1 and 30 ms. As expected, only diagonal peaks were observed and no cross peaks were detected. The loss of intensity of the fast diagonal peak at $T_2 \sim 10$ ms at a mixing time of 30 ms is remarkable. This was due to the fact that longitudinal relaxation is taking place during the mixing time, during which the magnetization was stored along the z axis.

Once the measurement procedure was tested, relaxation–relaxation exchange experiments were conducted for different mixing times for all samples (Fig. 4.8). The numbers of echoes used in the first encoding period was increased logarithmically in 64 steps so that the maximum first encoding time covered the same range as the second encoding time at the end. The mixing times used between the two T_2 encoding times were varied from 0 ms up to 1 s for the FH31 sand, up to 55 ms for the Selhausen soil, and up to 10 ms for the Merzenhausen soil.

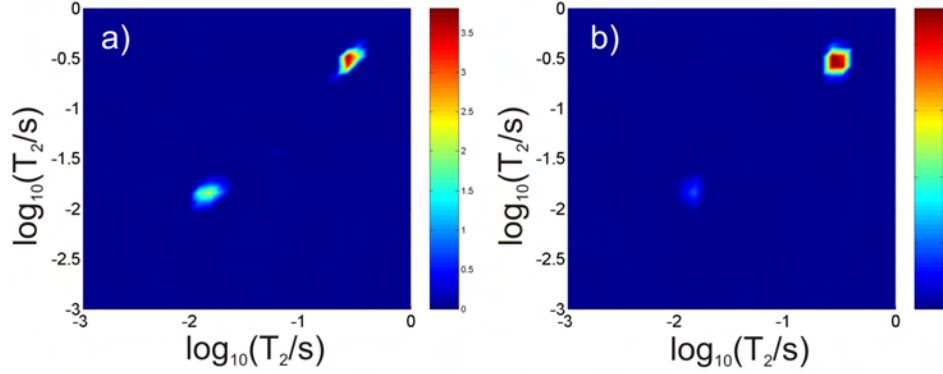


Figure 4.7: Test of the relaxation exchange method : transverse relaxation–relaxation (T_2 - T_2) correlation experiments of two aqueous CuSO_4 solutions measured with a) 1 ms and b) 30 ms mixing times. The colors represent the intensity of the signal (arbitrary units).

Each line of the two-dimensional data matrix was scanned eight times, and the repetition time between scans was set to five times the longitudinal relaxation time T_1 . The experiments lasted between 2 and 30 min depending on the duration of the mixing time and the repetition time. In the corresponding exchange distributions, molecules residing in their original environment gave rise to signals along the diagonal, while those changing their environment during the mixing time produced off diagonal peaks. Summarizing the results of the T_2 - T_2 correlation maps for the systems with zero mixing time (Fig. 4.8a–c), all signals appear as diagonal elements and the projections (not shown here) to the coordinate axes are similar to the one-dimensional distributions shown in Fig. 4.2. As expected, no exchange is observed for zero mixing time.

For the FH31 sand, the exchange of water molecules is barely observed at mixing times of both 200 and 700 ms by the presence of very weak cross peaks (Fig. 4.8d and g). The relaxation distribution function of sand was quite homogeneous, with only a small fraction of narrow pores, as evident from Fig. 4.2a and confirmed by T_1 relaxometry data (Pohlmeier 09). This is why exchange processes in the sand were caused mainly by molecular movement within pores of the same

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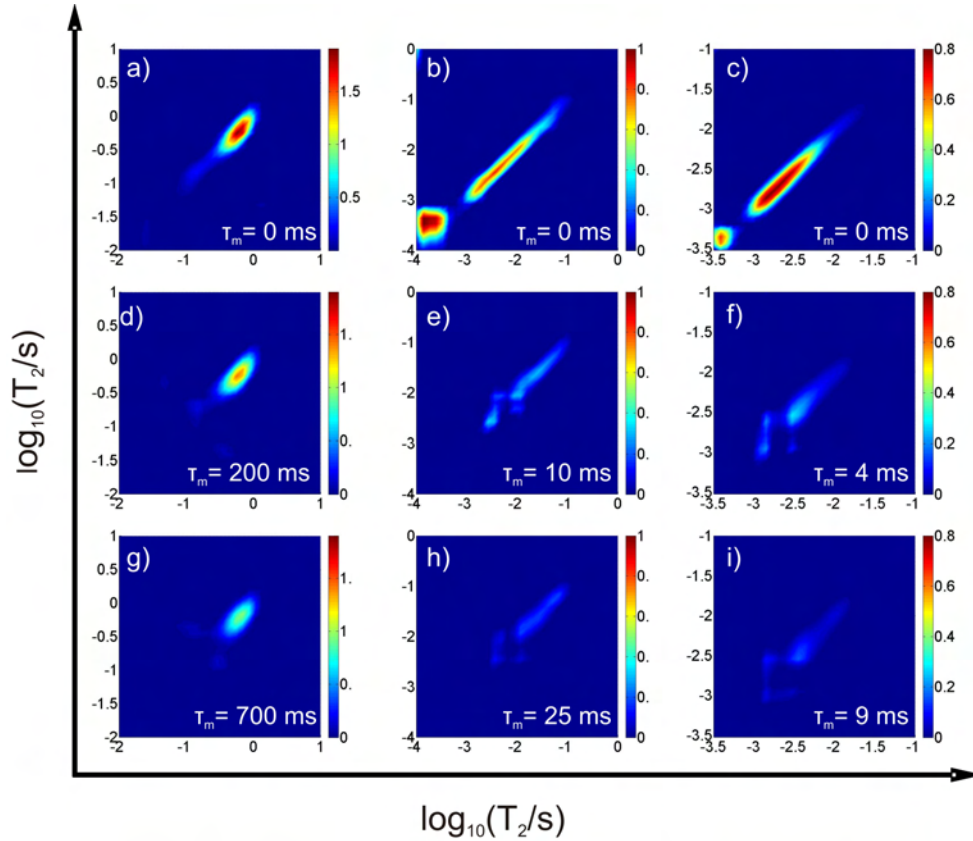


Figure 4.8: Two-dimensional transverse relaxation–relaxation (T_2 – T_2') correlation spectra obtained for the different soil types : a) FH31 fine sand, b) Selhausen silt loam, and c) Merzenhausen silt loam at mixing time $t_m = 0$ ms; d) FH31 fine sand at $t_m = 200$ ms; e) Selhausen silt loam at $t_m = 10$ ms; f) Merzenhausen silt loam at $t_m = 4$ ms; g) FH31 fine sand at $t_m = 700$ ms; h) Selhausen silt loam at $t_m = 25$ ms; and i) Merzenhausen silt loam at $t_m = 9$ ms. The colors represent the intensity of the signal (arbitrary units).

size class.

Figures 4.8e and h present the distributions for the Selhausen soil at mixing times of 10 and 25 ms. The observed values of T_2 are widely distributed and range between 1 and 50 ms, showing up to four components. Unlike the results for the FH31 sand, cross peaks appear at mixing times of 10 and 35 ms, indicating strong exchange between the shortest components and weak exchange to sites with longer relaxation times. These cross peaks identify migration of water molecules between relaxation environments. It is also noted that the distribution matrix is rather asymmetric, which is a strong indication of multi-site exchange (Landeghem 10). The fastest component on the diagonal completely vanishes at long mixing times (Fig. 4.8h) due to T_1 relaxation effects.

Like the Selhausen soil, the Merzenhausen soil shows a multi-modal distribution with short T_2 components ranging from 0.5 to 10 ms (Fig. 4.8c, f, and i). As in the Selhausen soil, cross peaks between the shortest components arise already at a mixing time of 2 ms (data not shown here) and became stronger at a mixing time of 4 ms (Fig. 4.8f). At a mixing time of 9 ms, the shortest diagonal peaks vanish due to a short T_1 relaxation of about 2 ms (Pohlmeier 09), and exchange with the longer components becomes observable. In addition, with increasing mixing times, the two-dimensional spectra become slightly asymmetric, which indicates the onset of multi-site exchange.

For all samples, the evolution of each T_2 component integral over the mixing time was tried to be plotted, but problems occur when trying to determine each component limits in the 2D correlation map. In the case of these soils we cannot talk any more about relaxation sites, because we have a distribution of relaxation sites, and the theory presented in Chapter 3 needs to be extended. This work is now in progress at another research group within this transregional research project TR32 (Mohnke 10).

4.3 Unsaturated soils

1D experiments

In case of the unsaturated soils, differences can be observed, as for all of these three samples, the long T_2 components vanish with decreasing the saturation level. This fact is evidenced very well in Fig. 4.9, where for each soil type the 1D transverse relaxation distribution is shown at different saturation levels. In case

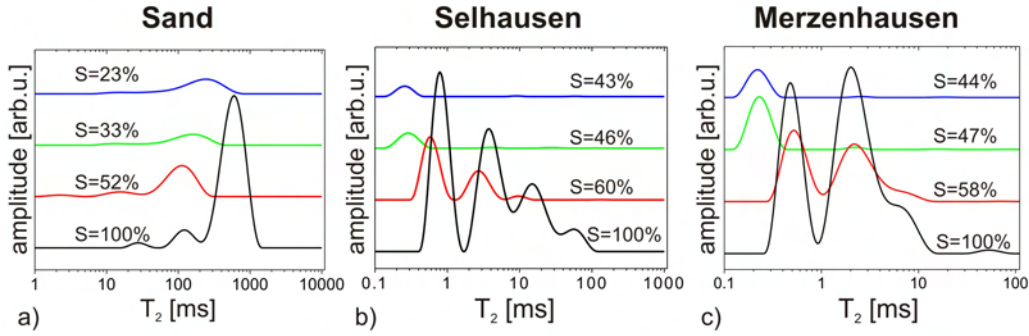


Figure 4.9: 1D relaxation distributions for different saturation levels for each soil type. a) Sand FH31, the T_2 values are shifting to shorter values for lower saturation levels. b) Selhausen silt loam, the relaxation distribution components present at fully saturated soil are vanishing with decreasing saturation level. Only the fastest component remain. c) Merzenhausen silt loam, the same behavior as for Selhausen soil is noticed.

of the sand FH31, the T_2 values are shifting to shorter values for lower saturation levels. The amplitude is decreasing dramatically as the hydrogen spins are getting less and less. This is in agreement with the data of Stangaciu (Stingaciu 09). For the Selhausen silt loam, the relaxation distribution components present at fully saturation level are vanishing with decreasing saturation level. Only the fastest component remain at a very low amplitude, which is shifting slightly to even faster values. For the Merzenhausen silt loam, the same behavior as for Selhausen soil is evidenced.

The same dependence of $1/T_2$ on t_E^2 was tested for the unsaturated samples too. For sand FH31, these dependence is shown in Fig. 4.10. Fig. 4.10a presents

the behavior of relaxation distribution functions for different t_E and Fig. 4.10b the dependence of $1/T_2$ with t_E^2 . In Fig. 4.10a a shift of the longer T_2 component towards shorter values is noticed. Plotting the $1/T_2$ against t_E^2 , a linear dependence for the logarithmic mean T_2 value and for the long T_2 value is noticeable. The field gradient was estimated using eq. 4.1 to be 0.414 T/m for T_{2LM} and 0.385 for T_{2l} . This values is higher than the value obtained for the saturated sample.

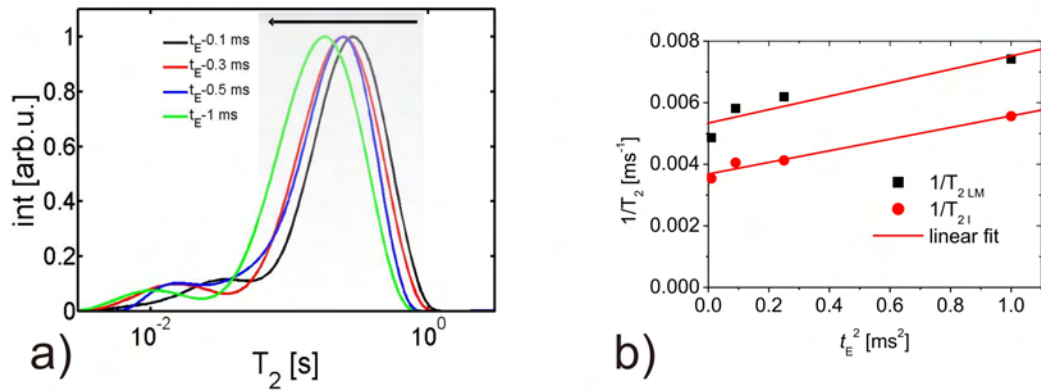


Figure 4.10: Partially saturated Sand FH31. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 .

In case of the Selhausen silt loam, the same behavior is observed as for the saturated case. Fig. 4.11a presents the behavior of relaxation distribution functions for different t_E and Fig. 4.11b the dependence of $1/T_2$ with t_E^2 . The majority of the spins are in the smaller pores trapped, which is influencing in a different way the relaxation distribution. In Fig. 4.11a all the different T_2 components are shifting to higher values with increasing the echo time. In case of the $1/T_2$ dependence with t_E^2 , a linear dependence is noticed only for the longest T_2 value, whereas the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 2.39 T/m for T_{2LM} . This might be incorrect as theoretically it should have a higher value than for the saturated case.

The Merzenhausen silt loam, Fig. 4.12, behaves in the same way as the Selhausen silt loam. Fig. 4.12a presents the behavior of relaxation distribution functions

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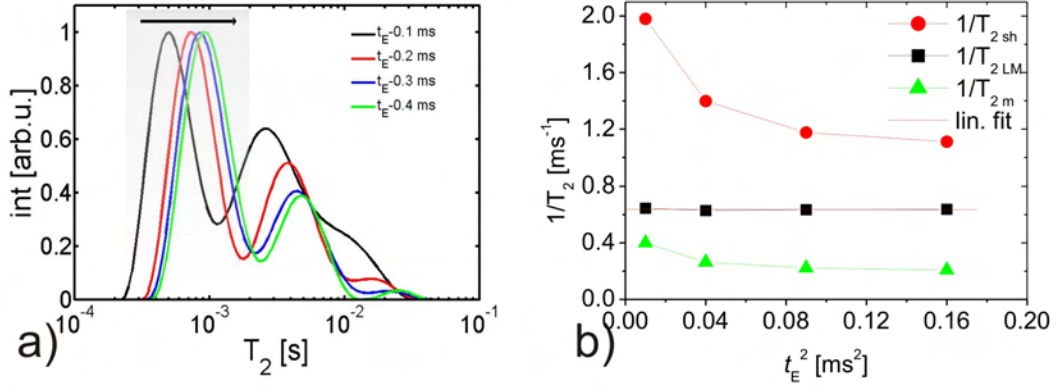


Figure 4.11: Partially saturated Selhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence on t_E^2 . The connecting lines serve as a guide to the eye.

for different t_E and Fig. 4.12b the dependence of $1/T_2$ with t_E^2 . In Fig. 4.12a it is noticeable how all the relaxation components shift to higher T_2 values. The $1/T_2$ dependence on t_E^2 , show linear dependence only for the longest T_2 value, and all the other components show an exponential behavior. The fastest decay rate is noticed again for the shortest T_2 component. The field gradient was estimated using eq. 4.1 to be 8.76 T/m for T_{2LM} .

The results obtained for the partially saturated samples are in good agreement with the results obtained for the saturated samples. The behavior of the silt loams is different from the one for sand, and the gradients extracted are even higher. This is a confirmation for the presence of thin films in the silt loams, that is getting more accentuated when these are not fully saturated.

2D experiments

The parameters used in obtaining the 2D relaxation exchange maps are listed in Table 4.3. These parameters as well as the 2D maps will show the comparison between the fully saturated and partially saturated soil sample.

For the FH31 sand, 2D relaxation exchange maps are plotted for two saturation levels and in each case for three different mixing times in Fig. 4.13. With

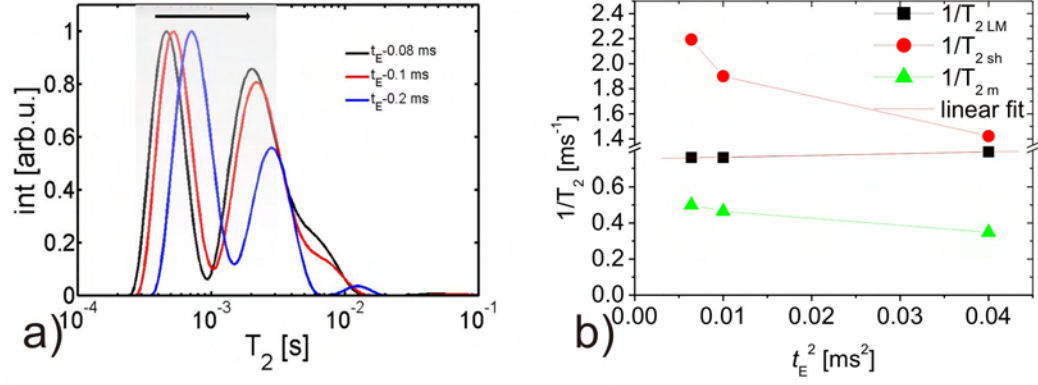


Figure 4.12: Partially saturated Merzenhausen silt loam. Influence of the echo time on the T_2 relaxation distribution. a) T_2 relaxation distributions for different echo times. b) $1/T_2$ dependence with t_E^2 . The connecting lines serve as a guide to the eye.

Table 4.3: NMR parameters used for the experiments and the obtained longitudinal as well as transverse relaxation time values.

Soil type and Saturation		NMR parameter			Relaxation time	
		t_E [ms]	nr. echoes	nr. scans	T_2 [ms]	T_1 [ms]
Sand	$S = 100\%$	0.5	7000	4	220/700	840/1650
FH31	$S = 23\%$	0.5	6500	12	88/330	160/750
Selhau	$S = 100\%$	0.2	600	8	1.3/7/40	21/114
sen	$S = 60\%$	0.2	300	24	0.85/4/14	1.4/11
Merzen	$S = 100\%$	0.1	500	8	0.3/2/8.6	4/16
hausen	$S = 58\%$	0.1	300	16	0.4/2/7	7

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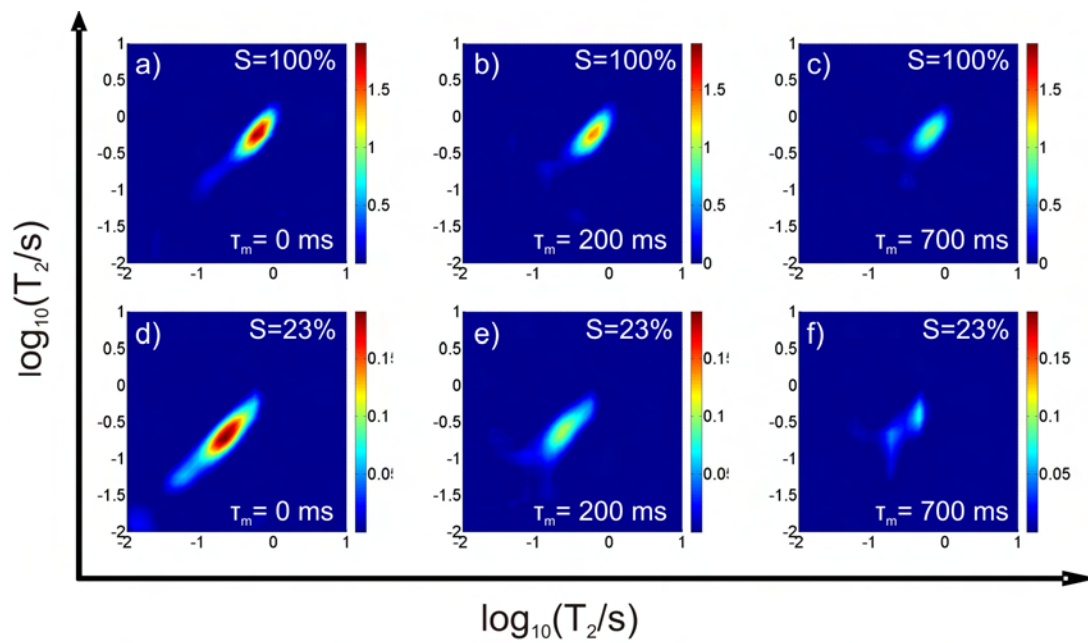


Figure 4.13: Two-dimensional T_2 - T_2 correlation spectra obtained for sand FH31 :a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 200$ ms; c) $S = 100\%$ and $t_m = 700$ ms; d) $S = 23\%$ and $t_m = 0$ ms; e) $S = 23\%$ and $t_m = 200$ ms; f) $S = 23\%$ and $t_m = 700$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

decreasing water content the relaxation times are shifted to faster values. Exchange processes are observed but weakly, which indicates the fact that water molecules move within pores of the same size class.

In the case of the silt loam Selhausen, the 2D relaxation exchange maps are plotted as well for two saturation levels and in each case for three different mixing times in Fig. 4.14. Different pore size classes are observable and moreover,

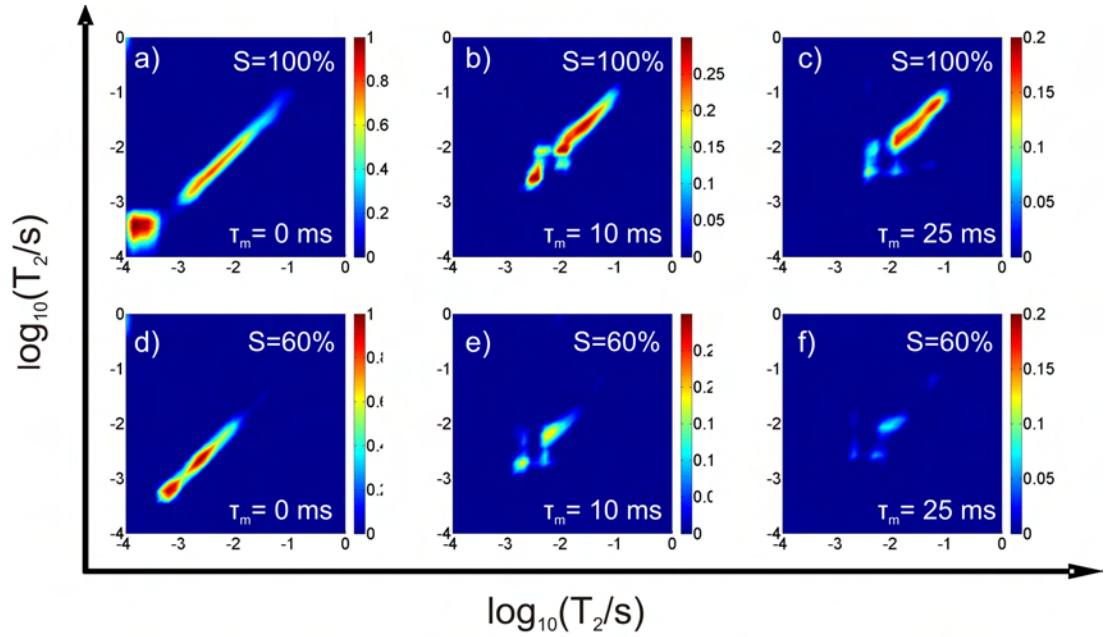


Figure 4.14: Two-dimensional T_2 - T_2 correlation maps obtained for Selhausen silt loam : a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 10$ ms; c) $S = 100\%$ and $t_m = 25$ ms; d) $S = 60\%$ and $t_m = 0$ ms; e) $S = 60\%$ and $t_m = 10$ ms; f) $S = 23\%$ and $t_m = 25$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

distribution of relaxation times are noticed. Water exchange occurs between the smallest pore classes. With decreasing water content first the more mobile water, with longer T_2 values vanishes, and at the same time the relaxation times shift to faster values.

For the Merzenhausen silt loam, the 2D relaxation exchange maps are plotted for two saturation levels and in each case for three different mixing times in Fig. 4.15. Water exchange occurs, as well as for the Selhausen silt loam, between the

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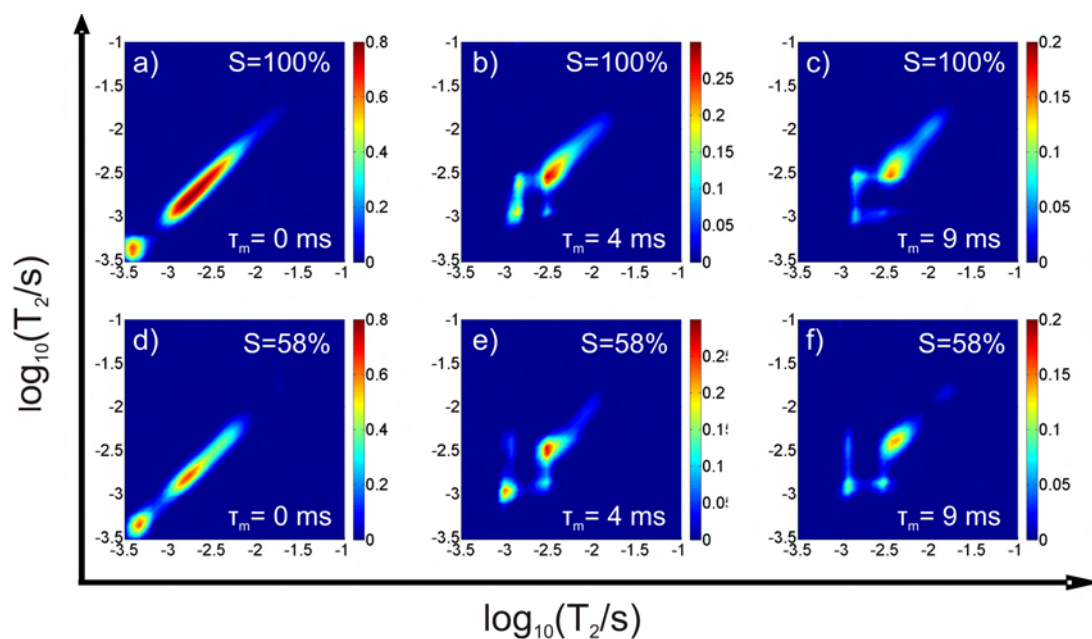


Figure 4.15: Two-dimensional T_2 - T_2 correlation spectra obtained for sand Merzenhausen silt loam :a) Saturation (S) $S = 100\%$ and $t_m = 0$ ms; b) $S = 100\%$ and $t_m = 5$ ms; c) $S = 100\%$ and $t_m = 9$ ms; d) $S = 58\%$ and $t_m = 0$ ms; e) $S = 58\%$ and $t_m = 5$ ms; f) $S = 58\%$ and $t_m = 9$ ms. The color bar was kept fixed in order to be able to compare the amplitudes.

smallest pores. Unsaturated samples present a shift in the relaxation times but only at saturations under 55 %. Water is predominantly lost from the big pores where the water is more mobile.

4.4 Conclusion

The results obtained from these experiments prove that two-dimensional exchange relaxometry methods are suitable to observe moisture migration in loose, natural, porous media like soils. For sand, only weak and slow exchange was found, while in more complex, textured soils, exchange of water molecules was observed for short mixing times between the fastest components and between slower components at longer mixing times. It is not surprising that, in such heterogeneous systems, the water molecules exchange between more than two sites as the mixing time is prolonged. By such two-dimensional experiments, the relaxation modes identified in one-dimensional experiments can be correlated and interpreted in terms of distances travelled by water molecules during the exchange experiment once the exchange rates are quantified by numerical simulation of the experimental exchange maps. This is in line with the assumption incorporated in the Brownstein Tarr model, where the observed relaxation modes can be assigned to pores of different sizes. In particular, it was shown that the existence of cross peaks identifies the motion of the water molecules between different pore size classes in a qualitative analysis. If exchange between two pore size classes is observed, water in these classes should be regarded as mobile, not immobile.

Depending on the soil texture the pore size distribution varies from a simple distribution, obtained for sand, to multi-modal distributions, obtained for the two different silt loams. A non linear behavior, in case of the silt loams, for the dependence of the relaxation rates with t_E^2 was observed which give us an idea about the texture and pore connectivity inside the sample. Water exchange is observable mostly within the small pores. Exchange within the big pores may occur, but due to the T_1 relaxation the exchange between large pores is hard to observe and the small components vanish at longer mixing times. Once the saturation level becomes lower the water from the bigger pores vanish and the T_2 values are shifting towards shorter values.

4. NATURAL POROUS SYSTEMS

Chapter 5

Complex porous systems

Concrete is a multi-length scale composite material first invented by the Romans. The name concrete comes from the Latin “concretus”, which means to grow together. Today, concrete has become ubiquitous — about five billion tones of concrete are used worldwide each year. The cement used mostly in today’s concrete is called Portland cement. The process to produce Portland cement was invented by Joseph Aspdin in the early 1800s in England. The name Portland may have been originally a marketing ploy, as Portland building stone was very popular in England at that time and Aspdin may have wanted people to favorably compare concrete made with his cement to the popular building stone (Garboczi 96).

It is important to remember that cement is the powder that reacts with water to form cement paste, a hard, solid material that forms the matrix for the concrete composite. The addition of sand (fine aggregates) that are up to a few millimeters in diameter forms mortar, and the addition of rocks (coarse aggregates) of up to a few centimeters in diameter forms concrete. It has been known that concrete is a porous material whose properties depend on its pore space. There are many different kinds of pores in concrete, ranging from the air voids that are entrapped in the mixing process, which can be quite large (up to a few millimeters in diameter) to the capillary pores, which are essentially the space occupied by the leftover water from mixing, down to the nanometer-scale pores that exist in some of the hydration products produced by the cement-water chemical reaction (Garboczi 96).

5. COMPLEX POROUS SYSTEMS

5.1 Introduction

In order to have a realistic view about how much an outdoor concrete element is exposed to the environment and its influence on the quality and durability of the material, an accurate knowledge of the time evolution of the local temperature and moisture distribution in the material is required. For this purpose, the German Federal Institute for Materials Research (BAM or Bundesanstalt für Materialforschung) leads several research projects throughout Germany in order to determine the moisture content both inside and at the surface of concrete. For moisture measurements they have used the MRE (multi-ring electrode) method developed at IBAC RWTH Aachen University (Brameshuber 08). It allows indirect moisture measurements through the electrolyte resistance of the concrete. In order to be able to convert correctly the resistance into moisture, numerous calibration tests need to be done. The most important error in the measurements arrives from the salt concentration. For a correct evaluation of the quality of the MRE measurements, NMR measurements were performed, where only the signal from the hydrogen atoms is recorded, which can be converted to moisture content.

The NMR measurements were executed with the Profile NMR-MOUSE[®] for on-site measurements as well as for laboratory measurements. On-site measurements were performed at the Gäubahntunnel, near Stuttgart and at the Hohenwarthe lock, near Magdeburg. The aim was to compare concrete with different composition, but also concrete that is exposed to different environmental conditions. For the laboratory studies, two concrete blocks with the same composition as the on-site concrete were studied.

5.2 Experimental

The Profile NMR-MOUSE[®] sensor used for the measurements was mounted on a computer-driven positioning device. The NMR sensor and the positioning device used were manufactured by ACT GmbH, Aachen, Germany. The spectrometer was a Kea spectrometer manufactured by Magritek Ltd, New Zealand, connected to a Dressler 1 kW rf amplifier. In order to acquire depth profiles, the distance between sensor and object was changed in appropriate steps and the sensitive

volume was shifted through the object along the depth direction. This method can be applied to all objects which contain hydrogen nuclei and are neither electrically conductive nor magnetic. Depending on the penetration depth, different sensors are available. The greater the penetration depth, the larger is the sensor. Standard depths are 3, 5, 10, and 25 mm. Since the magnetic field decreases with increasing distance from the sensor, the field strength of sensors with larger depth access is smaller than that of sensors for smaller depth access. At the same time, the intensity of the received signals decreases with increasing depth. This loss is balanced for sensors with high depth access by an increase of the sensitive volume in width and thickness. The NMR measurements at the two different sites as well as in the laboratory were performed with a Profile NMR-MOUSE® with 21 mm depth access at a proton resonance frequency of 13.86 MHz. The echo time was kept as short as possible for this penetration depth, 140 μ s. 16 echoes were acquired and each CPMG echo train was repeated 512 times. Each profile was recorded in about 1 to 2 hours, depending on the on-site conditions. For the laboratory measurements the conditions were kept similar, except the number of scans was doubled as we had more time available.

5.2.1 On-site measurements

The first of the on-site measurements was performed in August 2009 at the Gäubahntunnel. This is a 450 m long tunnel where the temperature and moisture has been monitored for 8 years with the MRE method. In October 2009, the lock measurements at Hohenwarthe took place. This lock is a double-chamber water-saving lock that achieves a difference in water elevation of 18.55 m. The monitoring of temperature and moisture has been ongoing since 2003 (Weise 10). The two experimental setups are shown in Fig. 5.1. The composition of the two concretes is similar, except for the fact that the aggregates that consist of sand, pebble and chippings are from the respective regions. With the help of mercury intrusion porosimetry the pore radius was determined. In the case of Gäubahntunnel concrete, the distribution is more monomodal with a peak maximum at 25 nm, a mean radius of 32 nm and a limiting radius of 50 nm. In the case of the Hohenwarthe concrete, a bimodal distribution is evidenced with the two

5. COMPLEX POROUS SYSTEMS

peak maxima at 3 nm and 7 to 16 nm, a mean pore radius of 7 to 9 nm and a limiting radius of 16 to 25 nm (Weise 10). These values indicate a much more compact concrete in the Hohenwarthe case. This is necessary as the concrete wall is occasionally under water.

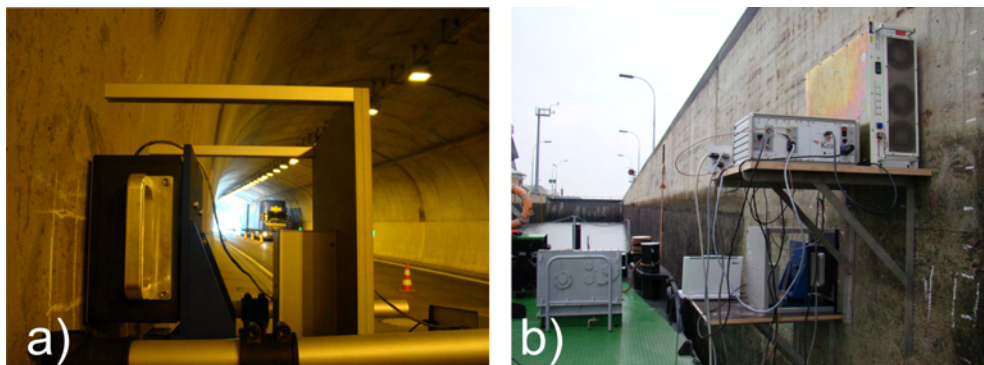


Figure 5.1: On-site measurements. a) The Gäubahntunnel setup, showing just the NMR-MOUSE positioned parallel to the measured point. b) The Hohenwarthe setup that needed to be fixed to the locks wall to avoid setup movements.

In case of the Gäubahntunnel, two NMR profiles were made: one at the entrance in the tunnel (marked as “Outside” in the Fig. 5.2 a and c) and another one at 120 m inside the tunnel (marked as “Inside” in the Fig. 5.2 b and d). To match the usually used units, the amplitude of the NMR signal was converted into grams of water per gram of dried cement. The conversion was done in the following way:

amplitude [g of w./g of dried c.] = (NMR signal / NMR signal of water) / density of concrete,

where the value for the density of concrete was evaluated before by the BAM to 2.1 g/cm³.

The results on the depth profiles and the extracted T_2 values for each depth are presented in Fig. 5.2. At each depth, a Carr Purcell Meiboom Gill (CPMG) measurement was performed. The real channel was recorded and fitted in order to extract the amplitude at zero echo time. Different fits were tried: single exponential, double exponential, Gaussian, etc. Ultimately, the best one was obtained

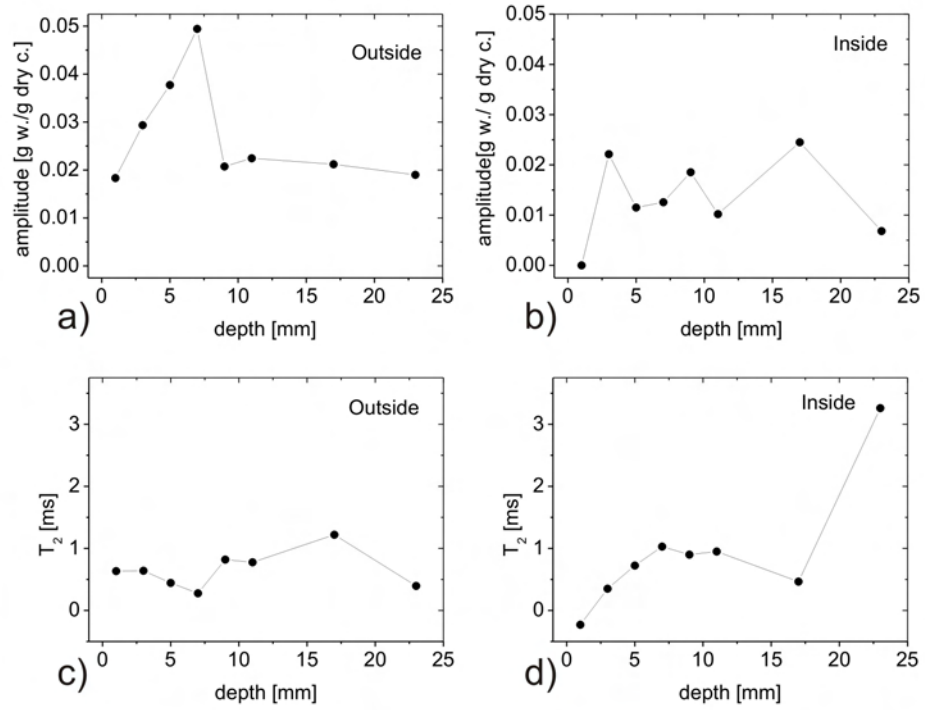


Figure 5.2: NMR depth profiles and extracted T_2 profiles for the two measured positions in the Gäubahntunnel. a) Depth profile for the outer measurement point at the entrance in the tunnel. b) Depth profile for the measurement point situated 120 m inside the tunnel. c) T_2 profile for the outer measurement point. d) T_2 profile for the inside measured point at 120 m inside the tunnel.

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for a so-called stretched exponential:

$$S(t) = \exp(-(1/b) * (t/T_2)^b) \quad (5.1)$$

The best value for b was found to be 1.2.

Comparing the two depth profiles obtained for the two different measurement points, differences can be observed. The amplitude at the entrance in the tunnel (marked as “Outside” in Fig 5.2a) is almost double for the first 7 millimeters as the amplitude for the profile measured 120 m inside the tunnel (marked as “Inside” in Fig 5.2b). After this point both amplitudes show a water content of up to 2%. One possible explanation for this could be that the wall at the entrance is more exposed to the outer environment and to the salt laid during the winter; these may induce cracks in the wall and water accumulation. The accumulation of salt, especially chloride, is confirmed by measurements performed in 2003 by BAM (Weise 10). The results show that the chloride percentage is five times bigger “Outside” than “Inside” but is not exceeding 1% out of the total concrete mass. This is the reason why the water content that was found through the NMR measurements is less than the value found by the MRE measurements. The NMR water content was with 1-1.5% less than the MRE values. In case of the extracted T_2 profiles, the values in both cases increase slightly with depth (Fig. 5.2c and d) which means that the pores are larger deeper in the sample. This can be understood as an effect of the CO_2 in air which leads to CaCO_3 formation, by a chemical reaction with the calcium ions of the C-S-H hydrates (Calcium Silicate Hydrate, components of every cement paste). This leads to a smaller porosity as some of the free space is now occupied by CaCO_3 (Ngala 97).

In the case of the Hohenwarthe lock, two profiles were planned: one in the lock wall part which is just exposed to ambient air, and the other one at a part which is occasionally under water (when the lock is full). One difficulty that was encountered during this on-site measurements was the reinforcement located 60 mm inside the wall. Because of this, the NMR-MOUSE was magnetically attracted and we expected alterations to the position of the sensitive slice and to the final results. After looking for a position where the magnetic attraction was the lowest, the measurements were performed. To match with the usual units and with the other on-site measurements, the amplitude of the NMR signal was

converted into grams of water per gram of dried cement, as shown above. Figure 5.3 exhibits the profile amplitudes and T_2 values of the measurements performed under water level (marked with blue in the figure) and above (marked with red in the figure). For each measurement, a Carr Purcell Meiboom Gill (CPMG) measurement was performed. Both channels were recorded and a fit using the two channels but a single relaxation rate was performed in order to extract the amplitude at zero echo time. The fitting was made again with the stretched exponential function, because it was the best fit obtained and for the consistency of the results. The function was mentioned before in equation 5.1. The best value for b was evaluated to 1.2.

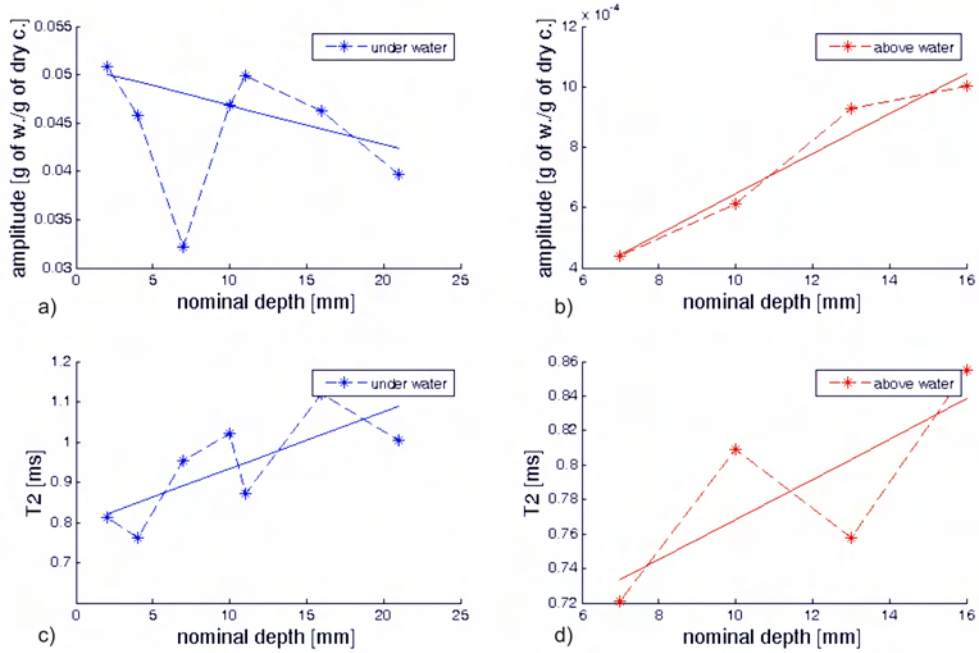


Figure 5.3: NMR depth profiles and extracted T_2 profiles measured at the Hohenwarthe lock. a) Depth profile measured for the part that is exposed to water. b) Depth profile for the part that is exposed just to ambient air and no water. c) T_2 profile for the part exposed occasionally to water. d) T_2 profile for the part that is exposed just to the ambient air.

For the measurements in the region of the lock wall that is occasionally under water, we can notice that the T_2 values increase steadily (Fig. 5.3c), whereas

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the amplitude slightly decreases (Fig. 5.3a). Only the third point in amplitude exhibits much lower amplitude. This is interpreted as heterogeneity of the material. Indeed, this concrete contains stones from 0 to 16 mm, and the maximum thickness of the sensitive layer which can be used at this depth is 150 μm which is probably not enough to average out all heterogeneities. The profile amplitudes measured in the region that is exposed to ambient air show a very low value (Fig. 5.3b). This low signal intensity may have occurred due to bad reference data that was used for normalizing the results. The measurements for obtaining the reference data were performed one hour before the lock wall measurements took place. Between these two experiments, the whole setup needed to be taken off and then put back as a ship needed to go through the lock. So the reference data amplitude had probably changed. However, we can state that the trend is good and the wall becomes more moist when we go deeper. In both cases, T_2 increases with the depth (Fig. 5.3d), which means that the pores are larger deeper in the sample. This trend was noticed also at the Gäubahntunnel measurements.

5.2.2 Laboratory measurements

For the laboratory measurements two concrete blocks were examined, which had the same composition as the concrete from the on-site measurements. These blocks were used also by the IBAC group and had inside electrodes at different depths (7 mm, 12 mm and 22 mm from the bottom surface) for the MRE measurements. In this way two types of measurements could be performed at the same time. The MRE experiments were monitored on a different PC as the NMR measurements. The cement blocks were named after the origin of the aggregates included in the composition, that is Gäubahn and Hohenwarthe. Figure 5.4a shows the container where the blocks were put in and kept in water. They were took out it just for the time of the experiment. The measurement setup is shown in figure 5.4b. All the block faces, except the bottom and the top, were covered with a watertight substance, so the water can ingress into the block only from the bottom through the capillary forces. The NMR experiments lasted between 45 and ninety minutes, depending on the water saturation level in the blocks.

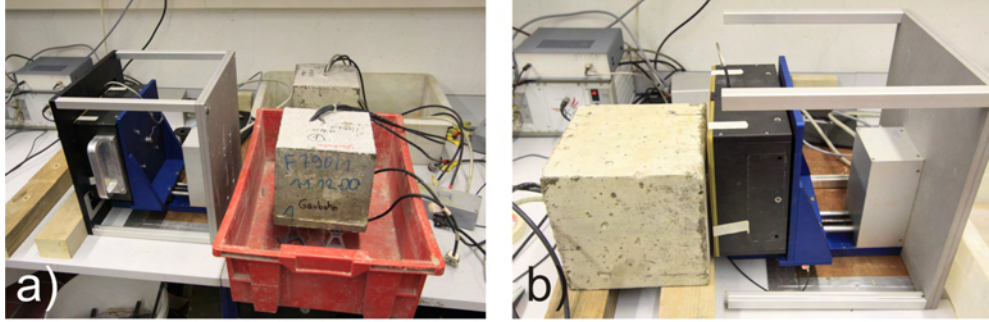


Figure 5.4: NMR experimental setup for the laboratory experiments. a) Preparation of the concrete blocks for immersing in water. b) NMR experimental setup.

The first step of the study was to perform the measurements on the dry blocks. These measurements were done for a depth of 17 mm, because one spacer was still between the magnets and the coil. Later on, for having a better comparison between the NMR and the MRE results, we took the spacer out and measured the water ingress up to 21 mm deep. Even though the differences between the pore diameters are not that big, the two blocks behaved differently, during the water ingress processes. The Hohenwarthe block was saturated in four days; after that, the drying process was observed during our study. At the end of the study (after 63 days), the Gäubahn block was still not saturated at a depth of 21 mm. To have a good comparison between the on-site measurements and the laboratory studies, first the Gäubahn block study will be presented and then the one for the Hohenwarthe block.

The Gäubahn block was very dry at the beginning, as seen from the black curve in Fig. 5.5a. After two days in water, the saturation level reached above 4% in the first two millimeters. A constant gradient of water content with depth can be observed, the amplitude at a depth of 21 mm being always lower than in the rest of the sample. The constant gradient is even more visible in the case of the T_2 depth profiles. The values in the first six millimeters are three times bigger than in the last five measured millimeters. We split the amplitude and T_2 depth profiles into six regions, where the amplitude as well as the T_2 values stay constant. These six regions are 0 to 3 mm, 4 to 7 mm, 8 to 11 mm, 12 to 15 mm, 16 to 19 mm and 20 to 21 mm. In this way we can plot the mean values

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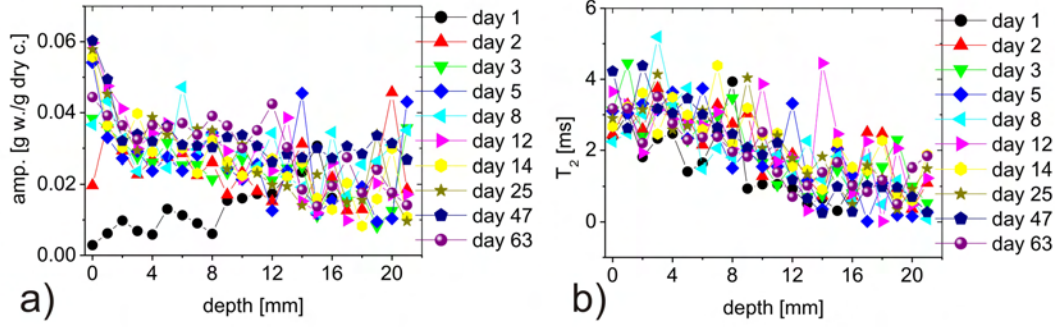


Figure 5.5: Water saturation levels and T_2 depth profiles for the Gäubahn block.
a) Water content for the different measurement days during the study. After 64 days, the amplitude still increased. b) T_2 depth profiles for the Gäubahn block.

over these six regions for each measurement day. The obtained plots are shown in Fig. 5.6. In Fig. 5.6a the evolution of the amplitude over the study days are plotted. Considering just the black curve, representing the mean amplitude value over the first three millimeters, it is clear that it reached saturation after 15 days. This means that the pores inside the block were saturated and had a 4% water content. Observing the other curves, it is clear that, deeper regions in the sample are less saturated. Additionally, lower proton signal leads to a poorer signal to noise ratio, so the points scatter more. The differences between the six regions are clearer in the mean T_2 evolution curves, where the T_2 value for the saturated region is three times higher than for the deepest measured region. The MRE measurements reveal the same water content.

Both plots in Fig. 5.6 are a function of time. In this way we can plot the mean amplitudes for each region against their mean T_2 values, to show the dependence of the saturation level with the T_2 values as mentioned by Halperin (Halperin 94). The results are plotted in Fig. 5.7a. All points for all depths are on the same line, except the points for the deepest 2 mm measured. A reason might be the bad signal to noise for this region or the fact that values are an average of two values, as the other points are obtained by averaging 4 points. The results of Halperin (Halperin 94) were obtained on a white cement setup and the linear dependence between the T_2 and the saturation level is going through the origin. In

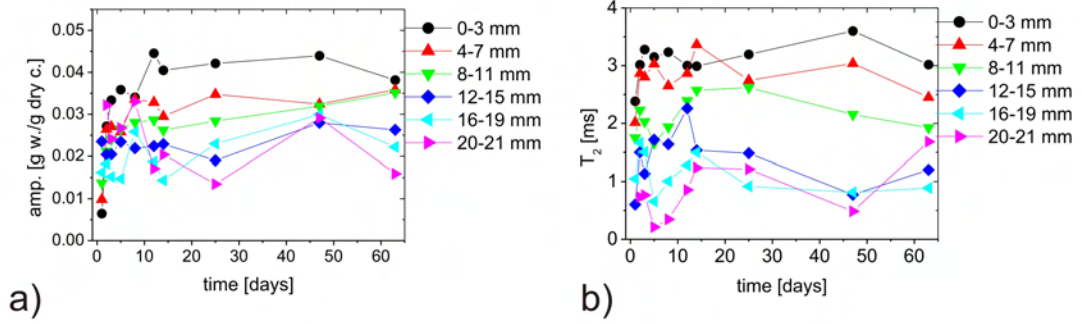


Figure 5.6: Evolution of the water saturation and T_2 values over the study time. a) For the first seven millimeters the saturation is reached already after 15 days, but for the last millimeters the pores are still not saturated with water. b) T_2 evolution for different regions of the measured depth during the study days. The values for the unsaturated region are clearly lower as for the saturated region.

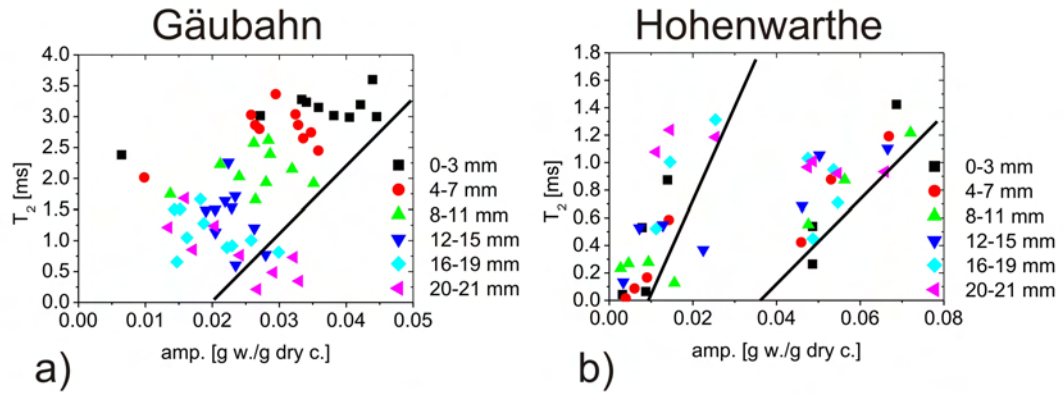


Figure 5.7: Linear dependence between the saturation level and the T_2 values. a) Dependence showed for the Gäubahn block. b) Dependence for the Hohenwarthe block. The linear trend is for both of the blocks noticeable but they have a different offset.

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our case an offset is observed. A reason for this discrepancy might be the material composition, as our sample is much more heterogeneous. In order to confirm the accuracy of our work, performed both on-site as well as in the laboratory, we took the T_2 values obtained at the on-site measurements and from those values we extracted the amplitudes from the correlation plot shown before in Fig. 5.7a. The comparison between the measured amplitude and the calculated one is represented in Fig. 5.8. The comparison was made for both on-site measured

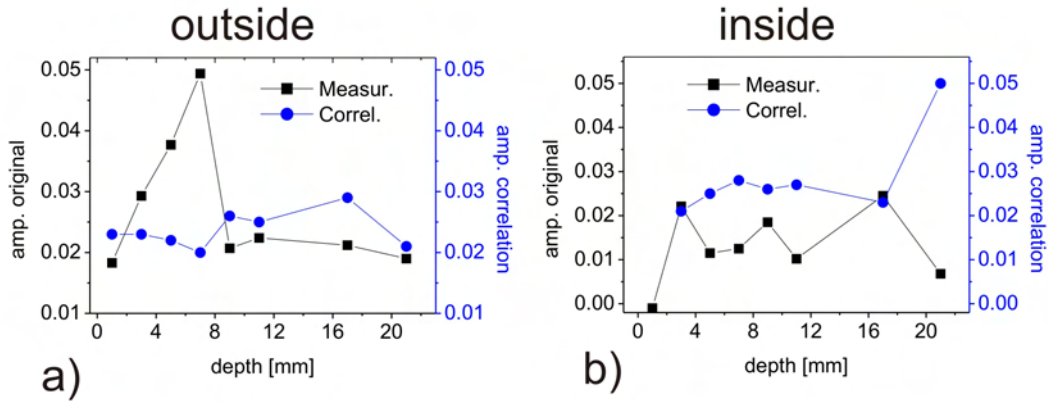


Figure 5.8: Measured and calculated values for the saturation levels. a) Comparison between the amplitudes measured and calculated at the entrance in the tunnel. b) Comparison between the amplitudes measured and calculated at 120 m inside the tunnel.

points, at the entrance into the tunnel (Fig. 5.8a) and 120 m inside the tunnel (Fig. 5.8b). The values obtained from the correlation are slightly higher than the on-site values. This is expected as we do not take into account the environmental effects and the poorer signal to noise ratio from the on-site measurements. And this values are closer to the values obtained by the MRE measurements. Exception are the first 7 mm at the entrance point, but this way to high values can be explained by the chloride problem that exists in this region (Weise 10).

The Hohenwarthe block looked at the first sight more homogeneous in consistency than the Gäubahn block. Then, during the measurements, the differences between the two blocks were even more clear when, after only four days, the saturation level was reached even in the deepest region of the block. Because of

this fast rate of water ingress and too few measurement points to make conclusive claims, we decided to monitor the water desaturation of the concrete block. Figure 5.9 shows the saturation as well as the desaturation of the Hohenwarthe block. In both cases the gradient that was seen at Gäubahn is not that high, and

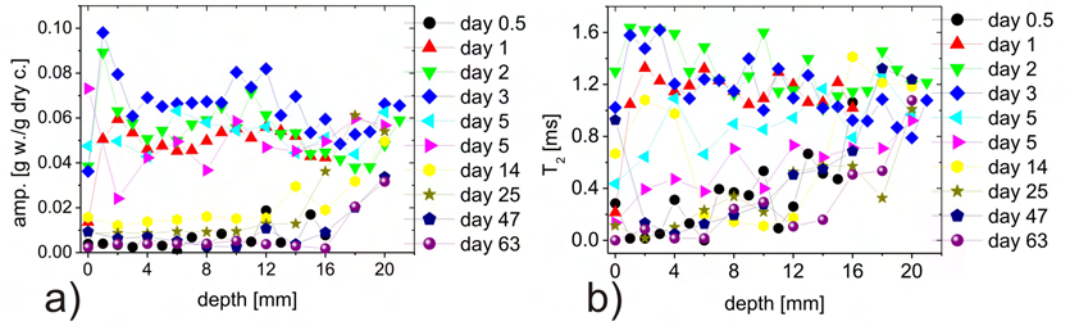


Figure 5.9: Water saturation levels and T_2 depth profiles for the Hohenwarthe block. a) Water content for the different measurement days during the study. After the fourth day the block was saturated, so the drying of the block was monitored. b) T_2 depth profiles for the Hohenwarthe block.

is more visible in the T_2 desaturation part. The water saturation is at around 5% which is slightly higher than for the Gäubahn block. However, the T_2 values are lower by a factor of two. This could be explained by the fact that the pore size of the Hohenwarthe block is much smaller than for the Gäubahn block. The T_2 depth profiles show also that the block was not completely desaturated during this time, the values at higher depth being still larger than those near the surface. If we divide now the depth profiles in the same six regions, as before for the other block, the mean values of the amplitudes and for T_2 over the study period can be obtained. These results are reported in Fig. 5.10. In Fig. 5.10a the evolution of the amplitude over the study days are plotted. For the first 15 mm of depth, the amplitude grows until the maximum water saturation is reached (in four days) and then it decreases to a minimum value, suggesting this region's desiccation and drying. Only the last two measured region did not dry out completely during this time. In case of the T_2 evolution curves (Fig. 5.10b), only the values for the

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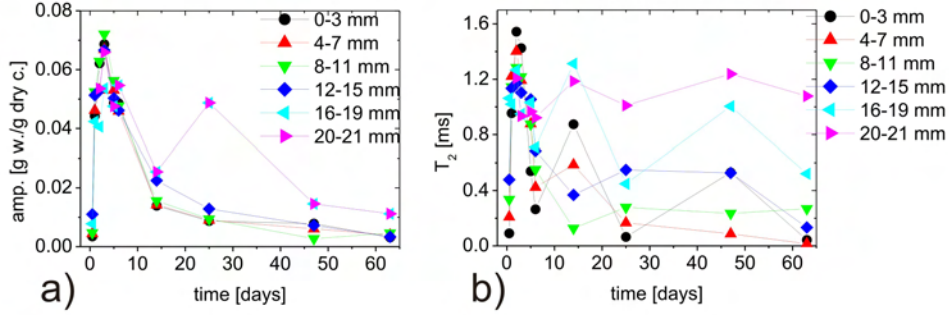


Figure 5.10: Evolution of the water saturation/desaturation and T_2 values over the study time. a) The fast saturation and the slow desaturation is visible. b) T_2 evolution for different regions of the measured depth during the study days.

deepest measured region is still increased. These observations are in accordance with the MRE results, where the maximum water content was found to be 5.8 %.

Regarding the plot of the dependence of the saturation level with the T_2 values, only the desaturation part was taken into account. This result is shown in Fig. 5.7b. The dependence show a different behavior than the one for Gäubahn. It looks like we would have two dependencies. This is maybe because we have a bimodal pore distribution as mentioned before. Gäubahn has just one pore size. As we have two pore sizes, we have also two different surface relaxivities and this influence in different ways the relaxation mechanism. Both linear dependencies present an offset. The reason for this might be again, as for Gäubahn too, the sample heterogeneity. The accuracy of our work was studied again, and we made the same correlation as before for the other concrete block. For this comparison we took into consideration the linear dependence line with the highest offset from Fig. 5.7b. The comparison between the measured amplitude and the calculated one is represented in Fig. 5.11. The correlation was performed again for both on-site measured points, the region which is flooded with water named as “under” in Fig. 5.11a and the region that is exposed only to the environment named as “above” in Fig. 5.11b. In both cases the correlated amplitudes are much higher. This might be because we did not consider the right linear dependence from Fig 5.7b. The MRE water content for the “under” region is close to 6%. A significant

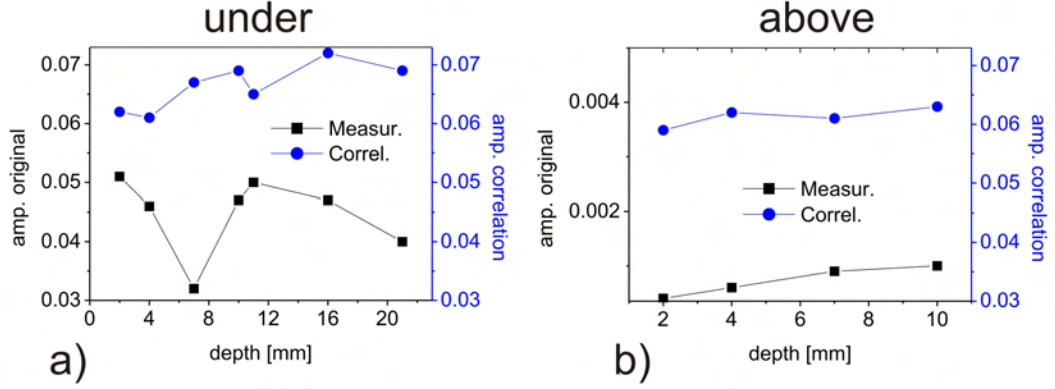


Figure 5.11: Measured and calculated values for the saturation levels. a) Comparison between the amplitudes measured and calculated for the region that is occasionally under water. b) Comparison between the amplitudes measured and calculated for the region that is exposed only to the environment.

difference can be noticed for the “above” case, where the difference is a factor of sixty. This can be explained by the reposition of the equipment between two measurements. As mentioned before, between the experiments the whole setup needed to be set up and taken down in order not to disturb the naval traffic. Nevertheless, this factor is consistent for all points and the trend is similar for both measured and calculated amplitudes.

5.3 Conclusion

Two different concrete types with different composition were examined on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process behaves completely differently. This study was relevant both for the industry partner, who could compare two different analytical techniques based on different methods, and for us in order to understand more about complex porous systems. The use of the NMR-MOUSE proved appropriate, especially for the on-site measurements, where the measurements could be performed quickly, non-invasively and without affecting traffic, both road and the naval.

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In the case of the on-site measurements, the tunnel and the lock measurements, differences were noticed both between the two different concrete types as well as between the points measured at each site. For the tunnel measurements, differences in profile amplitudes and T_2 values were observed, evidencing the influence of the environment on concrete quality and lifetime. At the lock site, differences were already expected, as one part of the concrete wall is exposed more often to water.

The laboratory measurements proved to be efficient and informative. Not only could the differences in the rate of water uptake be evidenced, but the results could be compared with the MRE method and showed a good agreement. During 63 days, the total duration of the study, the Gäubahn concrete block did not reach complete saturation while the Hohenwarthe block was saturated and subsequently almost dried out. It is impressive how a difference in pore size of less than one order of magnitude can influence so dramatically the water dynamics in the concrete blocks.

The complexity of the composition of these concrete blocks, poured and in active use prior to our study, makes it difficult to attribute cause to artifacts in our results. It would be interesting to make a new case study with experiments performed just on different types of cement pastes, and then modify these samples to include the fine sand particles and the aggregates, and in different percentages, to see which components influence the most.

Chapter 6

Cultural Heritage

Cultural heritage is a vast and fascinating field. It tells the story of our society, from the beginning until now, giving us insight into various ways of life. Starting with objects like monumental remains, the concept now includes new categories like ethnographic heritage (Unesco 11). In this chapter, we focus on the tangible heritage: physically extant cultural property like wall paintings from Herculaneum, master oil paintings of Rembrandt, or an unknown Cologne painter. Knowing the fact that tangible cultural heritage is unique and cannot be replaced, it is a big responsibility for conservators to maintain art objects in their original state. Therefore, the use of non destructive testing techniques is a must. Furthermore, in many cases the transport of the artwork is not possible and the use of portable equipment is required.

Single-sided NMR (the Profile NMR-MOUSE) has been employed for non-destructive testing of different types of objects of cultural heritage (Blümich 10a), including investigations of the structure of paintings (Proietti 05, Proietti 07), paper (Blümich 03, Casieri 04a, Proietti 04, Federico 10), bones and mummies (Rühli 07), porous materials (Proietti 06, Sharma 03, Blümich 05b, Blümich 08), and the moisture distributions of wood (Casieri 04b) and wall paintings (Proietti 05, Proietti 07, Blümich 10b).

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6.1 Ancient Roman wall paintings

6.1.1 Introduction

Herculaneum is an ancient city located in the bay of Naples. The city gained its name from the Greeks and became a Roman municipium under Titus Didius in 89 BC. In 62 AD an earthquake destroyed parts of the city and in 79 AD it was completely buried under volcanic material by the eruption of the volcano Vesuvius. The city was at that time a summer residence for wealthy Romans. The best known residence is the Villa of the Papyri, so named due to a library of 1800 papyrus scrolls found in the Villa during the 18th century. Excavations of Herculaneum started in 1737 and they are still continuing today. They have led to a wealth of information about the Roman life in general, but have also brought deterioration and general decay to the site and its precious wall paintings and mosaics. In 2001, the Herculaneum Conservation Project was launched in order to address and halt the serious decay conditions found at the site (Project 07).

Two major types of wall paintings can be differentiated, fresco and secco. In the case of a fresco, the paint pigment is mixed with water and applied on a fresh (not dried) lime mortar. For this technique a binder is not required, as the pigment will sink into the mortar. On the other hand, in case of a secco, the paint pigment is applied on a dry mortar or plaster. In this case the pigments need a binder (usually egg, glue or oil) in order to attach to the wall (Procacci 69).

Moisture is a key concern in Herculaneum, especially in the conservation of wall paintings and mosaics. Moisture triggers the leaching of salt ions from the ground, from new consolidation building as well as from depositions of a polluted atmosphere or from biological activity which are then transported through the walls by capillary forces. Transported salts can accumulate in areas of water evaporation and, once saturation is reached, they precipitate. Salts generally become concentrated on the surface of paintings as efflorescences or below the surface as sub-efflorescences. The highest damage is caused when salts precipitate below the paint layer as it causes the painting to flake and disintegrate. Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is one of the most damaging salts present in deteriorated wall paintings, and most wall painting restoration procedures have been developed to remo-

ve this salt followed by the reconsolidation of the wall (Matteini 87, Matteini 99). It is now known that the organic consolidants used in early conservation treatments cause more damage in the long term because they occlude the natural porosity of a wall painting and interfere with salt transport, causing salt precipitation where it otherwise would not occur. Therefore, the use of inorganic substances, such as barium hydroxide or ammonium oxalate for the consolidation of mural paintings is favored over treatments with primarily organic substances (Matteini 99). Moreover, the identification of past restoration efforts is equally as important as the assessment of the state of conservation and the effectiveness of new conservation treatments.

The experimental results about the stratigraphy of decorated walls in ancient Herculaneum are presented here. Depth profiles were recorded from the Mosaic of Neptune and Amphitrite, and from the wall paintings located in the House with the Black Room and in the Villa of the Papyri. These measurements were made to identify the type of information available from unilateral NMR with the NMR-MOUSE and their significance to the conservation and restoration of mosaics and wall paintings.

6.1.2 Experimental

The NMR measurements at the excavation site were conducted with a Profile NMR-MOUSE[®] with 21 mm depth access at a proton resonance frequency of 13.86 MHz. The measurements of the Mosaic of Neptune and Amphitrite and of the wall paintings at the Villa of the Papyri were recorded with a 5 mm spacer between the sensor and the rf coil to reduce the depth access to 16 mm for the benefit of improved sensitivity. For the measurements in the Black Room, a 15 mm spacer was introduced to gain even higher sensitivity at the expense of the depth access being reduced to 6 mm. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer was a Kea spectrometer manufactured by Magritek Ltd, New Zealand, connected to a Dressler 1 kW rf amplifier. The measurement parameters of all experiments are summarized in Table 6.1. Complementary to the NMR measurements, reflectance Fourier transform

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Table 6.1: Acquisition parameters for depth profiles for the measured points

Parameters	Mosaic of Neptune and Amphitrite	House of the Black Room	Villa of the Papyri
Maximum depth [mm]	16	6	16
Pulse width [μ s]	21	6.5	19
Echo time [μ s]	140	145	140
Number of echoes	16	16	256/64
Number of scans	512	256	32/64
Time/profile [min]	80	60	90

mid-infrared spectroscopy (FTIR) and X-ray fluorescence (XRF) experiments were performed. Recently, reflectance FTIR and XRF analysis has been shown to be very valuable for the in situ, non-destructive analysis of the organic material present on wall paintings (Miliani 07b, Rosi 09, Miliani 07a). These techniques were applied in situ with portable instrumentation by the group of Prof. Eleonora Del Federico. Some results are presented in this chapter, more results can be found in the paper (Haber 11).

6.1.3 On site studies

6.1.3.1 Mosaic of Neptune and Amphitrite

The mosaic at the house of Neptune and Amphitrite (Fig. 6.1a) is one of the most famous ancient Roman mosaics, still preserved in its original setting on the south wall of a triclinium (dining room) of a middle-class household. The room is decorated with wall paintings with garden scenes, a nymphaeum, creating an illusion of the outdoors.

The mosaic was examined twice in 2009, once in March and once in October, in order to observe the climate influence over the moisture distribution in the mosaic. A mosaic is a decorative piece of art, where on a mortar bed cubiform, colored glass pieces (called tesseræ) are assembled, such that a picture or a pattern is created. Two regions can be identified in the profiles, one from 0 to 6 or 7 mm depth and one at greater depth. They are assigned to the tesseræ with some

mortar between them and to the mortar embedding the tesserae, respectively. Points 1 and 2 were measured in March, and all other points including point 1 were measured in October (Fig. 6.1a,b). Point 1 was measured in both March and October to investigate the reproducibility of the measurements and the influence of seasonal moisture variation. The locations of the measurement points were chosen depending on the composition and color of the tesserae and their height from the bottom of the mosaic, as both may have an impact on the moisture content.

The depth profiles shown in Fig. 6.1b compare the results from March and October. The two profiles obtained for point 1 overlap almost identically in the region of the tesserae from 0 to 7 mm, but they differ at higher depths: the amplitude of the profile measured in the fall is lower than that of the profile measured in the spring. The reason for this difference is that the wall dried during the hot summer. This interpretation is supported also by the fact that points 1 and 2 show identical profiles for the mortar bed in the March measurements, being also at the same height from the ground (Fig. 6.1a). The profiles from all the points, except point 1, overlap well in the region of the tesserae (Fig. 6.1c). Differences are detected only at higher depths and are characteristic of the mortar bed. Point 6 is the highest one measured in the mosaic and shows the lowest signal in the mortar region. Point 7, on the other hand, is the lowest and shows the highest mortar signal. Points 3 and 4 are positioned at the same height and overlap in the mortar region. In summary, the higher the measurement position, the lower the profile amplitude in the region of the mortar bed. This is a strong indication that the source of moisture is located on the ground.

Regarding the T_2 profiles (Fig. 6.1d), they look very different from the amplitude profiles. They show the highest T_2 value at the interface between the tesserae and the mortar bed with a broad maxima. While the amplitude in the regions of the tesserae at point 1 is very intense, the corresponding T_2 values are very low. Together with the fact that the first 7 mm of the profile did not change during more than 6 months is a strong indication that the ^1H signal does not derive from water but from an organic coating at the surface, possibly an undocumented early consolidation treatment. Also noteworthy is the extreme behavior of the amplitude and T_2 profiles at point 7. The signal amplitude increases strongly with depth

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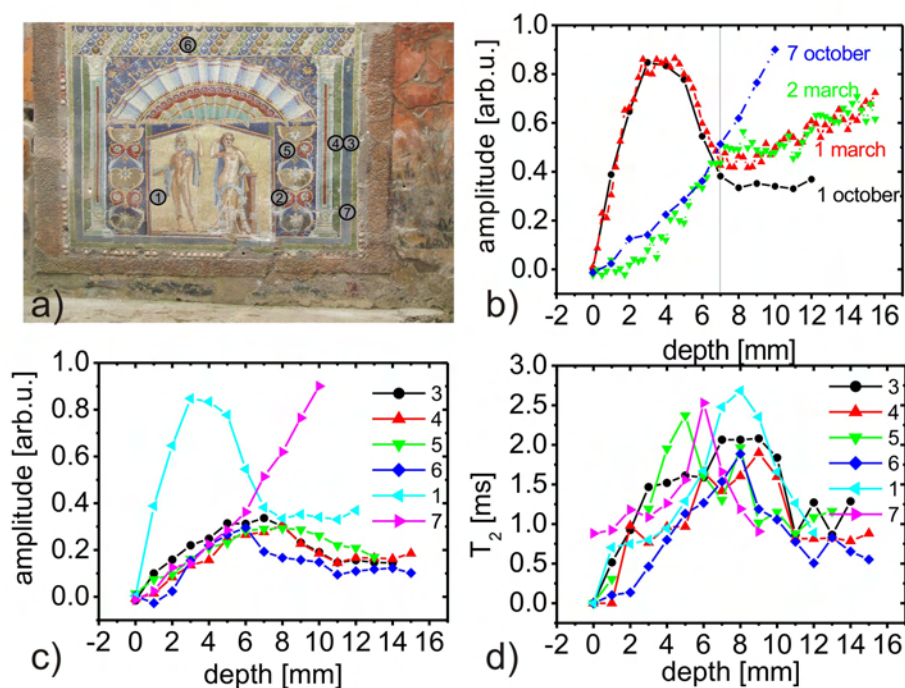


Figure 6.1: Mosaic of Neptune and Amphitrite. a) The mosaic showing the position where the NMR depth profiles were recorded. Point 2 was measured just in March, while point 1 was measured both in March and October. b) Depth profiles for point 1 (black and red), recorded more than 6 months apart. The first parts of the profiles overlap identically while inside the mortar bed they differ. The mortar signal from point 2 (green) overlaps with the mortar signal from point 1, both measured in March, evidencing the same moisture conditions for this section of the mortar bed. The depth profile for point 7 (blue) being at the lowest point measured reports an even higher moisture content of the mortar bed. The line indicate the boundary between tesserae and mortar. c) Depth profiles for all the points measured in October. The moisture content in the mortar decreases with the increasing height above the floor. d) T_2 profiles for all measured points.

(Fig. 6.1b), while the T_2 profile exhibits a narrow peak at the interface between tesserae and mortar bed. This indicates high moisture deep inside the wall and possibly a beginning detachment of the tesserae from the mortar bed.

6.1.3.2 The Black Room

Across the street from the House of Neptune and Amphitrite is the back entrance to the House of the Black Room, one of the most luxurious houses in Herculaneum. The House owes its name to a large main room that is decorated with pillars and chandeliers painted with a black background in the third Pompeian style. The room was excavated between 1933 and 1961. Although not documented, it is believed that all wall paintings excavated during that time period were treated with a mixture of beeswax and paraffin as a preventative conservation measure to avoid further damage. Over time, changes in relative humidity, the use of modern cement for restoration of the walls, and air pollution have caused damaging salts to leach from the walls and precipitate as efflorescence either on the surface or just below the paint layer. This efflorescence has caused significant damage, including delamination and pulverization of the paintings. The presence of earlier conservation treatments containing an organic coating have contributed to this decay. The north-western wall of the main room (marked as wall B in the figures), behind which a modern Portland cement wall was built, has been damaged the most by the salt, with its paint layer delaminated in many areas. Salt efflorescence on this wall appears especially around the area behind which a new wall was built.

Recent conservation efforts involved the removal of the organic consolidants from early restorations and application of test treatments in small areas of the cleaned walls. The three painted walls labeled A, B, and C (Fig. 6.2a,b) were analyzed in order to record depth profiles at different positions identified by Prof. Mauro Matteini, a professor involved for several years in applying different inorganic conservation treatments in Herculaneum. On the walls A and B (Fig. 6.2a), measurements were made at points where the organic fixative treatment was still present and at points where it had been removed. On wall C (Fig. 6.2b), ammonium oxalate and barium hydroxide inorganic treatments were tested. To maximize the sensitivity for the measurements at natural proton content, the depth profiles

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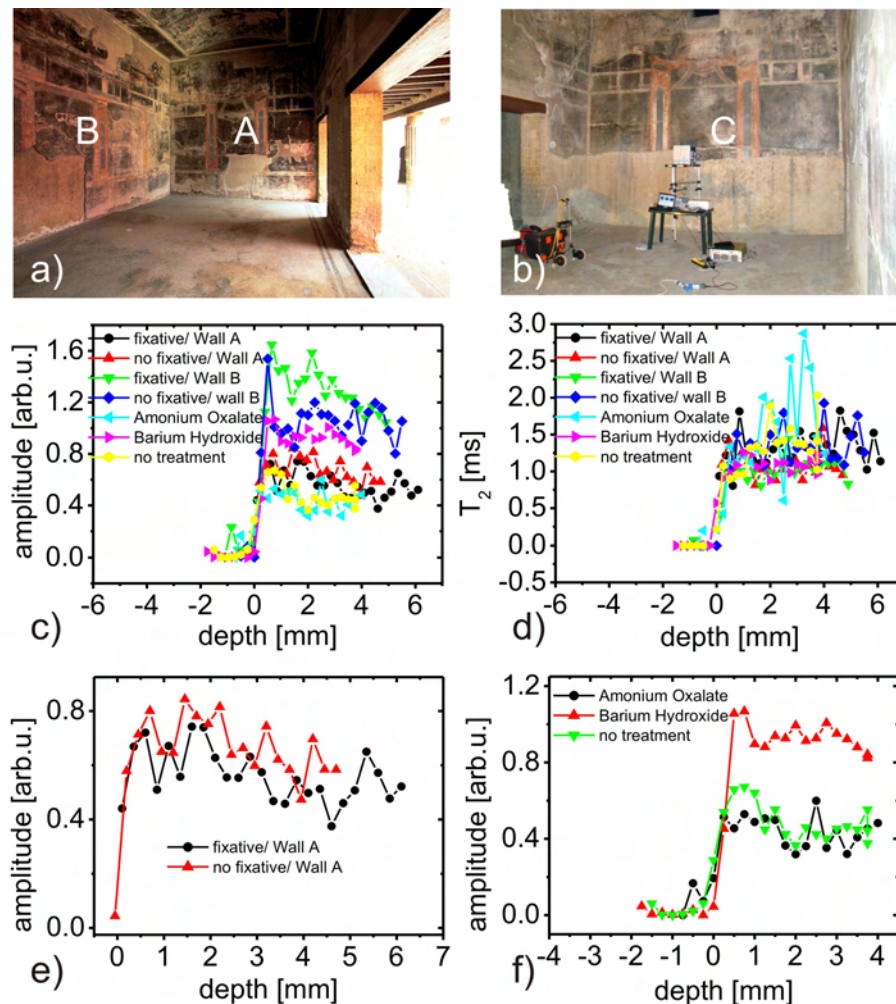


Figure 6.2: NMR measurements in the Black Room. a) The measurement points were on walls A and B. b) Wall C, where sections have been treated with barium hydroxide and ammonium oxalate. The experimental setup is seen in the foreground. c) Depth profiles for all measured points. Depending on the wall and the conservation technique, differences in moisture content are found. d) T_2 profiles for all measured points. All points show mostly the same T_2 value. e) Depth profiles for wall A showing the proton content with and without fixative on the wall painting. f) Depth profiles for wall C showing the difference in proton content for the treatments with barium hydroxide and ammonium oxalate and their comparison with the proton content for an untreated region from the same wall.

were taken with all spacers placed in the magnet. This resulted in a maximum accessible depth of 6 mm.

The proton density profiles (Fig. 6.2c) differ in the absolute signal amplitudes but are similar in shape: they show the highest signal at the surface, a peak at a depth of about 2 mm, with the signal amplitude decreasing with increasing depth. The T_2 profiles (Fig. 6.2d) show little variation between different positions which is mainly attributed to the low signal-to-noise ratio of the measurement. The profiles acquired on wall A (Fig. 6.2e), both for the area which is coated with organic consolidants and the area where this organic coating was removed, do not differ significantly in signal amplitude.

The NMR depth profiles from wall B show a proton density significantly higher than in depth profiles from walls A or C. This observation is consistent with the existence of visible salt layers at the surface, due to the hindered moisture exchange by the modern concrete wall (Fig. 6.2c). For this wall, with the exception of the first two profile points at the surface, the amplitudes of the profiles measured at locations with organic consolidant treatment and with the consolidant removed are quite different. There, the amplitude is high and the same for both profiles suggesting incomplete removal of the organic treatment. The profiles of wall C (Fig. 6.2f) present the proton density amplitude from a point where the older organic consolidant was removed (no treatment) and from the points with inorganic conservation treatments (ammonium oxalate, barium hydroxide) are all very different, although they are positioned really near one to each other. The point where the organic consolidant was removed shows a profile similar to that at the point treated with ammonium oxalate. However, in the first millimeter, the higher signal is given by the cleaned area, probably due to incomplete removal of the older organic treatment. At larger depths, the amplitude of the area treated with the barium hydroxide displays the highest intensity and, therefore, is the area with the highest moisture content. Although the higher moisture content in the barium hydroxide treated area may suggest that the treatment with barium hydroxide is inferior to the treatment with ammonium oxalate, it is not the moisture content but the moisture volume flux which should be assessed to rank the effectiveness of the treatments. These observations are consistent with laboratory studies of moisture in mock-up mortar beds presented later in this chapter and

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with the reflectance FTIR spectra obtained by the Pratt Institute research group, led by Prof. Eleonora Del Federico (Haber 11).

6.1.3.3 The Villa of the Papyri

The Villa of the Papyri was first explored from 1750 to 1765 through a network of tunnels burrowed into the volcanic rock by Karl Jacob Weber, a Swiss architect and engineer, under the patronage of Charles III of Spain. From his detailed drawings an accurate map of the Villa could be established. Today, a large part of the Villa remains underground. In 1826, a library of about 1800 papyrus scrolls was discovered, giving the Villa its name. It is believed that the house belonged to a wealthy and learned Roman noble man, probably Lucius Calpurnius Piso Caesoninus, father-in-law of Julius Caesar and consul in 58 BC. Recent excavations have revealed a small room with brightly colored wall paintings in a great state of conservation. Therefore, this room is of major interest to conservators, conservation scientists, archaeologists, and historians as its surfaces remain as they were left in 79 AD. It is in this room and on the above terrace where our NMR measurements were taken.

As a benefit to our NMR measurements, this room is kept moist at 96 % relative humidity to direct the moisture migration into the walls and keep salts from crystallizing. Two of its walls were examined (Fig. 6.3a). In all, four positions were studied in detail, three inside the room and one on the outside wall (Fig. 6.3b) facing the terrace above that room. Interestingly, two types of stratigraphy can be identified, each being represented by two different profiles (Fig. 6.3c,d). One type was measured at the points A and B on a wall dating from the time before the earthquake of 62 AD (Fig. 6.3a), and the other on a wall rebuilt after the earthquake (without any wall paintings) and the outside terrace wall (Fig. 6.3b). The amplitude profiles look similar, though with an offset in the depth, which reveal the layer structure of the mortar bed. The fabrication of these layers has been described by Vitruvius (Pollio 99) and more recently in many scientific studies of wall fragments found in different locations (Donner 69, Bearat 96).

The profiles of the older wall (A and B) are thicker and more intricate than those of the newer wall (C) and the terrace wall (D). This may be because either

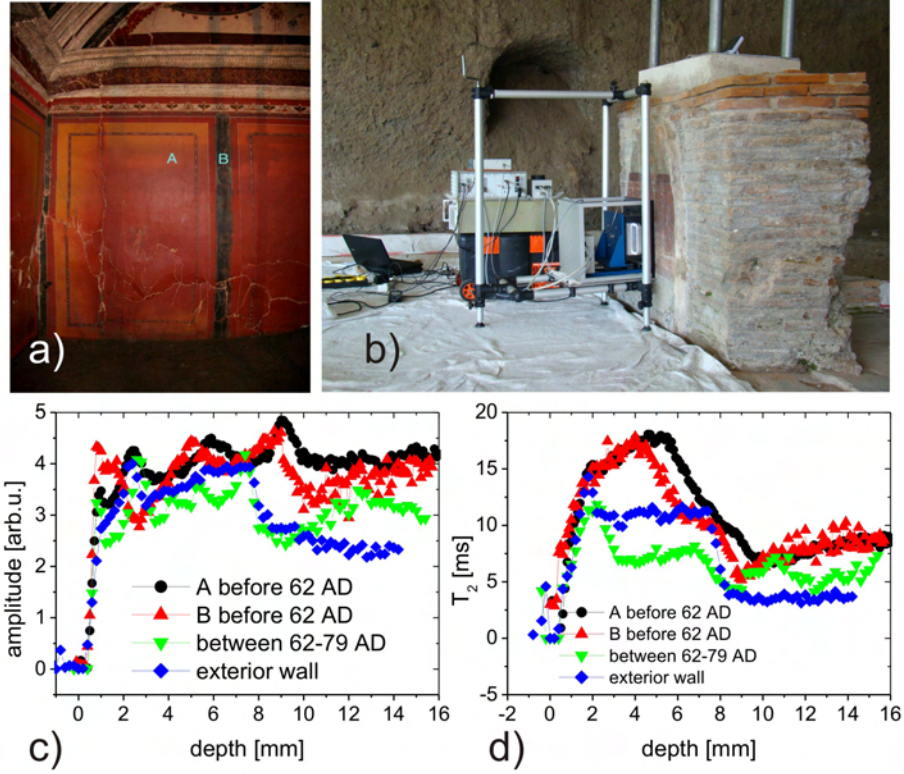


Figure 6.3: NMR measurements at the Villa of the Papyri. Location of the measured points dating from before 62 AD, when the earthquake took place. b) Experimental setup at the exterior wall on the terrace. c) Depth profiles at the four measured positions identifying different mortar layers and two different types of layer structures. The profiles for A and B differ from the other two. d) T_2 profiles for all measured points. Different T_2 values are assigned to different mortar materials

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the more recent method of preparing the mortar for the wall painting was less expensive, as much reconstruction work had to be completed after the earthquake, or the refined technique of crafting the more intricate mortar layers was lost. This interpretation suggests that either the outside wall of the terrace (D) may have been reconstructed after the earthquake, or that the technique used for outside walls had been applied for the reconstructed wall inside (C).

The NMR depth profile results clearly suggest that two different techniques were employed, which is in agreement with the XRF studies of the walls recorded by the research group of Prof. Del Federico (Haber 11). Comparing the XRF spectra of the red pigments used for the restored and the original paintings shows that the restorers used cheaper red pigment as the spectrum of the restored wall shows a high iron content suggesting the use of iron oxide pigment to replace the highly expensive, precious and intensely colored cinnabar (HgS), which is inferred by the presence of mercury and sulfur lines in the second spectrum. The overlay of the XRF spectra of both walls is reported in figure 6.4a.

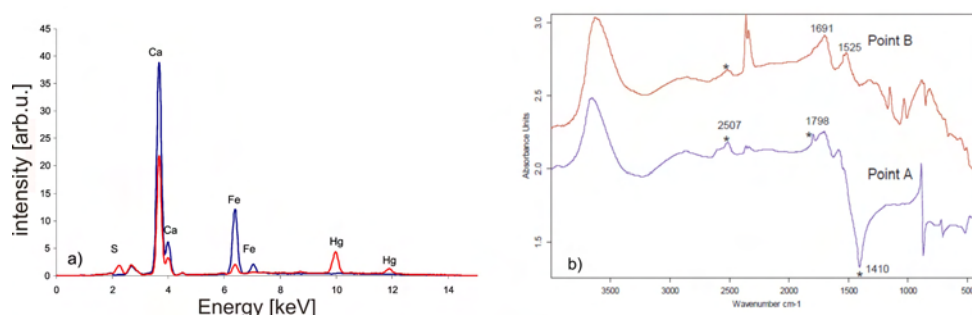


Figure 6.4: XRF and FTIR measurements at the Villa of the Papyri. a) Red: XRF spectrum of the red pigment on the wall built before 62 AD showing the presence of mercury (Hg) and sulfur (S) indicating that the red pigment used was cinnabar. Blue: XRF spectrum of red pigment on the wall built after 62 AD, showing the absence of mercury and sulfur and the high content of iron (Fe), suggesting an iron red pigment such as hematite or red ochre. b) FTIR spectra. Point B reveals the possible presence of organic binder and oxalates, known to be degradation products of organic substances. Point A reveals stronger calcium carbonate bands, suggesting that the technique in this case is fresco. Data by E. del Federico and collaborators from Pratt Institute.

6.1 Ancient Roman wall paintings

It should be mentioned that the two depth profiles recorded at the positions A and B differ in their moisture content in the first 3 mm. Profile A has been measured at a blank section of the painted wall, while profile B was taken on the painted vertical decoration. It is speculated that the decoration is a secco so that the moisture uptake from the humid air is different in the two sections. This is also consistent with FTIR results of both surfaces (Fig. 6.4b).

Following Vitruvius (Pollio 99), the mortar bed for wall paintings consists of three mortar layers prepared with sand followed by the application of three lime plaster layers prepared with marble dust. The painting is applied on the last layer while still wet. Vitruvius recommends that the marble plaster layers be prepared with decreasing particle size towards the surface and to press them with a trowel to improve the packing density (Pollio 99). The packing of the layers coupled with the use of different aggregates (sand vs. marble dust) is likely to give rise to differences in the pore sizes across the mortar-layer structure. We postulate that the last three layers give rise to the top three peaks observed up to a depth of 9 mm, beneath which the sand-mortar layers follow.

Similar to the amplitude profiles, the T_2 profiles can also be grouped into the same two pairs, first are the ones obtained for position A and B and the second the other two (Fig. 6.4d). The T_2 profiles from points A and B differ only in the thickness of the second mortar layer, shown by the shift of the end of the profile of point B to the left when compared to the profile of point A. It is possible that the mortar with the second types of profiles was prepared with a different aggregate or mixture of aggregates. Based on the T_2 profiles at least two types of mortar layers can be identified. One type was used for the layers between 0 and 8 mm and another one for the deeper layers. However, the layer between 6 and 8 or 9 mm (for the wall reconstructed between 62 - 79 AD) may indicate an inconsistency to Vitruvius' rule because the measured T_2 values are smaller, there could be smaller particles or lower moisture content in the inner layers, especially in the case of the interior walls in the newly excavated room which is kept at above 95 % RH. Furthermore, a period of rain during the days before the measurements cannot be dismissed and could also account for the higher moisture content in the surface layers of the outer wall. The T_2 profile from the interior wall reconstructed between 62 - 79 AD shows high T_2 values in a thin surface layer, a

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lower T_2 plateau between 3 and 8 mm depth and even lower values further down. This general trend is followed by the profile from the terrace wall; however, the initial peak is broader and the T_2 plateau is higher which is likely to be due to the different moisture content of the outside wall.

Given the good signal-to-noise ratios of the data measured at the Villa of the Papyri, the measured CPMG decays could be inverted by inverse Laplace transformation to produce distributions of relaxation times T_2 (Fig. 6.5). This

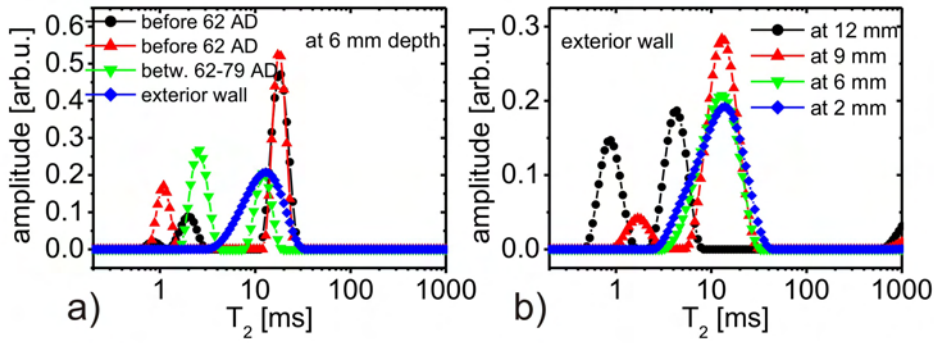


Figure 6.5: Distributions of relaxation times obtained by inverse Laplace transformation of T_2 decays. a) T_2 distributions for all measured points at 6 mm depth. b) T_2 distributions at different depths for the exterior wall.

is a more refined analysis than fitting the measured signal with a model decay function, such as a monoexponential decay, to extract a single relaxation time T_2 (cf. Fig. 6.3d). The T_2 distributions at a depth of 6 mm at the four positions reveal the similarities of the mortar structures identified in the depth profiles above. Two peaks are observed, one at short relaxation times and one at long relaxation times. The one at short times can be assigned to bound water confined to small pores, while the one at long relaxation times is assigned to more mobile water that partially fills larger pores. At a depth of 6 mm, this peak is basically identical for both positions A and B of the original wall and very similar for the restored wall and the outside wall (Fig. 6.5a). The peaks at short times seem to vary in an uncorrelated way, which may be due to the noise of the measurement data. The change of the T_2 distribution with depth reports the moisture content

and pore structure in the different layers (Fig. 6.5b). All peaks at large T_2 arise at the same position (up to a depth of 9 mm), indicating similarity of the mortar material across that range and a change towards larger depth.

The use of different types of mortar can also be seen by visual inspection of mortar pieces broken off the walls (Fig. 6.6a). A section from the Villa of

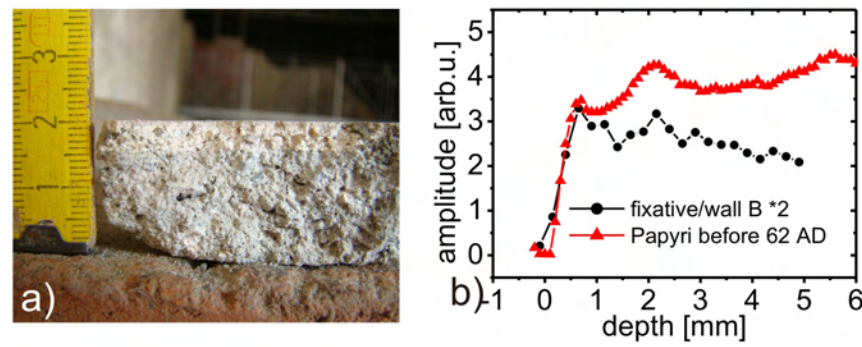


Figure 6.6: Mortar layer comparison. a) Photo of a mortar fragment from a fresco of the Villa of the Papyri showing the layer structure. b) Comparison of depths profiles recorded from walls in the Villa of the Papyri and the Black Room for the first 6 mm.

the Papyri clearly shows a whitish first layer about 4 mm thick, followed by a somewhat darker layer up to a depth of 9 mm and slightly lighter colored mortar beyond that; this matches the profiles A and B from the older wall inside the Villa of the Papyri. As profiles ranging to up to 16 mm depth were not acquired from the much dryer walls of the Black Room, only the initial parts of the profiles from the Villa of the Papyri (point A) and the Black Room can be compared (Fig. 6.6b). The initial contours match, showing, that at least the outer mortar layers onto which the paintings were drawn are similar.

6.1.4 Laboratory studies

For a better understanding of the moisture flow rate inside the wall paintings, mockup samples were prepared by the group of Prof. Eleonora del Federico (Pratt Institute, New York) and shipped to Aachen to be characterized by low field

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NMR. The mockup samples were prepared with one or more mortar layers, and different conservation treatments, found also in the Black Room, were applied. One of these samples is shown in Figure 6.7. Already upon arrival in Aachen, the pigment layer of the sample was flaking off. A reason for this might be an incorrect concentration of components used for preparing the mortar layers and the pigment layer on top. Initial attempts to measure the samples dry yielded

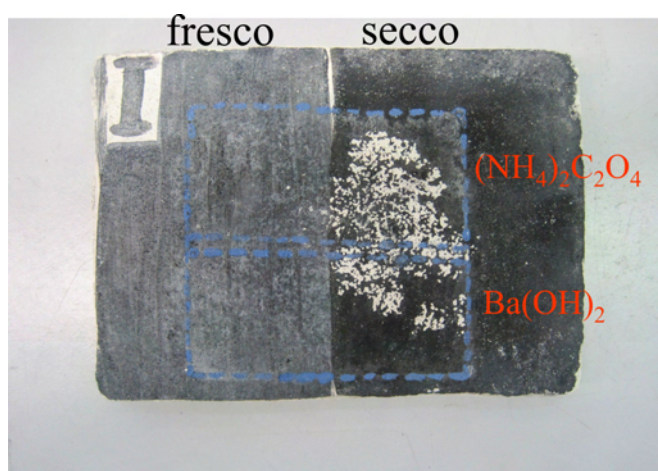


Figure 6.7: Mock-up sample. The mortar layers were applied with different techniques (fresco and secco) and different conservation treatments were applied on the pigment layer; $\text{Ba}(\text{OH})_2$ is barium hydroxide and $(\text{NH}_4)_2\text{C}_2\text{O}_4$ is ammonium oxalate. The pigment layer in the secco region was flaking off already before the measurements.

no signal. Consequently, they were put for 3 hours into a climate temperature chamber at 90 % relative humidity; still no signal could be recorded. Therefore, the samples were saturated with water and the drying process over 15 hours was observed through depth profile measurements. In order to have a better informative value of the moisture transport, the signal amplitude over a region of 1 mm was averaged for each sample and plotted against the drying time. The measurements were performed with the Profile NMR-MOUSE[®] with 5.4 mm depth access at a proton resonance frequency of 17.1 MHz. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer

hardware was a Minispec spectrometer manufactured by Bruker Biospin (Rheinstetten, Germany). The 90° pulse width was $5 \mu\text{s}$, the echo time was $45 \mu\text{s}$ and 512 scans were repeated for each depth position. The averaged signal amplitudes in a range of 1 mm for different samples with different conservation treatments are plotted in figure 6.8.

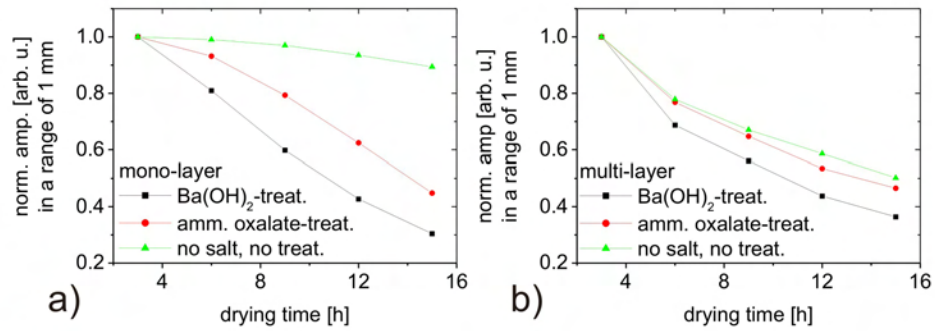


Figure 6.8: Fresco drying experiments. a) Drying process of the mono-layer mortar for different conservation treatments. b) Drying process of the multi-layer mortar for different conservation treatments.

Figure 6.8a shows the results for mono-layered fresco samples. It is noticeable that the ammonium oxalate treated region dries more slowly than the barium hydroxide region. This means that the moisture transport in the barium hydroxide treated frescoes is much higher than in the ammonium oxalate treated frescoes. The region that was not treated shows the slowest drying. But this is not relevant for wall paintings exposed to the environment as the wall paintings in Herculaneum, because there is salt in the air (being near to the sea) and the wall paintings will absorb the salt from the ambience. The same trend was observed also for the multi-layer fresco models. Here the salt-free, treatment-free region dries out much faster than in the mono-layer structure, but it is still slower than the ammonium oxalate treatment and the barium hydroxide treatment. In the first 5 hours of drying it behaves exactly the same as the ammonium oxalate treatment. But still the barium hydroxide treatment shows faster drying. As a comparison between the mono-layer structures and the multi-layer structures the ammonium

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oxalate conservation treatment shows the most stable behavior. The barium hydroxide treatment shows much faster drying in the mono-layer structure than in the multi-layer structure, as well as the salt-free and treatment-free region.

Along with the drying experiments, diffusion experiments were performed, in order to get a better understanding of the moisture transport. The diffusion experiments were executed with the same Profile NMR-MOUSE[®], having a constant magnetic field gradient of 20 T/m. The pulse sequence used is a stimulated echo experiment with a constant gradient. The pulse sequence was presented before in figure from the first chapter. The time between the first two 90° pulses is 180 μ s and the signal from 1024 scans was averaged. The diffusion coefficients obtained for the ammonium oxalate treatment are similar in the mono-layer and the multi-layer sample (7.8×10^{-10} m²/s and 7.7×10^{-10} m²/s). In case of barium hydroxide treatment differences in diffusion coefficient between mono-layer and multi-layer are observed (8.5×10^{-10} m²/s compared to 7.5×10^{-10} m²/s). By comparison of the two diffusion coefficients obtained for the mono-layer fresco model, a higher diffusion coefficient is obtained for the barium hydroxide treatment than for the ammonium oxalate treatment. This means that the molecular mobility in the ammonium oxalate treated region is lower than in the region treated with barium hydroxide, and thus was the moisture transport slower. These measurements are in good agreement with the results obtained for the drying experiments. In the case of multi-layer fresco models, the values are not that different, but they show that the diffusion coefficient for the ammonium oxalate treatment is higher than the diffusion coefficient for the barium hydroxide treatment. This is not in agreement with the drying experiments. A reason for this disagreement could be the inhomogeneity of the mortar layers or that the sensitive volume during the measurements was at the interface between two different mortar layers having different compositions.

In order to be able to decide which treatment is more effective, simple moisture studies or diffusion measurements are not sufficient because, the effectiveness is given by the overall flux of the liquid through the mortar. This flux is a product between diffusion coefficient and concentration variation. This means, a good estimation of treatment effectiveness is made when both, diffusion and moisture content, measurements are performed. Of course these complex measurements

will increase the measurements time as well on-site as in the laboratory, but the results will be more conclusive.

6.2 Old master paintings

6.2.1 Introduction

Rembrandt Harmensz van Rijn, a 17th century painter, is considered the greatest painter of the Netherlands and generally one of the greatest painters who ever lived (Gombrich 08). He is best known for his enormous number of self-portraits and for biblical illustrations. With the help of his self-portraits it is easy to make an autobiography. His sincere way of painting himself left us the story of his life, from his early years when he was a successful and wealthy master until his old age when he was a respected but lonely artist. Rembrandt made over his lifetime over 40 self-portraits, which is an astonishing number in art society generally. Rembrandt was also known for his etching technique. It has the same principle as printing letters, but here he used copper plates, which he then covered with wax and drew upon with a needle. The copper plate was placed in acid and, where the needle removed the wax, the acid attacked the copper. In this way only the printing of the drawing from the copper plate was needed (Gombrich 08).

6.2.2 Studied paintings

Within a collaboration with the Wallraf-Richartz-Museum & Fondation Corboud in Cologne, we had the opportunity to analyze the Rembrandt self-portrait as Zeuxis that belongs to this museum. The measurements were made in the museum, under the observation of the conservators. It is one of the last self-portraits of the great artist, showing him as an old man. The aim of the study was to analyze the stratigraphy of the painting and the NMR properties of the measured points, which is an oil painting on canvas. The conservators from the museum made a detailed evaluation of the painting and noticed that the painting suffered during his formation modifications (Schaefer 12). Another challenge in this project was to determine the age of painting and to confirm the results of the conservators

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with the help of the NMR measurements. Complementary NMR measurements were performed in order to enhance our age dating technique on another master painting dating from the 14th and 15th century made by an anonymous painter originally from Cologne.

The NMR measurements were executed with a Profile NMR-MOUSE[®] with 5.4 mm depth access at a proton resonance frequency of 17.1 MHz. The measurements of both paintings were recorded with a 2 mm spacer between the sensor and the rf coil, this increased the sensitivity at the expense of the depth access being reduced to 3.4 mm. The sensor was mounted on a computer-driven positioning device. The NMR sensor and the positioning device were manufactured by ACT GmbH, Aachen, Germany. The spectrometer hardware was a Minispec spectrometer manufactured by Bruker Biospin (Rheinstetten, Germany). The measurement parameters of all experiments are summarized in Table 6.2. Complementary to

Table 6.2: Acquisition parameters used in the NMR depth profile experiments for the measured points

Parameters	Rembrandt	Anonymous painter
Pulse width [μ s]	5	5
Echo time [μ s]	36	46
Number of echoes	16/32	50
Number of scans	256	256
Time/profile [min]	20	50

the NMR measurements, infrared reflectography (IRR), X-ray, UV-fluorescence and other invasive measurement results are documented.

6.2.2.1 Rembrandt

As mentioned before, this painting is one of the last Rembrandt's self-portraits. It has been part of the Wallraf-Richartz-Museum & Fondation Corboud Museum since 1936. A stir is caused by the fact that the painting presents numerous deviations between the incident light photography and the X-ray photography (Fig. 6.9). In the X-ray photography, Rembrandt looks more sad, the eyebrows and forehead are painted differently, he has longer hair, and the person to the left

is missing. The question is, where the changes made by him, or after his death by his students. It is also possible that he assigned the changes during his lifetime to a student, fact that explains the high number of self-portraits existing worldwide, but not attributed to him. Invasive techniques, like cross-sections from the right



Figure 6.9: Rembrandt self-portrait. a) Incident light photography, where the six points measured by NMR are pointed out. b) X-ray photography, where the differences are clearly evidenced (Pilz 07).

area of the background, evidence a very complex structure of up to seven layers of paint over the quartz-containing primer. In the X-ray photograph the mouth is closed. The current visible opening of the mouth is based on red and brown color jobs that diffusely describe a form of a smile (Pilz 07).

NMR depth profiles were recorded in the 6 different positions marked in figure 6.9a. The results of the depth profiles, in which the mean of the second and third echoes were taken into account are shown in Fig. 6.10a. In Fig. 6.10b the T_2 values over the depth are represented for all the measured spots. The depth profiles evidence differences in the stratigraphy as well as in paint thickness. The depth profile from the forehead (point 3) is the thinnest confirming that this paint layer is the thinnest. In contrast, the depth profile of the scarf (point 6) is the widest one, meaning that this paint layer is the thickest. The upper lip (point 2) and the forehead (point 3) profiles are the lowest ones in amplitude intensity in

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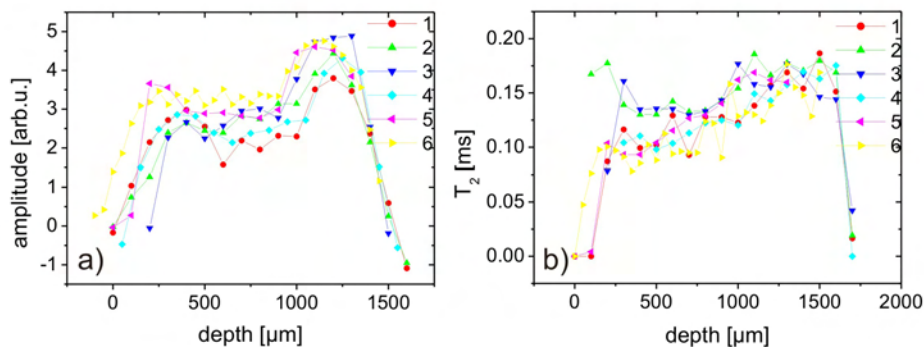


Figure 6.10: NMR depth profiles and T_2 values over the depth for all the measured points a) Depth profiles evidencing the paintings stratigraphy. Three main layers can be observed, the new canvas over a range of about 500 μm , the old canvas being around 600 μm thick and the paint layer at the surface 200 to 400 μm thick. b) T_2 depth profiles for each measured point. The T_2 values corresponding to the new canvas are slightly higher than that of the old canvas. Differences in the paint layer T_2 values are noticeable.

the region of the paint layer. The paint layer corresponding to the mouth center (measurement point 1) is thicker and more intense than the one corresponding to the upper lip (point 2).

The T_2 depth profiles underline the stratigraphy observed in the depth profiles: the T_2 values corresponding to the new canvas are higher than the values for the old canvas, and even differences in the T_2 values corresponding to the paint layers in different measured points are evidenced. The T_2 values in the paint layers corresponding to the upper lip and forehead are much higher than the other paint layers. For a better understanding and visibility, each of the depth profiles was plotted separately together with the corresponding T_2 depth profile in Fig 6.11. The amplitude corresponding to the paint layer is highest for the scarf measured point (Fig. 6.11f) and the background on the left side (Fig. 6.11e). This suggests that these paint layers are newer than the other ones, observation underlined by the X-ray photography, in which no other person except Rembrandt is visible on the painting, and the conservators' report, in which they remark that the paint layer in the region of the scarf is applied differently. Moreover, the depth profile

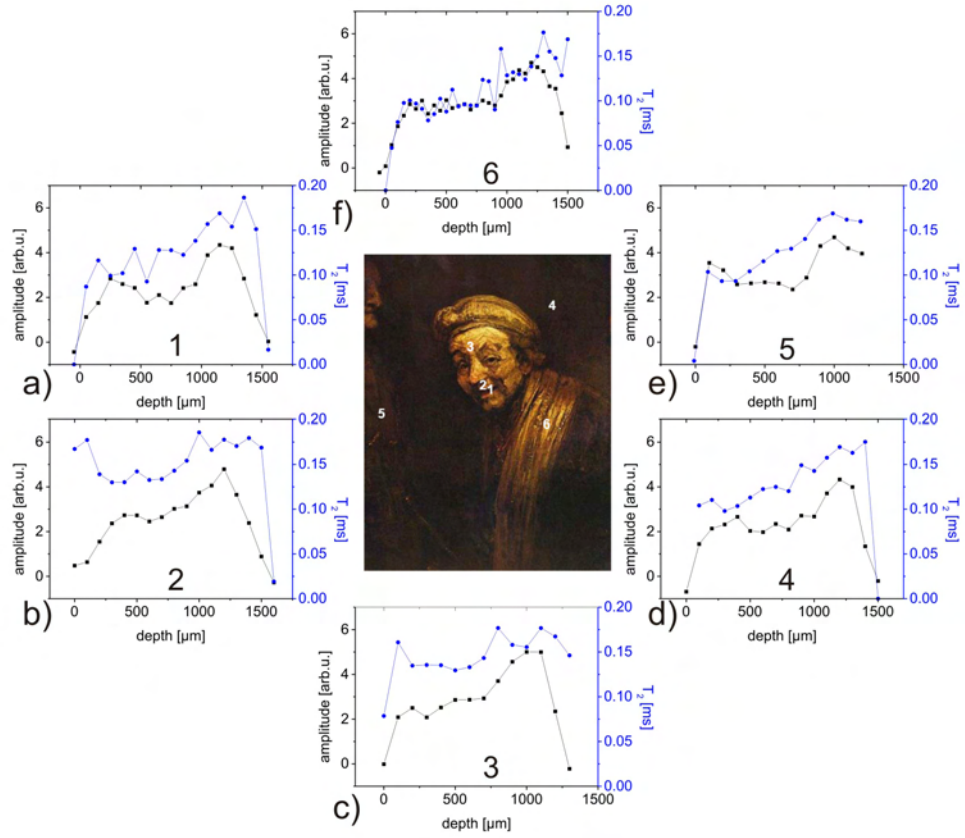


Figure 6.11: NMR depth profiles (black) and T_2 depth profiles (blue) for each of the measured points. Comparison between the depth profiles and T_2 profiles for the a) mouth, b) upper lip, c) forehead, d) dark background upper right, e) dark background middle left, and f) scarf.

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for the scarf is the broadest, revealing a thick paint layer. The higher T_2 values for the upper lip and forehead paint layer is a hint for the younger age of the paint layer in that region: the drier the paint layer, the lower the T_2 value. Of course, the T_2 value is influenced also by the composition of the paint. It may contain some paramagnetic elements which will lower the T_2 value. For example, the signal amplitude of the scarf paint layer is high compared with the other ones, but the T_2 values are low.

Along with the depth profiles and T_2 experiments, T_1 saturation recovery experiments were performed at the paint layer depth for each point. These experiments, together with the CPMG experiments, were made to investigate the aging of the binder. Similar experiments were reported in (Presciutti 08), where naturally aged old master paintings were compared to artificially aged tempera paints (Fig. 6.12a). The authors reported much lower relaxation times for natural aged master paintings. The T_1 relaxation times were between 3 and 10 ms,

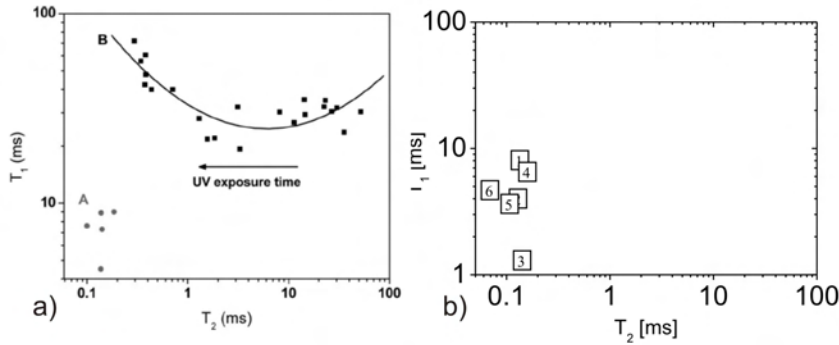


Figure 6.12: T_1 values compared with T_2 values for oil paintings a) Comparison for T_1 and T_2 values for naturally aged master paintings (point A) and artificially aged oil paintings (Presciutti 08). b) Comparison for T_1 and T_2 values for each measured point (1 to 6) for the Rembrandt self-portrait.

and the T_2 relaxation times were between 0.08 and 0.3 ms. In case of the Rembrandt self-portrait measurements the values for both relaxation times are in the same region. A challenge was to learn if it is possible to determine the age of a painting only by a T_1 NMR measurement. Several measurements on different old master paintings were already performed in our research group, and from these

measurements a T_1 contrast aging map was established (Fig.6.13). To obtain

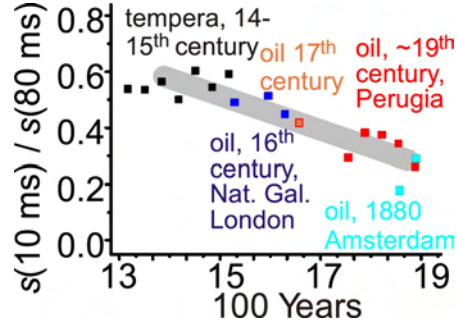


Figure 6.13: Natural aging by T_1 weighted data. Different oil and tempera paintings, measured over the years in our working group, age dated by the T_1 weighted data where the normalized signal at 10 ms is divided by the signal at 80 ms recorded in a T_1 saturation recovery experiment.

these T_1 weighted data, the signal from the saturation recovery measurement is first normalized, then the signal at 10 ms is divided by the signal at 80 ms. The results of the aging values for the Rembrandt are shown in Table 6.3. The va-

Table 6.3: T_1 weighted data for each measured position

Position	$S(10)/S(80)$	Position	$S(10)/S(80)$
1	0.469	4	0.38
2	0.6	5	0.68
3	0.67	6	0.54

lues for the T_1 weighted data presented in Table 6.3 raises some questions. If we compare the values for the T_1 weighted data in Fig. 6.13 and knowing that the Rembrandt self-portrait was made around 1668, the expected values should be between 0.37 and 0.5. But in the table only two of the measured points are in that range, point 1 and 4. One of the reasons why this is so could be the poor signal to noise ratio, due to lack of measurement time. Another reason could be the chemical composition of the paint: it may contain some compounds that are influencing the measurements. A third reason could be the fact that the several paint layers were not parallel and that our T_1 measured signal had a component from a different paint layer and this could influence the T_1 weighted data values.

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Another reason could be the fitting of the data. T_1 data was fitted biexponentially for the Rembrandt points, but all other data from the aging map was obtained by mono exponential fits of the T_1 data. The second component could come from the organic solvent vapor to which once the painting had been exposed. For the available data at least position 1 and 4 agree in this T_1 data with the aging curve.

6.2.2.2 Anonymous Cologne painting

The art of the 13th and 14th century was strong under the influenced of religion (Gombrich 08). In this period the big cathedrals were built all over Europe and the paintings illustrate scenes from the Bible. The painting we examined comes from Cologne region by an anonymous artist and is dated on 1330. It shows in the upper part Saint Elisabeth donating clothing to poor people and in the lower part nursing a sick man.

In order to test the aging curve, we performed depth profiles and T_1 measurements on this oil painting on wood. We chose this painting because it has fewer and flatter paint layers than the Rembrandt self-portrait, so the paint layers can be resolved much better with depth profiling. Another important reason is the difference of about 100 years between the first version and the second version of the painting (Fig. 6.14). In this way we can determine the age of the two different versions and fill the aging curve with another masterpiece.

The depth profiles recorded at the museum site are outlined in Figure 6.15, and the parameters used for the measurements are in Table 6.2. Because the measurements took place in the museum's laboratory (unlike the Rembrandt measurements, which were done in the exhibition room), the measurement time for one profile is almost three times longer. Depth profiles were recorded at the forehead of Jesus and the left part and right part of Saint Elisabeth's face. Both sides of St. Elisabeth's face were measured because a newer conservation treatment, mentioned in the archives, was applied on the right side of the face. Similarities between the stratigraphy of the measured points are noticeable and several layers are evidenced. The most intense signal amplitude is attributed to the wood on which the oil painting lies; this appears between 1500 and 2000 μm . The canvas produces a NMR signal between 1300 and 1500 μm . On the canvas, a



Figure 6.14: Anonymous Cologne painting a) Incident light photograph showing Saint Elisabeth dressing poor people (top) and caring for sick people (bottom). b) Mapping of the first version and the second version of the painting, with a difference of 100 years between them.

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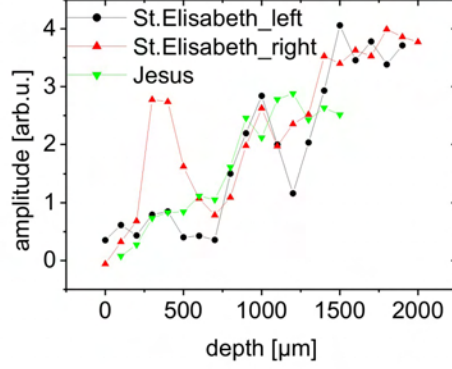


Figure 6.15: Measured depth profiles of Jesus' face and left and right of Saint Elisabeth face. The face of Jesus belongs to the first version of the painting and the face of Saint Elisabeth to the second version. Differences in stratigraphy and signal intensity are observed.

300 μm thick ground layer is applied. The region of interest is the 400 μm thick paint layer, then the varnish covers the whole surface of the painting. The signal intensity in the paint layer does not really differ from one profile to the next, which is understandable because the same paint color was applied. In order to resolve the difference in age, T_1 measurements were performed at the paint layer depth. Again, because we had much more time available to do our measurements, T_1 measurements were made at several depths. T_1 weighted data were obtained in the same way as for the Rembrandt measurements. The results, presented in Table 6.4, show clearly that different paint layers from the first version and the second version are from different ages. The T_1 weighted data for St. Elisabeth are around 0.58, corresponding to 14th century master paintings, and the T_1 weighted data for Jesus are around 0.7, corresponding to the 13th century. Of course, one of the T_1 weighted data values is much too high, but this is one reason why this project is still ongoing, to have a more accurate aging map.

Recent studies revealed that the positioning device present a hysteresis curve in its back and forth movement when it is not in the vertical position. This may induce an uncertainty in the results and in the aging map. It might be that the T_1 experiments were not performed exactly in the paint layer, that the sensitive slice

Table 6.4: T_1 weighted data for the paint layer depth for each measured profile.

Depth [μm]	St. Elisabeth left part	St. Elisabeth right part	Jesus
800			0.68578
900	0.58064		
1000		0.58779	0.74768
1100		0.55345	
1200	0.5881		

was in the paint layer part but also in the canvas layer and the signal recorded is from the spins in the paint layer as well as in the canvas. This could induce errors in the T_1 weighted data and make the paint layer appear to be older than it really is. However, the results obtained for the T_1 weighted data in the mouth region are in the correct age time. Moreover, with the results obtained on the other oil painting a better accuracy, lower than 100 years, for the T_1 weighted data and the aging map was obtained. These two facts lead us to the conclusion that the corrections made on the painting, near the mouth region, are less than 100 years away from the paintings original version.

6.3 Conclusions

A non-invasive investigation of ancient wall paintings and mosaics was performed using the NMR-MOUSE®. For the measurements at the Mosaic of Neptune and Amphitrite, differences between the tesserae and the mortar bed could be evidenced and strong indications were found for an impregnation of the tesserae with an organic substance. Different conservation treatments and their influence on the moisture content in wall paintings in the Black Room were evidenced and the efficacy of each of the conservation treatments was estimated. The study shows that the conservation treatment with ammonium oxalate leads to lower moisture content of the wall, while the treatment with barium hydroxide leads to higher moisture content.

For the first time, depth profiles of untreated, recently excavated wall paintings

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were recorded in the Villa of the Papyri, which identified two different building techniques. The results of all of these measurements are expected to help identify suitable conservation strategies and to detect undocumented restoration efforts of the past. These studies demonstrate that NMR is a suitable tool to analyze and characterize the layer structures of different wall paintings and mosaics. Measurements taken in the same point but at different time periods demonstrate the accuracy and reproducibility of the measurements. Moreover, in situ XRF studies coupled with reflectance FTIR spectra are in agreement with the NMR-MOUSE data and complemented the studies of the wall surfaces. It is shown that the NMR-MOUSE measurements coupled with in situ FTIR and XRF studies make a powerful, non-destructive methodology to access the conservation treatments and the techniques of wall paintings.

Further laboratory studies, to elucidate the mobility of the proton spins in terms of diffusion measurements, were in good agreement with the on-site measurement results. It was shown that treatment with ammonium oxalate leads to lower mobility of the proton spins inside the mortar, while treatment with barium hydroxide leads to higher mobility of the proton spins.

For the old master paintings, both of them being oil paintings, the stratigraphy could be elucidated. T_1 weighted data for the Rembrandt and the anonymous Cologne master paintings revealed the age of the painting and the aging map was filled with new data points. This shows that different stages of oil aging, or different aging conditions, can be discriminated by the NMR method. This indicates that NMR has the potential to differentiate between old and recently restored sections in a painting, thus raising the hope of identifying genuine and forged paintings.

Chapter 7

Conclusions

In this thesis, mobile, low-field NMR techniques were used to study porous system, from simple to complex structures, and objects of cultural heritage. It was shown that NMR relaxometry can be used to study the fluid dynamics inside a porous system. A simple theoretical model for multi-site relaxation exchange NMR permitted us to extract exchange kinetic parameters when applied on a model porous systems. It provided a first step towards the study of more complex systems, where continuous relaxation distributions are present, such as soil systems or building materials. The single-sided NMR sensor proved to be a useful tool for on-site measurements. This is very important also in the case of the cultural heritage objects, as most of the objects cannot be moved out of their environment.

At the beginning of this thesis, a general analytical solution for multi-site exchange NMR was established, with a focus on a simple two-site exchange problem. In order to understand the simple systems, experimental results were compared to results obtained from simulations for a simple two-site system made of water saturated silica particles. T_1 - T_2 and T_2 - T_2 exchange correlation maps were compared. Moreover, the integrals of each relaxation distributions and their cross peaks were compared for different mixing times. The overlapping of values obtained by simulation and experiments is remarkable. Complementary experiments were made at different temperatures for the same system and compared to simulations. The activation energy was extracted from the dependence of the exchange rate on temperature. These 2D exchange experiments were tested further on a

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more complex system, a water saturated aluminium oxide system, which showed a more complex relaxation-site distribution (as predicted in the simulations). The restricted diffusion of the water molecules inside this porous system was evidenced by a D - T_2 correlation map.

The efficiency of these 2D correlation experiments was proved furthermore on natural porous systems, on saturated and unsaturated soil systems. The results obtained for all three soil types proved that 1D and 2D NMR relaxometry methods are suitable to observe moisture migration in these types of porous media. Depending on the soil texture the pore size distribution varies from a simple relaxation distribution, obtained for sand, to multi-modal distributions, obtained for the two different silt loams. Water exchange is observable mostly within the small pores. Exchange within the big pores may occur, but the T_1 effect from the small pores is bigger and the small components vanish at longer mixing times. Once the saturation level becomes lower the water from the bigger pores vanishes and the T_2 values are shifting towards shorter values. All this results give us an insight about the pore connectivity and texture.

In case of the complex concrete samples, two different concrete types with different composition were examined on-site as well as in the laboratory. Because of their different composition, especially the aggregate composition, the water ingress in the saturation process did behave totally different. The use of the NMR-MOUSE showed to be appropriate, especially for the on-site measurements, where the measurements could be performed fast and non-invasive. In case of the tunnel measurements differences in profile amplitudes and T_2 values were observed, evidencing the influence of the environment onto concrete quality and lifetime. At the lock site, these differences were already expected, as one part of the concrete wall is exposed more often to water. The laboratory measurements proved to be efficient and informative. Not only the differences in water uptake rate could be evidenced, also the results could be compared with the MRE results and conclusions could be drawn. It is impressive how a difference in pore size, which was not even one order of magnitude, can influence so hardly the water dynamics in the concrete blocks.

Regarding the measurements on the cultural heritage objects, a non-invasive investigation of ancient wall paintings and mosaics was performed using the NMR-

MOUSE[®] in Herculaneum, Italy. For the measurements at the Mosaic of Neptune and Amphitrite, differences between the tesserae and the mortar bed could be evidenced. Different conservation treatments and their influence on the moisture content in wall paintings in the Black Room were evidenced and the efficacy of each of the conservation treatments was estimated. For the first time, depth profiles of untreated, recently excavated wall paintings were recorded in the Villa of the Papyri, which identified two different building techniques. Furthermore, laboratory studies were performed, to elucidate the mobility of the proton spins in terms of diffusion measurements in mock-up samples with different surface conservation treatments. In case of the old master paintings, the stratigraphy could be elucidated. T_1 weighted data for the Rembrandt and the anonymous Cologne master paintings could be used to make the painting dating method more accurate and the aging map was filled with new data points.

The results presented in this thesis, demonstrate that low-field NMR is a suitable tool to analyze different porous systems and characterize the layer structures of different cultural heritage objects. As a next step the numerical simulations could be expanded to study systems with continuous relaxation distributions and compared, for example, with the results that we obtained on the soils systems. A more detailed study of the soils systems, where clay and silt would be added gradually to sand, would help to understand how this components are influencing pore connectivity and texture. Complementary 2D correlation experiments, like D - T_2 correlation experiments, would lead to a better understanding of water transport between pores. This study could be applied also in the case of the concrete block. With the help of the on-site measurements the seasonal influence on the concrete could be monitored and compared with the MRE results. As for the cultural heritage objects, on-site diffusion and 2D correlation measurements would lead to a better understanding on moisture transport inside the wall painting.

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Observation of exchange of micropore water in cement pastes by two-dimensional T_2 - T_2 nuclear magnetic resonance relaxometry

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The first detailed analysis of the two-dimensional (2D) NMR T_2 - T_2 exchange experiment with a period of magnetization storage between the two T_2 relaxation encoding periods (T_2 -store- T_2) is presented. It is shown that this experiment has certain advantages over the T_1 - T_2 variant for the quantization of chemical exchange. New T_2 -store- T_2 2D ^1H NMR spectra of the pore water within white cement paste are presented. Based on these spectra, the exchange rate of water between the two smallest porosity reservoirs is estimated for the first time. It is found to be of the order of 5 ms^{-1} . Further, a careful estimate of the pore sizes of these reservoirs is made. They are found to be of the order of 1.4 nm and 10–30 nm, respectively. A discussion of the results is developed in terms of possible calcium silicate hydrate products. A water diffusion coefficient inferred from the exchange rate and the cement particle size is found to compare favorably with the results of molecular-dynamics simulations to be found in the literature.

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I. INTRODUCTION

Nuclear magnetic resonance (NMR) spin-lattice (T_1) and spin-spin (T_2) relaxation-time measurements have been shown to be a powerful tool for the characterization of liquids in porous media [1]. The fast exchange model of relaxation is often invoked and provides a powerful means of characterizing the pore surface-to-volume ratio. According to this theory, the observed relaxation rate is the weighted average of that for fast relaxing nuclei in molecules associated with pore surfaces and that for more abundant, but more slowly relaxing nuclei associated with molecules in the bulk liquid filling the pore [2]. In other work, NMR pulsed field gradient measurements of self-diffusion (D) of liquids in porous media and, in particular, of restricted diffusion, have been shown to be equally important for the characterization of pore structures [3]. More recently, two-dimensional relaxation correlation experiments, first suggested in the early 1980's [4], have come to the fore [5] following the availability of robust methods of performing the requisite 2D inverse Laplace transform [6]. In the usual form of these experiments, a measurement of T_1 made in a first measurement interval is correlated with a measurement of either T_2 or D in an immediately following second interval. In particular, T_1 - D experiments have been shown to be well suited to the separation of signals from oil and water in reservoir rocks [7]. Others have applied the methods to the characterization of emulsions and foods [8,9]. The T_2 - T_2 experiment is much less common and to date has not been part of this resurgence. However, it was first proposed more than a decade ago by Lee *et al.* [10]. The experiment is made useful by the inclusion of a magnetization storage period between the T_2 encoding periods, T_2 -store- T_2 , during which the magnetization is held along the longitudinal direction and during which exchange can occur.

Cement-based materials are a further significant class of porous materials. Understanding nanoscale pore-water interactions is important for both new build with fresh materials and for the repair of existing structures. With new build, for instance, knowledge of the degree of cure of cement is critical to the construction timeline. With regard to repair, it is accepted that nearly all forms of degradation of cement-based materials are ultimately linked to the availability of water and its transport through the porous structure.

Previous NMR relaxometry studies of cements have largely focused on simple measurements of T_1 and T_2 [11–13] and/or D [14,15]. These measurements have generally shown three or more distinct relaxation reservoirs that are generally associated with chemically combined water (the shortest T_2) through water in the cement gel structures (CSH) and onto water in capillary pores and microcracks (the longest T_2). While the broad picture remains constant across these several reports, numerous factors conspire to leave considerable uncertainty in the overall interpretation. Amongst these are the variety of materials and water-to-cement ratios used; the varying cure regimes adopted and hence the varying degree of cure and saturation of porosities encountered; the different NMR spectrometer frequencies available; and the acknowledged difficulty of inverting spin-relaxation data to yield relaxation rates.

In an earlier paper [16] we showed that T_1 - T_2 correlation experiments are valuable in helping to understand pore-water interactions in cement paste. Key findings included the following:

- (i) confirmation of a series of discrete-sized porosity reservoirs;
- (ii) the first observation of an off-diagonal peak linking relaxation of ^1H in the two smallest porosity reservoirs. This was attributed to water exchange between the gel and capillary structures;
- (iii) the observation that $T_1/T_2 = \alpha$ is a constant greater than 1 across the full spectrum of relaxation times for water in different-sized pore structures;

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(iv) new analysis that extended a model of surface relaxation by a two-dimensional random walk of nuclei across a pore surface passed paramagnetic relaxation sinks [17]. This analysis satisfactorily explained the constant α , and, by measurement of its frequency dependence [18], enabled an estimate of the pore surface residency time and the pore surface 2D walk hopping time, both consistent with previous measurements inferred from methodologically more complex field-cycling NMR experiments [19,20].

It is the purpose of this paper to carry this work forward, both in terms of the methodology and its analysis, and in terms of the interpretation of results in cements. For the first part, this paper focuses on analysis of the T_2 -store- T_2 exchange experiment. Although in our previous paper we presented the first use of a T_2 -store- T_2 exchange experiment for studying cement pastes, the application was not fully explored and only very preliminary results were included. It is now shown that this experiment has considerable advantages compared to a T_1 - T_2 experiment for studying materials such as cements. New calculations show that 2D spectra with strong off-diagonal peaks can be observed unambiguously with a T_2 -store- T_2 experiment in cases where there is ongoing exchange. Measurement of the relative intensity of the off-diagonal features as a function of the storage interval length permits an estimate of the exchange rate to be made. In the case of cement pastes this goes on to allow an estimate of water diffusivity within pore structures that can be compared to theoretical modelling.

II. MATERIALS AND METHODS

Small samples of white Portland cement were carefully hand mixed in the water-to-cement ratio 0.4 by mass, cast into 10 mm NMR tubes, gently vibrated on a vibration table to remove entrapped air, and allowed to cure unsealed for the first few hours. As soon as the samples solidified, a small quantity of saturated $\text{Ca}(\text{OH})_2$ aqueous solution (simulated pore-water) was poured onto the samples to allow continued cure underwater. This solution was temporarily removed during NMR measurements but replaced immediately thereafter. Notwithstanding this methodology, it is likely that a significant fraction of the open *capillary* porosity remained devoid of water since it is known that it is extremely hard to manufacture fully saturated samples. Samples were stored at 20 °C.

Samples were investigated with the T_2 -store- T_2 exchange experiment, defined by the pulse sequence

$$P_{90}-(\beta^i \tau_{\text{echo}}-P_{180}-\beta^j \tau_{\text{echo}}-\text{echo})_m-P_{90}-\tau_{\text{delay}}-P_{-90}-(\beta^j \tau_{\text{echo}}-P_{180}-\beta^i \tau_2-\text{echo})_n-\tau_{RD}.$$

This experiment consists of three time intervals: t_1 , t_2 , and t_3 . During the first interval a Carr-Purcell-Meiboom-Gill (CPMG) pulse train consisting of m inversion pulses and echoes is applied. In our implementation, the inversion pulse gap increases throughout the sequence according to a geometric sequence ($2\beta^i \tau_{\text{echo}}$, $i=0, 1, \dots, m-1$ and $\beta=\text{constant}$) in order to span efficiently the entire range of relaxation times from tens of microseconds to seconds. At the end of

the sequence, at the time of the m_{th} echo, a 90° pulse is applied so as to store the magnetization along the z axis. Phase cycling is used between averages so as to store along both $+/-z$ (see theory). The magnetization is stored for the period $\tau_{\text{delay}}=t_2$. Thereafter, the magnetization is recovered with a further 90° pulse. During the third period t_3 , a second CPMG sequence of n inversion pulses similarly spaced in geometric progression is applied and the series of echo intensities is recorded. The experiment is repeated for $m=1, 2, \dots, n$ with repetition delay τ_{RD} . In this manner, a square two-dimensional relaxation data set is acquired.

Measurements were made at room temperature on a 20 MHz ^1H NMR frequency Maran benchtop spectrometer (Oxford Instruments Molecular Biotoools Ltd, Oxon, UK). The 90° pulse length was 2.5 μs . In a typical experiment, the inversion pulse gap interval of the CPMG sequence increased from 60 μs to 0.2 s over $n=32$ echoes so that the T_2 range up to circa 1 s was well explored. However, signal from the chemically combined water (with a T_2 of only a few microseconds) was not seen. The storage interval t_2 was kept constant in any one experiment and was of the order of 0–30 ms. As many as 128 averages of each echo train were recorded with a repetition time of 1 s in an experiment lasting circa 2.5 hours.

The 2D data sets were carefully phased prior to analysis. None of the data sets showed a noticeable baseline offset. Relaxation time distributions were extracted from the data by 2D Laplace inversion using the algorithm proposed by Venkataramanan *et al.* [21] using software running on a desktop computer. All subsequent data plots are as output by this algorithm except that intensities have been normalized to the number of averages recorded.

III. THEORY

A. T_2 -store- T_2 experiment

We consider two reservoirs of magnetization, M^a and M^b , corresponding to ^1H (water) in pores of different sizes. There is exchange between the reservoirs. The system of coupled differential equations governing the relaxation of the magnetization in the reservoirs is well known [22], viz.,

$$\begin{aligned} \frac{dM_a}{dt} &= -k_a M_a + k_b M_b + R_a (M_a^{\text{eq}} - M_a), \\ \frac{dM_b}{dt} &= -k_b M_b + k_a M_a + R_b (M_b^{\text{eq}} - M_b), \end{aligned} \quad (1)$$

where $k_{a,b}$ is the exchange rate, $R_{a,b}$ is the nuclear spin-lattice or spin-spin relaxation rate as appropriate, $M_{a,b}^{\text{eq}}$ is the equilibrium magnetization for each reservoir for the relevant process, and t is time. For spin-lattice relaxation $M_{a,b}^{\text{eq}}=M_{a,b}^0$, the overall equilibrium value, while for spin-spin relaxation $M_{a,b}^{\text{eq}}=0$. The condition of detailed balance requires that $k_a M_a^0 = k_b M_b^0$. Conveniently, under conditions of detailed balance, the equations can be written in the form

$$\begin{bmatrix} d(M_a - M_a^{eq})/dt \\ d(M_b - M_b^{eq})/dt \end{bmatrix} = \begin{bmatrix} -R_a - k_a & k_b \\ k_a & -R_b - k_b \end{bmatrix} \begin{bmatrix} M_a - M_a^{eq} \\ M_b - M_b^{eq} \end{bmatrix}. \quad (2)$$

The solutions to Eq. (2) are also well known [22] and have been presented previously in the notation of this work by us. [16] In order to treat the T_2 -store- T_2 experiment, the time is divided into three periods, t_1 , t_2 , and t_3 , corresponding to the three sequential elements of the experiment. The starting conditions are $M_{a,b}(0) = M_{a,b}^0$ and the magnetization $M_{a,b}(t_1)$ is calculated using spin-spin relaxation parameters [Eq. (10) of Ref. [16]]. $M_{a,b}(t_1)$ becomes the starting conditions for the second period. Hence, using spin-lattice relaxation parameters, the magnetization at the end of the second period, $M_{a,b}(t_1, t_2)$ is calculated. Likewise, $M_{a,b}(t_1, t_2)$ is the starting condition for period t_3 so that, finally, reverting to spin-spin relaxation parameters, $M_{a,b}(t_1, t_2, t_3)$ is calculated. The observed magnetization M is the sum of M_a and M_b . The final solution is given by

$$s_{1,2}^{+/-} = -\frac{R_a^{1,2} + k_a + R_b^{1,2} + k_b}{2} \pm \frac{1}{2} \sqrt{(R_a^{1,2} + k_a + R_b^{1,2} + k_b)^2 - 4[(R_a^{1,2} + k_a)(R_b^{1,2} + k_b) - k_a k_b]}, \quad (7)$$

$$A_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_a^{1,2} + k_a}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \quad (8)$$

$$B_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_b^{1,2} + k_b}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \quad (9)$$

$$F_{1,2} = \frac{k_a}{s_{1,2}^- - s_{1,2}^+}, \quad (10)$$

and

$$G_{1,2} = \frac{k_b}{s_{1,2}^- - s_{1,2}^+}. \quad (11)$$

Expressions for C_4 , C_5 , and C_6 are found by interchanging a and b throughout C_1 , C_2 , and C_3 , respectively. Finally, $R_a^1 = 1/T_{1a}$, for example, is the spin-lattice relaxation rate for reservoir a .

The solution can be rewritten as a sum of products including factors $\exp(s_2^{+/-}t_1)\exp(s_2^{+/-}t_3)$ and other terms not including these factors. The amplitudes of terms involving factors of $\exp(s_2^{+/-}t_1)\exp(s_2^{+/-}t_3)$ correspond to the intensities of the diagonal, P^{++} and P^{--} and off-diagonal, P^{+-} and P^{-+} peaks in the 2D Laplace inversion of the magnetization. The other terms are an unwanted distraction and are conveniently removed by phase cycling. Experimentally, the magnetization is stored along the $+z$ and $-z$ axes during t_2 . Theoretically, the calculation is repeated using $-M_{a,b}(t_1)$ at the start of the

$$M(t_1, t_2, t_3) = C_1(t_3)[C_2(t_1, t_2) + C_3(t_1, t_2)] + C_4(t_3)[C_5(t_1, t_2) + C_6(t_1, t_2)], \quad (3)$$

where

$$C_1(t_3) = (A_2^- - F_2)\exp(s_2^+t_3) + (A_2^+ + F_2)\exp(s_2^-t_3), \quad (4)$$

$$C_2(t_1, t_2) = M_a^0 + [A_1^- \exp(s_1^+t_2) + A_1^+ \exp(s_1^-t_2)] \times [-M_a^0 + (A_2^- M_a^0 - G_2 M_b^0)\exp(s_2^+t_1) + (A_2^+ M_a^0 + G_2 M_b^0)\exp(s_2^-t_1)], \quad (5)$$

$$C_3(t_1, t_2) = -G_1[\exp(s_1^+t_2) - \exp(s_1^-t_2)] \times [-M_b^0 + (B_2^- M_b^0 - F_2 M_a^0)\exp(s_2^+t_1) + (B_2^+ M_b^0 + F_2 M_a^0)\exp(s_2^-t_1)], \quad (6)$$

and where

second period. The results of the two variants are summed whereupon the unwanted terms cancel to zero leaving just those wanted. The resultant amplitudes are

$$P^{++} = \exp(s_1^+t_2)\{(A_2^- - F_2)[A_1^-(A_2^- M_a^0 - G_2 M_b^0) - G_1(B_2^- M_b^0 - F_2 M_a^0)] + (B_2^- - G_2)[B_1^-(B_2^- M_b^0 - F_2 M_a^0) - F_1(A_2^- M_a^0 - G_2 M_b^0)]\} + \exp(s_1^-t_2)\{(A_2^+ + F_2)[A_1^+(A_2^+ M_a^0 - G_2 M_b^0) + G_1(B_2^+ M_b^0 - F_2 M_a^0)] + (B_2^+ - G_2)[B_1^+(B_2^+ M_b^0 - F_2 M_a^0) + F_1(A_2^+ M_a^0 - G_2 M_b^0)]\}, \quad (12)$$

$$P^{--} = \exp(s_1^+t_2)\{(A_2^+ + F_2)[A_1^-(A_2^+ M_a^0 + G_2 M_b^0) - G_1(B_2^+ M_b^0 + F_2 M_a^0)] + (B_2^+ + G_2)[B_1^-(B_2^+ M_b^0 + F_2 M_a^0) - F_1(A_2^+ M_a^0 + G_2 M_b^0)]\} + \exp(s_1^-t_2)\{(A_2^+ + F_2)[A_1^+(A_2^+ M_a^0 + G_2 M_b^0) + G_1(B_2^+ M_b^0 + F_2 M_a^0)] + (B_2^+ + G_2)[B_1^+(B_2^+ M_b^0 + F_2 M_a^0) + F_1(A_2^+ M_a^0 + G_2 M_b^0)]\}, \quad (13)$$

$$P^{+-} = P^{-+} = \exp(s_1^+t_2)\{(A_2^+ + F_2)[A_1^-(A_2^- M_a^0 - G_2 M_b^0) - G_1(B_2^- M_b^0 - F_2 M_a^0)] + (B_2^+ + G_2)[B_1^-(B_2^- M_b^0 - F_2 M_a^0) - F_1(A_2^- M_a^0 - G_2 M_b^0)]\} + \exp(s_1^-t_2)\{(A_2^+ + F_2)[A_1^+(A_2^- M_a^0 - G_2 M_b^0) - G_2 M_b^0] + G_1(B_2^- M_b^0 - F_2 M_a^0) + (B_2^+ + G_2)[B_1^+(B_2^- M_b^0 - F_2 M_a^0) - F_2 M_a^0] + F_1(A_2^- M_a^0 - G_2 M_b^0)\}. \quad (14)$$

In the event that $T_1 = T_2$, then the off-diagonal peak intensities evaluate to zero. However, experimental work suggests

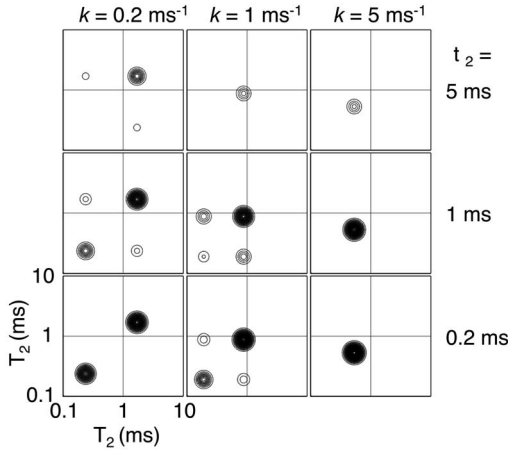


FIG. 1. Calculated 2D T_2 -store- T_2 spectra for (left to right) slow, medium, and fast exchange rates and (bottom to top) short, intermediate, and long storage times. Calculation parameters are given in the text. The plots are on a common scale with the contours spaced at 1.5% of the total magnetization intensity.

that, in cements as in some other porous media, $T_1 \neq T_2$. In our previous work on cements at a Larmor frequency of 20 MHz we observed two principal relaxation reservoirs of comparable volume differing in T_1 by a factor of approximately ten each with T_2 a factor of 4 smaller. The common T_1/T_2 ratio of 4 is discussed further in the next section. Here we focus on how the four peak intensities vary with the storage time t_2 for different exchange rates.

We have performed numerical calculations with $M_a^{eq} = M_b^{eq} = 0.5$ and $T_{2a} = T_{1a}/4 = 0.25$ ms and $T_{2b} = T_{1b}/4 = 2.5$ ms. For reservoirs of equal size, $k_a = k_b = k$. Figure 1 shows simulated 2D correlation spectra for each of “slow,” “medium,” and “fast” exchange rate ($k = 0.2, 1.0$, and 5.0 ms $^{-1}$, respectively) with a “short,” “intermediate,” and “long” storage interval ($t_2 = 0.2, 1$, and 5 ms, respectively). The terms slow and long etc. are defined relative to the difference in the spin-lattice relaxation rates. The calculated spectral peaks are delta functions: they have been artificially broadened for the sake of pictorial clarity. The contour plots are all shown on the same vertical scale, relative to a zero baseline, and are directly comparable.

With slow exchange and only a short storage interval [Fig. 1 (bottom left)], then the spectrum consists essentially of two strong diagonal peaks P^{++} and P^{--} , of comparable amplitude and located close to the intersections $(T_2, T_2) = (0.25, 0.25)$ and $(2.5, 2.5)$ where T_2 is expressed in ms, as expected. In the limit $k \rightarrow 0$ and $t_2 \rightarrow 0$ the peaks are of equal amplitude and centered exactly on these intersects. With increasing exchange rate (bottom center) the two diagonal peaks start to drift and change in relative intensity such that one becomes dominant in the expected fashion. Additionally, two off-diagonal peaks P^{+-} and P^{-+} appear as the signature of the ongoing exchange. With fast exchange (bottom right), only one diagonal peak is seen at a location $(0.45, 0.45)$, determined by the average T_2 relaxation rate. The off-diagonal peaks are no longer seen. As the storage time increases (from the bottom to the top of Fig. 1), so the integrated intensity across the four peaks declines due to spin-lattice relaxation

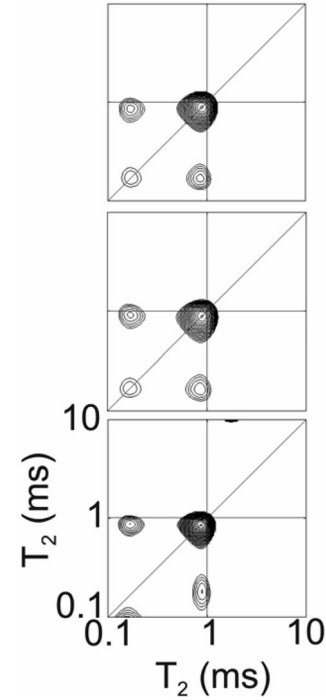


FIG. 2. Laplace inversions of simulated NMR data sets with (top) no noise, (middle) 2% white noise, and (bottom) 2% white noise plus 1% baseline offset. These plots have the same initial parameters as the center plot of Fig. 1 ($k = 1$ ms $^{-1}$, $t_2 = 1$ ms).

losses. This is particularly true for the lower diagonal peaks that have short T_2 and hence correspondingly short T_1 . It is also true, however, that the fraction of the spectral intensity in the off-diagonal peaks increases.

In order to assess how the Laplacian inversion affects data, we have simulated NMR signals and added noise and systematic errors and subjected these to the 2D inverse Laplace transform for comparison with Fig. 1. The simulated signals were generated with $k = 1$ ms $^{-1}$, $t_2 = 1$ ms, $T_2^a = 0.25$ ms, $T_2^b = 2.5$ ms, and $T_1/T_2 = 4$. In the absence of added noise, it can be seen, with reference to Fig. 2 (top), that the positions of the peaks obtained by this method are almost identical to those obtained by direct numerical simulation [Fig. 1 (center)]. A comparison of the integrated peak intensities with the direct numerical simulations reveals an absolute error of 3% in the intensity of the off-diagonal peaks. This translates to a relative error of 15% in the fraction of the intensity appearing in the off-diagonal peaks. The inclusion of 2% white noise [Fig. 2 (middle)], made a marginal further difference to the inversion. However, the addition of 1% baseline offset [Fig. 2 (bottom)], distorted the plot, forcing the P^{--} peak to lower values of T_2 .

Figure 3 shows the calculated intensity within the off-diagonal peaks I_{OD} , relative to the total signal intensity I_{TOT} ,

$$\frac{I_{OD}}{I_{TOT}} = \frac{P^{+-} + P^{-+}}{P^{++} + P^{-+} + P^{+-} + P^{--}}, \quad (15)$$

plotted as a function of storage (t_2) and exchange time ($\tau = 1/k$) for $T_{1a} = 4T_{2a} = 1$ ms and $T_{1b} = 4T_{2b} = 7.1$ ms. It is clear

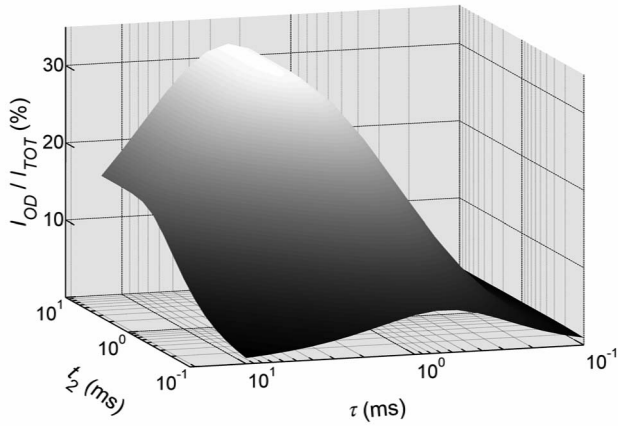


FIG. 3. A surface plot showing the fraction of the observed spectral intensity in the off-diagonal peaks as a function of exchange time $\tau=1/k$, and storage time t_2 . The calculation parameters are given in the text.

that, under favorable conditions, as much as 30% of the magnetization can appear in the off-diagonal peaks.

B. Surface relaxation

The observed spin-spin relaxation time T_2^{OBS} can be related to a pore surface-to-volume ratio (S/V ; pore size) through the well-known equation

$$\frac{1}{T_2^{OBS}} = \frac{\lambda S}{V} \frac{1}{T_2^S}, \quad (16)$$

where T_2^S is the relaxation time for ^1H in molecules adsorbed on the pore surface and λ is the thickness of a surface layer. In Eq. (16), the contribution of the bulk spin-spin relaxation rate has been eliminated due to its very small value. A key issue is to evaluate T_2^S . In our previous work [16,17,19], a model for surface relaxation was proposed and shown to be applicable. The model supposes biphasic fast exchange between liquid molecules that are transiently adsorbed on the pore surface and molecules in the bulk pore liquid phase. The adsorbed molecules relax as they undergo 2D translational diffusion in the dipolar field of the electronic spins fixed at the pore surface. The frequency dependence of this relaxation has been carefully evaluated. It depends on two correlation times, namely, the correlation time τ_m of surface diffusion events and a finite residence time τ_s of the protons on the surface. In most cases $\tau_m/\tau_s \ll 1$. In the case of T_2 , the model analysis yields the result

$$\frac{1}{T_2^S} = \frac{4}{9} \gamma_I^2 \gamma_S^2 \hbar^2 S(S+1) \left\{ J_L^0(0) + \frac{1}{4} J_L^0(\omega_I - \omega_S) + \frac{3}{4} J_L^1(\omega_I) + \frac{3}{2} J_L^1(\omega_S) + \frac{3}{2} J_L^2(\omega_I + \omega_S) \right\}, \quad (17)$$

where γ_S and γ_I are the gyromagnetic ratio of the electron and proton, respectively, S is the spin of the paramagnetic impurities (Fe^{3+}) embedded in the pore surface, and ω_S and ω_I are the electron and proton resonant frequencies, and where

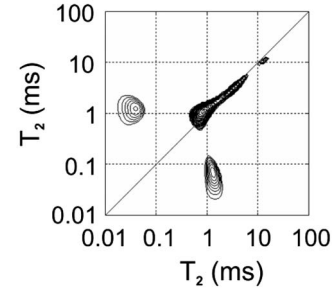


FIG. 4. Experimental T_2 -store- T_2 2D correlation spectrum for a white cement paste with water-to-cement ratio 0.4 and three days old. The storage time is 10 ms. The spectrum, which shows a strong upper diagonal peak and two off-diagonal peaks, is qualitatively comparable to the central part of Fig. 1.

$$J_L^{0,1,2}(\omega) = \frac{3}{40} \frac{\pi \sigma_S}{\delta^4} \tau_m \ln \left[\frac{1 + \omega^2 \tau_m^2}{\left(\frac{\tau_m^2}{\tau_s^2} \right) + \omega^2 \tau_m^2} \right]. \quad (18)$$

Here, σ_S is the surface density of paramagnetic impurities and δ is the distance of minimum approach of the water ^1H to the paramagnetic (Fe^{3+}) ions. The result can be simplified using the inequality $\omega_S \gg \omega_I$ ($\omega_S = 658.21 \omega_I$). In the case of spherical pores, $S/V = 6/d$, where d is the pore diameter and $\lambda = x\epsilon$ where x is the number of adsorbed water molecules of diameter ϵ in the surface layer. For planar pores, such as might better characterize CSH gel structures, $S/V = 2/d$, where d is the pore width.

A similar analysis has been performed for T_1 from which the ratio T_1/T_2 is calculated. For a fixed Larmor frequency, the value achieved for T_1/T_2 is not universal but depends critically on the ratio τ_m/τ_s . In particular, T_2/T_1 reaches the value 4 when $\tau_m/\tau_s = 10^{-3}$. What is important in our 2D T_1 - T_2 data used to underpin the modeling in the previous section, is the strong correlation of all the discrete pore size distribution features along the line $T_1/T_2 = 4$. Given that paramagnetic impurities are expected to be distributed uniformly throughout the solid matrix of the sample, this provides strong supportive evidence for the surface diffusion model of proton nuclear spin relaxation in cements. Further support comes from the predicted frequency dependence of this ratio, which agrees well with experimental results [18].

IV. EXPERIMENTAL RESULTS

A. 2D correlation experiments

Figure 4 shows the results of an experiment to measure T_2 -store- T_2 correlations in a white cement sample 4 days old. In this case, the storage interval was 10 ms. The spectrum shows a clear diagonal ridge with a primary peak P^{++} at circa $(T_2, T_2) = (1, 1)$ and two strong off-diagonal peaks P^{+-} and P^{-+} at circa $(0.05, 1)$ and $(1, 0.05)$, respectively. There is no evidence for the second lower diagonal feature P^{--} and so, superficially at least, the spectrum is comparable to that in the upper left of Fig. 1.

Figure 5 shows a second example. In this case the storage time is reduced to 1 ms. Now, overall, there is considerably

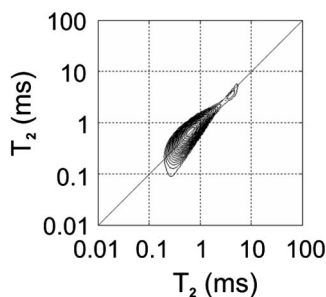


FIG. 5. Experimental T_2 -store- T_2 2D correlation spectrum for a white cement paste with water-to-cement ratio 0.4 and four days old. The storage time is 1 ms. Compared to Fig. 3, the contour spacing is $10\times$ greater: the maximum intensity is almost $10\times$ as great. The spectrum, which shows a strong diagonal ridge, is qualitatively comparable to the lower left/central parts of Fig. 1.

more intensity in the spectrum (notice the expansion of the contour interval). However, it is located predominantly along the diagonal. The off-diagonal features are not seen. The spectrum is more comparable to the lower left of Fig. 1. The slight asymmetry in this spectrum could be due to one of several causes including ongoing hydration of the sample during the measurement introducing systematic error or failure of the Laplace inversion.

To check the efficacy of the pulse sequence, a T_2 -store- T_2 experiment with a store time $t_2=5$ ms was performed on a four-day-old sample of white cement. The experiment was repeated with the first 90° pulse of the sequence suppressed. In the latter case, the signal intensity echo showed no signal arising from unwanted coherences visible above the baseline noise. Equally, the inverse Laplace transform spectrum showed no clear structural features. The ratio of the integrated intensity of the spectra between the two experiments was $<0.4\%$, well below the amplitude associated with the exchange peaks where they occur. We are therefore confident that the observed exchange peaks are not an artifact of the pulse sequence implementation.

Figure 6 shows the fractional intensity in the off-diagonal peaks as a function of storage time for all the white cement samples of age four days and water-to-cement ratio 0.4 that have so far been surveyed in our laboratory. The circles are for data recorded in the experiments discussed here. The triangles are for earlier preliminary experiments where the signal-to-noise ratio was not as good. Also, the cement powder used was from a different (older) batch and the sample storage was not as carefully controlled as here. As expected, it is found that the fractional intensity in the off-diagonal features increases with storage time. However, since the overall signal intensity is declining, so the measurement error also increases. The error bars have been obtained as follows. The standard deviation of the baseline signal in the tail of the raw data was calculated. Gaussian white noise of the same standard deviation was generated artificially and added to the raw data. The modified data was inverted and the peak intensities and ratios recalculated. The process was repeated several times and the standard deviation of the resultant peak intensities and ratios was thus calculated and taken as an estimate of the experimental error. It is noted that even after

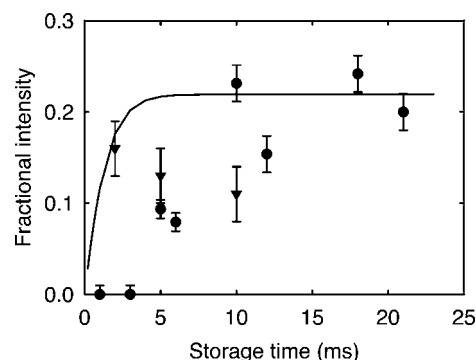


FIG. 6. The integrated intensity in the off-diagonal peaks of measured 2D spectra as a fraction of the total intensity as a function of storage time for white cement paste samples four days old with water-to-cement ratio 0.4. The solid line is a numerical calculation according to the theory developed in Sec. III A. The symbols are data for two series of experiments. The parameters are given in the text, Sec. IV A.

consideration of the errors in this way, there remains considerable scatter in the data. There are at least two potential reasons for this. First, at four days, the cement is still hydrating and the microstructure rapidly developing during the course of the measurement. This is in strong contrast to other porous media often studied by these methods, such as rock cores. This could lead to systematic error. Moreover, when two measurements at different storage times are made on the same sample, the measurements are necessarily for differently aged samples. If two samples are used, then they may be significantly different due to the mechanical mixing and natural sedimentation and bleeding of the cement. Second, as shown in Sec. III A, noise in the data and particularly a residual baseline offset can systematically affect the Laplace inversion and hence the measured peak intensity ratio.

The solid line in Fig. 6 is an attempt to fit the theory of Sec. III A to the data. A least-squares analysis has not been possible due to the very large number of unknown parameters and the fact that the sample almost certainly contains a distribution of pore sizes. The fit shown is based upon two reservoirs of equal volume, as justified below, with T_2 relaxation times of $250\ \mu\text{s}$ and $1.8\ \text{ms}$, respectively and corresponding T_1 relaxation times four times longer. The exchange rate k is $0.2\ \text{ms}^{-1}$. The analysis tends to overestimate the rate of increase in the off-diagonal magnetization with increased storage time. This is not readily overcome by varying the parameters within acceptable limits: better fits can only be achieved by using unrealistically long T_1 and T_2 times. However, without improved data, which our experience shows is very difficult to obtain, this discrepancy is not readily resolved.

Finally, Fig. 7 shows the total spectral intensity and the fraction of this intensity contained within the bottom left corner spectral features (i.e., that corresponding to the smallest and next-smallest pore-size porosity reservoirs) as a function of the sample age for white cement pastes with a water-to-cement ratio of 0.4. These data are extracted from T_1 - T_2 correlation experiments of the sort previously described rather than T_2 -store- T_2 experiments because the T_1 - T_2 ex-

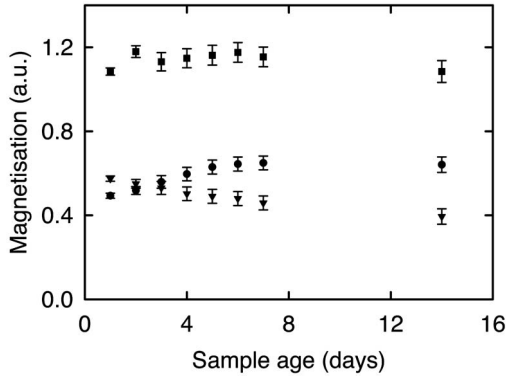


FIG. 7. The signal intensity (squares) and intensity within the fraction representing the smallest (circles) and next-smallest (triangles) pore-size porosity reservoirs as a function of curing time for a white cement paste with a water-to-cement ratio of 0.4. The data are evaluated from T_1 - T_2 correlation experiments of the sort previously described, rather than T_2 -store- T_2 correlations to minimize the complications of exchange averaging.

periment is better for quantization of the intensity of the lower diagonal peak corresponding to the smallest pores. It is seen that the total magnetization remains constant within experimental error throughout the curing period from 1 to 14 days. This suggests that the total water-filled porosity is not varying greatly. Over 95% of the magnetization is found in the spectral features corresponding to reservoirs of the two smallest pores. Of this magnetization, the fraction occurring in the smallest gel pores increases from circa 45% to 62% over the same period. This is consistent with the growth of CSH gel phase. The two small pore reservoirs are of approximately equal magnitude after three days, thereby justifying the use of reservoirs of equal volume in the previous analysis.

B. Measurement of the pore size

Table I gives estimates of the parameters required to calculate the principal characteristic pore sizes in the measured cements using Eq. (16)–(18) along with the origin of these estimates. The experimentally measured values of the two principal relaxation reservoirs T_2^{OBS} are circa 300 μ s and 2.0 ms. If the smaller pores are planar as seems reasonable for CSH gel structures, then the water layer is calculated to be 1.0+/- 0.3-nm-thick. The larger pore shape is less certain. If it is planar, then the most likely size is 6+/-2 nm; if spherical then the diameter is 19+/-6 nm. The significance of these sizes is discussed in Sec. V B. Here we focus on the measurement accuracy. The uncertainties in the above analysis are based on the uncertainties in the parameters given in Table I, including T_2^{OBS} , but excluding δ , the distance of closest approach of the water 1 H to the ferric ions and x , the effective number of water molecules in the surface layer. The analysis is sensitive to both of these values. Careful inspection of the intermediate, values leading to these results suggests two related difficulties. The first is that $2x\epsilon$, the combined thickness of the two surface layers on either side of a planar pore exceeds the pore thickness. The second is that the surface relaxation time $T_2^s=520 \mu$ s exceeds the smallest value of $T_2^{OBS}=300 \mu$ s. Both results are physically unrealistic. If the calculation is repeated for smaller values of $\delta=0.20$ nm and a monosurface layer $x=1$, then the pore size is increased only slightly, as the two changes largely cancel. The small planar pore spacing evaluates to 1.6+/-0.3 nm thick, for instance. However, a physically realistic value of $T_2^s=100 \mu$ s is found and the surface layers fit within the pore. A distance of minimum approach $\delta=0.2$ nm between the proton and the ferric ions on the surface is greater than half of the water molecule diameter. However, it is known that water has extensive hydrogen bonding capabilities and may exchange protons with other surface sites. It may be

TABLE I. Best estimates of the parameters required to calculate the pore size from the observed NMR relaxation times.

Parameter	Best estimate	Uncertainty	Source
σ_s	$2.5 \times 10^{11} \text{ Fe}^{3+} \text{ cm}^{-2}$	$\pm 20\%$ (model dependent)	Calibrated ESR experiment on this cement following method of Ref. [19]
δ	0.3 nm	+20/-100%	See text, Sec. IV B; Ref. [19]
τ_m	1.3 ns	± 0.3 ns	Field cycling NMR, Ref. [19]
τ_s	13 μ s	$\pm 5 \mu$ s	T_1 - T_2 NMR correlations, Refs. [16,18]
x	1-3	see text, Sec. IV B	Refs. [23,24]
ϵ	0.28 nm	$\pm 5\%$	Ref. [25]
T_2^{OBS}			
Large	2000 μ s	$\pm 10\%$	This work in the short t_2 limit and T_1 - T_2 correlations, Ref. [16]
Small	300 μ s	$\pm 10\%$	

have as both a Lewis acid or base and generally coordinates to most metal ions. Hence it is very plausible that there are labile-OH proton species at such a small distance from the ferric ions. We have studied elsewhere [26] the anomalous surface dynamics of water in nanoporous silica glasses and have considered in detail the local geometry of minimal approach of a proton species to a ferric ion.

V. DISCUSSION

A. 2D correlation experiments

The T_2 -store- T_2 experiment has been shown to give 2D relaxation spectra from white cement pastes with considerable information content. The storage period is critical to the experiment. Without it ($t_2=0$) the experiment collapses into no more than a conventional CPMG experiment, albeit with the data collected and presented in an unusual fashion. As such, the inverted 2D spectrum is expected to be entirely diagonal and to contain no additional information over its 1D counterpart. This has been confirmed experimentally by the appropriate analysis of a 1D data set although the trivial result is not shown here. The t_2 interval breaks the inherent symmetry of the experiment and yields off-diagonal peaks in the presence of exchange so long as $T_1 \neq T_2$. With small storage time, the fractional intensity in the off-diagonal elements is small. As t_2 gets longer, so the relative intensity in these peaks increases. However, the overall integrated intensity in the spectrum decreases due to spin-lattice relaxation losses occurring during t_2 . The T_2 -store- T_2 experiment has two significant advantages over the T_1 - T_2 experiment, which was the focus of our previous work. First, the new experiment more clearly identifies and correlates measurements in two intervals t_1 and t_3 that are separated in time by a third interval, t_2 . It is therefore more straightforward to consider that there is an exchange period during which the ^1H are moving between reservoirs. Second, the off-diagonal peaks can be assigned unambiguously to exchange. If there is no exchange, then there are no off-diagonal peaks. In the case of the T_1 - T_2 experiment, theory shows that the off-diagonal peaks are antisymmetric. However, the fast Laplace inversion algorithm does not detect the negative peak and leaves the ambiguity that the sole surviving positive off-diagonal peak may be due to a further ^1H reservoir with $T_1 \neq T_2$, perhaps a solid component. As an aside, we note that we previously suggested on the basis of preliminary results [16] that the off-diagonal peaks in the T_2 -store- T_2 experiment become asymmetric as t_2 increases due to the T_1 weighting. This is now seen as erroneous from both the theoretical and experimental points of view. Equation (3) shows that the off-diagonal peaks should be symmetric; equally, the further and improved data such as is shown in Figs. 4 and 5 suggest that this is true. The T_2 -store- T_2 experiment is not always to be preferred over T_1 - T_2 . Empirically, we find that the T_1 - T_2 experiment is better for the quantization of the intensity of the lower diagonal peak; hence this experiment was used to generate the data needed for Fig. 7.

B. Measurement of the pore size

The calibration of the pore sizes suggests that the two primary relaxation reservoirs seen both comprise rather small

pores. The smallest pore size is circa 1.0–1.7 nm. This is consistent with estimates to be found in the literature of the CSH interplanar spacing, which range from 0.9 to 1.4 nm [27]. However, the second significant reservoir suggests a size ranging from 7 to 30 nm, dependent on pore shape and details of the surface water layer. This is much smaller than the expected size of capillary pores. We therefore now suggest that the two observed reservoirs may be attributed to alternate gel structures. One possibility is that the smaller pore-size reservoir is associated with the inner product, the larger with the outer product, both created as the material hydrates. An alternate possibility is that the smaller-sized pores are associated with the intersheet spacing within CSH and the larger with the spaces between the packets of CSH sheets comprising the gel, but still much smaller than the normally accepted capillary porosity. This discussion will be expanded in a future discussion of results obtained from specific and otherwise (e.g., by electron microscopy, etc.) well-characterized CSH phases such as alite. For the time being we make just the following observations. The lack of directly observed capillary water in larger pores is not inconsistent with the methodology of preparation of the samples. The samples are exposed to air for the first 12 h of cure, and although they are subsequently stored underwater, any open capillary porosity is likely to remain unfilled except at the sample surface.

Figure 7 suggests that the total volume of water-filled porosity within our samples is not changing with time. This is consistent with the known sample history where, although the samples are cured underwater from age ≈ 12 h onward, the mobile water content is unlikely to be changing significantly. However, Fig. 7 further shows that the distribution of water between the different internal reservoirs is changing. There is a clear increase in the fraction of water associated with the smallest calcium silicate hydrate (CSH) pores during the first 14 days. This is consistent with a growth of CSH.

C. Water diffusivity

Finally, the data presented in Fig. 5 suggest an empirical exchange time of circa 5 ms between the two porosity reservoirs. The particle-size distribution of the cements used in this study is strongly centered on a radius $r=7.5 \mu\text{m}$. A typical length scale between inner and outer product within a hydrating grain will therefore be $r/2 \approx 4 \mu\text{m}$. A diffusion coefficient can be calculated from this time ($t=5$ ms) and distance ($a=r/2=4 \mu\text{m}$) using the Einstein equation

$$D = \frac{\langle a^2 \rangle}{6t}. \quad (19)$$

The result is $5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$. In this simple analysis, issues such as tortuosity have not been considered and will be significant. Moreover, it is not clear exactly what the length scale corresponds to, since it is not yet clear exactly where the water is moving from and to. Nonetheless, the value of D obtained is remarkably similar to that calculated by Kalinichev *et al.* [28] using molecular dynamics simulations for water diffusing in the pore space above the internal pore

surface of tobermorite, another mineral phase closely related to CSH. They obtain $6.0 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$.

We have previously pointed out that a surface diffusion coefficient based on a length derived from the size of a water molecule ($a = \epsilon = 0.3 \text{ nm}$) and our previous measurements of a surface hopping time ($t = \tau_m = 1 \text{ ns}$) yield a diffusivity¹ $D = 2.3 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$. This value is in good agreement with the value of $5.0 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ obtained by Kalinichev *et al.* for water diffusing on the internal pore surfaces of tobermorite, again by molecular dynamics simulation. This double coincidence of agreement between diffusivities clearly warrants further investigation.

VI. CONCLUSION

We have presented the first detailed analysis of the NMR T_2 -store- T_2 2D correlation-relaxation experiment and shown that it has certain advantages over the T_1 - T_2 variant previously studied. Experimental results for curing white cement

pastes have revealed new information about the water dynamics within the cement nanopores. The principal pore sizes of two porosity reservoirs have been carefully estimated and attributed to different structures with the CSH gel. For the first time, the exchange rate between these structures has been estimated. It is circa 5 m s^{-1} . From this, the water diffusivity within the CSH gel has been calculated and found to broadly correspond to the results of molecular-dynamic simulations in related systems to be found in the literature.

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¹Note that the factor 6 needs to be replaced by a factor 4 in the Einstein equation for a 2D walk.

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Analytic theory of two-dimensional NMR in systems with coupled macro- and micropores

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Two-dimensional (2D) nuclear magnetic resonance (NMR) experiments involve a sequence of longitudinal (T_1) and transverse (T_2) measurements. When such experiments are applied to porous media, they are believed to provide new and important information regarding diffusive coupling between distinct pore subpopulations. However, we show in this paper that, in many cases of interest, this is simply not true. One often encounters systems in which the one-dimensional (1D) T_1 and T_2 processes are each controlled by just two distinct decay modes. If these modes form a complete set, then one can derive analytic formulas that describe, exactly, the 2D NMR measurements. Therefore, for such systems, the 2D measurements bring *no additional information* over that which is already present in the 1D results. Our predictions agree quite well with numerical results based on the microporous grain-consolidation (μ – GC) model.

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I. INTRODUCTION

Two-dimensional (2D) nuclear magnetic resonance (NMR) experiments have recently been applied to a wide range of fluid filled porous media including reservoir rocks [1–4], cements [5], catalysts [6], and surfactants [7]. These measurements provide a correlation between the initial (encoding) and final (detection) part of the experiment. Of particular interest is the T_2 – T_1 – T_2 (also known as T_2 –storage– T_2) protocol in which the spins are first rotated to be perpendicular to the static magnetic field, $\mathbf{B}_0$, where they undergo transverse (T_2) relaxation for a variable amount of time, τ_1 . Next, they are rotated to be parallel to $\mathbf{B}_0$ where they undergo longitudinal (T_1) relaxation for a fixed amount of storage time, t_s . Finally, the spins are rotated back to the transverse plane and undergo transverse relaxation for a second variable amount of time, τ_2 [3,5,8]. Typically for each value of the storage time, t_s , the NMR data are processed to produce lifetime distributions $\mathcal{P}(t_s; T_2^a, T_2^b)$ where $T_2^{a(b)}$ refers to a decay time relevant to the first (second) T_2 part of the experiment.

A T_2 –storage– T_2 example, based on microporous grain-consolidation (μ – GC) simulations [8] described below, is shown in Fig. 1 together with the corresponding 1D T_2 distribution. In the 2D cross-plot off-diagonal peaks are clearly evident. These peaks have been interpreted as providing new information regarding diffusive coupling between distinct spin populations (e.g., micro- and macropores in the μ – GC model) [8]. Indeed, by monitoring the amplitude of these off-diagonal peaks as a function of the storage time, t_s , several authors have estimated exchange rates or coupling parameters [3–5]. While this approach is intuitively appealing, we will show that the behavior of the off-diagonal peaks can sometimes be predicted from the one-dimensional (1D) lifetime distributions. In such cases no *new* information is contained in the T_2 – T_2 cross-plot. Thus, we maintain, the very significant complexity of doing the 2D experiments as opposed to the more conventional 1D experiments (CPMG for T_2 plus inversion recovery for T_1 [1–7]) is often unwarranted.

We consider here the simple but important examples of 2D NMR, namely those for which the relevant 1D relaxation can be described as the sum of two exponential functions. (This includes, as a subset, the two-site exchange model which has often been used to interpret 2D NMR data [5,8].) The analytic theory we develop allows us to predict, with no additional information, the integrated intensity of each of the peaks appearing in two-dimensional cross-plots of the kind shown in Fig. 1. Therefore, for such systems, the 2D measurement brings no additional information over that which is already present in the 1D results. Equally important, our analysis points the way forward if we are to understand the value of 2D NMR in more complicated systems. We present numerical results on a system in which three modes are needed to describe the 1D lifetime distributions. Our analysis can be extended to such systems and should allow us to identify which features of the 2D distributions contain information that is not already present in the 1D spectra.

II. BASIC CONSIDERATIONS

A. Analysis of 1D and 2D NMR measurements

In both 1D and 2D NMR, the evolution of the magnetization (magnetic moment per unit volume), $M(\mathbf{r}, t)$, is described by the diffusion equation [9]

$$D_0 \nabla^2 M(\mathbf{r}, t) = \frac{\partial M(\mathbf{r}, t)}{\partial t} \quad (1)$$

subject to a partially absorbing boundary condition at the interface between the pore fluid and the solid grain

$$[D_0 \hat{\mathbf{n}} \cdot \nabla M(\mathbf{r}, t) + \rho M(\mathbf{r}, t)]_{\text{interface}} = 0. \quad (2)$$

Here D_0 is the diffusion coefficient for the bulk fluid and the parameter ρ describes interface absorption. (In Eq. (1) the effect of bulk relaxation has already been factored out [9].) This Brownstein-Tarr formalism ignores the effects of field inhomogeneities due to magnetic susceptibility contrasts. In practical terms this means the theory may be limited to low field NMR [9].

The value of ρ is generally larger for transverse relaxation than for longitudinal relaxation but, in both cases, is assumed to be constant over the entire interface. Equations (1) and (2)

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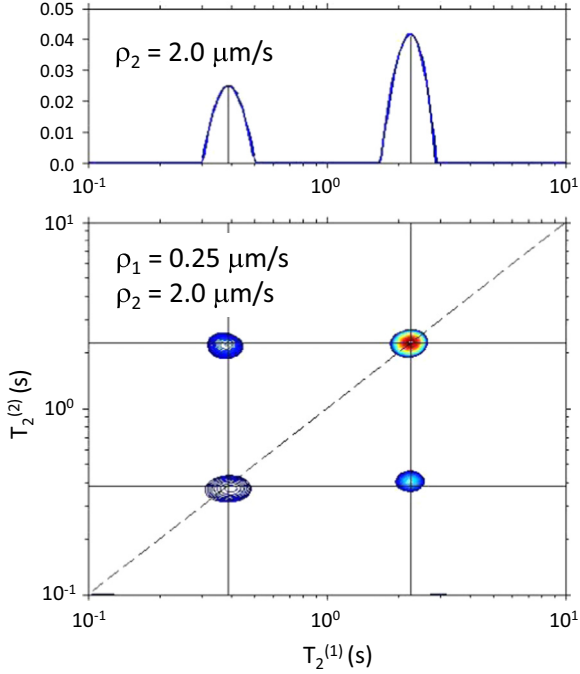


FIG. 1. (Color online) One- and two-dimensional lifetime distributions obtained from our numerical simulations. The storage time, $t_s = 3$ s, and location of the peaks in the 2D distribution are seen to be directly related to locations of their 1D counterparts. The assumed surface relaxivity values are indicated.

can be solved directly for very simple geometries or by random walk simulations for more complex pore models. In a recent paper, we applied variable step size random walk techniques to the μ – GC model to study the T_2 -storage- T_2 process in a system with two quite different pore sizes [8]. Our purpose was to see if our results could be understood in terms of the empirical two-site exchange model, which is often employed to interpret 2D NMR data. The calculations presented there were limited to the case in which there was no relaxation during the storage period but were sufficient to expose a number of significant limitations of the two-site exchange model.

Because the controlling equations (1) and (2) are derived from a self-adjoint operator, we know that the eigenmodes of the system form a complete and orthonormal set [10]. Several authors have extended the formalism originally developed by Brownstein and Tarr [9] to the analysis of 2D NMR [8,11,12]. From the outset one must consider two sets of eigenfunctions and eigenvalues. We denote these as $\{\mathbf{q}_n(\mathbf{r})\}$ and $\{\lambda_n\}$ for transverse decay and $\{\mathbf{Q}_n(\mathbf{r})\}$ and $\{\mu_n\}$ for longitudinal decay. The orthonormality condition is simply

$$\begin{aligned} \mathbf{q}_i \cdot \mathbf{q}_j &= \delta_{ij} \quad i, j = 1, 2, \dots, \\ \mathbf{Q}_k \cdot \mathbf{Q}_l &= \delta_{kl} \quad k, l = 1, 2, \dots, \end{aligned} \quad (3)$$

where the inner product of two functions is $\mathbf{f} \cdot \mathbf{g} = \int_{V_P} \mathbf{f}(\mathbf{r}) \mathbf{g}(\mathbf{r}) d\mathbf{r}$.

In the 1D case the total magnetic moment can be represented as a series expansion. For T_2 decay,

$$M_2(t) = M_0 V_P \sum_{n=1}^{\infty} A_n e^{-\lambda_n t}, \quad (4)$$

where M_0 is the initial magnetization and V_P is the volume of the pore space. We have $A_n = (\mathbf{m}_0 \cdot \mathbf{q}_n)^2 / V_P$, where $\mathbf{m}_0(\mathbf{r})$ is a function equal to unity in the pore space and zero elsewhere. Similarly, the T_1 decay function is

$$M_1(t) = M_0 V_P \sum_{n=1}^{\infty} B_n e^{-\mu_n t}, \quad (5)$$

where $B_n = (\mathbf{m}_0 \cdot \mathbf{Q}_n)^2 / V_P$. It is understood that T_1 is conventionally monitored with an inversion-recovery procedure in which the quantity actually measured is $M_0 V_P - 2M_1(t)$, but we will refer to $M_1(t)$ as the T_1 decay function.

The theoretical description of the T_2 -storage- T_2 experiment begins, as in Eq. (4), with an expansion of $M(\mathbf{r}, t)$ in terms of $\{q_n, \lambda_n\}$ during the encoding period $0 \leq t \leq \tau_1$. $M(\mathbf{r}, \tau_1)$ is then the initial condition for an expansion, covering the storage period, based on the longitudinal eigenfunctions and eigenvalues, $\{\mathbf{Q}_l\}$ and $\{\mu_l\}$. [Because $M(\mathbf{r}, \tau_1)$ will in general not be spatially uniform, its projection onto the longitudinal eigenfunctions is non-trivial.] Finally, $M(\mathbf{r}, \tau_1 + t_s)$ is the initial state for the transverse basis function expansion between $t = \tau_1 + t_s$ and $t = \tau_1 + t_s + \tau_2$, where τ_2 is the length of the detection period. The result is [11,12]

$$M(\tau_1, \tau_2 : t_s) = \sum_{m,n} \mathcal{M}_{nm}(t_s) e^{-[\lambda_n \tau_1 + \mu_m \tau_2]}, \quad (6)$$

where

$$\mathcal{M}_{nm}(t_s) = M_0 \sum_l (\mathbf{m}_0 \cdot \mathbf{q}_n)(\mathbf{q}_n \cdot \mathbf{Q}_l)(\mathbf{Q}_l \cdot \mathbf{q}_m)(\mathbf{q}_m \cdot \mathbf{m}_0) e^{-\mu_l t_s}. \quad (7)$$

From Eq. (7) we see, by inspection, that the matrix is symmetric: $\mathcal{M}_{nm}(t_s) = \mathcal{M}_{mn}(t_s)$. Furthermore, each coefficient of $e^{-\mu_l t_s}$ is positive for the diagonal elements, M_{ii} . Finally, we note a useful sum rule which follows from the completeness of the set $\{\mathbf{q}_n\}$:

$$\sum_{n,m} \mathcal{M}_{nm}(t_s) = M_0 \sum_l (\mathbf{m}_0 \cdot \mathbf{Q}_l)^2 e^{-\mu_l t_s} \equiv M_1(t_s). \quad (8)$$

Thus, if one sums over all elements of the storage matrix (integrates over all the decay times in the contour plot) the result must lie on the T_1 decay curve. This result provides a powerful check on experimental data and, as we shall see, on the results of our own numerical simulations.

It is worth pointing out that if ρ_1 and ρ_2 are small enough, no matter how complicated the geometry as long as the pore space is connected, then there is only one decay rate each for T_1 and for T_2 and there are no off-diagonal elements in the 2D experiment. (There is only the one diagonal peak.) Moreover, if ρ_1 and ρ_2 are large enough such that the slow diffusion limit applies for T_1 and for T_2 there will be multiple decay rates for each but there will be no off-diagonal elements in a T_2 -storage- T_2 experiment simply because the normal modes for T_1 and for T_2 are identical. So this off-diagonal effect

can happen only if at least one of T_1 and T_2 are intermediate between fast and slow diffusion, however that is defined.

The above equations are quite general. A tremendous simplification of this formalism occurs if we can make two related assumptions, viz (1) that two modes are sufficient to describe both the T_1 and T_2 decay spectra and (2) that any relevant function of position within the pore space can be expressed as a linear combination of these two orthonormal basis functions (e.g., $\mathbf{q}_1$ and $\mathbf{q}_2$ or, equivalently, $\mathbf{Q}_1$ and $\mathbf{Q}_2$). [In general, continuum models of porous media (like the GC or μ - GC models) have a countably infinite number of normal modes [9]; in most cases only a few contribute to any NMR signal. However, it is possible to quantify these higher order modes by means of direct numerical computations on model porous media [13].] These assumptions allow us to derive an analytic expression for the matrix elements $\mathcal{M}_{nm}(t_s)$ as we now demonstrate.

Because $(\mathbf{f} \cdot \mathbf{g})^2 \leq (\mathbf{f} \cdot \mathbf{f})(\mathbf{g} \cdot \mathbf{g})$ for any $\mathbf{f}, \mathbf{g}$ pair and because $|\mathbf{m}_0| = \sqrt{\mathbf{m}_0 \cdot \mathbf{m}_0} = \sqrt{V_P}$, we may define an angle ψ via

$$\mathbf{m}_0 \cdot \mathbf{q}_1 = \sqrt{V_P} \cos(\psi) = \sqrt{V_P A_1}. \quad (9)$$

The orthonormality and completeness of $\mathbf{q}_1$ and $\mathbf{q}_2$ implies $\mathbf{m}_0 = (\mathbf{m}_0 \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{m}_0 \cdot \mathbf{q}_2)\mathbf{q}_2$ which, in turn, implies $(\mathbf{m}_0 \cdot \mathbf{q}_1)^2 + (\mathbf{m}_0 \cdot \mathbf{q}_2)^2 = \mathbf{m}_0 \cdot \mathbf{m}_0 = V_P$. We then have $|\mathbf{m}_0 \cdot \mathbf{q}_2| = \sqrt{V_P} \sin(\psi) = \sqrt{V_P A_2}$. Similarly, we may define the angle θ by writing $\mathbf{Q}_1$ in terms of $\mathbf{q}_1$ and $\mathbf{q}_2$:

$$\mathbf{Q}_1 = \cos(\theta)\mathbf{q}_1 + \sin(\theta)\mathbf{q}_2. \quad (10)$$

Since $\mathbf{Q}_1 \cdot \mathbf{Q}_2 = 0$, we have

$$\mathbf{Q}_2 = \pm[\sin(\theta)\mathbf{q}_1 - \cos(\theta)\mathbf{q}_2]. \quad (11)$$

Substituting these results into Eq. (7), the matrix elements $\mathcal{M}_{ij}$ can be written as

$$\begin{aligned} \mathcal{M}_{11}/(M_0 V_P) &= \cos^2(\psi)[\cos^2(\theta)e^{-\mu_1 t_s} + \sin^2(\theta)e^{-\mu_2 t_s}], \\ \mathcal{M}_{22}/(M_0 V_P) &= \sin^2(\psi)[\sin^2(\theta)e^{-\mu_1 t_s} + \cos^2(\theta)e^{-\mu_2 t_s}], \\ \mathcal{M}_{12}/(M_0 V_P) &= (1/4)\sin(2\psi)\sin(2\theta)[e^{-\mu_1 t_s} - e^{-\mu_2 t_s}]. \end{aligned} \quad (12)$$

The six integrals implied by Eq. (7) have been reduced to two parameters, ψ and θ .

Because the longitudinal decay can also be expressed in terms of just two distinct normal modes, we can define an angle χ in terms of the projection of the initial uniform magnetization onto the T_1 basis functions:

$$\cos(\chi) \equiv \mathbf{m}_0 \cdot \mathbf{Q}_1 / \sqrt{V_P}. \quad (13)$$

In view of Eq. (10), the angles ψ , θ and χ are related by

$$\cos(\chi) = \cos(\theta)\cos(\psi) + \sin(\theta)\sin(\psi) = \cos(\theta - \psi). \quad (14)$$

In a similar fashion we may write, utilizing Eq. (11),

$$\sin(\chi) = |\mathbf{m}_0 \cdot \mathbf{Q}_2 / \sqrt{V_P}| = |\sin(\theta - \psi)|. \quad (15)$$

Either of Eq. (14) or (15) implies

$$\theta = \psi \pm \chi. \quad (16)$$

The foregoing analysis lends itself to a simple visual depiction, which we show in Fig. 2. Consider any one of the four

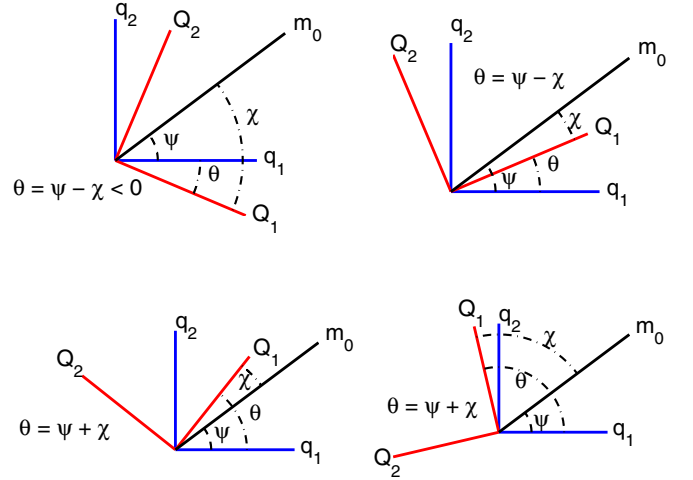


FIG. 2. (Color online) Schematic plot of the possible relationships between the two orthogonal modes for T_2 decay, $\mathbf{q}_{1,2}$, the two orthogonal modes for T_1 decay, $\mathbf{Q}_{1,2}$, and the initial uniform magnetization, $\mathbf{m}_0$, within the context of the present theory. We imagine that the orientation of $\mathbf{Q}_{1,2}$ to $\mathbf{q}_{1,2}$ is continuously rotated counterclockwise in going from the upper-left picture to the lower-right.

subplots. The T_2 normal modes, $\mathbf{q}_1$ and $\mathbf{q}_2$, are orthonormal. Any point on the graph represents a function which is a linear combination of these two functions, the coefficients of which are the coordinates on the graph. In particular, for porous media being considered in this article the initial uniform magnetization, $\mathbf{m}_0$, can be represented in terms of its coordinates along these two axes, thus defining the angle ψ as shown. Similarly, each of the two functions $\mathbf{Q}_1$ and $\mathbf{Q}_2$, which are mutually orthonormal, may also be represented as points in the $\mathbf{q}_1, \mathbf{q}_2$ plane. This is the basic assumption of the article. Their coordinates define the angle θ , as shown. The four subplots indicate how the picture evolves from left to right, top to bottom, as θ is continuously increased. It also shows each possibility of Eq. (16). Depending upon which side of $\mathbf{m}_0$ the vector $\mathbf{Q}_1$ lies, either of the $\pm$ signs in Eq. (16) may hold, in principle. In most cases of interest $\rho_1 \leq \rho_2$ and we believe that the minus sign in this equation is appropriate. When $\rho_1 = 0$ the mode $\mathbf{Q}_1$ corresponds to uniform magnetization, so $\chi = 0$ and $\theta = \psi$. When $\rho_1 = \rho_2$ one has $\mathbf{Q}_i \equiv \mathbf{q}_i$, so $\theta = 0$. For the most common situation encountered experimentally we have $0 < \rho_1 < \rho_2$, and therefore $0 < \theta < \psi$ which implies $\theta = \psi - \chi$.

Our algorithm, then, consists of the following steps: Fit the T_2 decay to the sum of two simple exponentials, with amplitudes $A_1 = \cos^2(\psi)$ and $A_2 = 1 - A_1$, and decay rates λ_1 and λ_2 . Solve for the angle $\psi = \cos^{-1}(\sqrt{A_1})$. Likewise, fit the T_1 decay to the sum of two simple exponentials, with amplitudes $B_1 = \cos^2(\chi)$ and $B_2 = 1 - B_1$, and decay rates μ_1 and μ_2 . Solve for the angle $\chi = \cos^{-1}(\sqrt{B_1})$; take $\theta = \psi - \chi$ and compute the storage matrix elements using Eqs. (12).

In the very special case that the two subsystems are unconnected, as happens for example if there is a barrier between them [8], then the amplitudes for both M_1 and

M_2 decay are simply the respective macro- and micropore volumes, i.e., $A_i = B_i$ $i = 1, 2$. In this very special case, then, $\psi \equiv \chi$ and $\theta \equiv 0$. The result according to Eqs. (12) is that $M_{12} \equiv 0$ and the two diagonal elements are simply the respective single exponential 1D M_1 decays, as one would intuitively expect.

Even if the 1D NMR relaxation experiments can be represented accurately as a sum of two exponentials it is possible for our theory of 2D experiments to fail. Consider any subplot in Fig. 2. Let us imagine that there is a third mode each for T_1 and T_2 relaxation, $\mathbf{Q}_3$ and $\mathbf{q}_3$, respectively. In Fig. 2 each points in a direction perpendicular to the plane of the figure. In this situation neither of these modes contributes to their respective 1D decays nor does either contribute to the 2D results. If, however, we imagine that the triplet $\{\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3\}$ is rotated rigidly around the vector $\mathbf{m}_0$ by an amount β , keeping the angle χ fixed and maintaining their mutual orthonormality we see that the T_1 decay is still represented as a sum of two exponentials because this third mode does not couple to the initial magnetization; the third mode is still orthogonal to the initial magnetization, $\mathbf{m}_0 \cdot \mathbf{Q}_3 \equiv 0$, for any value of β . This, then, would be a situation in which each 1D relaxation can be represented as a sum of two exponential decays but the storage matrices, $\mathcal{M}_{ij}$, require three exponentials because $\mathbf{Q}_3 \cdot \mathbf{q}_j \neq 0$. The sum rule, Eq. (8), still holds, however, and it has only two terms, the coefficients of the third exponential having canceled out.

These ideas can also be applied to the T_1 - T_2 protocol in which spins undergo longitudinal decay up to a switching time, T_s , and are then tipped into the transverse plane for an additional time τ . One has

$$M(T_s; \tau) = \sum_{n,m} \mathcal{M}_{nm}^{T_1, T_2} e^{-\mu_n T_s} e^{-\lambda_m \tau}, \quad (17)$$

where $\mathcal{M}_{nm}^{T_1, T_2} = M_0(\mathbf{m}_0 \cdot \mathbf{Q}_n)(\mathbf{Q}_n \cdot \mathbf{q}_m)(\mathbf{q}_m \cdot \mathbf{m}_0)$. Although the results could be presented in terms of the angles ψ and χ , they can more simply and succinctly be written in terms of the T_1 and T_2 amplitudes directly as

$$\frac{\mathcal{M}_{nm}^{T_1, T_2}}{M_0 V_p} = B_n A_m + (2\delta_{nm} - 1)\sqrt{A_1 A_2 B_1 B_2}. \quad (18)$$

In Ref. [12] very general arguments were used to conclude that negative off-diagonal matrix elements are to be expected in the analysis of T_1 - T_2 data. Equation (18) provides a specific example of this. Of the two off-diagonal elements, $\mathcal{M}_{12}^{T_1, T_2}$ and $\mathcal{M}_{21}^{T_1, T_2}$, one is negative and the other is positive. (A trivial exception occurs when $A_1 = B_1$, in which case both off-diagonal elements are zero.) The accuracy of the above equations is being pursued and the results will be presented elsewhere [15].

B. Numerical calculations

We confirm the validity of the theory by means of a comparison against a series of numerical simulations based on the μ -GC model [8] a schematic of which is shown in Fig. 3. The model consists of a periodic arrangement of overlapping grains each of which is comprised of smaller grains. Our calculations are based on variable step size random walks as described in detail in [8]. We have employed the

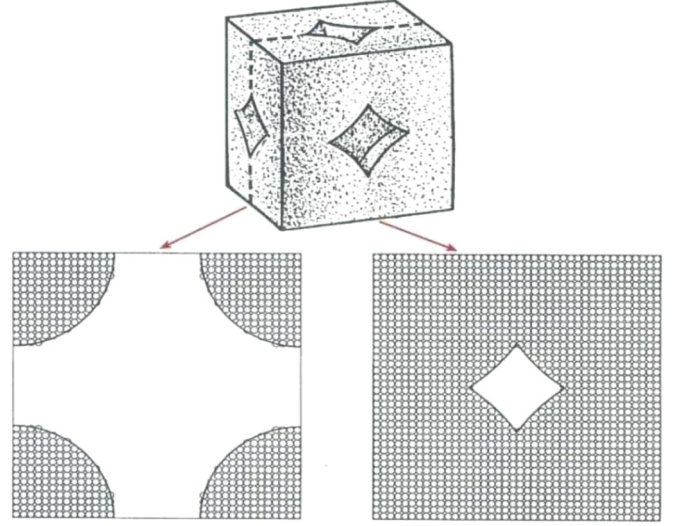


FIG. 3. (Color online) A microporous version of the grain consolidation model in which the large grains are comprised of smaller grains whose centers lie on a lattice constructed by subdividing the original unit cell. The solid curves separate the macroporous and the microporous regions.

same geometry as in [8] but with varying values of the surface relaxivities, ρ_1 and ρ_2 . The probability that a walker will be removed from the count when it hits the pore-grain interface is simply related to the values of these physical parameters.

For most of the calculations presented here the pore space is divided into two distinct, but coupled, populations whose dimensions differ by roughly a factor of 20 (see Table I of Ref. [8]). We see, however, that the two peaks in the T_2 distribution shown in the upper panel of Fig. 1 are separated by less than a factor of 10. Before going further, let us explain why this is the case. It is straightforward to estimate the position of the short-time peak in the distribution since the micrograins are simply smooth overlapping spheres. The pore volume to surface area ratio for this part of the pore space is $0.839 \mu\text{m}$ and the corresponding lifetime can be estimated as $T_2 = V_p/(S\rho_2) = 0.42$ s. This is what we see in Fig. 1. If the macropores had smooth surfaces, their V_p/S ratio would be $15.2 \mu\text{m}$ and the corresponding lifetime would be 7.6 s, which is well above the value seen in Fig. 1. Of course, the macropores do *not* have smooth surfaces; the interface between the two pore populations is controlled by the large amount of surface area associated with the micrograins. This interface is where the diffusive coupling takes place; its presence is thus quite apparent in just the T_2 distribution.

TABLE I. Amplitudes and decay times deduced by fitting two exponentials to numerical simulations of $M(t)$. It is understood that when a specific value of ρ is used for the storage part of the simulation, the entries for λ_i now represent μ_i and that for A_1 represents B_1 .

$\rho(\mu\text{m/s})$	0	0.25	0.5	1.0	2.0
$1/\lambda_1(\text{s})$	∞	8.2	5.05	3.3	2.25
A_1	1.0	0.924	0.825	0.735	0.620
$1/\lambda_2(\text{s})$	3.5	1.7	1.2	0.61	0.42

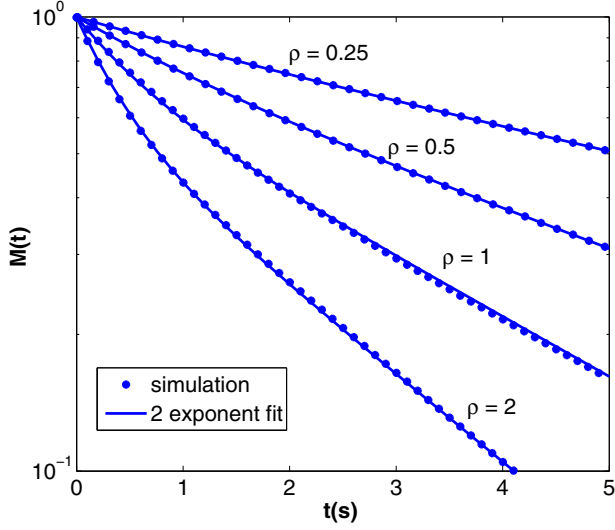


FIG. 4. (Color online) Comparison of the results of numerical simulations of the 1D decays, $M_i(t)$ $i = 1, 2$, for different values of the surface relaxivity, ρ , in $\mu\text{m/s}$.

A necessary condition for the success of this theory is that each of the 1D measurements, $M_1(t)$ and $M_2(t)$, must be accurately represented as a sum of two exponential functions. To test this part we have performed a number of simulations, assuming different values of relaxivity, ρ , whose results are shown in Fig. 4. We see that each decay is well represented as a sum of two exponentials. The fitting parameters are collated in Table I and are plotted in Fig. 5. The decay rate μ_2 for the case where $\rho = 0$ was determined by means of a separate simulation, in which walkers were initially released only in the macropore and the number of walkers remaining in the macropore as a function of time was monitored [8].

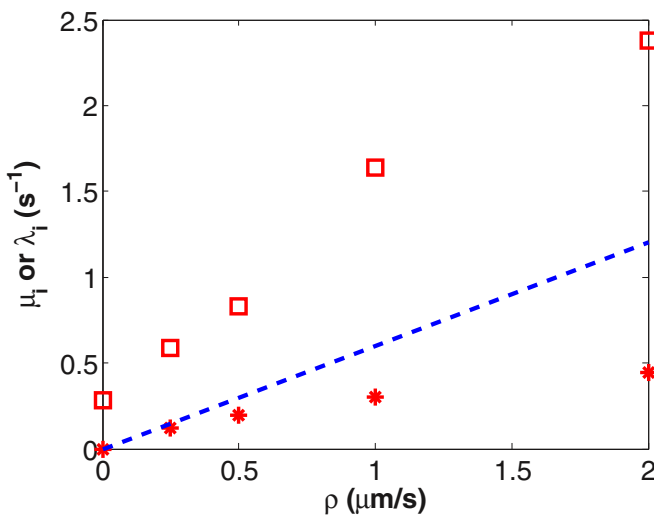


FIG. 5. (Color online) The two decay rates deduced from our 1D simulations as a function of surface relaxivity. For the 2D simulations the decay rates for $\rho = 2 \mu\text{m/s}$ are appropriate to the T_2 decay parts; the others are appropriate to T_1 decay, i.e., to the storage part of the simulations. The dashed line is Eq. (19) using the known surface-to-pore-volume ratio.

On quite general grounds [9] the slowest relaxation rate in the limit of small ρ is

$$\lim_{\rho \rightarrow 0} \mu_1(\rho) = \rho(S/V_P), \quad (19)$$

where (S/V_P) is the total surface area to pore volume ratio of the pore system. From the geometry of our model we have determined that $(S/V_P) = 0.603 (\mu\text{m})^{-1}$. Equation (19) is plotted as a dashed line in Fig. 5 using this value. In this limit, the slowest mode, $\mathbf{q}_1$ or $\mathbf{Q}_1$ as the case may be, corresponds to uniform magnetization throughout the entire pore system. The normalization condition, Eq. (3), requires that this value is equal to $\frac{1}{\sqrt{V_P}}$ throughout the entire pore system.

We note that the second decay rate does *not* approach zero as $\rho \rightarrow 0$ but it tends to a constant, which constant is directly proportional to the bare diffusion constant. That is because this mode corresponds to a distribution which has one sign in the middle of the macropore and the opposite sign in the micropores. This fact follows immediately from the orthogonality condition, Eq. (3), on $\mathbf{Q}_2$ which reads

$$0 = \mathbf{Q}_1 \cdot \mathbf{Q}_2 \equiv \frac{1}{\sqrt{V_P}} \int_{V_P} Q_2(\mathbf{r}) dV. \quad (20)$$

Relaxation for this mode occurs strictly by the diffusion between the macro- and the micropores so as to even out the imbalance. Equation (20) is the reason why a separate simulation [8] was needed in order for us to deduce the decay rate of this $\mathbf{Q}_2$ mode; it does not couple to an initial condition of uniform magnetization. In general, then, except by accident the ratio of the decay rates of the two modes is essentially never equal to the ratio of the pore sizes (20 in this article) no matter the value of the surface relaxivity, ρ . Diffusive coupling is an integral part of each of the modes. The slowest mode does *not* correspond to magnetization exclusively in

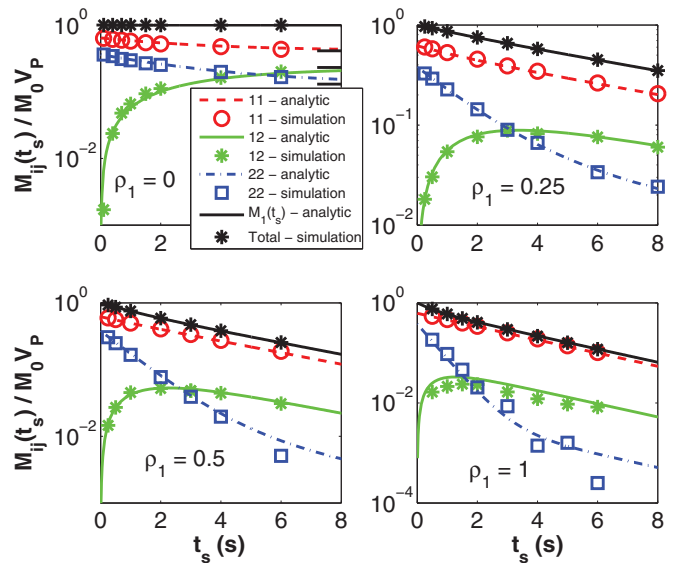


FIG. 6. (Color) The results of numerical simulations of the 2D storage matrix, $M_{ij}(t_s)$, in the micro grain consolidation model are compared against the predictions of the analytic model, Eqs. (12). The values assumed for ρ_1 , in $\mu\text{m/s}$, are indicated. In all cases we used $\rho_2 = 2 \mu\text{m/s}$.

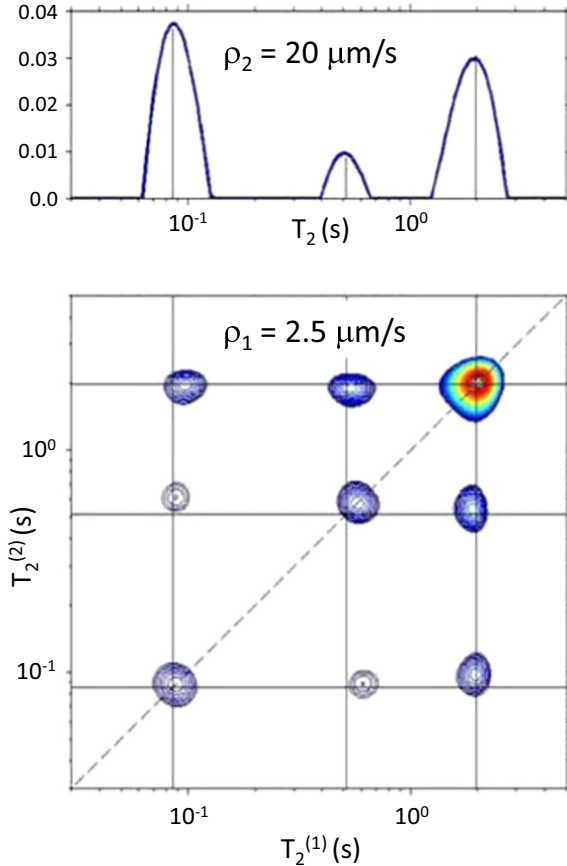


FIG. 7. (Color online) Numerical results for the 2D storage matrix, $M_{ij}(t_s)$, in a micrograin consolidation model in which the sizes of the micro- and macrograins have been increased by a factor of 2 relative to our previous calculations. The storage time is $t_s = 2$ s the values of ρ_1 and ρ_2 are given in the figure.

the macropores; the second mode does *not* correspond to magnetization exclusively in the micropores. Rather, each mode is a distribution throughout the entire pore space.

Numerical simulations of T_2 -storage- T_2 were carried out using the parameters $\rho_2 = 2 \mu\text{m/s}$ and $\rho_1 = 0, 0.25, 0.5$, and $1.0 \mu\text{m/s}$. A large number of random walkers (typically 10^6) are launched randomly and uniformly within the pore space. For the initial part of the simulation, $t \leq \tau_1$, relaxation at the pore-grain interface is controlled by ρ_2 . At the first switching time, τ_1 , the surviving walkers continue but with a surface relaxation rate, ρ_1 . Then at the second switching time, $\tau_1 + t_s$, the surface relaxivity reverts to ρ_2 . We processed the time

domain data using an inverse 2D Fredholm integral equation solver to generate two dimensional T_2 - T_2 contour plots, one of which is shown in Fig. 1. We integrated the volume under each peak in order to determine M_{ij} for each value of the storage time, t_s . The results are shown in Fig. 6 along with the analytic results of Eqs. (12) using as input data the parameters listed in Table I. We see that the theory works quite well for this geometry and for these choices of relaxivity values [14]. We also see that the simulation data obeys the sum rule Eq. (8).

In principle, this comparison could have been made based on just the time domain data using the definition, Eq. (6) and the model, Eqs. (12). This would have involved the analysis of a large number of 1D data sets (one for each value of τ_1 and t_s). We employed the 2D Fredholm equation solver as a matter of convenience; its use is not essential.

Obviously, there are many conditions under which it will not be possible to express both of the 1D decays as the sum of two exponentials. In such cases we want to understand what features of the 2D NMR spectrum contain information that cannot be derived from the 1D distributions. An example can be seen in the μ - GC model if we enlarge the unit cell size by a factor of two and increase the values of ρ_1 and ρ_2 as shown in Fig. 7. In this case the 1D decays can be written as the sum of three exponentials and it is possible to extend the present theory [15]. In more complicated systems, our analysis will inevitably breakdown and we will be in a position to identify those features of the 2D spectra that truly contain new information about diffusive coupling.

III. CONCLUSIONS

In summary, if the 1D decay functions, $M_1(t)$ and $M_2(t)$, can be represented accurately as the sum of two exponentials, we have developed an analytic theory for the results of T_1 - T_2 and T_2 -storage- T_2 experiments. The only inputs to the theory are the decay rates and amplitudes of the two 1D measurements. The validity of the theory has been confirmed by comparison against numerical modeling in a mixed pore size system. While the present analysis is based on the diffusion equation and fluid filled porous media, our results apply equally to measurements on surfactants and catalysts where the evolution of the magnetization is governed by a different self-adjoint operator. The only other requirement is that the initial magnetization and the final amplitude measurement are described by the same function, $\mathbf{m}_0$.

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Modeling two-dimensional magnetic resonance measurements in coupled pore systems

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We present numerical simulations of a two-dimensional (2D) nuclear magnetic resonance process, T_2 -storage- T_2 , on a simple mixed porosity system, the micrograin consolidation (μ GC) model. The results of these calculations are compared with predictions based on the analytic two-site exchange model, for which we have independently established numerical values for all the input parameters. Although there is qualitative and semiquantitative agreement between the two models, we identify specific instances where the two-site model fails to properly describe the combined effects of relaxation and diffusion. Generally, these instances occur when a gradient in magnetization within the large pores of the μ GC model is established during the initial phase of the 2D process. The two-site model assumes that the magnetization is spatially uniform within each of its subpore systems and thus cannot describe such effects.

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I. INTRODUCTION

Two-dimensional (2D) nuclear magnetic resonance (NMR) relaxation experiments have been applied to a wide range of fluid-filled porous media including petroleum reservoir rocks [1–7], bead packs [8], cements [9,10], and gypsum plaster [11]. In addition, they have been used to study catalysts [12], surfactants [13], liquid crystals [14,15], and other materials of interest [16,17].

Two-dimensional NMR relaxation measurements provide a correlation between the initial (encoding) and final (detection) parts of the experiment. The encoding and detection sequences may involve longitudinal T_1 or transverse T_2 , decay. An important class, including T_2 - T_2 [6,10,18] and T_1 - T_1 [19] correlations, incorporates a user-controlled storage interval t_s between the encoding and detection intervals. Typically, the experimental data are processed to produce a 2D distribution of lifetimes. In the resulting cross plot off-diagonal peaks may be observed; these have been interpreted, heuristically, in terms of diffusive exchange between distinct spin populations. Indeed, by varying t_s , exchange rates or coupling parameters have been estimated by several authors [6–8,10,19]. While the well-defined storage time is appealing for qualitative interpretation, there is a continuing controversy as to the link between the off-diagonal peak amplitudes and t_s .

In this paper we focus on the T_2 - t_s - T_2 correlation in model systems with distinct pore subpopulations. Our calculations proceed in the time domain; we follow the spin decay from the initial encoding sequence through the storage period and into the detection sequence. Our results are easily displayed as 2D distributions of relaxation lifetimes and we can directly monitor the dependence of the off-diagonal peaks amplitudes on the storage time t_s . The first model we work with is a two-site exchange model introduced by Zimmerman and Brittin [20] and by McConnell [21]. The model is based on several critical assumptions, most important that each spin population, in isolation, exhibits just a single relaxation mode. Physically, this would be consistent with very weak surface relaxation as discussed by Brownstein and Tarr [22]. However, individual pores in real samples may exhibit multiple

relaxation modes [22]. Such behavior can arise for a variety of reasons, for example, from intermediate or large values of surface relaxation or from magnetic susceptibility contrast inducing intrapore field gradients [6,23]. We turn next to the micrograin consolidation (μ GC) model. This model is based on an explicit three-dimensional pore geometry that idealizes the mixture of macropores and micropores found in many carbonate grainstones [24]. Such textures are important in oil reservoirs and have been studied experimentally using the T_2 - T_2 correlation [7].

The purpose of our analysis is to clarify the interpretation of T_2 - t_s - T_2 measurements. Since the value of 2D measurements is related to the way that they describe coupling between distinct pore populations, it will be useful to modulate this coupling directly in our calculations. This is easily done in both the two-site exchange and μ GC models and we make extensive use of this feature. In the first case we can turn off (i.e., set to zero) the coupling parameters during any of the three stages of the measurement. Similarly, in the μ GC model we can introduce a *barrier* between the macropores and micropores. As in the two-site exchange model, the barrier can be put in place during the spin encoding, spin storage, or spin detection parts of the calculation. With the ability to monitor the coupling, we show that while the amplitude of the off-diagonal peaks in the T_2 - T_2 map is initially proportional to t_s , it is not true that these peaks are wholly controlled by diffusive interaction during the storage period. Diffusion within a given pore may already impose a magnetization gradient even during the first (encoding) phase of the T_2 - t_s - T_2 protocol.

Section II of this paper has three parts, a brief review of NMR diffusion and surface relaxation, a description of the experimental protocol, and a summary of the equations underlying the two-site exchange and μ GC models. In Sec. III the results of our calculations are presented. In many respects the two-site exchange and μ GC models yield correlations that are quite similar. However, when we modulate the diffusive coupling some interesting differences between the two models are revealed. In Sec. IV we analyze our results based on the normal modes of the diffusion equation. We show that there

is a significant simplification in the formalism if one assumes that the system can be approximated as a linear combination of only two different basis functions. One of the problems in applying the two-site exchange model to real experimental data is that it is difficult to unambiguously determine values for the parameters that define the model. Here, because we have control over the μ GC geometry, we are able to deduce all the parameters cleanly. We then compare the predictions of the two-site model directly against our simulations of this 2D process. We conclude (Sec. V) by summarizing the relationship between the two-site exchange and μ GC models. Under certain conditions the two-site exchange model can be relied upon to estimate exchange rates. However, more generally, direct solutions of the diffusion equation in specific geometries are required, as is done in a variety of different contexts with existing 3D microtomography techniques [25].

II. BASIC CONSIDERATIONS

A. Diffusion and magnetic relaxation

Before considering the 2D NMR measurement, it will be useful to summarize the equations that describe magnetic resonance in fluid-saturated porous media. We picture a system initially prepared with a uniform magnetization $M(\mathbf{r}, t = 0) = M_0$ in the fluid phase. Here M_0 is the initial magnetic moment per unit pore volume in the sample. The evolution of $M(\mathbf{r}, t)$ is then controlled by a combination of molecular diffusion within the pore space and absorption at the pore-grain interface. The basic equations are discussed in detail by Brownstein and Tarr [22]. The evolution of the magnetization [magnetic moment per unit volume $M(\mathbf{r}, t)$] is described by the diffusion equation

$$D_0 \nabla^2 M(\mathbf{r}, t) = \frac{\partial M(\mathbf{r}, t)}{\partial t} \quad (1)$$

subject to the boundary condition on the interface between the pore fluid and the solid grain

$$[D_0 \hat{\mathbf{n}} \cdot \nabla M(\mathbf{r}, t) + \rho M(\mathbf{r}, t)]_{\text{interface}} = 0. \quad (2)$$

[We have set the bulk fluid relaxation equal to zero, without loss of generality, because this effect can be added simply by the substitution $M(\mathbf{r}, t) \rightarrow e^{-t/T_B} M(\mathbf{r}, t)$, where T_B is either the longitudinal or transverse relaxation time, as the case may be [22].]

Here D_0 is the diffusion coefficient for the bulk fluid and the parameter ρ describes the interface absorption strength. In the present analysis the values of ρ will be assumed constant over the entire interface [26]. Given these equations, the total magnetic moment of the sample can be represented as a standard eigenfunction-eigenvalue expansion

$$M(t) = M_0 V_P \sum_{n=1}^{\infty} A_n e^{-\lambda_n t}, \quad (3)$$

where V_P is the volume of the pore space. Each decay rate λ_n is an eigenvalue pertaining to Eqs. (1) and (2); each amplitude A_n is given in terms of the associated eigenmode $\mathbf{q}_n \equiv q_n(\mathbf{r})$. These are defined via solutions to the eigenvalue-eigenvector problem

$$D_0 \nabla^2 q_n(\mathbf{r}) + \lambda_n q_n(\mathbf{r}) = 0, \quad n = 1, 2, 3, \dots, \quad (4)$$

subject to the same boundary condition (2). The set $\{\mathbf{q}_n\}$ forms a complete and orthonormal basis for continuous functions of position within the pore space, properties that we shall use later. The amplitudes in Eq. (3) are

$$A_n = \frac{(\mathbf{m}_0 \cdot \mathbf{q}_n)^2}{V_P}, \quad (5)$$

where $m_0(\mathbf{r})$ is a function that is equal to unity in the pore space and zero elsewhere. Our notation is that the inner product of two functions is the usual

$$\mathbf{f} \cdot \mathbf{g} \stackrel{\text{def}}{=} \int_{V_P} f(\mathbf{r}) g(\mathbf{r}) d^3 \mathbf{r}, \quad (6)$$

where $f(\mathbf{r})$ and $g(\mathbf{r})$ are any two functions of position within the pore space. Equations (1)–(6) are discussed in detail in Ref. [22].

In general, the eigenfunctions $q_n(\mathbf{r})$ and eigenvalues λ_n depend on the value of ρ and on the shape of the pore-grain interface. In the two most familiar experiments one looks at the decay of either longitudinal (1) or transverse (2) magnetization. The above equations describe both measurements although, typically, $1.5 < \rho_2/\rho_1 < 2$.

A word on notation is in order: For the purposes of equations that involve the inner product it is convenient to represent a state function in bold type, e.g., $\mathbf{f}$. However, this represents a continuous function of position within the pore space, e.g., $f(\mathbf{r})$. So $\mathbf{f}$ and $f(\mathbf{r})$ are the same thing. Later, in Sec. IIC we introduce the two-site exchange model in which any state of the system is represented by a two-dimensional vector $\mathbf{f} = (f_A, f_B)$. Here $\mathbf{f}$ represents a function that takes on the constant value f_A in pore A and the constant value f_B in pore B . The inner product between two so-defined state vectors $\mathbf{f}$ and $\mathbf{g}$ still follows from the same definition (6): $\mathbf{f} \cdot \mathbf{g} = f_A g_A V_A + f_B g_B V_B$, where V_A and V_B are the volumes of pore A and pore B , respectively. The equations that we presented above, as well as those we develop in Secs. IIB and IV B, are equally applicable to the formalism in which the magnetization is described by continuous functions or to that in which it is described by two-dimensional vectors. In the latter case the integration over the pore space becomes trivially simple.

B. The 2D measurement protocol

It will be useful to summarize the experimental protocol to which our calculations relate. The first T_2 interval comprises a standard Carr-Purcell-Meiboom-Gill (CPMG) series of N_1 spin echoes [27] (or see [28]). The duration of this preparation portion is $\tau_1 \equiv N_1 t_e$, where t_e is the temporal separation of individual spin echoes. At the end of τ_1 , a 90° radio frequency (rf) pulse is applied to store the encoded spins on the longitudinal axis for a time t_s , during which the spins undergo longitudinal relaxation. The spins are then restored to the transverse plane by another 90° rf pulse and a second CPMG sequence of N_2 echoes is used to monitor the decay of magnetization. The duration of this second T_2 interval is $\tau_2 \equiv N_2 t_e$. Data (spin echo amplitudes) are acquired only in this latter portion of the experiment, but a 2D data matrix is constructed by varying the number of echoes N_1 in the first T_2 interval. In this manner, the experiment is sensitive to the decay of magnetization in the first T_2 interval via the amplitude of the

first echo in the second T_2 interval. The entire rf pulse sequence is illustrated in Fig. 1(a). We denote the magnetization at an arbitrary time in the experiment by $M(\tau_2, \tau_1; t_s)$.

A 2D relaxation time correlation is obtained by numerical inversion of the 2D Fredholm integral that describes the expected form of the data; methods for inverting NMR relaxation and diffusion correlations are reviewed elsewhere [29]. The relevant integral equation is

$$M(\tau_2, \tau_1; t_s) = \iint K(\tau_2, T_2^{(1)}, \tau_1, T_2^{(2)}) \times P(T_2^{(1)}, T_2^{(2)}; t_s) d \log_{10} T_2^{(1)} d \log_{10} T_2^{(2)} + e(\tau_2, \tau_1), \quad (7)$$

where K is the kernel function (expected form of the data), P is the required distribution, and e is experimental noise. The different T_2 axes are denoted by $T_2^{(2,1)}$; $T_2^{(1)}$ is associated with the τ_2 data dimension for consistency with the familiar one-dimensional lifetime distribution. For the T_2 - T_2 experiment

$$K(\tau_2, T_2^{(1)}, \tau_1, T_2^{(2)}) = \exp(-\tau_2/T_2^{(1)}) \exp(-\tau_1/T_2^{(2)}). \quad (8)$$

To solve Eq. (7) for the distribution $P(T_2^{(1)}, T_2^{(2)}; t_s)$ we employ Tikhonov regularization [30]; a stable solution is obtained in the presence of noise by forcing the resultant distribution to be non-negative, smooth, and bounded by finite values of $T_2^{(1,2)}$ [31–34].

The theoretical description of this experiment follows along the lines of the derivation of Eqs. (3) and (5). Between $t = 0$ and $t = \tau_1$ the magnetization is given by expressions analogous to Eqs. (3) and (5) using the transverse eigenfunctions and eigenvalues $\{\lambda_n, \mathbf{q}_n\}$. Then $M(\mathbf{r}, \tau_1)$ is the initial condition for an expansion based on the longitudinal eigenfunctions and eigenvalues $\{\mathbf{Q}_l\}$ and μ_l . [Note that, because $M(\mathbf{r}, \tau_1)$ will in general not be spatially uniform, its projection onto the longitudinal eigenfunctions is nontrivial.] Finally, $M(\mathbf{r}, \tau_1 + t_s)$ is the initial state for the final transverse basis function expansion between $t = \tau_1 + t_s$ and $\tau_1 + t_s + \tau_2$. The result is [31,33]

$$M(\tau_1, \tau_2; t_s) = \sum_{m,n} \mathcal{M}_{nm}(t_s) e^{-[\lambda_n \tau_1 + \lambda_m \tau_2]}, \quad (9)$$

where

$$\mathcal{M}_{nm}(t_s) = M_0 \sum_l [\mathbf{m}_0 \cdot \mathbf{q}_n][\mathbf{q}_n \cdot \mathbf{Q}_l][\mathbf{Q}_l \cdot \mathbf{q}_m] \times [\mathbf{q}_m \cdot \mathbf{m}_0] e^{-\mu_l t_s}. \quad (10)$$

From Eq. (10) we see, by inspection, that the matrix is symmetric: $\mathcal{M}_{nm}(t_s) = \mathcal{M}_{mn}(t_s)$. (Here and for the remainder of this article, l, m, n are indices relating to the specific normal modes of the problem and are not to be confused with echo numbers, mentioned earlier.) In practical terms we take it that any well-defined narrow peak in the experimentally determined distribution $P(T_2^{(1)}, T_2^{(2)}; t_s)$ at values $T^{(n)}$ and $T^{(m)}$, say, represents a broadened δ function in the space of the eigenvalue decay rates $\lambda_n = 1/T^{(n)}$ and $\lambda_m = 1/T^{(m)}$.

As mentioned earlier, the T_1 normal modes form a complete set, which we write in dyadic notation as

$$\sum_l \mathbf{Q}_l \mathbf{Q}_l = I \quad (11)$$

and similarly for the T_2 modes

$$\sum_l \mathbf{q}_l \mathbf{q}_l = I. \quad (12)$$

[When Eqs. (11) and (12) are applied to continuous functions of position, $I = \delta(\mathbf{r} - \mathbf{r}')$; for discrete basis sets, $I = \delta_{ij}$.] Moreover, the basis functions are orthonormal:

$$\mathbf{Q}_n \cdot \mathbf{Q}_m = \delta_{nm} \quad \forall n, m \quad (13)$$

and

$$\mathbf{q}_n \cdot \mathbf{q}_m = \delta_{nm} \quad \forall n, m, \quad (14)$$

where δ_{nm} is the Kronecker delta. These general properties allow one to derive the limiting behavior for short storage times $\mu_l t_s \ll 1$. All the exponentials in Eq. (10) are equal to unity, the sum on l collapses because of Eq. (11), and the sum on m collapses because of Eq. (14) with the result that

$$M(\tau_1, \tau_2; t_s \rightarrow 0) = M_0 \sum_n |\mathbf{q}_n \cdot \mathbf{m}_0|^2 e^{-\lambda_n(\tau_1 + \tau_2)}. \quad (15)$$

However, this is just Eq. (3) with $t = \tau_1 + \tau_2$. If the storage time is short enough the magnetization distribution does not change during storage and one has simple T_2 decay; the off-diagonal elements of $\mathcal{M}_{nm}$ vanish identically.

C. Models

We will compare calculations based on two models that can be used to describe the experiment represented in Fig. 1. Throughout this paper two assumptions are made to simplify our analysis. First, we ignore both longitudinal and transverse relaxation in the bulk fluid. As noted above, this effect can easily be added after calculating the surface controlled relaxation. Second, we set $\rho_1 = 0$ so there is no relaxation during the storage period. This maximizes the number of

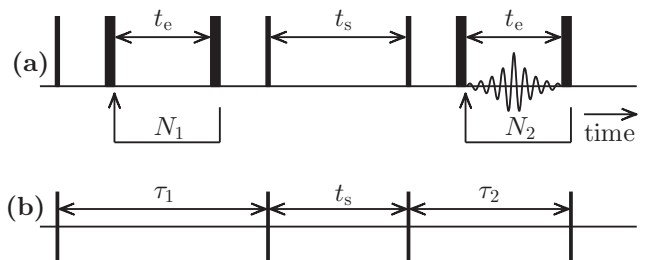


FIG. 1. Schematic of the T_2 - T_2 exchange measurement: (a) the rf pulse sequence diagram for the NMR experiment; thin and thick vertical lines represent 90° and 180° rf pulses, respectively. N_1 echoes are generated in the first T_2 encoding interval and N_2 echoes are monitored in the second T_2 interval. The spins are stored for a time t_s between the two T_2 intervals. The timing diagram (b) indicates the corresponding parts of our calculations. The duration of the first T_2 interval is τ_1 and the duration of the second T_2 interval is τ_2 . For a given t_s , τ_1 is varied on a logarithmic scale and the data are then processed.

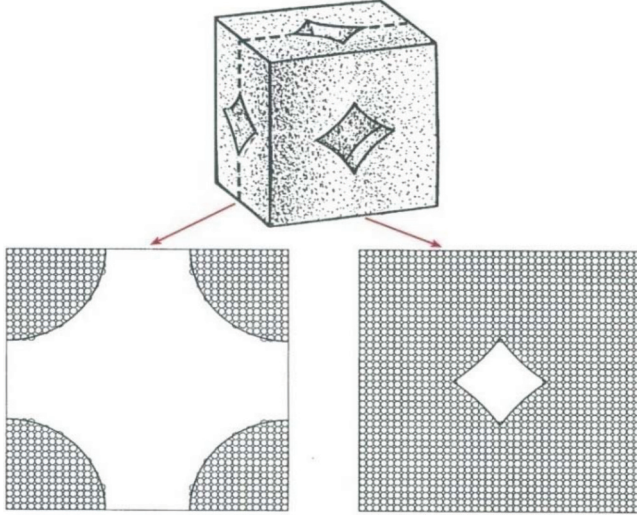


FIG. 2. Microporous version of the grain consolidation model in which the large grains are comprised of smaller grains whose centers lie on a lattice constructed by subdividing the original unit cell. The solid line separating the macropores and micropores represents a synthetic barrier that can be switched on and off during the random walk simulation.

spins that survive to the data acquisition segment and are available for coupling between distinct spin populations. This assumption simplifies our numerical simulations, but does not change the qualitative features of our results.

The μ GC model, shown in Fig. 2, is based on a three-dimensional periodic packing with a bimodal pore-size distribution [24,35]. Here large grains, each of which is itself porous, are packed on a simple cubic lattice. The large grains are comprised of smaller (possibly) overlapping grains centered on a lattice constructed by dividing the original unit cell into smaller cells. The scale factor governing this division controls the degree of mismatch in the model's pore sizes. If the large grains overlap, the μ GC model has pathways through the large open pores and (roughly) parallel pathways through the microporous grains. For the calculations described in this paper the edge of the large cube is taken to be $200 \mu\text{m}$ and the scale factor is 20, corresponding to small cells with an edge length of $10 \mu\text{m}$. The radius of the large grains R_M and that of the small grains R_m are given in Table I. Also given in the table are the corresponding macroporosities and microporosities (ϕ_M and ϕ_m , respectively). [The intrinsic porosity within the microporous region is $\phi_\mu = 0.180$ and $(1 - \phi_M)\phi_\mu = \phi_m$.] In Refs. [24,35] we have shown how variable step size random walks can be applied to the μ GC model to simulate the solution of Eqs. (1) and (2). These same techniques can be applied to the 2D measurement sequence described above. As in Ref. [35], the diffusion coefficient D_0 assigned to the fluid that fills the pore space is $1 \mu\text{m}^2/\text{ms} = 10^{-5} \text{ cm}^2/\text{s}$.

TABLE I. Parameter values for the simulations based on the μ GC model.

R_M	$125 \mu\text{m}$
ϕ_M	0.149
R_m	$6.1 \mu\text{m}$
ϕ_m	0.153
ϕ_{total}	0.302
D_0	$10^3 \mu\text{m}^2/\text{s}$
ρ_1	0
ρ_2	$2 \mu\text{m}/\text{s}$

The second model, which can be treated analytically, describes two spin reservoirs coupled by diffusive exchange. We picture two spin populations (in our case the macropores and the micropores) contributing signal volumes S_A and S_B , which are coupled to each other. Here $S_{A,B}$ represents the difference from the equilibrium values of the magnetic moment in A or B ; the total out-of-equilibrium magnetic moment is simply $M(t) = S_A(t) + S_B(t)$. In the absence of coupling, each population is characterized by a single rate for longitudinal (transverse) decay rate $R_A^{(2)} = 1/T_A^{(2)}$ and $R_B^{(2)} = 1/T_B^{(2)}$. [In the language of Eq. (3), a single mode describes the magnetization decay in the absence of coupling.] In the language of Ref. [22], each subsystem is assumed to be in the so-called fast diffusion limit in the sense that $\rho a/D_0 \ll 1$, where a is a characteristic spatial dimension of each subsystem. This assumption guarantees that the magnetization within each subsystem is spatially uniform at any time.

Analytic expressions for the time dependence of S_A and S_B during the $\tau_1 \rightarrow t_s \rightarrow \tau_2$ measurement sequence are derived by solving the coupled equations

$$\begin{pmatrix} \frac{dS_A}{dt} \\ \frac{dS_B}{dt} \end{pmatrix} = \begin{pmatrix} -R_A - \kappa_A & \kappa_B \\ \kappa_A & -R_B - \kappa_B \end{pmatrix} \cdot \begin{pmatrix} S_A \\ S_B \end{pmatrix}, \quad (16)$$

where the parameters κ_A and κ_B describe the effective exchange rates between the two reservoirs. When the magnetization densities in the two pores are equal $S_A/V_A = S_B/V_B$ (V_A and V_B are the volumes of the pores) exchange from $A \rightarrow B$ occurs at the same rate as $B \rightarrow A$. This condition of detailed balance implies $\kappa_A V_A \equiv \kappa_B V_B \stackrel{\text{def}}{=} k$.

Equation (16) describes a heuristic two-site model with potential applicability to a variety of situations. Here we explore its applicability to the evolution of spin populations in the microporous μ GC model. Thus we envision that the parameter k is related to the diffusivity D_0 , though the latter parameter does not appear explicitly in the model.

The magnetization in the coupled system is evaluated from the eigenvalues of Eq. (16) [9,33]:

$$\lambda_{\pm} = \frac{R_A + R_B + k(u_A + u_B) \pm \sqrt{(R_A - R_B)^2 + 2k(R_A - R_B)(u_A - u_B) + k^2(u_A + u_B)^2}}{2}, \quad (17)$$

where $u_{A,B} = 1/V_{A,B}$. For each eigenvalue $\lambda_{\pm}$ there is an associated eigenvector $(S_A^{\pm}, S_B^{\pm})$. We have found it convenient to express our results in terms of the magnetization $\mathbf{m}^{\pm} = (m_A^{\pm}, m_B^{\pm}) = (S_A^{\pm}/V_A, S_B^{\pm}/V_B)$. From Eq. (16) we may use either of the following expressions for the (unnormalized) normal modes:

$$\begin{aligned}\mathbf{m}^{\pm} &= [k, (R_A V_A + k - \lambda_{\pm} V_A)], \\ \mathbf{m}^{\pm} &= [(R_B V_B + k - \lambda_{\pm} V_B), k].\end{aligned}\quad (18)$$

The inner products that appear in Eqs. (5) and (10) are given by Eq. (6), which becomes

$$\mathbf{f} \cdot \mathbf{g} = f_A g_A V_A + f_B g_B V_B, \quad (19)$$

where $f_{A,B} = S_{A,B}/V_{A,B}$ is the excess magnetization density in pore *A* or *B* for mode *f* and similarly for mode *g*. The normalized normal modes for the two-site exchange model are therefore

$$\mathbf{q}_{1,2} = \frac{\mathbf{m}^{\mp}}{\sqrt{\mathbf{m}^{\mp} \cdot \mathbf{m}^{\mp}}}. \quad (20)$$

Similarly, the normal modes for the storage part $\mathbf{Q}_{1,2}$ are given by the same procedure, but with $R_A = R_B = 0$.

If there is no coupling between the two pore systems ($k = 0$) these decay rates become simply R_A and R_B . It will be useful to look at the limiting behavior of the eigenvalues as a function of the coupling parameter, k . For small k

$$\lim_{k \rightarrow 0} \lambda_{\pm}(k) = \begin{cases} R_A + k/V_A + O(k^2) \\ R_B + k/V_B + O(k^2), \end{cases} \quad (21)$$

while for large k

$$\begin{aligned}\lim_{k \rightarrow \infty} \lambda_{+}(k) &= k(u_A + u_B) + \frac{R_A u_A + R_B u_B}{u_A + u_B} + O(1/k), \\ \lim_{k \rightarrow \infty} \lambda_{-}(k) &= \frac{R_A V_A + R_B V_B}{V_A + V_B} + O(1/k).\end{aligned}\quad (22)$$

A useful estimate of the crossover value k_{xc} separating the weak coupling regime and the strong coupling regime is given by the intersections of either of the two pairs of asymptotes in Eqs. (21) and (22). As it happens, these intersections occur at the same value of k , which is

$$k_{xc} = \frac{|R_A - R_B|}{u_A + u_B}. \quad (23)$$

The eigenvalues defined by Eq. (17) and their limiting values (21) and (22) are plotted as functions of k in Fig. 10 for specific choices of the parameters $R_{A(B)}$ and $V_{A(B)}$.

Our purpose is to clarify the interpretation of T_2 - t_s - T_2 measurements by comparing calculations based on the two-site exchange and μ GC models. Since the value of 2D measurements is related to the way that they describe coupling between distinct pore populations, it will be useful to modulate this coupling directly in our calculations. This is easily done in both models. In the two-site exchange model we can turn off (i.e., set to zero) the coupling parameters $\kappa_{A(B)}$ during any of the three stages of the measurement. Similarly, in the μ GC model we can introduce a *barrier* (Fig. 2) between the macropores and micropores. As in the two-site exchange model, the barrier can be put in place during the spin preparation τ_1 , spin storage t_s , or data acquisition τ_2 parts of the measurement sequence.

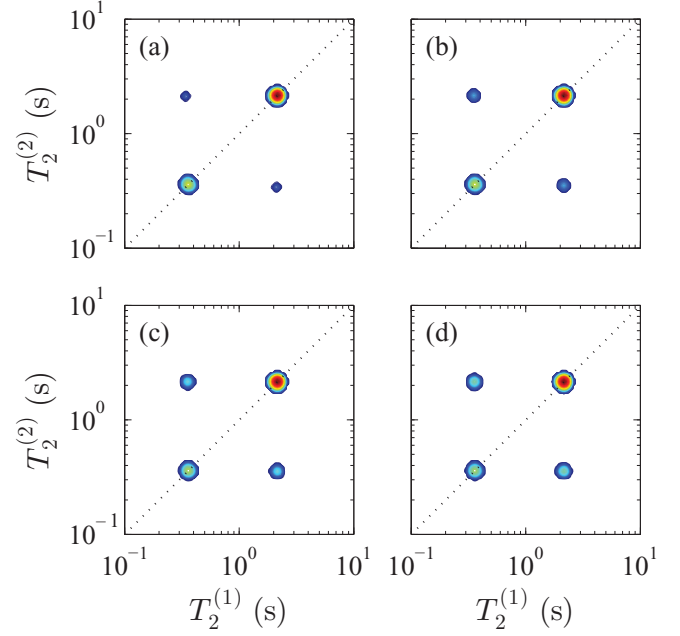


FIG. 3. (Color online) The T_2 - T_2 correlation plots generated using the two-site exchange model. The set of parameter values is given in Table II. Exchange is allowed during all three intervals in the calculation. The storage times are (a) $t_s = 100$ ms, (b) 400 ms, (c) 700 ms, and (d) 1000 ms. The dotted line indicates the diagonal $T_{2(1)} = T_{2(2)}$ in each plot. The apparent relaxation rates are $\lambda_1 = 0.44 \text{ s}^{-1}$ and $\lambda_2 = 2.82 \text{ s}^{-1}$.

III. CALCULATIONS

In Figs. 3 and 4 we show representative calculations based on the two-site exchange and μ GC models. The T_2 - T_2 cross plots are shown for four values of the storage time t_s . Here the parameter values used in the two-site exchange model have been chosen so that the results approximate those of the μ GC model. We describe how this is done in Sec. IV A. The parameter values are listed in Table II. In these calculations the model parameters have been adjusted to give roughly equal volumes in the two pore populations and, as noted above, we have set $\rho_1 = 0$. While the storage of the spins during t_s is perfect, there is nevertheless significant rearrangement of the spin density because the boundary condition at the pore-grain interface has changed. [As expected, calculations with $\rho_1 \neq 0$ (e.g., $\rho_1 = \rho_2/2$) yield results that are similar to those shown on Figs. 3 and 4, but the off-diagonal peaks are less pronounced.]

Generally, the diagonal peaks in the T_2 - T_2 cross plots are thought to represent the lifetime spectrum within distinct pore populations and the off-diagonal peaks represent diffusive coupling between them. Indeed, the off-diagonal peaks are often referred to as *exchange* peaks and are believed to represent diffusion (between the two populations) that takes place during the storage period t_s . It is well-established [31,33] that this is not really true; according to Eq. (10), the off-diagonal peaks are controlled by the degree to which the normal modes in the storage period are different from those in the initial T_2 decay. In principle, one may have off-diagonal peaks corresponding to different modes within the same single pore. In the very special case that the T_2 modes are describable

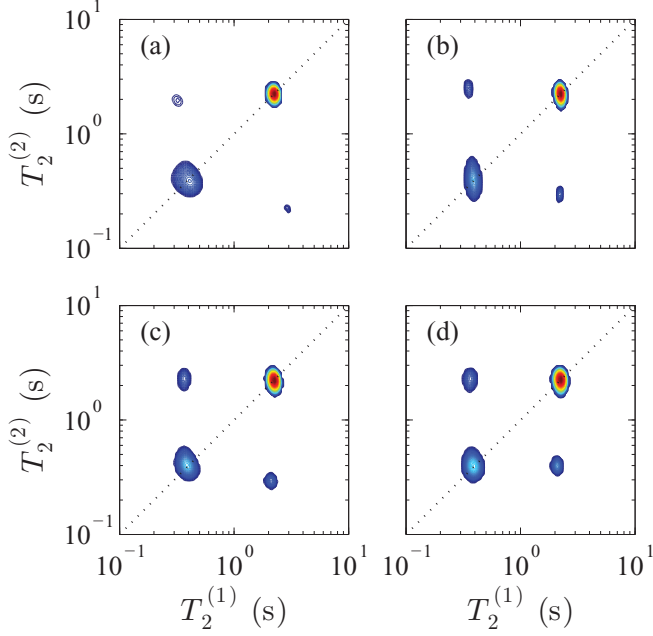


FIG. 4. (Color online) Four T_2 - T_2 cross plots based on the μ GC model. We have taken $\rho_1 = 0$ and four values of t_s : (a) $t_s = 100$ ms, (b) 400 ms, (c) 700 ms, and (d) 1000 ms. The dotted line indicates the diagonal $T_{2(1)} = T_{2(2)}$ in each plot. In these calculations the model parameters have been arranged so that the microporosity $\phi_m = 0.16$, the macroporosity $\phi_M = 0.14$, and the transverse surface relaxation parameter $\rho_2 = 2.0 \mu\text{m/s}$.

as being weakly coupled in the sense of Eqs. (21), the term “exchange” may have some approximate validity, but even so it is really just a manner of speaking. In physical terms, of course, diffusive exchange between the two pore populations is taking place over the entire measurement sequence. To better understand what is happening, it will be useful to turn the coupling on and off during the different stages of the model calculations. As we noted above, this can be done by setting the coupling parameter $\kappa \rightarrow 0$ in the two-site exchange model and by imposing a barrier between the macropores and micropores in the μ GC model.

In Figs. 5 and 6 four distinct protocols are employed to compare calculations based on the two-site exchange and μ GC models. In both sets of results the value of t_s is fixed at 700 ms and the value of $\rho_1 = 0$. In Figs. 5(a) and 6(a) coupling is turned off for the entire measurement and, as expected, there are no off-diagonal peaks. In Figs. 5(b) and 6(b) coupling is allowed during τ_1 and τ_2 but is turned off during t_s . Here we see that the μ GC and two-site exchange models lead to quite different results. If the off-diagonal peaks

TABLE II. Parameter values for the two-site exchange model as applied to the μ GC model.

$V_A (\mu\text{m}^3)$	1.28×10^6
$V_B (\mu\text{m}^3)$	1.12×10^6
$R_{2A} (\text{s}^{-1})$	2.26
$R_{2B} (\text{s}^{-1})$	7.46×10^{-2}
$R_{1A} = R_{1B}$	0
$k (\mu\text{m}^3 \text{s}^{-1})$	5.40×10^5

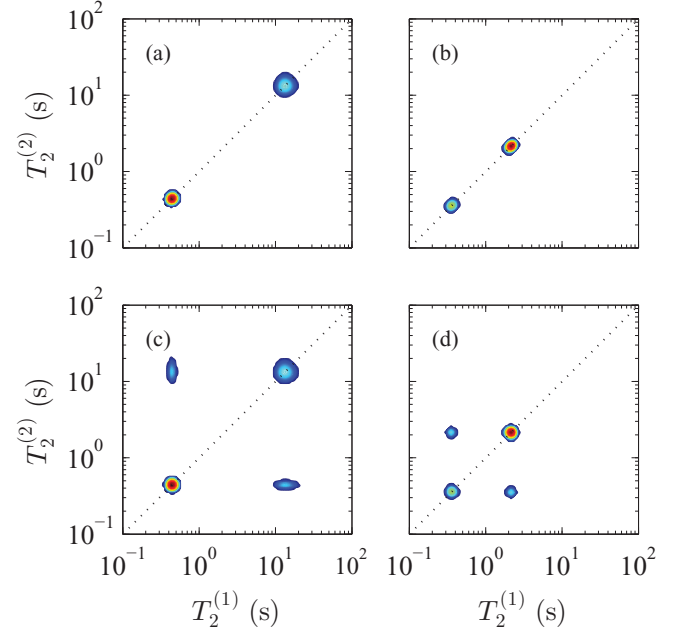


FIG. 5. (Color online) Four T_2 - T_2 calculations based on the two-site exchange model with t_s is held fixed at 700 ms. The values of k are (a) $k = 0$ at all times, (b) $k = 0$ during t_s , (c) $k = 0$ during τ_1 and τ_2 , and (d) $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$ at all times. [Panel (d) is identical to Fig. 3(c).] All other parameters are listed in Table II. Note the qualitative agreement between these results and those in Fig. 6 except for (b).

really describe diffusion between the pores during t_s then we need to understand why they are still present in the μ GC calculations. In Figs. 5(c) and 6(c) coupling is allowed only during t_s and in Figs. 5(d) and 6(d) coupling is allowed over the entire measurement (as in Figs. 3 and 4). For both Figs. 5(c) and 6(c) and Figs. 5(d) and 6(d) the two models are in good qualitative agreement.

We note that both models share one qualitative feature: When the coupling is turned on there appears to be a much more significant reduction in the longer-lived lifetime than in the shorter one. For the two-site exchange model there is a very simple explanation, which we defer to Sec. IV A. For the μ GC model the spins in the macropore see an enhanced avenue for decay into the micropores when the barrier is removed, thereby reducing the lifetime of this mode. Most of the spins in the micropores are buried deep inside the micrograin and never see the macropore before they decay. Hence this mode’s relaxation time is largely unaffected.

In Sec. IV the behavior of the two-site exchange and μ GC models will be analyzed in detail. It is not difficult, however, to understand qualitatively what is going on. The two-site exchange model builds in the assumption that both pore populations are in the fast diffusion limit discussed in Sec. II C. Thus the magnetization is uniform within each pore between $t = 0$ and $t = \tau_1$ and then remains uniform within each pore during $\tau_1 \leq t \leq \tau_1 + t_s$. By contrast, in the μ GC model the value $\rho_2 = 2 \mu\text{m/s}$ corresponds to an intermediate case, viz., the small pores are in the fast diffusion limit but the large pores are not fully in that limit ($\rho_2 a/D_0 \approx 0.1$). Thus, as indicated in Fig. 7, between $t = 0$ and $t = \tau_1$ the

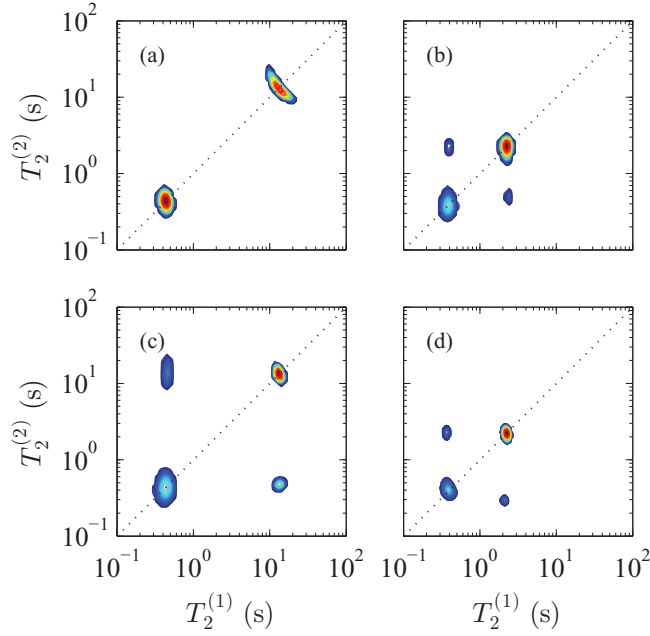


FIG. 6. (Color online) Four T_2 - T_2 simulations based on the μ GC model with t_s held fixed at 700 ms: (a) a barrier blocks coupling between the macropores and micropores for all times, (b) the barrier is in place only during t_s , (c) coupling of the two populations is allowed only during t_s , and (d) coupling is allowed at all times. [Panel (d) is identical to Fig. 4(c).] As before, the value of $\rho_2 = 2.0 \mu\text{m/s}$.

magnetization in the large pores builds up significant spatial gradients. The spin density will be largest near the center of the pore and smallest near the interface with the micropores where relaxation is taking place. During the interval $\tau_1 \leq t \leq \tau_1 + t_s$ diffusion *within* the large pores will make the magnetization somewhat more uniform (i.e., smaller in the center of the pore and larger at the interface with the micropores). Thus $M(\mathbf{r}, \tau_1)$ and $M(\mathbf{r}, \tau_1 + t_s)$ will have distinctly different representations in terms of the transverse eigenfunctions. This difference is a consequence of intrapore diffusion within the macropores rather than interpore diffusion and it is responsible for the off-diagonal peaks in the T_2 - T_2 cross plot.

To test this explanation, we have done two calculations based on the μ GC model. First, one might suppose that with $\rho_1 = 0$ and a barrier in place, the magnetization decay for $t \geq \tau_1 + t_s$ would simply be identical to that obtained from a T_2 calculation without any waiting period. From the results

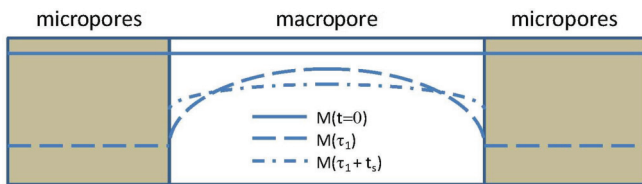


FIG. 7. (Color online) One-dimensional representation of a macropore coupled on the left and right to microporous regions. The magnetization profiles at $t = 0$, $t = \tau_1$, and $t = \tau_1 + t_s$ are indicated schematically. Because diffusion between the macropores and micropores is forbidden between $t = \tau_1$ and $\tau_1 + t_s$, the macropore magnetization at the interface *increases* during this period.

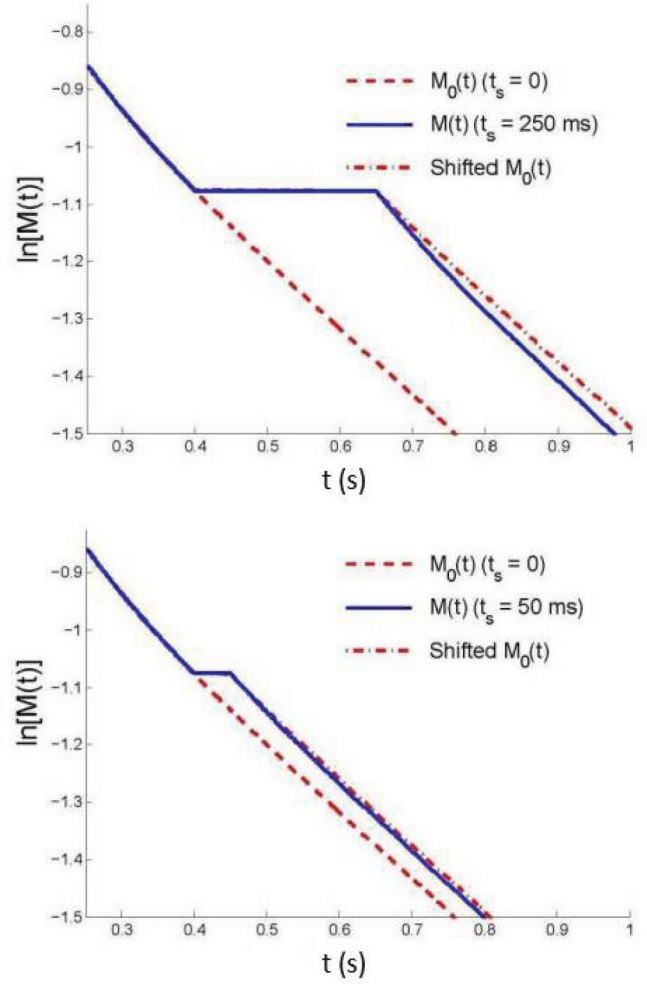


FIG. 8. (Color online) Two simulations (with different values of t_s) based on the μ GC model are compared. The dashed curves are simply the magnetization decay with coupling allowed at all times and no storage period. The solid curves are the magnetization decay with a barrier in place during t_s and storage times of 250 and 50 ms. Also shown (as dot-dashed curves) are the dashed curves shifted by t_s for $t \geq \tau_1 + t_s$. With $t_s = 250$ ms, we see a faster decay rate beginning at $t = \tau_1 + t_s$. This is a consequence of the increased magnetization at the interface pictured in Fig. 7. When $t_s = 50$ ms there is not enough time for the magnetization at the interface to adjust and there is no increase in the decay rate.

shown in Fig. 8, we see that this is not the case (although it is nearly true as $t_s \rightarrow 0$). Because the magnetization density at the interface has increased during the waiting period, the decay rate at $t = \tau_1 + t_s$ is larger than the prediction of the $t_s = 0$ calculation. (The dependence of this effect on the value of t_s will be discussed in Sec. IV C.) Second, we have repeated our calculations (with the barrier in place during t_s) with progressively smaller values of ρ_2 . In Fig. 9 we see that as ρ_2 decreases and the magnetization within the large pores becomes more uniform, the strength of the off-diagonal peaks is greatly reduced. Of course, in the limit $\rho_2 \rightarrow 0$ the macropeaks and micropeaks on the diagonal merge and we have only a single peak whose location is determined by the model's net surface area [24]. The entire pore system has moved into the fast diffusion limit, during the T_2 process.

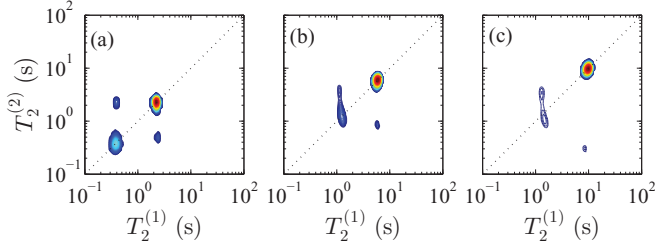


FIG. 9. (Color online) Three T_2 - t_s - T_2 simulations based on the μ GC model with progressively smaller values of ρ_2 : (a) $\rho = 2 \mu\text{m/s}$, (b) $\rho = 0.4 \mu\text{m/s}$, and (c) $\rho = 0.2 \mu\text{m/s}$. In all cases a barrier blocks coupling between the macropores and micropores during t_s . As the value of ρ_2 decreases the entire pore system moves further into the fast diffusion limit, the magnetization becomes spatially uniform at any time, and the off-diagonal peaks in the cross plot vanish.

At any instant of time during the process the magnetization becomes spatially uniform throughout the entire system.

IV. DISCUSSION

Here we examine in some detail the connection between the two-site exchange model and the results of our calculations based on the μ GC model. Is the former a useful way to understand the latter? The first question to address is whether the eigenvalues (the decay rates) can be made to match in the two systems. Second, we investigate the dependence of the matrix components of the magnetization on the storage time t_s . Finally, we look at the effects of modulating the interaction of the macropores and micropores.

A. Eigenvalues

The eigenvalues that define the behavior of the two-site exchange model are given by Eq. (17). For the T_2 part of the decay process in the μ GC model we determine the values of R_A and R_B by setting them equal to the decay rates when there is a barrier in place, i.e., when $k = 0$. For the case where $\rho_2 = 2.0 \mu\text{m/s}$ these decay rates are seen, in Fig. 6(a) or 6(c), as the reciprocals of the T_2 times where the peaks are located. From these peaks we get $R_{2A} = 1/0.4425 = 2.26 \text{ s}^{-1}$ and $R_{2B} = 1/13.4 = 0.0746 \text{ s}^{-1}$. It is straightforward to compute the volume of each of the pore subsystems in the μ GC model: $V_A = 1.28 \times 10^6 \mu\text{m}^3$ and $V_B = 1.12 \times 10^6 \mu\text{m}^3$.

There is only one remaining free parameter k in the two-site exchange model. As far as we know, there is no direct way of computing this parameter directly so we determine it by requiring that the decay rates in the coupled system match. (See the final paragraph of Sec. IV B.) In Fig. 10 we plot Eq. (17) as a function of k . The dot-dashed horizontal lines are the decay rates determined from the simulations when there is no barrier in place [Fig. 6(b) or 6(d)]. If the two-site exchange model is to have any validity for the μ GC system the former curves should intersect the latter rates at the same value of k in the two cases and we see that they basically do. We have $k = 4.96 \times 10^5 \mu\text{m}^3/\text{s}$ for the lower curve and $k = 5.80 \times 10^5 \mu\text{m}^3/\text{s}$ for the upper one. Considering the approximate way in which we have determined the decay rates from the μ GC simulations, for which there is appreciable uncertainty in the determination of the decay rates from the decay spectrum,

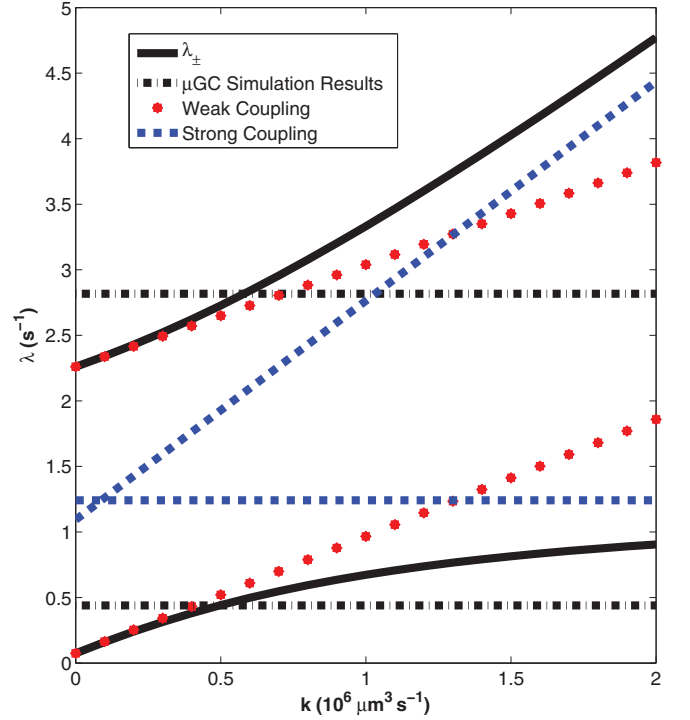


FIG. 10. (Color online) Decay rates for the two-site exchange model. The solid lines show the dependence of the lifetimes T_2 derived from Eq. (17) as a function of the coupling parameter k . The other parameters of the model are determined from the results of the μ GC simulations shown in Fig. 6(a) or 6(c), when the barrier is in place, and are given in Table II. The dot-dashed horizontal lines are the decay rates determined from the μ GC simulations in the absence of the barrier [Fig. 6(b) or 6(d)]. The intersections of these lines with the curves from Eq. (17) determine the values of k implied for each of the two modes. That these two values of k are nearly the same lends credence to the two-site exchange model as applied to the μ GC model. The dotted lines and the dashed lines are the small- k and large- k limits, respectively [Eqs. (21) and (22)].

we consider these values of k to be substantially the same. Of course, there are inherent uncertainties in this comparison. For example, the values derived from the simulations depend on (i) where exactly the barrier is placed in the simulations, (ii) the value of ρ_2 , and (iii) the details of the processing algorithm employed to determine the decay spectrum. Notwithstanding, we have found that as long as the values of ρ_2 are not too large, the two-site exchange model is able to reproduce the two decay rates with approximately the same value of k . We shall use the average value $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$. The numerical values that we have determined for all the two-site exchange parameters relevant to the μ GC model are shown in Table II. These values were used to generate Figs. 3 and 5.

The dotted and dashed lines in Fig. 10 are, respectively, the small- and large- k limits described by Eqs. (21) and (22). We note in passing that the coupling in the present μ GC calculations is fairly weak in the sense that the curves shown in Fig. 10 are fairly close to Eqs. (21) at the points of intersection, i.e., $k < k_{xc}$.

We note also that we can give a simple explanation of the observation we made in Sec. III about the effect of coupling on the lifetimes that are apparent in Fig. 5. By removing

the barrier ($k = 0$ is changed to $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$) the change in the decay rates for each of the two modes is about the same in absolute terms. However, this represents a much larger percentage change for the slower relaxing mode than it does for the faster one and this is what we see on the logarithmic axes of Fig. 5.

B. Matrix components of the magnetization

Suppose that there are only two relevant modes in the storage period and two for the T_2 part. This is rigorously true for the two-site exchange model and approximately true for the μGC model. In this section we explore the consequences of assuming it to be true for the μGC model.

At first glance Eq. (10) would lead one to think that there are six different inner products that must be evaluated in order to develop a theoretical prediction of the three functions $\mathcal{M}_{nm}$, but this is not true. We show that there are at most two independent parameters, which reduce to one for the case at hand, where we assume that there is no surface relaxivity during the storage period. Our convention in the following is that subscript 1 refers to the mode with the smaller decay rate in each case and 2 refers to the one with the larger decay rate.

Because of the completeness relation (11), one has the following result for the off-diagonal component of Eq. (10) with $n = 1$ and $m = 2$:

$$\mathcal{M}_{12}(t_s) = M_0 V_P C_{12} [e^{-\mu_1 t_s} - e^{-\mu_2 t_s}], \quad (24)$$

where

$$C_{12} = [\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_2 \cdot \mathbf{m}_0][\mathbf{q}_1 \cdot \mathbf{Q}_1][\mathbf{Q}_1 \cdot \mathbf{q}_2] / V_P \\ = -[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_2 \cdot \mathbf{m}_0][\mathbf{q}_1 \cdot \mathbf{Q}_2][\mathbf{Q}_2 \cdot \mathbf{q}_2] / V_P. \quad (25)$$

Equation (24) exhibits the very general property that initially each off-diagonal component of $\mathcal{M}_{nm}$ increases linearly with time t_s and then reaches a maximal value before decaying to zero, as t_s is further increased. The simplicity of the two-component system lends itself to a simple geometrical interpretation. Because $[\mathbf{f} \cdot \mathbf{g}]^2 \leq [\mathbf{f} \cdot \mathbf{f}][\mathbf{g} \cdot \mathbf{g}]$ for any $\mathbf{f}, \mathbf{g}$ pair and because $|\mathbf{m}_0| = \sqrt{\mathbf{m}_0 \cdot \mathbf{m}_0} = \sqrt{V_P}$, where V_P is the total pore volume, we may define an angle ψ via

$$\mathbf{m}_0 \cdot \mathbf{q}_1 = \sqrt{V_P} \cos(\psi). \quad (26)$$

Because orthonormality and completeness of $\mathbf{q}_1$ and $\mathbf{q}_2$ imply $\mathbf{m}_0 = (\mathbf{m}_0 \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{m}_0 \cdot \mathbf{q}_2)\mathbf{q}_2$, which in turn implies $[\mathbf{m}_0 \cdot \mathbf{q}_1]^2 + [\mathbf{m}_0 \cdot \mathbf{q}_2]^2 = \mathbf{m}_0 \cdot \mathbf{m}_0 = V_P$, we have

$$\mathbf{m}_0 \cdot \mathbf{q}_2 = \sqrt{V_P} \sin(\psi). \quad (27)$$

Similarly, we may define the angle θ via

$$\mathbf{Q}_1 \cdot \mathbf{q}_1 = \cos(\theta), \quad \mathbf{Q}_1 \cdot \mathbf{q}_2 = \sin(\theta). \quad (28)$$

That is

$$\mathbf{Q}_1 = \cos(\theta)\mathbf{q}_1 + \sin(\theta)\mathbf{q}_2. \quad (29)$$

Because $\mathbf{Q}_1 \cdot \mathbf{Q}_2 = 0$ we must have

$$\mathbf{Q}_2 = \pm[-\sin(\theta)\mathbf{q}_1 + \cos(\theta)\mathbf{q}_2]. \quad (30)$$

Therefore,

$$C_{12} = \frac{1}{4} \sin(2\psi) \sin(2\theta). \quad (31)$$

Specializing to the case where the T_1 relaxivity vanishes, $\rho_1 \equiv 0$, one of the normal modes, corresponding to a uniform magnetization, does not decay at all:

$$\mu_1^0(\rho_1 = 0) \equiv 0. \quad (32)$$

The corresponding normal mode is

$$\mathbf{Q}_1(\rho_1 = 0) \equiv \mathbf{m}_0 / \sqrt{V_P}. \quad (33)$$

Now Eq. (24) reads

$$\frac{\mathcal{M}_{12}(t_s)}{M_0 V_P} = \frac{1}{4} \sin^2(2\psi) [1 - e^{-\mu_2^0 t_s}], \quad (34)$$

where we have used the fact that $\theta \equiv \psi$ when $\rho_1 = 0$ because of Eq. (33). Likewise, we may derive similar expressions for the diagonal elements. They are

$$\frac{\mathcal{M}_{11}(t_s)}{M_0 V_P} = \cos^4(\psi) + \frac{1}{4} \sin^2(2\psi) e^{-\mu_2^0 t_s}, \quad (35)$$

$$\frac{\mathcal{M}_{22}(t_s)}{M_0 V_P} = \sin^4(\psi) + \frac{1}{4} \sin^2(2\psi) e^{-\mu_2^0 t_s}. \quad (36)$$

At this level of approximation, all elements of $\mathcal{M}_{ij}$ are positive definite; as far as we know, there is no general proof that this must be so.

That all the elements of $\mathcal{M}_{nm}$ asymptotically approach constant values as $t_s \rightarrow \infty$ is a consequence of the assumption that the surface relaxivity vanishes for the storage period $\rho_1 \equiv 0$ and is independent of the assumption that there are only two relevant modes. The storage mode corresponding to uniform magnetization density $\mathbf{Q}_1 \equiv \mathbf{m}_0 / \sqrt{V_P}$ does not decay at all. Consequently, we have

$$\frac{\mathcal{M}_{nm}(t_s \rightarrow \infty)}{M_0 V_P} = A_n A_m, \quad (37)$$

where $A_{n,m}$ are given by Eq. (5). From Eq. (15) we have

$$\frac{\mathcal{M}_{nm}(t_s \rightarrow 0)}{M_0 V_P} = A_n \delta_{nm}. \quad (38)$$

Equations (34)–(38) will prove to be useful for us when we interpret our numerical results for the μGC model.

Finally, we note that the sum of all matrix elements is independent of storage time when $\rho_1 = 0$:

$$\sum_{n,m} \mathcal{M}_{nm}(t_s) = M_0 V_P. \quad (39)$$

This result is easily proved from the definition (10), using the completeness property of the T_2 modes (12). It is straightforward to demonstrate that Eqs. (34)–(36) satisfy Eq. (39).

We emphasize that Eqs. (24)–(39) all follow from two assumptions. (i) There are only two relevant modes in the problem ($\mathbf{q}_1$ and $\mathbf{q}_2$) and every quantity or function of interest ($\mathbf{m}_0, \mathbf{Q}_1, \mathbf{Q}_2$) may be expressed as a linear combination of those two. (ii) One of the modes in the storage period corresponds to uniform magnetization and does not decay at all. At this level of approximation the six different inner products implied by Eq. (10) are all expressible in terms of a single parameter ψ , which in turn is evaluated by means of its definition (26).

In order to proceed further, we need a specific model with which we evaluate ψ and μ_2^0 . Specifically, for the two-site

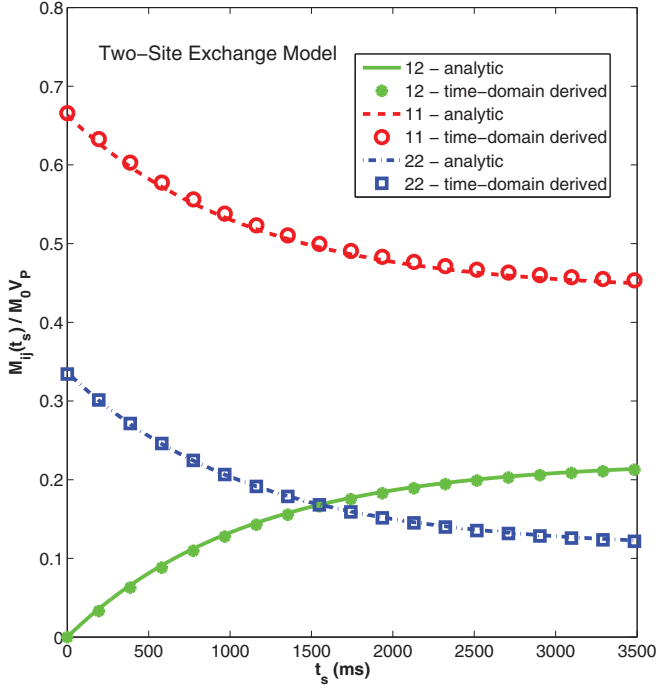


FIG. 11. (Color online) Strengths of the 2D peaks for the two-site exchange model in Fig. 3 plotted as a function of t_s . The symbols represent values numerically derived by processing time-domain data using the 2D Fredholm integral equation. The solid curves are Eqs. (34)–(36) using the parameter values listed in Table II.

exchange model we have

$$\begin{aligned}\mu_1^0 &= \lambda_-(R_A = 0, R_B = 0) = 0, \\ \mu_2^0 &= \lambda_+(R_A = 0, R_B = 0) = k(u_A + u_B),\end{aligned}\quad (40)$$

which follow from Eq. (17). Mode 1, which has a decay rate = 0, corresponds to a uniform magnetization, viz.,

$$\mathbf{Q}_1(R_A = 0, R_B = 0) \equiv \mathbf{m}_0 / \sqrt{V_P}. \quad (41)$$

The initial magnetization in turn is simply $\mathbf{m}_0 = (1, 1)$.

In Fig. 11 we plot the values of $\mathcal{M}_{ij}$ appropriate to the two-site exchange model. For each value of t_s we have generated time-domain data that we processed as if they were experimental data in the contour plots such as those shown in Fig. 3. We numerically integrate the volume under each peak as an estimate of the relevant component $\mathcal{M}_{ij}(t_s)$. These values are plotted with the relevant symbols in Fig. 11. Using the same input parameters as in Table II we plot Eqs. (34)–(36) as solid curves. Here the relevant relaxation time is given by the second of Eqs. (40): $1/\mu_2^0 = [k(u_A + u_B)]^{-1} = [\kappa_A + \kappa_B]^{-1} = 1108$ ms. We evaluate the angle ψ from its definition (26) using Eqs. (18) and (20); the initial magnetization is simply $\mathbf{m}_0 = (1, 1)$. For this parameter set we find $\psi = 0.619$ rad = 35.5° . That the strictly analytic curves agree so well with the numbers determined from the processed time-domain data provides confidence that this processing scheme will provide accurate results when we apply it to the results for the μ GC data.

To what extent do Eqs. (34)–(36) describe the evolution of the magnetization elements that we see in our μ GC simulations? We have already seen that consistency of the

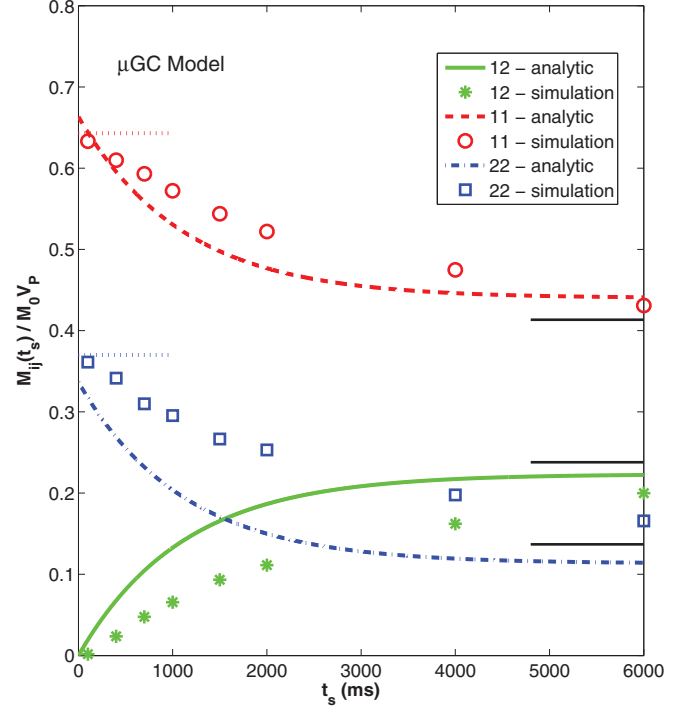


FIG. 12. (Color online) Strengths of the 2D peaks for the μ GC model in Fig. 4 plotted as a function of t_s . The symbols represent values numerically derived by processing time-domain data using the 2D Fredholm integral equation. The solid curves are Eqs. (34)–(36) using the parameter values given in Table II. The dotted lines indicate the values of the diagonal elements of the simulations at $t_s = 0$ from which the long-time asymptotic values are predicted according to Eq. (37). These values are indicated with solid horizontal lines. It is apparent that the simulation results are approaching these values in the limit of long storage times.

eigenvalues in the simulations with the predictions of the two-site exchange model determines the parameter set listed in Table II. With these values of the parameters, Eqs. (34)–(36) are replotted in Fig. 12 along with the results of the time-domain simulations based on the μ GC model, processed as before. The dotted lines indicate the values of the two diagonal elements implied by the simulations at $t_s = 0$. Using these values we indicate the asymptotic values of the simulations as predicted by Eq. (37). It is clear that the simulation data are consistent with those predictions.

We have several comments. First, the initial values predicted by the two-site exchange model for $\mathcal{M}_{11}(0)$ and $\mathcal{M}_{22}(0)$ are in rough agreement with those that are deduced from the simulation data. This is equivalent to saying that the two-site exchange model does a reasonably good job of describing ordinary T_2 decay (3) for the μ GC model. Because of Eq. (37) the behavior in the limit of long storage times is also reasonably well described by the two-site exchange model. Where the model fails is in its estimation of the time scale for equilibration, namely, the predicted decay rate $\mu_2^0 = k(u_A + u_B)$ is much larger than what we actually observe. This is a major shortcoming of the two-site exchange model in the present context. This in turn is due to the hard-wired assumption in the two-site exchange model that each subsystem is strictly in the fast diffusion limit. In reality,

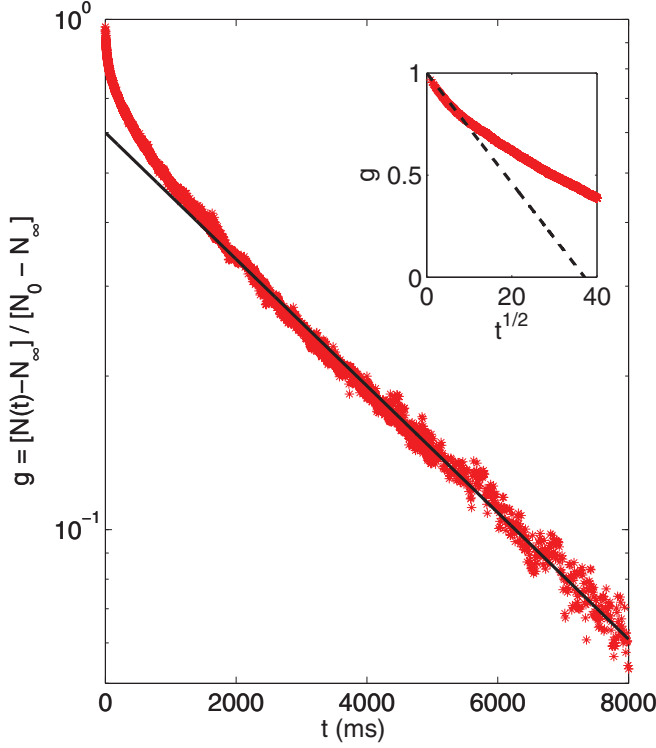


FIG. 13. (Color online) Number of walkers $N(t)$ in the macropore space monitored as a function of time for a μ GC simulation in which walkers start only in the macropore. The surface relaxivity vanishes: $\rho_1 \equiv 0$. A uniform density is achieved at long times because of diffusion. The solid line represents a decay time of 3.5 s. In the inset the data are plotted vs $\sqrt{t}$ and the dashed line corresponds to Eq. (42).

during the first T_2 decay period gradients in the magnetization are established across the entire unit cell and they relax much more slowly during the storage period than the second of Eqs. (40) implies.

In order to correctly estimate this rate we have done a simple diffusion simulation in which $\rho = 0$. We randomly launch walkers throughout the macropore only and we monitor the number of walkers within the macropore as a function of time (Fig. 13). This quantity decays from its initial value N_0 to an equilibrium value $N_\infty = N_0 V_B / (V_A + V_B)$, where V_A and V_B are the micropore and macropore volumes, respectively. Although we see several distinct decay regimes, the slowest decay rate corresponds to a lifetime of 3500 ms. We suppose that this corresponds to the dominant mode that we observe during the storage part of the T_2 - t_s - T_2 simulations in Fig. 12. Accordingly we have replotted Eqs. (34)–(36) against the numerical data for the μ GC model in Fig. 14 using this relaxation time, i.e., $\mu_2^0 = 1/3.5 \text{ s}^{-1}$. The parameter ψ in Eqs. (34)–(36) is of course unchanged because its value is set by an analysis of the coupling of the initial uniform magnetization to the T_2 modes, via Eq. (26). Although agreement is not perfect, it is much improved over that in Fig. 12.

We conclude that for this sort of μ GC geometry with the sizes approximately equal to $100 \mu\text{m}$ and relaxivity values approximately equal to $1 \mu\text{m/s}$, which are equal to those we have used in our calculations, the two-site exchange model

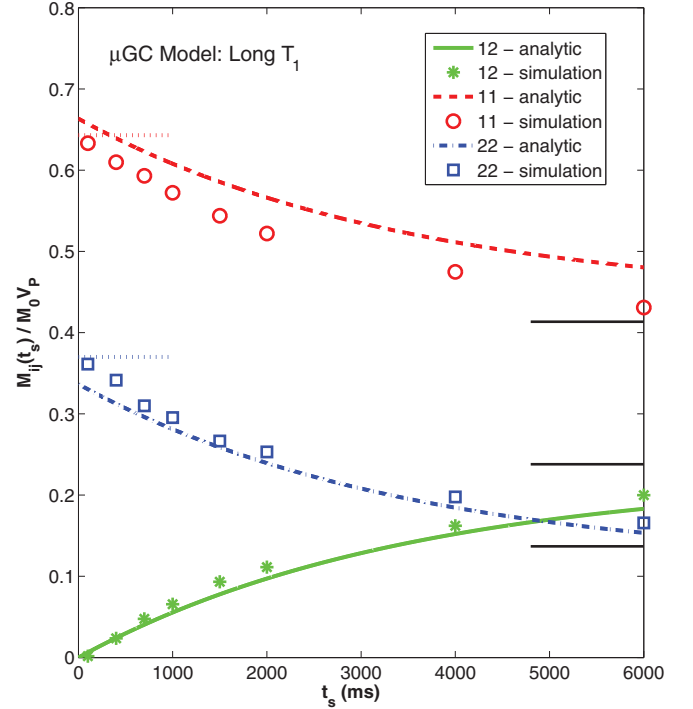


FIG. 14. (Color online) Same as for Fig. 12 except that the decay time has been changed to $T_1 = 3500 \text{ ms}$, as described in the text.

works reasonably well in describing the T_2 decay. Within the context of the two-site exchange model, the system is in the weak coupling limit $k < k_{xc}$. It fails during the storage part because we have set $\rho_1 \equiv 0$. Now the two-site exchange model seems to be in the strong coupling limit because $k \gg k_{xc} \equiv 0$. The only mechanism for decay of magnetization imbalance is via diffusion, which the two-site exchange model attributes solely to diffusion between macropores and micropores. The results of our simulations show that intrapore diffusion is clearly the limiting process in such a case. The strong suggestion of our results thus far is that if a more realistic value of $\rho_1 \approx \rho_2/2$ were used, the results of the numerical simulations would be reasonably well described by the two-site exchange model. An analysis of this case is beyond the scope of the present paper.

In connection with Fig. 13, there is an additional observation we should make. We said above that there is no direct relation between the μ GC simulations and the coupling parameter k of the two-site exchange model. One might have thought that the initial rate at which walkers diffuse into the micropores in this simulation could be interpreted as an effective exchange rate. According to Eq. (16), $\frac{dS_B}{dt}|_{t=0} = -(k/V_B)S_B|_{t=0}$, where we identify $S_B(t)$ with $N(t)$. In fact, we see that the initial slope in Fig. 13 is infinite. At short times the decay varies as $t^{1/2}$ and is well described by the equation

$$\frac{N(t)}{N_0} = 1 - \frac{S_B}{V_B} \frac{2\phi_\mu}{\alpha_\mu^{1/2} + \phi_\mu} \sqrt{D_0 t}, \quad (42)$$

where N_0 is the initial number of walkers in the macropore volume V_B , S_B is the surface area bounding the macropore, ϕ_μ is the intrinsic porosity of the micropores, and $\alpha_\mu \approx \phi_\mu^{-1/2}$

is the corresponding tortuosity. Equation (42) follows from the approximation that the micropores are describable as a continuum having a diffusivity D_0/α_μ and a porosity ϕ_μ . At short times the curvature of the interface between the macropore and this continuum may be neglected; the solution to the one-dimensional heat equation given by Carslaw and Jaeger [36] may be integrated over the coordinate perpendicular to the interface to give Eq. (42). Equation (42) is reminiscent of a result derived by Mitra *et al.* [37] in a similar but different context.

C. Effect of barriers to coupling

A more striking demonstration of the shortcomings of the two-site exchange model was presented in Figs. 6(b) and 8. There $M(\mathbf{r}, \tau_1)$ and $M(\mathbf{r}, \tau_1 + t_s)$ are not identical and so the off-diagonal peaks are not zero as would be implied by the two-site exchange model. Within the context of Eq. (10) we may write the expression for the off-diagonal element when there is a barrier in place during storage as

$$\begin{aligned} \mathcal{M}_{12}(t_s) = & M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_1][\mathbf{Q}_1 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0] \\ & + M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_2][\mathbf{Q}_2 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0] \\ & + M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_3][\mathbf{Q}_3 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0]e^{-\mu_3 t_s} \\ & + \dots \end{aligned} \quad (43)$$

Here $\mathbf{Q}_1$ represents the storage mode corresponding to uniform magnetization in the macropore. Because $\rho_1 = 0$ its decay rate is zero. Similarly, $\mathbf{Q}_2$ represents uniform magnetization in the micropores, whose decay rate is also zero. All the normal modes relevant to the storage period are determined by solutions to the diffusion equation with Neumann boundary conditions on the pore walls. Further, $\mathbf{Q}_3$ is the mode with the smallest decay rate μ_3 contributing to the relaxation pictured schematically in Fig. 7. There is no counterpart to it in the two-site exchange model. The lowest nontrivial decay rate may be estimated as $\mu_3 = D_0(2\pi/L)^2$, where L is the diagonal distance from one grain surface across the macropore to the other: $L = \sqrt{3}(200) - 2(125) = 96 \mu\text{m}$. This implies a decay time $1/\mu_3 = 120 \text{ ms}$. Thus, for storage times t_s significantly less than this value the magnetization does not have enough time to redistribute and contribute to $\mathcal{M}_{12}$, whereas for longer times it does. This is just the behavior we observe in our results depicted in Figs. 6(b) and 8.

V. CONCLUSION

The T_2 - t_s - T_2 measurements are an important class of 2D NMR experiments and have been used to study a wide range of complex materials. The general features of T_2 - T_2 cross plots can be modeled either by analytic calculations or by random walk simulations. In the limit of weak surface relaxation (often referred to as the fast diffusion limit) off-diagonal peaks in the cross plot are the consequence of diffusive exchange during the storage time t_s . This is the case for materials like cements, where all the pores are in the fast diffusion limit and are well mixed when exchange occurs [9,10]. More generally, for intermediate values of surface relaxation, the off-diagonal peaks are related to both intrapore and interpore diffusion and the interpretation of the measurement is more complicated. This is most likely the case in carbonate rocks (and some sandstones), where macropores are 10–100 μm in diameter, while the micropores are in the 0.1–1.0 μm range.

In order to clarify the physics underlying the T_2 - t_s - T_2 measurement in mixed pore-size systems, we presented here the results of numerical simulations for the μGC model. This model consists of a periodic arrangement of macrograins (and pores) in which each macrograin is itself porous on a length scale 10–100 times smaller than that of the macropores. It was because we used a variable step size algorithm that we were able to simulate accurately processes that occur on vastly different time scales. In order to simplify our analysis, we limited our attention to the case where, during the storage period, the surface relaxivity ρ_1 vanishes. The resulting T_2 - T_2 contour plot, at each value of the storage time t_s , was well described in terms of a simplified 2×2 matrix $\mathcal{M}_{ij}(t_s)$. We compared the results of $\mathcal{M}_{ij}(t_s)$ deduced from our simulations against those predicted directly from an analytic two-site exchange model, for which we independently deduced the values of all the relevant parameters. Although the general features of $\mathcal{M}_{ij}(t_s)$ are captured by the two-site model it does not accurately describe the time over which the matrix elements decay from their initial values to their asymptotic values. Within the context of the two-site model this decay time is simply related to the exchange parameter k . Conversely, if one were to attempt to estimate k from the observed relaxation time constant, one would deduce quite the wrong answer. The issue is that the two-site model attributes relaxation solely to interpore coupling, whereas the relaxation seen during the storage period in our simulations is due to both interpore and intrapore diffusion.

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Two-Dimensional NMR of Diffusion Systems

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We consider diffusion in porous media with well-connected pore space for which isolated-pore models are insufficient. Explicit pore-to-pore exchange parameters were introduced in recent 2D NMR experiments. However, such parameters capture only certain aspects of the interpore spin dynamic which, for single-fluid saturated media, are wholly determined by diffusion. Here, we develop a theoretical approach suitable for a quantitative description of such 2D NMR taking a full account of the underlying diffusion modes. We use simple models of one pore and two coupled pores to demonstrate the rich behavior of 2D NMR.

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Connectivity of the pore space in sedimentary rocks plays a critical role in the flow of crude oils for petroleum extraction. Interconnection between different classes of pores in cements and concretes is critical for controlling its mechanical and chemical properties. A recent development of 2D NMR [1,2] has shown an exciting possibility of characterizing the diffusion of fluid from one pore to another in such systems. This is a significant progress beyond the conventional relaxation measurement (e.g., T_2), often employed to probe the pore geometry, as the latter does not probe the connectivity explicitly. The new experiments use T_1 - T_2 and T_2 - T_2 correlation spectroscopy [3,4] and found that the resulting 2D spectra show significant off-diagonal signals, which are interpreted as evidence of diffusion between different pores. The theoretical understanding of such experiments remains incomplete as the diffusion within pores is not treated on equal footing with the interpore diffusion that is described via a phenomenological hopping parameter. As the interpore coupling strengthens, the division of the pore space into intra- and interpore begins to lose grounds as does any interpretational scheme based on such division. Here we provide a general theoretical framework for understanding such 2D NMR for a porous media, which takes full account of the diffusion-based spin dynamics. We demonstrate that such a framework provides means to clarify and understand aspects of these experiments in a way free from the limited notion of an isolated pore and its variants.

The differential equation governing the evolution of the longitudinal (T_1) or the transverse (T_2) component of nuclear spin magnetization carried by diffusing molecules is the well-known Torrey-Bloch equation [5]:

$$\frac{\partial}{\partial t} m(\mathbf{r}, t) = D \nabla^2 m(\mathbf{r}, t) - \mu m(\mathbf{r}, t), \quad (1)$$

where D is the bulk diffusion constant and μ is the bulk spin relaxation rate. m is the magnetization deviation from its equilibrium. The detected signal is $M(t) = \int m(\mathbf{r}, t) d\mathbf{v}$. Suppressing the bulk relaxation, as it merely adds a constant decay rate, a solution can be found in the general

form:

$$m(\mathbf{r}, t) = \sum_{n=0}^{\infty} a_n \phi_n(\mathbf{r}) e^{-t/T_n}, \quad (2)$$

where ϕ_n and T_n are eigenfunctions and eigenvalues of the Helmholtz equation

$$\phi_n(\mathbf{r})/T_n + D \nabla^2 \phi_n(\mathbf{r}) = 0, \quad (3)$$

with the mixed boundary condition at the pore surface: $D \hat{n} \cdot \nabla \phi_n + \rho \phi_n = 0$ for $\mathbf{r} \in \Sigma_p$ where $\hat{n}$ is the unit vector normal to the pore-matrix interface Σ_p and ρ is the surface relaxivity. The eigenfunctions are normalized by $(1/V_p) \int d\mathbf{v} \phi_n^2 = 1$ where V_p is the pore volume. In the following, all volume integration should be understood to be over V_p , which is taken to be $V_p = 1$. The T_n and ϕ_n can represent the diffusion eigenvalues and eigenfunctions of either T_1 or T_2 processes depending on the experiment and will be denoted by $T_{1,n}$ and $\phi_{1,n}$ or $T_{2,n}$ and $\phi_{2,n}$, respectively. The two eigensystems are generally different as the surface relaxivity ρ assumes different values for the longitudinal (ρ_1) and the transverse (ρ_2) dynamics [6].

Using the completeness and orthogonality of the eigenmodes, the amplitude of each mode in the initial magnetization distribution $m(\mathbf{r}, 0)$ is given by

$$a_n = \int \phi_n^*(\mathbf{r}) m(\mathbf{r}, 0) d\mathbf{v}. \quad (4)$$

Brownstein and Tarr [7] investigated the diffusion eigenmodes for *isolated (closed)* pores of elemental geometries such as sphere and rectangles, and an extensive body of literature exists that has applied their basic techniques to various systems. Ryu has further developed a quantum mechanics analogy that relates diffusion dynamics in an extended pore system to the spectroscopy in a disordered environment [8]. In general, the Helmholtz equation Eq. (3) is of fundamental importance in quantum mechanics, electromagnetics, and acoustics. The inverse problem of determining the geometry of the enclosing space from the measured eigenspectrum is at the heart of numerous

applications, and has stimulated much theoretical [9] and experimental [10,11] research.

We use the diffusion eigensystem to build a theoretical framework for experiments such as the $T_1 - T_2$ correlation shown schematically as

$$\underbrace{\pi - \tau_1}_{T_1 \text{ relaxation}} - \underbrace{\frac{\pi}{2} - t_e/2 - [\pi - t_e]_N}_{T_2 \text{ relaxation}}. \quad (5)$$

The first 180° pulse inverts the nuclear magnetization from its initial distribution initiating the T_1 relaxation. During this period, the magnetization dynamics are described by T_1 modes; thus, the magnetization at the end of the τ_1 period is

$$m(\mathbf{r}, \tau_1) = \sum_{p=0}^{\infty} a_p \phi_{1,p}(\mathbf{r}) e^{-\tau_1/T_{1,p}}, \quad (6)$$

and the coefficient is determined by magnetization right after the π pulse, $m(\mathbf{r}, 0)$. Since the rf tipping angle is uniform on the pore scale, we assume $m(\mathbf{r}, 0) = -1$.

After a period of time τ_1 , a CPMG pulse sequence [12,13] is applied to rotate the magnetization from the axis along the external field to the transverse plane and record the transverse relaxation process. Neglecting the relaxation during rf pulses, the magnetization for the echo at τ_2 time (after the $\pi/2$ pulse)

$$m(\mathbf{r}, \tau_1, \tau_2) = \sum_{q=0}^{\infty} b_q \phi_{2,q}(\mathbf{r}) e^{-\tau_2/T_{2,q}}, \quad (7)$$

where $b_q = \int \phi_{2,q}^*(\mathbf{r}) m(\mathbf{r}, \tau_1) d\mathbf{v}$. The detected signal is an integral of $m(\mathbf{r}, \tau_1, \tau_2)$ with the rf coil sensitivity, which is assumed to be unity. Then the T_1 - T_2 correlation spectrum is obtained through two-dimensional Laplace inversion [14] of experimental data $m(\mathbf{r}, \tau_1, \tau_2)$ with respect to τ_1 and τ_2 and its amplitude in the T_1 - T_2 domain will correspond to

$$S_{12}(T_1, T_2) = \sum_{p,q} \langle f | \phi_{2,q} \rangle \langle \phi_{2,q} | \phi_{1,p} \rangle \langle \phi_{1,p} | i \rangle \times \delta(T_1 - T_{1,p}) \delta(T_2 - T_{2,q}), \quad (8)$$

using the standard bra-ket $\langle | \rangle$ of quantum mechanics for a compact and explicit accounting of the sequential order of operations. Note that $\langle \phi | \psi \rangle = \int d\mathbf{v} \phi^*(\mathbf{r}) \psi(\mathbf{r})$ and $|i\rangle$ and $|f\rangle$ represent the initial and the final distributions, which are both assumed to be uniform here. Experimentally, one may employ arbitrary distributions to obtain the S_{12} spectrum weighted differently.

There are two major features of the spectrum S_{12} . One originates from the modes with $p = q$ called diagonal features and others with $p \neq q$ called off-diagonal features. Since the longitudinal and transverse eigensystems are only different by surface relaxivity, the T_2 and T_1 modes are generally similar, particularly for higher modes.

Thus the diagonal peaks are almost always stronger than the off-diagonal ones, which represent the extent of the *nonorthogonality* of T_2 and T_1 modes. Note also that if $\rho_2 = \rho_1$, no off-diagonal features should be expected in $T_1 - T_2$ experiments. Similarly, if both T_2 and T_1 are in a fast diffusion limit, e.g., $d\rho_{1,2}/D \ll 1$ (d is pore size), T_2 and T_1 modes are essentially the same and no off-diagonal features will be excited. We emphasize that this is the case regardless of the geometry of the pore space. A consequence made clear by this analysis is that, even if the pores are interconnected, for systems with $\rho_1 = \rho_2$, the $T_1 - T_2$ correlation spectrum will show no off-diagonal peaks. And conversely, in a single pore with different ρ_1 and ρ_2 , off-diagonal peaks may be observed.

We now examine these notions with an example of one-dimensional pore of length L . Figure 1 shows its $T_1 - T_2$ spectrum, with $L\rho_1/D = 0.5$, $L\rho_2/D = 5$, and $L^2/D = 1$ s. We define $K_1 \equiv \rho_1/D$ and $K_2 \equiv \rho_2/D$, and they are the inverse of the length scale that characterizes the strength of the surface relaxation. The two peaks indicated by the two arrows are the diagonal signals, due to the two terms $\langle \phi_{1,0} | \phi_{2,0} \rangle$ and $\langle \phi_{1,1} | \phi_{2,1} \rangle$, respectively. Other peaks are off-diagonal signals and the corresponding eigenmodes (p, q) are indicated in the brackets.

When both LK_1 and LK_2 are small, $1/T_{1(2),0} \approx \rho_{1(2)}/L$, $1/T_{1(2),p>0} \approx D(p\pi/L)^2$, the two sets of eigenmodes (T_2 and T_1) are essentially identical and the cross terms become very small. When both LK_1 and LK_2 grow very large, $1/T_{1(2),p} \approx D[(p+1/2)\pi/L]^2$, the two sets of eigenmodes (T_2 and T_1) become progressively more independent of $\rho_{1(2)}$ and the cross terms again become small. When LK_1 and LK_2 are near unity, the eigenfunctions become sensitive to $K_{1(2)}$, and the cross peak gains weight (if $K_1 \neq K_2$), as shown in Fig. 2.

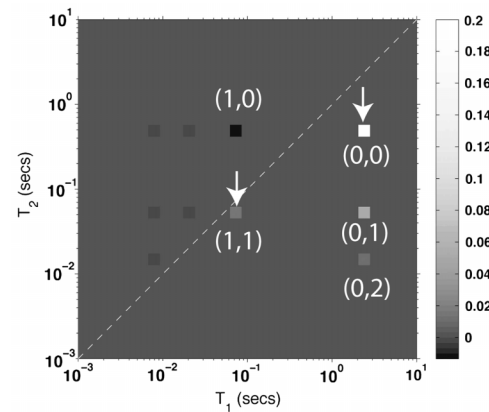


FIG. 1. T_1 - T_2 spectrum for a one-dimensional pore of length L with $L\rho_1/D = 0.5$, $L\rho_2/D = 5$, and $L^2/D = 1$ s. The relevant eigenmodes are indicated for some of the peaks. The dashed line is a guide. Note that the cross peaks above the diagonal ($T_2 > T_1$) show a negative sign owing to the structure of the eigenfunctions.

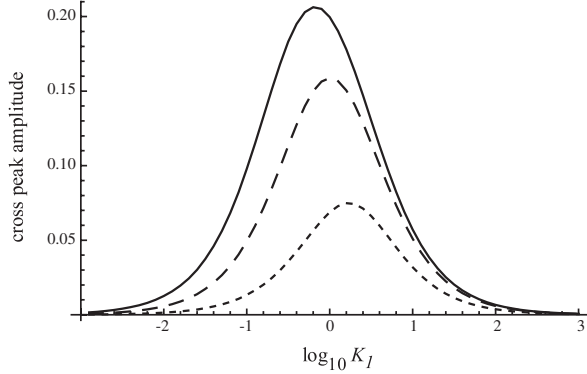


FIG. 2. The overlap integral $\langle \phi_{1,0} | \phi_{2,1} \rangle$ for a range of K_1 and $K_2 = 10K_1$ (solid line), $K_2 = 5K_1$ (dashed line), and $K_2 = 2K_1$ (short-dashed line).

Another type of experiments uses a T_1 relaxation period Δ sandwiched between two T_2 relaxation periods of duration $\tau_2^a = nt_e$ and $\tau_2^b = mt_e$ each:

$$\underbrace{\frac{\pi}{2} - \left[\frac{t_e}{2} - \pi - \frac{t_e}{2} \right]_n}_{T_2 \text{ relaxation } (T_2^a)} - \frac{\pi}{2} - \Delta - \underbrace{\frac{\pi}{2} - \frac{t_e}{2} - [\pi - t_e]_m}_{T_2 \text{ relaxation } (T_2^b)} \quad (9)$$

This sequence rotates the magnetization at the end of the first T_2 period to be parallel to a static field $\mathbf{B}_0$ by the $\pi/2$ pulse at the end of the period. The magnetization is then allowed to relax for time Δ , a T_1 process. Finally, spin echoes are detected by the CPMG train in the second T_2 relaxation period. Within the framework described above, it is straightforward to write down the evolution of the echo amplitudes for this experiment:

$$M(\tau_2^b, \Delta, \tau_2^a) = \sum_{p,q,r} \langle f | \phi_{2,p} \rangle e^{-\tau_2^b/T_{2,p}} \langle \phi_{2,p} | \phi_{1,r} \rangle \times e^{-\Delta/T_{1,r}} \langle \phi_{1,r} | \phi_{2,q} \rangle e^{-\tau_2^a/T_{2,q}} \langle \phi_{2,q} | i \rangle. \quad (10)$$

Thus, Laplace inversion will result in a 2D spectrum:

$$S_{22}(T_2^b, \Delta, T_2^a) = \sum_{p,q,r} \langle \phi_{2,p} | \phi_{1,r} \rangle \langle \phi_{1,r} | \phi_{2,q} \rangle \times e^{-\Delta/T_{1,r}} \langle f | \phi_{2,p} \rangle \langle \phi_{2,q} | i \rangle \times \delta(T_2^a - T_{2,p}) \delta(T_2^b - T_{2,q}). \quad (11)$$

In contrast to the S_{12} , S_{22} now has a user controlled parameter Δ that can be used to manipulate the weight distribution among the modes. For example, when $\Delta = 0$, $\sum_r |\phi_r\rangle e^{-\Delta/T_{1,r}} \langle \phi_r| \rightarrow 1$ and thus

$$S_{22}(T_2^a, 0, T_2^b) = \sum_{p,q} \delta_{p,q} \langle f | \phi_{2,p} \rangle \langle \phi_{2,q} | i \rangle \times \delta(T_2^a - T_{2,p}) \delta(T_2^b - T_{2,q}), \quad (12)$$

where only diagonal signals survive. When Δ is long, the slowest T_1 mode ($r = 0$) dominates the evolution and the

weight distribution will be affected by the nonorthogonality $\langle \phi_{2,p} | \phi_{1,0} \rangle$ between the T_1 and T_2 modes.

An example of a T_2 - T_2 single-pore spectrum is shown in Fig. 3. Notice that all peaks are positive and the off-diagonal peaks show a symmetric pattern as is expected from Eq. (11). Peaks on both sides of the diagonal line corresponding to $(p, 0)$ and $(0, p)$ up to $p = 3$ are visible in the plot. The lower panel of Fig. 3 shows the Δ dependence of the off-diagonal peak $(0, 1)$ showing a rapid increase near $\Delta \sim 0$ and a later decay with the time constant of the lowest mode, $T_{1,0} = 2.34$ s. The initial increase is dominated by the $\phi_{1,1}$ mode with a time constant of $T_{1,1} = 0.092$ s. This example with a single pore clearly shows where the main features of the 2D NMR spectrum originate from. The presence of the off-diagonal peaks is due to nonorthogonality of T_1 and T_2 modes, which arises from $K_1 \neq K_2$.

Next we examine the spectral effects of interpore connectivity. Recently, T_1 - T_2 and T_2 - T_2 experiments have been used as a probe for the coupling between different pores [1,2]. Off-diagonal signals were observed in these experiments and interpreted as the signature of the pore-to-pore diffusion exchange. A heuristic time constant τ_{ex} was defined to describe the diffusion between the pores as hopping or chemical exchange.

Here we numerically compute the eigenmodes of Eq. (1) in a coupled two-pore system [15] and show that, in the

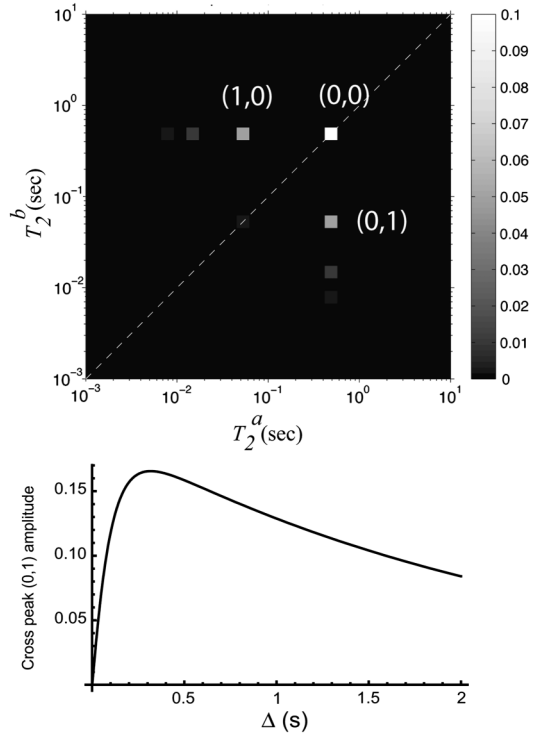


FIG. 3. Top: S_{22} calculated for the 1D-pore model with $LK_1 = 0.5$, $LK_2 = 5$, and $\Delta = 1$ s. Bottom: The amplitude of the off-diagonal peak $(0, 1)$ as a function of Δ .

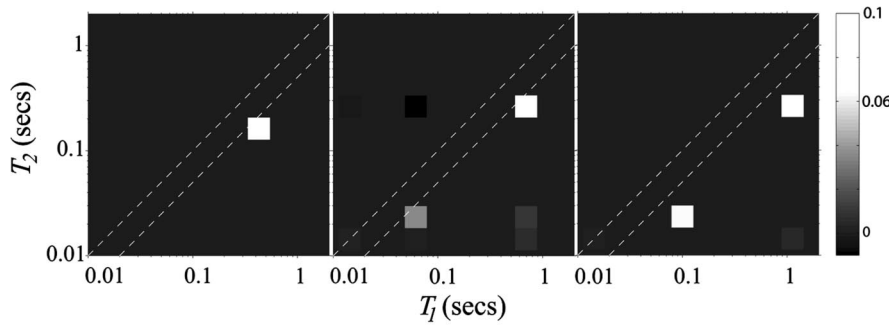


FIG. 4. The 2D NMR T_1 - T_2 spectra of coupled pores for different coupling lengths: $L_C/L_A = 10^{-2}$ (left), 5 (middle), and 10^6 (right). $L_A K_1 = 0.5$, $L_A K_2 = 2$, and $L_A^2/D = 1$ s.

T_1 - T_2 experiment, the off-diagonal signal can appear in specific diffusion regimes and that it is sensitive to the diffusive coupling. In this model, the diffusion flux between the two pores (A and B) is controlled by an effective coupling length L_C and is given by $\Delta m/L_C$ (Δm is the magnetization difference across the interface). The size ratio of the two pores was chosen to be $L_A:L_B = 1:0.121$ [15]. In Fig. 4, three regimes of the interpore coupling are shown: $L_C/L_A = 10^{-2}$, 5, and 10^6 . At $L_C/L_A = 10^{-2}$ (left panel), the two pores are fully coupled and the diffusion dynamics are similar to a single pore of size $L_A + L_B$; thus, the structure of the 2D spectrum is a single peak. For $L_C/L_A = 10^6$ (right panel), the two pores are uncoupled and the diffusion dynamics of the two pores are independent. Thus, the 2D spectrum is simply the superposition of the two pores, primarily the two peaks corresponding to the relaxation modes in each pore, respectively. The peak at $T_1 \sim 1$ s is from pore A and the peak at $T_1 \sim 0.1$ s is from L_B . A very weak off-diagonal peak at $T_1 \sim 1$ s and $T_2 \sim 0.01$ s is present due to the diffusion within pore A similar to Fig. 1; however, it may not be easily seen in the figure given the scale.

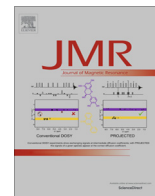
In the intermediate coupling case for L_C/L_A between 1 and 10, the diffusion dynamics are more complex and the spectrum contains more pronounced off-diagonal peaks. As L_C increases, the single peak (left panel) splits into two near the diagonal, one moves to longer T_1 to that of the large pore, and one moves to shorter T_1 toward the small pore. Also, more interestingly, two cross peaks appear at two places: T_1 of L_A and T_2 of L_B (positive), and T_1 of L_B and T_2 of L_A (negative). The amplitude of these peaks grows to a maximum for $L_C/L_A \sim 1$ –10 and then decreases for larger L_C . At $L_C/L_A = 10^6$, the two pores are decoupled and thus cross peaks disappear.

Figure 4 shows that the interpore diffusive coupling can significantly accentuate the off-diagonal peaks that may not be visible in a single pore. However, for the appearance of these “exchange” cross peaks, just as for the case of a single pore, it is necessary to have $K_1 \neq K_2$.

In conclusion, we described a theoretical framework for the spin dynamics in a general pore system such as cements, rocks, food systems, and granular materials and applied it to a single-pore and two coupled-pores systems

in the context of two-dimensional NMR diffusion-relaxation experiments. This framework may accommodate all the relevant diffusion dynamics, such as surface relaxation, membrane permeability, and diffusion between pores. We demonstrated the richness of the diffusion dynamics that could be further explored for real materials with a direct numerical evaluation of eigenmodes [16] as well as random-walk simulations. Such 2D NMR is also important to polymer and colloid sciences [17,18].

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The robust identification of exchange from T_2 – T_2 time-domain features



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ABSTRACT

Two-dimensional spin–spin relaxation (T_2 – T_2) techniques have been developed to probe coupling between different environments such as diffusive coupling between small and large pores or chemical exchange with clays. In these studies, Numerical Laplace Inversion (NLI) is used to obtain two-dimensional T_2 – T_2 relaxation distribution spectrum from the T_2 – T_2 signal decays, and the off-diagonal peaks characterize coupling. Often, these coupling peaks are both weak and close to the diagonal and combined with the inherently ill-conditioned nature of the inversion, their presence is difficult to differentiate from inversion related artifacts and blurring. This manuscript presents a time domain based analysis to identify the presence of coupling that avoids the ambiguities present in T_2 – T_2 spectra. The approach utilizes the symmetric nature of the two-dimensional time domain data, where the presence of curvature along one of these symmetries gives an unambiguous indicator of coupling. Measurements on porous glass beads are used to verify the technique.

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1. Introduction

NMR relaxometry yields information about diffusion and molecular dynamics [1–3], and requires minimal field homogeneity unlike conventional NMR spectroscopy and imaging techniques [4]. Relaxometry thus enables a wide range of applications for NMR in small and portable devices. Characterization of NMR T_2 relaxation is a common core component in many applications due to both its natural measurement in any refocusing pulse sequence and broad utility in porous media analysis [5–13]. T_2 measurements are often used to analyze a wide variety of biological [14], geological [15] and synthetic materials [16–18] to estimate quantities such as the pore size distribution and fluid composition (e.g. oil and water content) [3,8,9,19,20]. T_2 relaxation measurements furthermore obtain these quantities in a safe and non-intrusive way [4] both in and outside of the laboratory. For example, a major commercial application of NMR relaxation measurements is to assess the pore size distributions and fluid content of subsurface rock formation in oil well logging [21,22].

Although one-dimensional T_1 and T_2 relaxation distributions can provide the physical properties of porous media, just as in one-dimensional NMR spectroscopy, they provide limited information to resolve multiple components and correlations between

different NMR properties. Multidimensional relaxometry techniques have often proven to be able to address these limitations. Techniques like diffusion– T_2 (D– T_2) [23–25], diffusion–diffusion (D–D) [26,27] and T_1 – T_2 [28,29] correlation techniques have been developed to better analyze pore geometry [27,30,31], identify exchange [1,12,32], and improve fluid identification [20,33,34]. Similar to these multi-dimensional low-field NMR techniques, T_2 can also be correlated to itself. The (T_2 – T_2) relaxation exchange measurement not only includes the information that 1D T_2 measurement offers, but also gives addition information on molecular dynamics, connectivity, as well as T_1 information [21,35]. It has been used to measure the convective flow in partially saturated soil samples [4], violations of the fast diffusion limit [37] and to study diffusional coupling between different pores or in chemical exchange [12,15,35].

For porous media,² the exchange peaks in the T_2 – T_2 spectra can be roughly thought of as the spins that diffuse between different pores. Technically, these peaks correspond to exchange between T_2 relaxation eigenmodes (which are not necessarily exactly the same as pores) and their mutual overlap with the T_1 relaxation eigenmodes [35,36]. This places further requirements on the relaxation properties to observe exchange, such as differing T_1 and T_2 surface relativities and the requirement that one of the pores is not in the fast diffusion limit [35]. Nonetheless, the exchange information T_2 – T_2 contains is not easily available from other multidimensional NMR relaxometry techniques. For instance, while T_1 – T_2 experiments

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² And assuming the absence of coherent motion (e.g. flow, convection, etc.).

contain the same information as T_2 – T_2 experiments [35], exchange information is far more simply shown by and analyzed from cross peaks in the T_2 – T_2 spectra.

T_2 – T_2 measurements of porous media will often detect signals consisting of multiple T_2 decays that correspond to different sample components. Numeric Laplace Inversion (NLI) is necessary to convert this multi-exponential two-dimensional decay data into a T_2 – T_2 spectrum. Then, a sample with a single T_2 relaxation component such as pure water would yield the same exponential decay along each of the two measurement dimensions and would correspond to a single peak along the diagonal of the T_2 – T_2 spectrum. Similarly, a sample with two distinct T_2 relaxation components should result in a T_2 – T_2 spectrum with two separate peaks on the main diagonal line and a pair of off-diagonal peaks arising from exchange between these two components as illustrated in Fig. 1. Saturated porous media typically have even more complex spectra. In natural samples, for instance, soil [21] or rocks, the distributions of pore sizes is rarely mono-disperse and instead form an essentially continuous distribution of pore sizes. Then, the corresponding diagonal T_2 – T_2 peaks will often overlap with each other and form a long continuous peak that reflects the pore size distribution. Furthermore, even in two component systems, the corresponding two diagonal peaks can easily blur together due to inherently limited resolution of NLI. All NLI algorithms must solve an ill-conditioned inversion, which means that a wide range of solutions exist that will fit the data within the noise. While there are a number of regularization techniques [38–41] that can give stable and reasonable spectra, the ill-posed nature of the inversion intrinsically limits the resolution of different T_2 values and large differences in signal strength, and makes the inversion's performance signal to noise ratio (SNR) dependent.

T_2 – T_2 applications in particular frequently push the resolution limits of NLI, where exchange peaks are often weak and adjacent to intense diagonal peaks. Then the optimization of the reconstruction faces the compromise where use of a conservative regularization parameter (allowing for a larger misfit of the spectra to the acquired data) broadens the diagonal-peak to the point of obscuring the presence of cross-peaks, while weakening regularization risks the introduction of artifact peaks that may be mistaken for legitimate cross peaks as the inversion is forced to fit noise in the spectrum [42]. This compromise often makes it difficult to resolve the presence of coupling in T_2 – T_2 correlation maps without unambiguity and worsens with low SNR data [43]. This limitation is intrinsic to T_2 – T_2 correlation maps, limiting the ability of the technique to characterize exchange in cases of strong coupling between distinct T_2 components. In this paper, we introduce a T_2 – T_2 analysis that can unambiguously identify the presence of exchange by directly analyzing the time domain data. In this way, the limitations of NLI that obscure the presence of cross-peaks are avoided to enable the robust identification of exchange.

2. Theory

The Carr–Purcell–Meiboom–Gill (CPMG) is a common pulse sequence [44–46] to measure T_2 relaxation (see Fig. 2a), and forms the building block for the T_2 – T_2 experiment. The CPMG sequence includes a series of refocusing pulse to minimize the additional signal decay due magnetic field inhomogeneity. It starts with a nominal 90° pulse to excite the magnetization into the transverse plane and is followed by a train of 180° pulses to repeatedly re-focus the signal. The spacing time between adjacent 180° pulse is called echo time, t_e , and corresponds to the spacing between the refocused signal. The timing between the initial 90° and 180° pulses is half the echo time and can be improved by shortening the delay to compensate for spin precession during the excitation pulse [47].

The T_2 – T_2 pulse sequence consists of three parts shown in Fig. 2b and essentially consists of two CPMG sequences joined by a mixing period. During the first encoding period τ_1 , any given spin population in a particular chemical environment will have its signal decay as in a 1D CPMG experiment, and its T_2 labels the initial spin population. Typically, no signal is acquired during this period. The second section is the mixing period, and consists of two 90° pulses to form a stimulated echo over the mixing time (τ_m). During the mixing period, the spin populations undergo T_1 relaxation in order to preserve the magnetization and to allow for mixing between different environments (T_2 components). For example, the spins that belong to a spin population A with a relaxation time T_{2a} during τ_1 can diffuse into spin population B where its transverse relaxation time would be T_{2b} . The final spin populations are measured during the third section of the sequence, the detection period (τ_2), as in a standard CPMG acquisition. To form a two-dimensional experiment, for each measurement the mixing time and acquisition times τ_2 are fixed, while only the value of τ_1 is increased by varying the number of echoes during the first period. This time domain data is collected in a two-dimensional matrix and an example of simulated T_2 – T_2 data is shown in Fig. 2d.

Assuming a static sample at equilibrium, the signal with respect to these T_2 periods can be shown to be symmetric. Following, the derivation of the T_2 – T_2 signal in [48] based on the Torrey–Bloch equation, the dynamics of the magnetization during each of the periods of the experiment when undergoing either T_1 and T_2 relaxation can be broken up into the sum of orthogonal relaxation modes $\phi_n(r)$ of magnetization, each with some characteristic decay time T_n .

$$m(r, t) = \sum_{n=0}^{\infty} a_n \phi_n(r) e^{-\frac{t}{T_n}} \quad (1)$$

These are the eigenfunctions and eigenvalues of the Helmholtz equation.

$$\frac{\phi_n(r)}{T_n} + D \nabla^2 \phi_n(r) = 0 \quad (2)$$

with relaxing boundary conditions $Dn \cdot \nabla \phi_n + \rho \phi_n = 0$, where the magnetization loss is proportional to the surface relaxivity ρ .³ In this framework, these relaxation modes are a function of pore geometry and surface relaxivity as these alter the boundary conditions. The relaxation modes for T_1 and T_2 differ when the pores exhibit different ρ where higher relaxivities force faster magnetization loss near the walls and cause greater curvature in the eigenmodes along with shifting the corresponding relaxation values to shorter T_n . Because the modes describing spin relaxation ϕ_n are orthogonal, these transitions can be computed by projecting the different eigenmodes basis onto each other. Thus, the initial state of the system is the projection of the initial magnetization (e.g. uniform after the excitation pulse) projected onto the basis of T_2 relaxation modes: e.g. $a_n = \int \phi_n^*(r) m_i(r, t_i) dv$ where the modes $\phi_n(r)$ have been normalized. Using bra-ket notation to denote these overlap integrals, the magnetization at the start of each section of the T_2 – T_2 experiment is

$$m(r, t) = \sum_{n=0}^{\infty} \phi_n(r) e^{-\frac{t-t_i}{T_n}} \langle \phi_n | m_i \rangle \quad (3)$$

where m_i is the initial magnetization and t_i is the initial time. For the T_2 – T_2 experiment, the initial excitation creates uniform magnetization in the pores ($|i\rangle$) which first undergoes T_2 relaxation ($T_2, |\phi_{2,a}\rangle$) during the preparation period labeled τ_1 . It is then transferred over to longitudinal magnetization during the mixing period where it undergoes T_1 relaxation ($T_1, |\phi_{1,a}\rangle$), labeled τ_m . Finally,

³ The term for bulk relaxation has been omitted, and can be reincorporated by multiplying by the corresponding exponential decay.

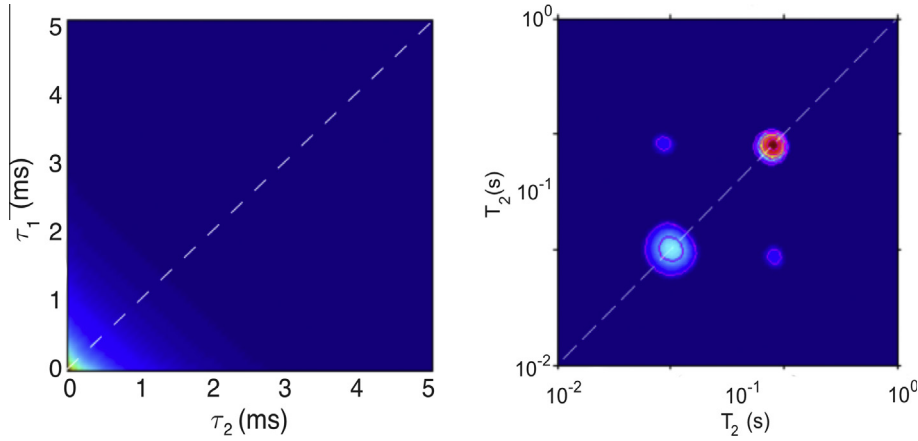


Fig. 1. Simulated T_2 - T_2 time domain data and its relaxation spectrum for two coupled T_2 populations with $T_{2a} = 0.1$ s and $T_{2b} = 1$ s. $t_e = 0.167$ ms.

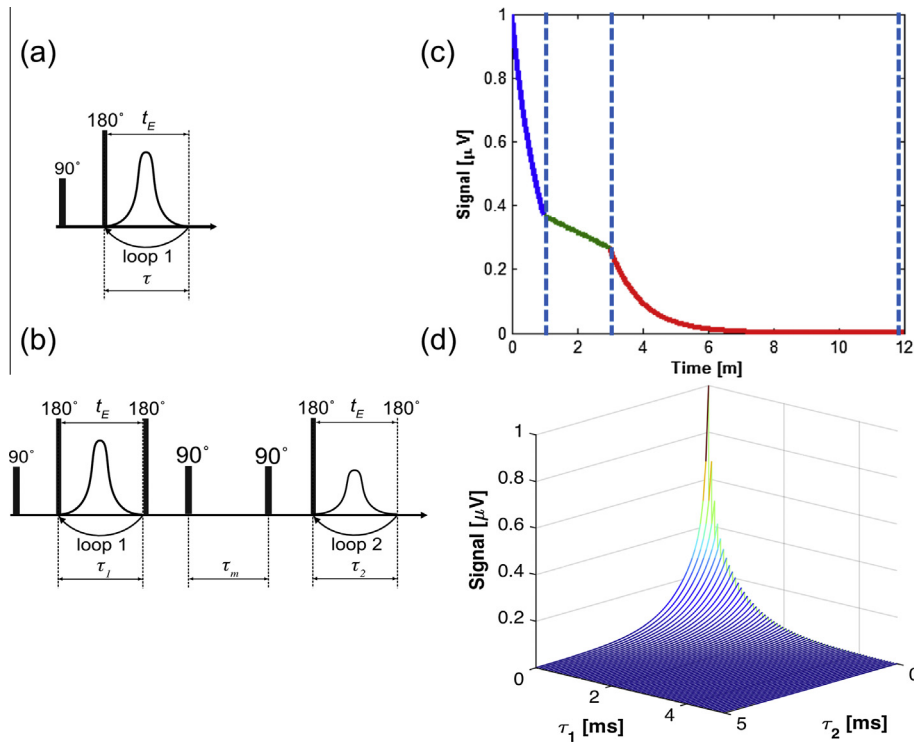


Fig. 2. (a) The CPMG pulse sequence and (b) the T_2 - T_2 pulse sequence. (c) The simulated signal attenuation acquired when collecting the signal for one τ_1 value. The blue line represents the signal undergoes T_2 relaxation in τ_1 period; the green line represents the signal undergoes T_1 relaxation during mixing time period, τ_m ; the red line represents the signal undergoes T_2 relaxation again during τ_2 period where it is detected and aggregated to form a two-dimensional measurement (d) where each line along the τ_2 direction is a single CPMG decay. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

during the detection period it again undergoes T_2 relaxation ($T_2, |\phi_{2,a}\rangle$) where the bulk magnetization ($|f\rangle$) is detected, labeled τ_2 . By repeatedly applying Eq. (3) for each period [43], the detected signal is

$$M(\tau_1^a, \Delta, \tau_2^b) = \sum_{p,q,r} e^{-\frac{\tau_1^b}{T_{2,p}}} e^{-\frac{\tau_1^a}{T_{1,r}}} e^{-\frac{\tau_2^a}{T_{2,p}}} \langle f | \phi_{2,p} \rangle \langle \phi_{2,p} | \phi_{1,r} \rangle \langle \phi_{1,r} | \phi_{2,q} \rangle \langle \phi_{2,q} | i \rangle \quad (4)$$

where the magnetization is the sum over all relaxation modes for each period. As the T_2 modes are identical between the two T_2 periods, $M(\tau_1^b, \Delta, \tau_2^a) = M(\tau_1^a, \Delta, \tau_2^b)$, the signal thus must be symmetric in the time domain data and in the T_2 - T_2 spectra.

The T_2 - T_2 time domain data is simulated/measured in a format of two-dimensional matrix with each dimension corresponding to

the CPMG times in the first and second T_2 encoding periods. This data stored in a two-dimensional matrix $M(\tau_1, \tau_2)$, and is then related to the T_2 - T_2 spectra, according to the exponential kernel applied along the τ_1 and τ_2 directions as shown in Fig. 2d.

$$S_1 = \exp\left(-\frac{\tau_1}{T_{2a}}\right) \quad (5)$$

$$S_2 = \exp\left(-\frac{\tau_2}{T_{2b}}\right) \quad (6)$$

The typical T_2 - T_2 inversion (NLI) of the signal using this kernel does not place the diagonal symmetry constraint on the data, and allows for an asymmetrical T_2 - T_2 spectrum. One way in which this symmetry can be incorporated is by modifying the signal equation to implicitly contain it. The advantage of this approach, over using

a constraint in the inversion, is that this modified signal equation will also reflect these symmetries. The diagonal mirror symmetry of the T_2 - T_2 spectra implies that any peak in spectra (T_{2a}, T_{2b}) must be paired with the same intensity peak mirrored about the diagonal (T_{2b}, T_{2a}) . Thus, the signal equation for the T_2 - T_2 spectrum can be written as the contribution from each pair of cross peaks, instead of as coming from individual values of (T_{2a}, T_{2b}) as shown in Fig. 1. The function for the pair of peaks is:

$$s(\tau_1, \tau_2) = 0.5e^{-\tau_1 R_{2a} - \tau_2 R_{2b}} + 0.5e^{-\tau_1 R_{2b} - \tau_2 R_{2a}} \quad (7)$$

where the relaxation times are represented as rates, $R_{2a} = 1/T_{2a}$, $R_{2b} = 1/T_{2b}$. Note that the on-diagonal peaks are still represented by this model when R_{2a} and R_{2b} are equal. As the principle axes corresponding to the symmetry of the T_2 - T_2 spectrum are along the diagonal $R_s = \frac{1}{2}(R_{2a} + R_{2b})$ and $R_d = \frac{1}{2}(R_{2b} - R_{2a})$, it would also suggest that the symmetry axes of the time domain signal would also correspond to the diagonals $\tau_s = \tau_1 + \tau_2$ and $\tau_d = \tau_2 - \tau_1$, as shown in Fig. 3(a) and (b). Eq. (7) can then be rewritten in these alternative coordinates, for a simplified expression

$$s(\tau_s, \tau_d) = e^{-R_s \tau_s} \times \cosh(\tau_d R_d) \quad (8)$$

This new kernel function is separable into two terms, one is an exponential decay in terms of τ_s and R_s , and the other one is a hyperbolic cosine function in terms of τ_d and R_d . Note that this simple decomposition is in terms of rates and that the corresponding decay time ($\frac{1}{R_{s,d}} \equiv R_{s,d}$) are not the simple sums and differences of the original T_2 times, but of their reciprocals ($\frac{1}{R_{s,d}} = \frac{1}{R_{2b}} \pm \frac{1}{R_{2a}}$).

Because of this separability, the decay along τ_s gives independent information on the mean relaxation rate $R_s = \frac{1}{2}(R_{2a} + R_{2b})$ of the exchange peaks, while variations along τ_d gives information about their off-diagonal separation $R_d = \frac{1}{2}(R_{2b} - R_{2a})$. The func-

tional form of the cosh function along τ_d gives an unambiguous indicator of exchange (cross-peaks) in the time domain data. Independent of R_d and τ_d , the cosh function (and any sum of them given only positive peaks), must increase with positive curvature as τ_d increases. The only exception is in the case that there is no exchange ($R_d = 0$) when the signal contribution is constant with respect to τ_d . Thus, if the acquired signal has curvature along τ_d , there must be legitimate off-diagonal components in the T_2 - T_2 spectrum. To illustrate, simulated data is plotted onto τ_s and τ_d direction (shown in Fig. 3(c) and (d)). In the τ_s slice at $\tau_d = 0$, R_s reflects the total magnetization decay, whereas the curvature along the τ_d cross-section reflects how far the cross peaks are away from each other. In the cross-sections in Fig. 3(c), the deeper curvature of the *cosh* function represents for larger R_d value and flatter curvature corresponding to a small R_d values. If R_d equals to zero, these two cross peaks in spectrum will be on top of each other and consists of a flat line in τ_d cross-sections.

3. Methods

Both simulation data and experimental data are used to verify the signal symmetry and time domain analysis. All the simulation data are calculated using Matlab® with two-site exchange kinetic model [43]. The two site kinetic model, instead of the full model [48], is selected for the simplified calculation of idealized T_2 - T_2 data. Our use of the kinetic model is limited to two sites because its implementation in [43] allows for asymmetry given greater than two sites. This asymmetry is possible in their model for three or more sites because it only assumes detailed mass balance. This looser prior allows for non-equilibria phenomena such as convection and flow that is not typical when analyzing most static porous media samples.

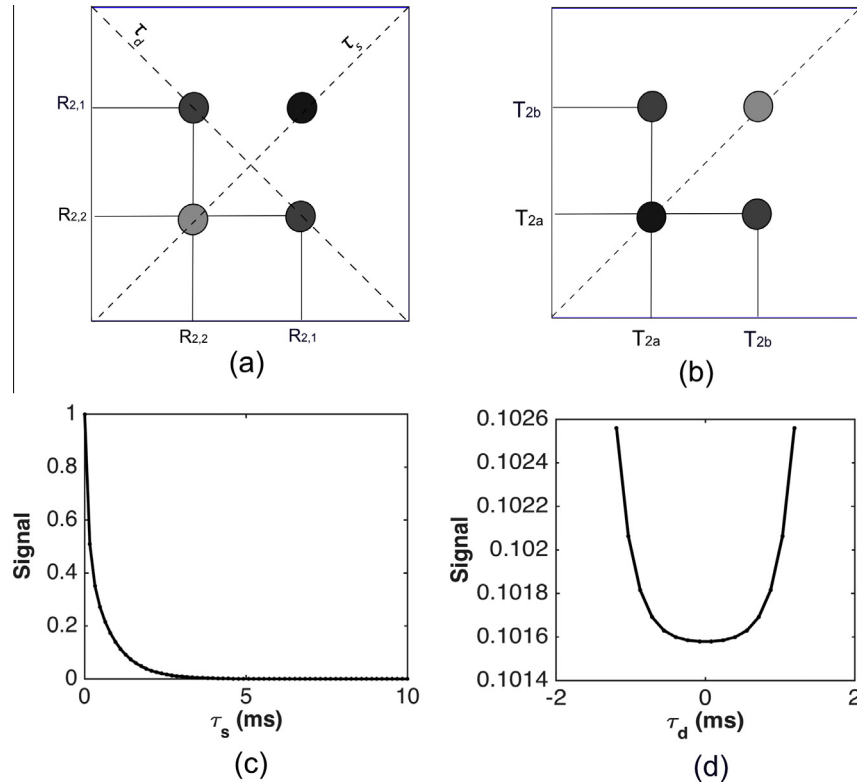


Fig. 3. (a) Schematic T_2 - T_2 spectrum of two coupled spin populations plotted according to the relaxation rates where $R_{2,1} = 1/T_{2a}$ and $R_{2,2} = 1/T_{2b}$. (b) Traditional T_2 - T_2 spectra. (c) Two site exchange simulation data, for the τ_s cross section at $\tau_d = 0$ ms. (d) τ_d cross section at $\tau_s = 1.2698$ ms. Note, the signal axis in (d) has been zoomed in for visualization of the curvature.

For the two site kinetic model, the initial magnetization can be represented as:

$$\mathbf{M} = \begin{bmatrix} M_a \\ M_b \end{bmatrix} \quad (9)$$

where M_a and M_b represent magnetization of the two T_2 populations. The relaxation matrix is:

$$\mathbf{R}_1 = \begin{bmatrix} 1/T_{1a} & 0 \\ 0 & 1/T_{1b} \end{bmatrix} \quad (10)$$

$$\mathbf{R}_2 = \begin{bmatrix} 1/T_{2a} & 0 \\ 0 & 1/T_{2b} \end{bmatrix} \quad (11)$$

And the kinetic matrix represents the exchange rate between spin pool A and spin pool B.

$$\mathbf{K} = \begin{bmatrix} k_a & -k_{ba} \\ -k_{ab} & k_b \end{bmatrix} \quad (12)$$

where k_a is the rate of spins leave and re-enter spin pool T_{2a} and k_{ab} is the exchange rate of spins go from pool A to pool B. The NMR signal is the superposition of all the magnetization components:

$$s(t) = \sum_i [M_i(t) - M_i^{eq}] \quad (13)$$

where M^{eq} is the initial equilibrium magnetization and $M_i(t)$ is the magnetization at time t . The magnetization follows the differential equation:

$$\frac{d}{dt} [\mathbf{M}(t) - \mathbf{M}^{eq}] = -(\mathbf{R} + \mathbf{K})[\mathbf{M}(t) - \mathbf{M}^{eq}] \quad (14)$$

where $\mathbf{R}$ is the relaxation matrix and $\mathbf{K}$ is the kinetic matrix describe the exchange frequencies between different spin pools. The signal adopts T_2 relaxation process, therefore $R_i = 1/T_{2i}$. The simulated time domain data, shown in Figs. 1 and 5, is calculated using Eq. (14) [43].

NMR measurements are performed on a 2 MHz NMR Rock Core Analyzer from Magritek®. For the T_2 - T_2 measurements, τ_2 and τ_1 have the same maximum value, 684.032 ms, with signal being recorded with linear spacing along the τ_2 direction with 4096 echoes and echo time equals to 0.167 ms and signal being recorded with logarithmic spacing along the τ_1 direction for 29 measurements. For each sample, three T_2 - T_2 experiments are done with the mixing times $\tau_m = 0, 5, 50, 100$ ms. For the exchange sample, 150 μm diameter porous glass beads were obtained from Milipore® with 3 μm diameter pores. The pore size and grain size were selected so to produce cross peaks in the T_2 - T_2 map. Two control samples were also used: (1) 150 μm diameter non-porous glass beads and (2) two simultaneously measured glass vials of deionized water with different concentrations of NiCl doping for T_2 relaxation times of 118 ms and 450 ms respectively. All glass beads samples are loose packed, settled by tapping the vial, and vacuum saturated with deionized water.

4. Results and discussion

A set of T_2 - T_2 measurements are performed on porous glass beads, where we expect to observe exchange between the inter- and intra-bead pores. In Fig. 4, the time domain data, T_2 - T_2 spectra and their τ_d slices are compared as the mixing time is changed over 0, 5, 50, and 100 ms. In the T_2 - T_2 spectrum (Fig. 4a), a narrow diagonal line indicates that there is no exchange and is consistent with the fact that a zero mixing time will not have time for exchange to occur. As the mixing time is increased the diagonal peak broadens, this is consistent with increasing exchange but it is also consistent with resolution loss due to the lower SNR at longer mixing times

due to T_1 relaxation and the SNR resolution dependence of NLI. In the two-dimensional time domain data (Fig. 4b), the red contour lines also evolve with mixing time, starting with no variation along τ_d , then increasing in curvature with longer mixing times. The evolution of the curvature in time domain data is clearly observed for the τ_d slice at $\tau_s = 85.04$ ms as shown in Fig. 4c. At zero mixing time, where exchange is impossible, the τ_d line is essentially flat, and at the 100 ms mixing time, the τ_d line curves indicating the presence of exchange.

The advantage of the time domain analysis (Fig. 4b and c) is that the indicator for exchange, curvature, cannot arise from noise. In contrast, the indication of exchange from broadening in the T_2 - T_2 spectra (Fig. 4a) can also occur due to greater regularization in the inversion to compensate for worsening SNR at longer mixing times.

A second feature of the T_2 - T_2 signal decay (Eq. (8)) is that, in the absence of exchange, the decay signal should purely be a function of τ_s , independent of τ_d , and should match the ordinary CPMG decay. A comparison between the first CPMG curve in T_2 - T_2 time domain data (τ_2 curve with $\tau_1 = 0.167$ ms) and the decay along τ_s diagonal ($\tau_d = 0$) are plotted in Fig. 5b. In Fig. 5a, the blue and the red lines, respectively illustrate the locations of these decay curves along τ_2 and τ_s . Along the τ_2 curve $\tau_s \approx \tau_2$ since τ_1 is small (0.167 ms or one echo). As predicted from the model (Eq. (8)), the decay curves show a very good agreement between CPMG decay and τ_2 slice plots (differing by less than the single echo noise amplitude) in the absence of exchange at zero mixing time.

Finally, to confirm that the curvature is indeed due to mixing between T_2 modes and not due to sequence artifacts at non-zero mixing times, the measurements are repeated for two control samples where mixing is impossible. In one control experiment, the sample consists of two separate bottles of NiCl doped water and so exchange between them is impossible. In the second, the sample consists of a non-porous mono-disperse glass-bead pack. Because it has only a single pore size, due to random packing of mono-disperse glass-beads, there are no other T_2 components to exchange with. In Fig. 6, the τ_d cross-sections of the time domain data are plotted and compared to the porous glass beads sample. Consistent with the absence of exchange, there is no significant curvature observed for either sample.

In all of these experiments, the τ_s - τ_d time domain analysis provides a clear, distinct indicator to the presence of exchange peaks in T_2 - T_2 data, and so provides a valuable complement to the analysis of the T_2 - T_2 relaxation spectra. This indicator appears regardless of the number of exchange peaks, but is only sensitive to their presence. By directly analyzing the time domain data, one avoids issues such as peak broadening and artifact peaks from NLI that can easily interfere with the identification of exchange peaks. Thus, the time domain analysis can act as a valuable tool to identify the presence of exchange and guide the subsequent interpretation of the T_2 - T_2 spectra. Application of the time-domain analysis however must also consider that, although the experimental data here agree with the modified kernel function (Eq. (8)), the method relies on the T_2 - T_2 signal having diagonal mirror symmetry and this property presumes a system in equilibrium. Otherwise, signal asymmetry may be possible as shown in recent work that assumes a weaker prior, mass-balance [43]. For characterizing most static samples, equilibrium is nonetheless a fair assumption and can be ensured by appropriate experimental design. The time-domain analysis may only not be potentially applicable in measurements dynamic phenomena such as flow and monitoring chemical reactions, and deviations in T_2 - T_2 signal symmetry can then be used as an indicator for non-equilibrium conditions.

Finally, as presented we only use a simple visual analysis to determine the presence of curvature and identify exchange. Quan-

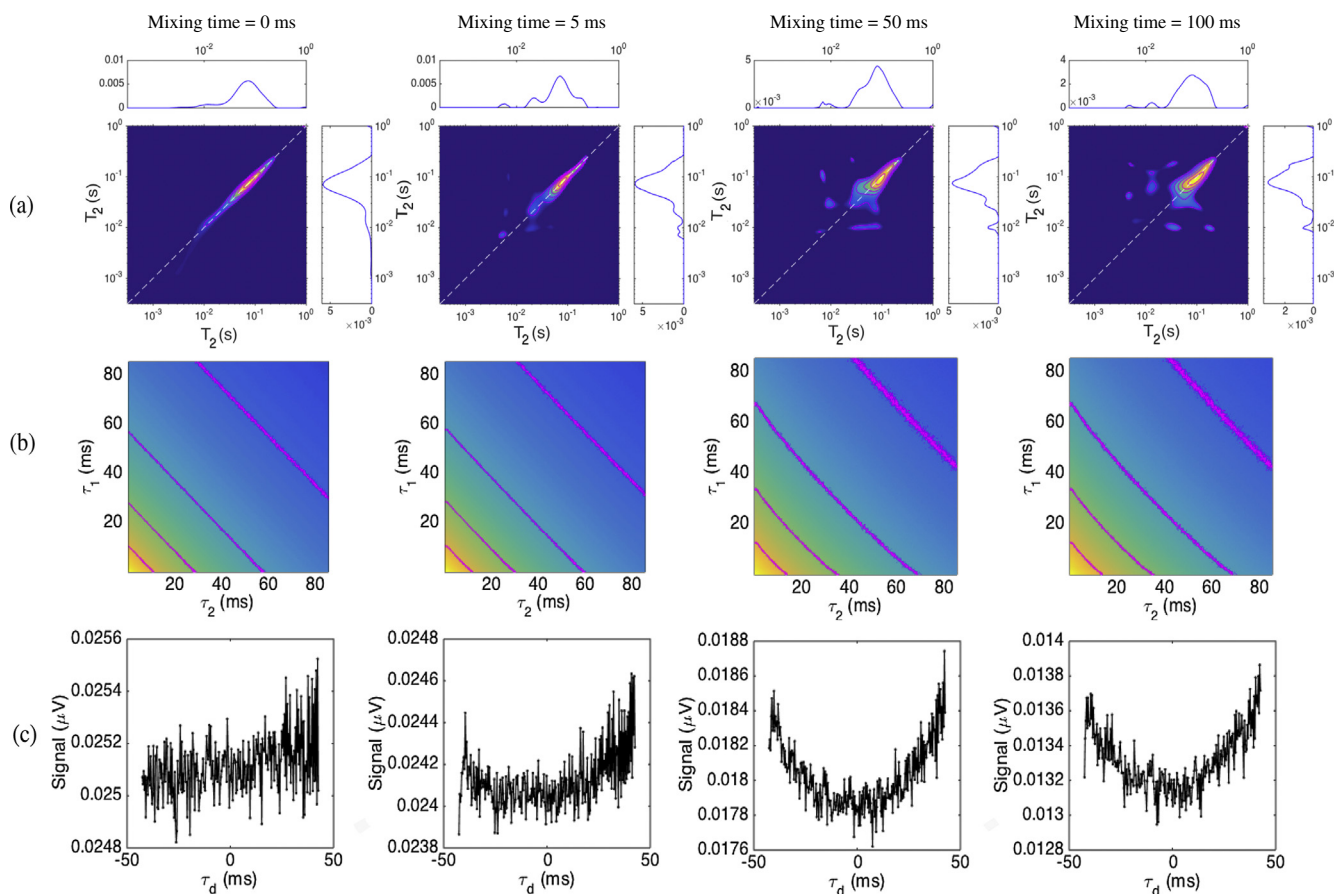


Fig. 4. (a) Numerical Laplace Inversion of T_2 – T_2 time domain data of a porous glass bead pack. All the experimental data presented in the figures above is logarithmic sampled in τ_1 . (b) T_2 – T_2 time domain data of porous glass beads, contour lines are marked red. Note the time domain data is zoomed for early experimental time for better signal intensity. (c) τ_d projection of T_2 – T_2 time domain data with $\tau_s = 85.04$ ms. The signal axes in (c) are zoomed to the signal in order to illustrate the curvature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

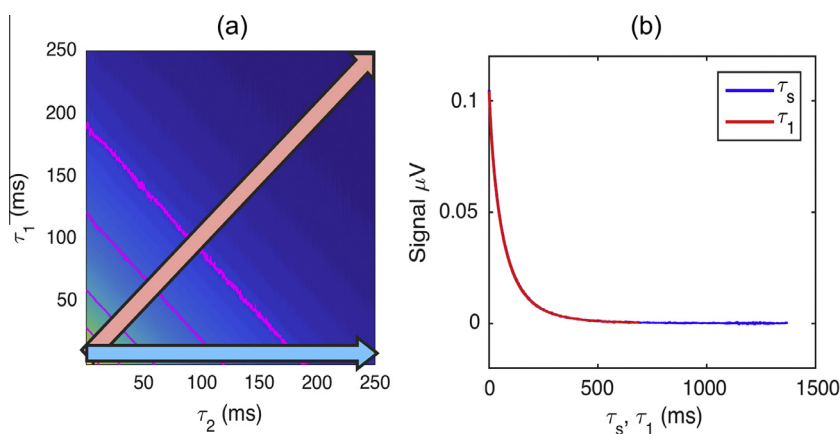


Fig. 5. (a) T_2 – T_2 time domain data of porous glass beads at zero mixing time. The blue arrow indicates the first CPMG curve in the data matrix ($\tau_1 = 0.167$ ms) and red arrow indicates the location of the τ_s signal decay at $\tau_d = 0$. Note: The figure is zoomed in for better signal intensity at early τ_1/τ_2 time. The corresponding signals are plotted in (b). The blue and red curves correspond to the blue and red arrows and the overlap is consistent with the absence of exchange at zero mixing time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

tification of the curvature should lead to further improvements. Applying a statistical test for curvature should greatly increase the analysis' sensitivity to identifying the presence exchange over a visual analysis, while fitting for the curvature may enable the estimation of cross peak position and area. Significant deviations

from a cosh functional form could also potentially flag the presence of multiple cross-peaks. Furthermore, our symmetric decomposition and alternative description of the T_2 decays as T_s – T_d or R_s – R_d may have additional applications when generating T_2 – T_2 spectra. The spectra from the T_s – T_d or R_s – R_d perspective may have

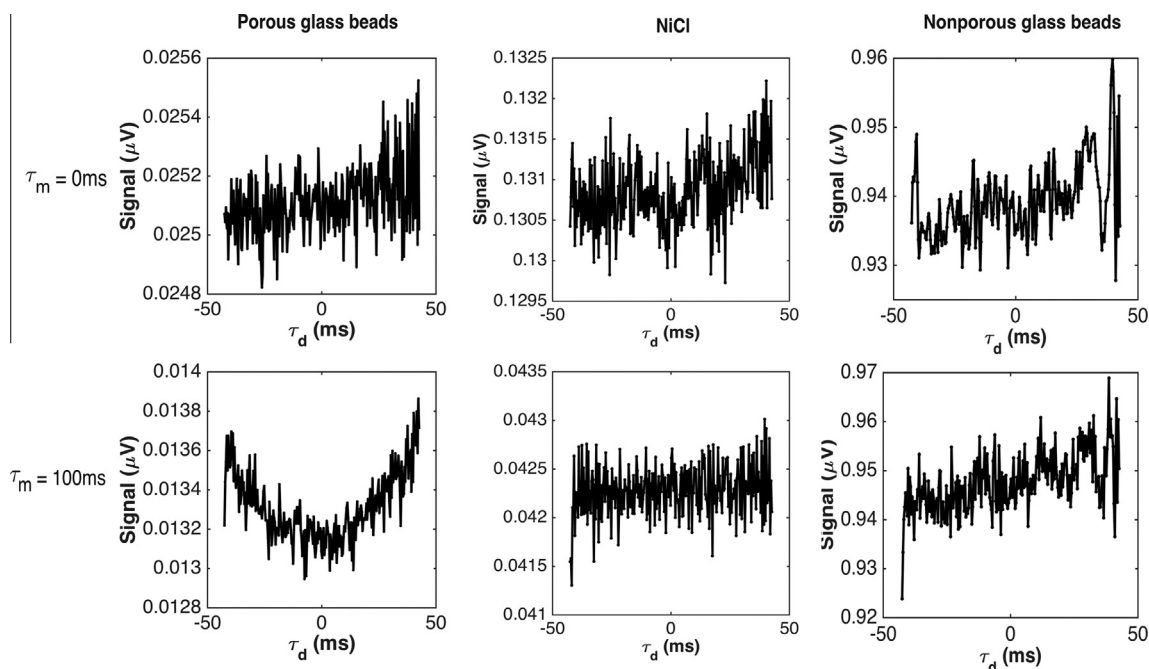


Fig. 6. τ_d projection of time domain data at $\tau_s = 42.75$. Note, all of the signal presented are normalized to the average of first 5 echoes at $\tau_1 = 0$. The signal axes have been zoomed into visualize the curvature.

advantages for interpretation, and its implementation would improve the resolution of the NLI by inherently constraining solutions to incorporate the T_2 – T_2 symmetry.

5. Conclusion

T_2 – T_2 experiments have the valuable capacity to infer exchange, providing information on molecular dynamics, connectivity and diffusional coupling between different pores. However, its traditional analysis requires that the T_2 – T_2 signal decay to first be converted into a two-dimensional T_2 – T_2 relaxation spectrum, an inherently ill-conditioned operation that can frequently hinder the unambiguous identification of exchange peaks. Instead, a direct analysis of the time domain signal can avoid inversion so to directly identify the presence of exchange. Curvature in the signal, when plotted as a function of the difference of the observation and encoding CPMG times ($\tau_d = \tau_2 - \tau_1$), provides an unambiguous indicator of exchange independent of the number of exchanging T_2 populations. This analysis also has the potential to improve quantification of exchange by directly fitting for off-diagonal peak area and position. The method, when applied to porous glass-bead samples, yields a clear indicator of the presence exchange even when the corresponding T_2 – T_2 spectral interpretations are ambiguous and yield only a moderate widening of the on-diagonal non-exchange peak that could also be due variations in the reconstruction parameters. Thus, we expect this time domain analysis to be a valuable tool to inform the T_2 – T_2 spectral analysis, and improve the T_2 – T_2 measurement's sensitivity to exchange.

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Original contribution

Characterization of porous media by T_2 - T_2 correlation beyond fast diffusion limitZhou Yu^a, Yan Zhang^{a,*}, Lizhi Xiao^{a,b}, Guangzhi Liao^a^a State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum, Beijing 102249, China^b Harvard SEAS-CUPB Joint Laboratory on Petroleum Science, Cambridge, MA 02138, USA

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ABSTRACT

Pore size distribution and surface relaxivity are two important properties of porous media such as rock samples and can be obtained by NMR methods. However, it is difficult to obtain these information beyond the fast diffusion limit. Here we present a new method to directly characterize the averaged pore size of a porous sample with a narrow pore size distribution. This method is based on the parallel plates pore model and the T_2 - T_2 correlation sequence. The pore size (a) - surface relaxivity (ρ) correlation maps were obtained using the non-negative least squares method. Three kinds of glass bead samples were measured and the averaged pore size and surface relaxivity were extracted.

1. Introduction

The nuclear magnetic resonance (NMR) method is a powerful tool for studying the structure of porous media and confined fluid. This technique can obtain many important petrophysical parameters such as porosity and pore size distribution, permeability, water saturation, and wettability, etc. [1]. It can be used for core analysis at low field, where there are two key parameters: One parameter is the pore size a , which characterizes the pore volume. Another parameter, the surface relaxivity ρ , reflects the boundary properties of pores and is dependent on the interaction between polarized nuclear spins and paramagnetic impurities at the fluid-solid interface [2]. The fast, intermediate and slow diffusion regime can be defined by pore size, surface relaxivity and diffusion coefficient. Generally, the validity of fast diffusion regime is assumed. Various 1D and 2D Laplace NMR methods were applied to extract pore information under this assumption. For instance, the pore length scale, surface-to-volume ratio (S/V) of porous materials can be obtained by measuring the spin relaxation [3]. Pore connectivity can be acquired by using 2D NMR correlation experiments [4,5]. The appearance of fast 2D inverse Laplace transformation algorithms has greatly accelerated the application of 2D Laplace NMR [6–8]. However, there are also conditions that the intermediate or slow diffusions regimes are applied, such as large pores or strong surface relaxation [3,9,10]. Consequently, the equation based on the fast diffusion assumption cannot be used to obtain pore structure and fluid information in these cases [11]. For example, experimental results of water-

saturated glass-bead packs indicated that the best fit between relaxation time and pore size was different from traditional linear relationships [3]. In 2D method, the off-diagonal peaks of the T_1 - T_2 spectrum with negative amplitude can be used to explore pore coupling [12]. Then, the time-domain data of T_1 - T_2 experiments were analyzed to identify the signature of diffusive coupling between different pores [13]. A similar analysis of the time domain data of T_2 - T_2 experiments were performed, which can unambiguously identify the presence of molecule exchange [14]. Furthermore, asymmetry cross peaks were also observed in T_2 - T_2 correlation maps when there are three T_2 peaks [15].

In this paper, we used T_2 - T_2 correlation sequence to study porous media beyond fast diffusion limit. First, we applied eigenmodes formalism to simulate the T_2 - T_2 signals based on the parallel plates pore model and the non-negative least squares method to obtain a - ρ correlation maps. Subsequently, we performed experiments on the glass bead samples saturated with water and extracted the averaged pore size and surface relaxivity information.

2. Method

2.1. Eigenmodes formalism and diffusion regimes

The evolution of magnetization in the presence of diffusion and relaxation phenomena is described by the Bloch-Torrey equation [16,17]. The solution to this equation without applied gradients was provided by Brownstein and Tarr using eigenmodes formalism in

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planar, cylinder and sphere pores [18]. It was assumed that the ρ of the pores was uniform and satisfied the boundary condition $(\hat{n} \cdot D \nabla m + \rho m)|_S = 0$, where $\hat{n}$ is the normal unit vector, D is self-diffusion coefficient, and $m(\vec{r}, t)$ is the magnetization density in the pore volume V . Combined with the initial condition $m(\vec{r}, 0) = m_0/V$, the general solution of $m(\vec{r}, t)$ can be written as the sum of eigenmodes,

$$m(\vec{r}, t) = \exp(-\mu t) \sum_{n=0}^{\infty} a_n \varphi_n \exp\left(-\frac{t}{\tau_n}\right), \quad (1)$$

where μ is the bulk spin relaxation rate, φ_n and τ_n are the normalized eigenfunctions and eigenvalues. According to the orthogonality of the eigenmodes, the expression of the eigenmode coefficient a_n is

$$a_n = \int m(\vec{r}, 0) \varphi_n^*(\vec{r}) dv. \quad (2)$$

The relaxation of magnetization can be derived by integration,

$$M(t) = M(0) \sum_{n=0}^{\infty} A_n \exp\left(-\frac{t}{\tau_n}\right), \quad (3)$$

where A_n is the relative strength of the different modes. In order to distinguish the multi-exponential relaxation in different regimes, there is a critical parameter $\kappa = \rho a/D$, which is used to define the three diffusion regimes: the fast diffusion regime ($\kappa \ll 1$), the intermediate diffusion regime ($1 < \kappa < 10$), and the slow diffusion regime ($\kappa \gg 10$).

When $\kappa > 1$ is applied, the contribution to the signal from the higher modes increases. Especially in the slow diffusion regime, the magnetization decay is dominated by the lowest mode eigenvalue that is about $a^2/D(\pi/2)^2$, so the signal $M(t)$ is related to a , not to ρ .

2.2. 2D NMR relaxation correlation

Here, we used the T_2 - T_2 correlation sequence proposed by Lee et al. [5] as shown in Fig. 1. This sequence consists of two CPMG sequences, where the first encoding period t_1 and the detection period t_2 , separated by a mixing time τ_{mix} .

Neglecting the relaxation during rf pulses, the evolution of the magnetization is

$$m(\vec{r}, t_1) = \exp(-\mu t_1) \sum_{n=0}^{\infty} a_n \varphi_{2,n} \exp\left(-\frac{t_1}{\tau_{2,n}}\right). \quad (4)$$

After the mixing time, τ_{mix} , where the magnetization relaxes according to the T_1 process, the second train of CPMG pulses in the period of t_2 is used for acquiring data. So the evolution of the CPMG echo signal is

$$m(\vec{r}, t_1, t_2) = \exp(-\mu t_2) \sum_{m=0}^{\infty} b_m \varphi_{2,m} \exp\left(-\frac{t_2}{\tau_{2,m}}\right), \quad (5)$$

where $b_m = \int m(\vec{r}, t_1, \tau_{mix}) \varphi_{2,m}^*(\vec{r}) dv$. Using the bra-ket notation $\langle | \rangle$ of quantum mechanics to define the scalar product as

$$\langle \phi | \psi \rangle = \int \varphi^*(\vec{r}) \psi(\vec{r}) dv, \quad (6)$$

the signal $M(t) = \int m(\vec{r}, t) dv$ can be written as $M(t) = \langle 1 | m(t) \rangle$, where $|1\rangle$ is the scalar field equal to one everywhere within the pore. Note that when $\tau_{mix} = 0$, the detected signal can be written as

$$M(t_1, t_2) = \sum_n \langle 1 | \varphi_{2,n} \rangle^2 \exp(-t_1/\tau_{2,n} - t_2/\tau_{2,n}). \quad (7)$$

To obtain the a - ρ correlation from measured T_2 - T_2 data, the following processing procedure was used. Given a pore size and a surface relaxivity, a 2D matrix $A(t_1, t_2)$ (according to Eq. (7)) can be calculated with the parallel plates pore model. The matrix was spliced into a 1D column in units of each row matrix.

$$\begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \Rightarrow \begin{bmatrix} A_{11} \\ \vdots \\ A_{1n} \\ \vdots \\ A_{mn} \end{bmatrix}. \quad (8)$$

Then, changing the pore size and surface relaxivity, the $A(t_1, t_2)$ were calculated in different diffusion regimes with q values. A 2D inversion coefficient matrix A was obtained by combining each column vector data.

$$A = \begin{bmatrix} A_{11,1} & A_{11,2} & \cdots & A_{11,q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n,1} & A_{1n,2} & \cdots & A_{1n,q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{mn,1} & A_{mn,2} & \cdots & A_{mn,q} \end{bmatrix}. \quad (9)$$

The measured data was converted into a 1D column matrix $b \in \mathbb{R}^{(m \times n) \times 1}$. The least squares fitting problem was thus established as follows.

$$\begin{bmatrix} A_{11,1} & A_{11,2} & \cdots & A_{11,q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n,1} & A_{1n,2} & \cdots & A_{1n,q} \\ \vdots & \vdots & \ddots & \vdots \\ A_{mn,1} & A_{mn,2} & \cdots & A_{mn,q} \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_q \end{bmatrix} = \begin{bmatrix} b_{11} \\ \vdots \\ b_{1n} \\ \vdots \\ b_{mn} \end{bmatrix}. \quad (10)$$

Finally, the a - ρ correlation was obtained from the $[d_1, d_2, \dots, d_q]$.

3. Simulation and experiment

Here parallel plates pore model was chosen to study how to use higher modes to characterize simple porous media beyond fast diffusion limit. First, simulations were done according to Eq. (7). When the mixing time is zero, longitudinal surface relaxivity ρ_1 has no influence on the signals. Therefore the relaxation relaxivity ρ was used to represent the transverse relaxation relaxivity ρ_2 . To investigate the different diffusion regimes, four pore models are given, where a were 50, 100, 250 and 400 μm , and ρ were 50, 50, 100 and 100 $\mu\text{m/s}$. In the T_2 - T_2 correlation sequence, the number of echoes in the first encoding period was logarithmically changed in 30 steps, and the number of echoes in the detection period was 50. In order to evaluate the effect of

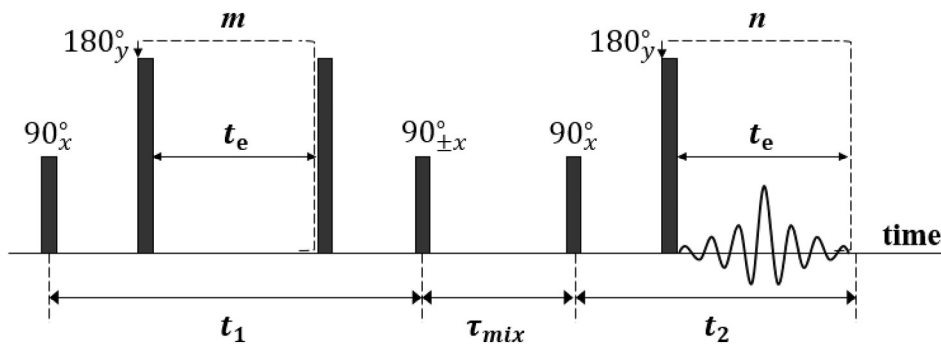


Fig. 1. A schematic of the T_2 - T_2 experiment pulse sequence. This sequence consists of two CPMG echo trains separated by a mixing time, τ_{mix} . The number of π pulses of the first CPMG is varied logarithmically from 1 to m . The two durations of T_2 interval are $t_1 = (m + 1)t_e$ and $t_2 = (n + 1/2)t_e$. In this work, τ_{mix} is set to minimum value (100 μs).

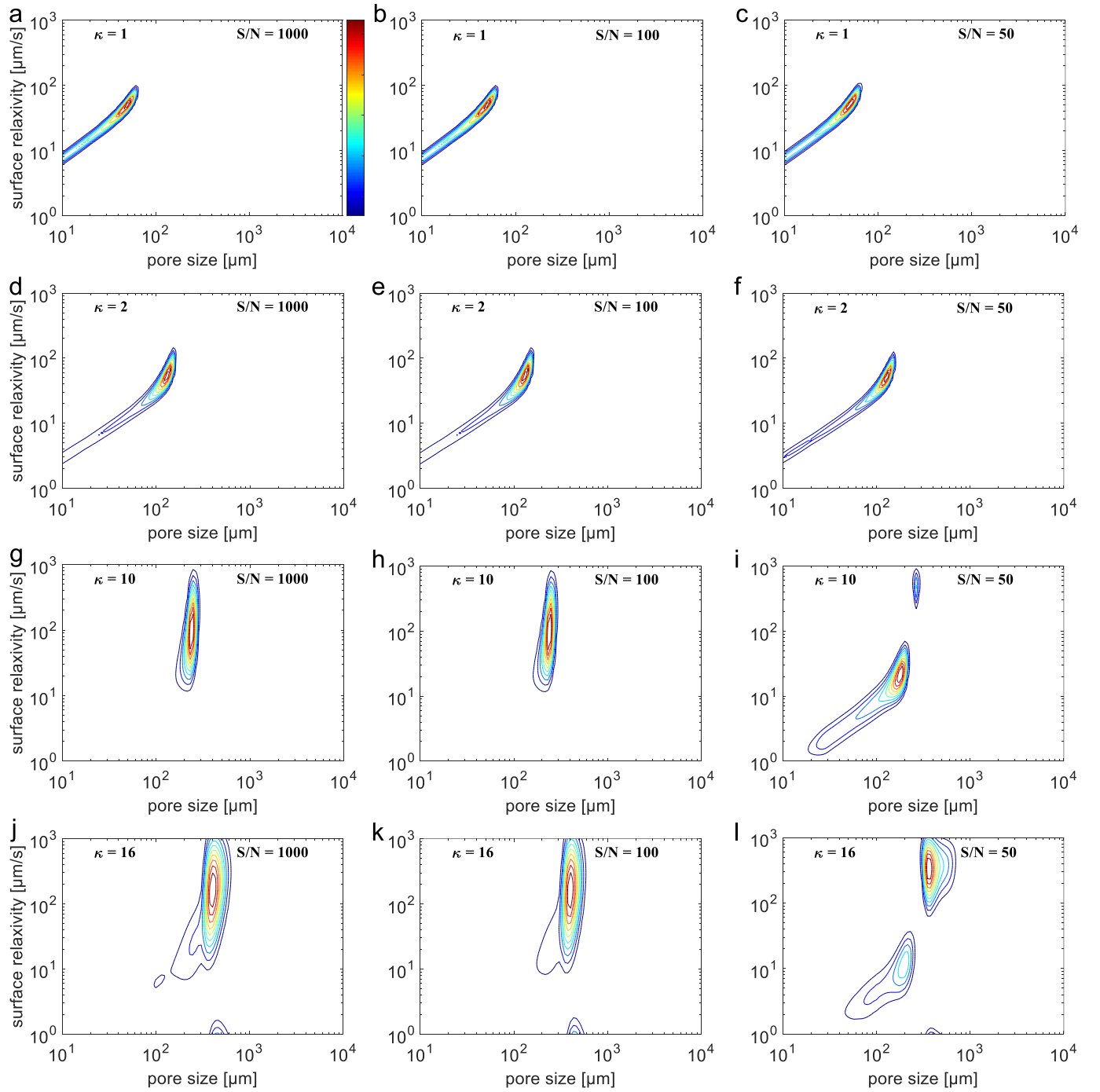


Fig. 2. Two-dimensional α - ρ correlation maps for four models with different S/Ns, which are 1000 (a, d, g, j), 100 (b, e, h, k) and 50 (c, f, i, l), respectively. (a–c) Pore size-surface relaxivity correlation maps for parameter setting $\kappa = 1$, $\rho = 50 \mu\text{m/s}$ and $a = 50 \mu\text{m}$. (d–f) Pore size-surface relaxivity correlation maps for parameter setting $\kappa = 2$, $\rho = 50 \mu\text{m/s}$ and $a = 100 \mu\text{m}$. (g–i) Pore size-surface relaxivity correlation maps for parameter setting $\kappa = 10$, $\rho = 100 \mu\text{m/s}$ and $a = 250 \mu\text{m}$. (j–l) Pore size-surface relaxivity correlation maps for parameter setting $\kappa = 16$, $\rho = 100 \mu\text{m/s}$ and $a = 400 \mu\text{m}$.

noise, three different signal-to-noise ratios (S/Ns) were examined. In all four cases, the self-diffusion coefficient ($D = 2.5 \times 10^{-9} \text{ m}^2/\text{s}$) and bulk relaxation time ($T_B = 3 \text{ s}$) were the same. To inverse the simulated data, the range of the pore size is set from $10 \mu\text{m}$ to $10^4 \mu\text{m}$ and surface relaxivity is set from $1 \mu\text{m/s}$ to $10^3 \mu\text{m/s}$.

In order to verify whether this method can be used in the actual samples, we performed experiments on water-saturated glass beads with a narrow pore size distribution, where the average diameter of the three samples were $100 \mu\text{m}$, $200 \mu\text{m}$ and $500 \mu\text{m}$, respectively. Before the three samples were measured, in order to obtain an appropriate

inversion model, the T_2 relaxation time and diffusion coefficient of the water sample were measured. Then, the T_2 - T_2 correlation experiments were performed using the pulse sequence shown in Fig. 1. The echo numbers in the first encoding period varied logarithmically in 50 steps, and the signals were detected in the second CPMG part. The mixing time τ_{mix} was set to $100 \mu\text{s}$ to minimize the T_1 relaxation effect. The experiments were implemented on a 2 MHz Rock analyzer (Oxford Instrument, UK). The pore size-surface relaxivity correlation maps were obtained by the non-negative least square inversion method.

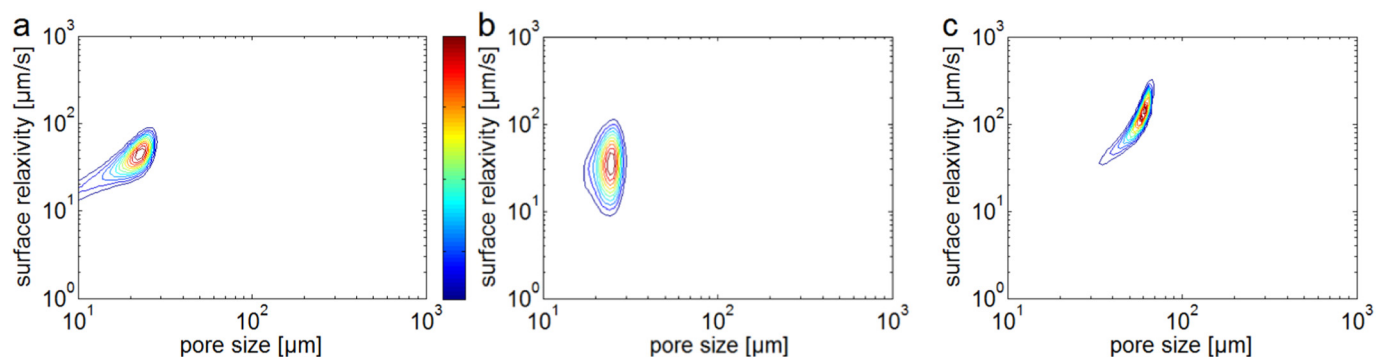


Fig. 3. Pore size-surface relaxivity correlation maps for three samples of glass beads with average diameter of (a) 100 μm , (b) 200 μm and (c) 500 μm .

4. Results and discussion

The simulated results were shown in Fig. 2, where three S/Ns were examined: 1000, 100 and 50. The influence of κ was also explored to examine the validity of this method at different diffusion regime. The averaged pore size and surface relaxivity were obtained from these correlation maps. In Fig. 2a–c, the peaks inversion from the data of $\kappa = 1$ were located at averaged pore size of 30–70 μm and the surface relaxivity 40–60 $\mu\text{m/s}$, which were consistent with the set value ($\rho = 50 \mu\text{m/s}$, $a = 50 \mu\text{m}$), and the results of different S/Ns were basically the same. It was noted that the position of the correlation peaks were along a slope line. This is because when κ small, the lowest mode dominated and only the value of a/ρ can be determined from the data. The correlation maps of the second model were shown in Fig. 2d–f. The parameters were set to $a = 100 \mu\text{m}$, $\rho = 50 \mu\text{m/s}$ and $\kappa = 2$. The peaks were more concentrated and the feature of slope line is weakened. From the position of peaks, the averaged pore size of 90–110 μm and surface relaxivity of 40–80 $\mu\text{m/s}$ were obtained. For the third model with parameter of $\kappa = 10$, $a = 250 \mu\text{m}$, $\rho = 100 \mu\text{m/s}$, the slope characteristic disappeared and the areas of the peak increases (Fig. 2g–i). Although the lowest mode still dominates, the relative intensities of the signals from the higher modes increases, resulting in a weaker correlation between the pore size and surface relaxivity. The peaks position were located at averaged pore size of 210–260 μm and surface relaxivity 50–200 $\mu\text{m/s}$, which were consistent with the set value ($\rho = 100 \mu\text{m/s}$, $a = 250 \mu\text{m}$). For S/N of 50, it was difficult to get accurate information about the averaged pore size and surface relaxivity. For the fourth model, the parameters were set to: $\kappa = 16$, $a = 400 \mu\text{m}$, $\rho = 100 \mu\text{m/s}$. The areas of the correlation maps became larger in Fig. 2j, k, where the peaks position were 350–420 μm , and surface relaxivity were 50–150 $\mu\text{m/s}$. In Fig. 2l, the peak is distorted, where S/N was 50. The distribution was broad for ρ and narrow for a with $\kappa = 10$ and $\kappa = 16$.

Based on the results of the simulation, we found that when κ was in the range of 1 and 10, the information of the averaged pore size and surface relaxivity could be effectively obtained. However, when κ was 16, an approximate estimate of the averaged pore size could be obtained, but the distribution of surface relaxivity was abroad and partially at the margins. When the maximum value of the surface relaxivity was increased to $10^4 \mu\text{m/s}$, the signal amplitude at the margins still existed. Therefore, this method is applicable to porous media with κ equal to 1 to 10 (intermediate diffusion regime).

Fig. 3 shows the experiment results, The self-diffusion coefficient and bulk relaxation time were first measured to inverse the data properly, where $D = 2.1 \times 10^{-9} \text{m}^2/\text{s}$ and $T_B = 2.5 \text{s}$. The averaged pore size of the three samples were estimated from Fig. 3a–c, which are 16–22 μm , 22–27 μm , and 55–65 μm , respectively. The obtained pore size distributions provide right information for the 100 μm , 200 μm and 500 μm diameter glass bead sample according to [19]. The averaged surface relaxivity of the three samples were obtained for the correlation

maps, which are 15–30 $\mu\text{m/s}$, 20–50 $\mu\text{m/s}$, and 100–160 $\mu\text{m/s}$, respectively. The precise value of the surface relaxivity will be determined in the future to check the accuracy of this method of measuring surface relaxivity. Furthermore, the κ values of the samples can also be estimated from the peak center of the correlation maps. For the first two maps, $\kappa < 1$, which means the fast diffusion regime is applied, and for the third map, $1 < \kappa < 10$, which means the intermediate diffusion regime is applied. This offers important information about the diffusion status of molecules.

5. Conclusion

In this work we used two-dimensional T_2 - T_2 correlation pulse sequence to record the relaxation information of porous media beyond fast diffusion limit. Based on the eigenmodes formalism, we simulated the signals of T_2 - T_2 correlation sequence in the parallel plates pore model. Pore size-surface relaxivity correlation maps were then obtained by non-negative least squares inversion. This method is less sensitive to surface relaxivity than the pore size, where a broad distribution was observed in surface relaxivity direction in the correlation maps. We also measured the T_2 - T_2 signals of water-saturated glass samples with a narrow pore size distribution and a uniform surface relaxivity, and directly extract the estimate of the averaged pore size and surface relaxivity. So this method is expected to extract internal pore information at intermediate diffusion regime such as carbonate rock.

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